Modern C: A Concise Introduction to C23

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Preface

C is a compact language, but mastering it is not the same as memorizing a compact set of keywords. The interesting questions begin where a program meets actual storage, input, timing, or another component: Who owns this buffer? How large is it? What happens when conversion overflows? Which operations are portable? What can another thread observe? A useful modern introduction should make those questions part of ordinary programming practice.

This book presents C23 through small programs and progressively more demanding examples. The chapters begin with stream processing and basic expressions, proceed to functions, arrays, structures, memory allocation and compilation, and then develop the correctness concerns that matter in systems software. Later chapters explore C23 additions, binary protocols, concurrency, callbacks, text encodings, and testing.

It is intentionally an **independent original textbook**. Its concise presentation and example-first progression acknowledge the value of classic C pedagogy, but this is not a third edition of *The C Programming Language* and is not affiliated with its authors or publisher. The examples, explanations, and exercises here are newly written.

Audience

The intended reader has some programming experience but need not have used C professionally. A reader comfortable with another language can move quickly through the opening material. A reader new to pointers and manual storage management should slow down, compile every complete example, and work through the boundary-case exercises.

The later chapters are especially relevant to systems and embedded engineers. They discuss host-versus-target portability, fixed-capacity data structures, interrupt-related limitations, and memory-ordering contracts. They do not assume that every MCU provides the full hosted C library, filesystem, heap, or threads.

What modern means

C23 is the 2024 publication of ISO/IEC 9899:2024, often described as the 2023 revision. Modern C encompasses both the updated language and a more explicit engineering discipline: prototypes rather than implicit declarations, documented ownership, checked sizes, controlled resource lifetimes, clean module interfaces, and tests that challenge failure paths.

Some C23 features have uneven implementation support. A compiler can recognize `-std=c23` without supplying every related library header or freestanding runtime facility. This manuscript therefore distinguishes language rules, implementation-defined properties, and deployment-specific services. It uses modern constructs where they clarify intent, but does not pretend one compiler flag can make every old target conform completely.

How to use the examples

Listings fall into two categories. A **complete program** has its own translation unit and entry point and may be compiled and run. An **illustrative fragment** shows a specific language rule or API contract and may omit enclosing declarations, headers, and supporting functions. The standalone programs are kept in the repository's `examples/` directory; the Makefile and test scripts document how they are built.

A useful study cycle is to predict the program's result, compile it with warnings enabled, run it on normal inputs, and then design adversarial boundary cases. For pointer-heavy code, write down the pointed-to object's owner and lifetime. For arithmetic, identify intermediate types and range limits before evaluation. For input processing, state whether a malformed record is rejected, partially accepted, or reported as incomplete.

Tests are evidence rather than proofs of universal correctness. A green hosted CI run is valuable, but cannot replace device-level validation of registers, DMA, interrupts, scheduling, and target libraries. Where a complete listing is not independently compile-tested, the manuscript should not claim otherwise.

Conventions

Code uses C23 spelling such as `bool`, `nullptr`, and empty initialization where appropriate. System-specific code is labeled as such. A pointer and length together describe a counted region only when the caller actually provides storage of that length; no signature can verify an arbitrary dangling pointer. An API may require a writable out-parameter and promise to leave it unchanged on failure. Such contracts are explained when introduced and reused consistently.

Array lengths and capacities are measured in elements unless explicitly described as bytes. Byte-oriented protocols specify octet widths when needed. Integer-overflow checks occur before potentially overflowing arithmetic. Sample code favors clarity and verifiable bounds over clever expression tricks.

An author's checklist for each chapter

A complete treatment should answer four questions. First, what problem does the feature solve? Second, what does the C language actually guarantee? Third, what assumptions does the example make about its caller or platform? Fourth, how would an engineer test the normal path and its boundaries? This revision was expanded with those questions in mind.

The manuscript remains a draft while technical review, compiler verification, layout checking, and indexing continue. The intended final length is roughly 200–230 typeset pages, but page count alone is not a substitute for coherent explanation.

Reading path

Chapters 1–8 establish the everyday language and program structure. Chapters 9–12 connect the modern standard to reliability and a larger example. Chapters 13–19 deepen testing, representation, semantics, concurrency, callbacks, text, and the preprocessor. The appendices are designed to be consulted while writing programs rather than read in one uninterrupted pass.

If a reader remembers one principle, it should be this: in C, **the contract is part of the program**. A short function is not complete until its types, bounds, ownership, and failure behavior are understood.

1. A First Look at C

A C program describes operations on values and storage. It can run with very little runtime support, but it gives the programmer correspondingly little protection from mistakes. We begin with programs small enough to understand without an elaborate vocabulary.

1.1 A complete program

```c
#include <stdio.h>

int main(void)
{
    puts("hello, world");
    return 0;
}
```

The `#include` directive makes declarations from a header available. `main` is the function from which a hosted program begins. The braces contain a block. `puts` writes a string followed by a newline. A return value of zero reports success to the execution environment.

Compile with `cc -std=c23 -Wall -Wextra -Wpedantic hello.c -o hello` and run `./hello`. The `cc` command may select a compiler too old for the C23 flag; a modern GCC or Clang may be named explicitly.

The program has a declaration, an expression statement, and a return statement. There is no implicit declaration of `puts`; its prototype comes from `stdio.h`. Unlike historical C, modern C requires function declarations that accurately describe the parameters.

1.2 Variables and arithmetic

```c
#include <stdio.h>

int main(void)
{
    for (int celsius = 0; celsius <= 100; celsius += 20) {
        double fahrenheit = 9.0 * celsius / 5.0 + 32.0;
        printf("%3d %6.1f\n", celsius, fahrenheit);
    }
}
```

The loop creates, tests, and increments `celsius`. In each iteration, `fahrenheit` exists within the loop body. A value like `9.0` is floating point; integer division such as `9 / 5` would discard the fractional part. The `printf` format is an agreement between the programmer and the arguments: `%d` expects an `int`; `%f` expects a floating-point argument (a `double` after promotion).

Types are not interchangeable merely because two values can both be printed as numbers. The compile-time type controls arithmetic, conversions, available operations, and storage requirements.

1.3 Characters and the input stream

```c
#include <stdio.h>

int main(void)
{
    int ch;
    unsigned long lines = 0;

    while ((ch = getchar()) != EOF) {
        if (ch == '\n')
            ++lines;
    }
    printf("%lu\n", lines);
}
```

`getchar` returns an `int`, not a `char`: it must represent every `unsigned char` value *and* the negative sentinel `EOF`. The parentheses in the loop make the assignment happen before the inequality test. The input is read until end-of-file or an input error; a production program checks `ferror(stdin)` to distinguish the two.

1.4 An array and a simple histogram

Suppose a program counts decimal digits in its input:

```c
#include <stdio.h>

int main(void)
{
    unsigned long count[10] = {0};
    int ch;
    while ((ch = getchar()) != EOF) {
        if (ch >= '0' && ch <= '9')
            ++count[ch - '0'];
    }
    for (int i = 0; i < 10; ++i)
        printf("%d: %lu\n", i, count[i]);
    return ferror(stdin) ? 1 : 0;
}
```

A C array is a contiguous sequence of elements. Indexes begin with zero. The array above has exactly ten positions; a value of ten is already out of bounds. Unlike many modern languages, C does not generally check array bounds at runtime.

1.5 Functions as named operations

```c
static int maximum(int a, int b)
{
    return a > b ? a : b;
}
```

A function packages an operation behind a name and an interface. `static` at file scope gives this function internal linkage: other source files cannot refer to it by this name. The conditional operator selects one of two expressions.

Exercises. 1.1 Change the temperature program to print from 100 down to 0. 1.2 Count tabs and spaces separately. 1.3 Count lines longer than a fixed limit without storing whole lines. 1.4 Explain what happens if the histogram test accidentally admits the character `':'`.

1.6 Counting words: states rather than lines

Counting lines required remembering only a number. Counting words requires remembering whether the previous character belonged to a word. A *state* is just the minimum information needed to interpret the next input.

```c
#include <stdio.h>
#include <stdbool.h>

int main(void)
{
    unsigned long words = 0;
    bool inside = false;
    int ch;

    while ((ch = getchar()) != EOF) {
        bool separator = ch == ' ' || ch == '\n' || ch == '\t';
        if (separator) {
            inside = false;
        } else if (!inside) {
            inside = true;
            ++words;
        }
```

```
    }
    if (ferror(stdin)) {
        fputs("input error\n", stderr);
        return 1;
    }
    printf("%lu\n", words);
}
```

The loop recognizes a transition from *outside* to *inside* a word. Notice that the definition of a word depends on the definition of separator. Here only three ASCII whitespace characters count. Changing to `isspace` requires `<ctype.h>` and care to pass only an `unsigned char` value or `EOF`.

A state-machine description is valuable beyond text processing. An embedded serial receiver similarly recognizes transitions among waiting, payload, and checksum states, and can process one byte at a time without allocating a complete input stream.

Trace a program

For input `"  one   two\n"`, write down `inside` and `words` after each character. Two adjacent separators must not count an empty word. End-of-file also must not count an extra word.

1.7 When the simplest program is not enough

The programs in this chapter fit in small files, but each has an implicit contract. A histogram counter might overflow on a sufficiently large input. An I/O operation may fail before reaching EOF. An unexpected input byte may not be meaningful under an intended encoding. A reliable C program deliberately records which of those conditions it will handle.

Three habits will recur throughout the book:

1. Carry buffer sizes explicitly.
2. Treat resource and I/O failures as ordinary control paths.
3. Use the type system and compiler diagnostics to expose assumptions rather than conceal them.

Further exercises

1.5 Modify the word counter to report separately how many words began with an ASCII digit. **1.6** Implement the line counter using a function `count_lines(FILE *stream)` that reports failure separately from a legitimate zero count. **1.7** Determine when `unsigned long` is wide enough for your intended input volume; document a policy for counter overflow. **1.8** Write a program that collapses runs of spaces to one space while preserving newlines.

1.8 Reading a program from the outside inward

A complete program has two interfaces. The first is its *source interface*: its declarations, definitions, and required headers. The second is its *execution interface*: how input arrives, what output means, and which exit status reports success. Keeping both visible makes the examples easier to reason about.

In a hosted C environment, execution starts at `main`. Either `int main(void)` or the conventional argument-bearing form `int main(int argc, char *argv[])` may be used. Parameters named `argc` and `argv` hold the command-line argument count and vector. The argument strings are accessible through pointers; they are not fixed-sized arrays embedded in the function. A freestanding embedded implementation may use a different entry point and need not provide all hosted I/O facilities.

An output statement has a return value for a reason. `puts` returns `EOF` on output error. In a quick hello-world demonstration we ignore that possibility; in a program that promises reliable output, we check it:

```
#include <stdio.h>

int main(void)
{
    if (puts("hello, world") == EOF)
        return 1;
    return fflush(stdout) == EOF ? 1 : 0;
}
```

This detects errors reported by these stream operations. It does not guarantee that bytes have reached nonvolatile storage.

1.9 Trace the loop, not the syntax alone

When studying an unfamiliar loop, create a three-column trace: the input just consumed, the current state, and the counters. For the word counter earlier in this chapter, consider the bytes `a`, space, space, `b`, newline. The transition on the first `a` increases the count; the two adjacent spaces do not; the `b` increases it once more. Writing this trace explains the algorithm better than memorizing the loop.

Now change the input to the empty stream. The loop body never executes, and the count remains zero. Change it to a stream containing only separators: the state never enters a word. Change it to a final word with no newline: the last word is still counted, because the increment happens at word entry rather than word exit. Such tests expose the program's true specification.

A byte read from `getchar` is always represented by an `int` return type. This permits every possible `unsigned char` value as well as `EOF`. If the value is assigned first to a plain `char`, some valid byte values may be confused with the sentinel on implementations with signed `char`.

1.10 Programs as small transformations

A useful first decomposition separates three questions: What is the input domain? What state must be retained? What result must be produced? The digit histogram's input domain is a byte stream; its state is ten counters; its output is a display of those counters. Its logic makes no assumptions about keeping the whole input in memory.

The same shape reappears in firmware: read a sample or message, update bounded state, and report or act on the result. The difference is not primarily a new language construct but the stronger need to bound time, memory, and error handling. Keeping simple algorithms streaming from the outset makes the later examples easier to adapt.

Further problems: Describe precisely what the word counter considers a separator. Add overflow checks to the histogram counters using `ULONG_MAX`. Change the line counter to distinguish a newline count from a count of human-readable lines that includes a final unterminated line.

1.11 The execution path from source file to program

The small examples hide several steps that are worth separating. A preprocessor handles directives such as `#include`; a compiler translates a translation unit into object code; a linker combines objects and libraries into an executable. On a hosted implementation, the runtime arranges for `main` to execute. None of those stages guarantees that an input file exists or that a terminal will accept output.

Consider a program split between `count.c` and `main.c`. The shared function declaration belongs in a header. If `main.c` is compiled without seeing the declaration, current C does not permit

inventing an implicit function declaration. That old convenience was dangerous: the compiler could assume an incompatible calling convention or return type. A correct C program states its interfaces.

A useful first experiment is to compile with `-c` and inspect the resulting object files without linking them. Then link those objects into an executable. Deliberately omit the object containing one function definition: the unresolved external symbol illustrates the boundary between compilation and linkage.

1.12 A state machine for byte-oriented input

The line and word counters suggest a general pattern. Each input byte causes a state transition, and the program emits results based on the resulting state. The same pattern recognizes a serial protocol whose messages begin with a marker, contain a length, and end with a checksum.

For a toy ASCII protocol, suppose lines are terminated by `'\n'` and may contain at most 16 ordinary bytes. A parser could maintain a buffer, a length, and one status flag. Every ordinary byte is accepted only if capacity remains. A newline terminates a record, and an overlong line is rejected rather than truncated into something that looks valid.

The key engineering step is to write the invariants before the loop: `0 <= length <= capacity`; no bytes are written beyond `buffer[length-1]`; failure has a documented outcome; and after rejecting a long line the parser must either drain to the next newline or stop processing. That final choice is a *resynchronization policy*, not a syntax issue.

Why not read the entire input file first? Streaming input requires bounded storage regardless of total input size, and it can process an indefinitely long stream. Its tradeoff is that the algorithm must remember enough state to handle records split across reads.

1.13 Three kinds of correctness

A counting program can fail in at least three distinct ways.

First, the algorithm can be logically wrong: perhaps two consecutive spaces are counted as a word. Second, it can invoke an operation the language does not define, such as accessing an array at index ten when its valid indexes are zero through nine. Third, it can encounter an ordinary external error: the input stream may fail, or the output may not be writable.

A test for one class does not necessarily reveal the others. Trace the state machine to reason about the first. Make bounds explicit and use compiler diagnostics and sanitizers to help investigate the second. Check I/O results to address the third.

1.14 Choosing types from contracts

For `getchar`, `int` is the correct return type because the function must represent `EOF` separately from ordinary character values. For an array length, `size_t` is the natural type because the standard library uses it to represent object sizes. For a fixed-width protocol field, use a suitable type such as `uint16_t` if the target provides it.

The preferred type depends on meaning rather than an aesthetic preference for unsigned or wide integers. A counter of unbounded input events can still overflow; using `unsigned long` does not make the data unlimited. Choose a maximum supported volume and decide what to do if it is exceeded.

Guided exercises

1.9 Write a table of the input states for the word counter: separator, word byte, and EOF. For each, state which variables change. **1.10** Propose a record parser with a 16-byte buffer and show the sequence of state changes for a 17-byte record. **1.11** Modify the histogram to report an error

before incrementing a counter already at `ULONG_MAX`. **1.12** Build a two-translation-unit program and explain exactly which step fails when a required object file is omitted.

2. Types, Expressions, and Control

The most important questions about an expression are not only what value it computes but which type it uses and whether every intermediate operation is defined.

2.1 Integer types and ranges

C provides `char`, `short`, `int`, `long`, and `long long`, each with signed and unsigned variants where applicable. The exact widths are implementation-dependent within standard constraints. Use `<stdint.h>` when a protocol requires an exact width such as `uint32_t`; these exact-width typedefs exist only when the implementation provides suitable types. `size_t` from `<stddef.h>` describes object sizes and array indexes.

```
#include <stdint.h>
#include <inttypes.h>
#include <stdio.h>
int main(void)
{
    uint32_t mask = UINT32_C(1) << 7;
    printf("%08" PRIx32 "\n", mask);
}
```

The formatting macros such as `PRIx32` are declared in `<inttypes.h>`, which is included above. Prefer unsigned integers for bit masks and modular arithmetic, not as a generic cure for negative values. Unsigned overflow wraps modulo a power of two; signed overflow has undefined behavior.

2.2 Conversion is an operation

```
int a = 7;
int b = 2;
double x = a / b;         /* 3.0, division occurred as int */
double y = (double)a / b; /* 3.5 */
```

Conversions occur at assignments, calls, returns, and within arithmetic. Integer promotions may change a narrow type to `int`. Mixed signed and unsigned arithmetic deserves particular suspicion: compare values only after choosing an appropriate common domain.

2.3 Conditions and loops

The forms `if`, `else`, `switch`, `while`, `do`, and `for` govern control flow. A controlling scalar expression is false if it compares equal to zero, true otherwise. Use braces even for short branches in code that will evolve.

```
unsigned gcd(unsigned a, unsigned b)
{
    while (b != 0) {
        unsigned remainder = a % b;
        a = b;
        b = remainder;
    }
    return a;
}
```

Each iteration reduces the second operand unless it is already zero. The termination argument is as useful as the program itself: a loop should have a quantity that progresses toward a bound.

2.4 Short circuit and sequencing

`&&` and `||` evaluate left to right and skip the right operand when its value cannot affect the result. Thus `p != nullptr && *p == 5` can test a pointer before dereferencing it, though `nullptr` requires C23 support. Avoid compact expressions that modify the same scalar multiple times without proper sequencing: `i = i++ + 1` is not a clever increment; it has undefined behavior.

2.5 Floating point

Floating-point values are approximations. A decimal fraction like 0.1 often cannot be represented exactly in binary. Equality comparisons can be appropriate for exact assignments and sentinels; they are often unsuitable as tolerance tests for numerical computations. Relative and absolute tolerances should be chosen from the problem's scale, not copied as universal constants.

2.6 C23 spelling and initialization

C23 recognizes `bool`, `true`, and `false` as keywords, and permits empty initialization:

```
struct point { int x, y; };
struct point origin = {};
bool ready = false;
```

An empty initializer gives the object an empty-initialization value (for this structure, both fields become zero). Check compiler support on older toolchains.

Exercises. 2.1 Implement absolute difference for unsigned integers without underflow. 2.2 Write `gcd` tests including zeros. 2.3 Explain why `int n = sizeof array;` can narrow an object size. 2.4 Determine which operations in a signed integer sum can overflow.

2.7 Integer conversion in practice

Consider `size_t`, an unsigned type used for object sizes. It is tempting to subtract two lengths and then test whether the result is negative:

```
size_t remaining = capacity - used;
```

If `used > capacity`, this subtraction does not produce a negative number; unsigned arithmetic wraps. Instead, establish the precondition before computing the difference:

```
if (used > capacity)
    return false;
size_t remaining = capacity - used;
```

A second trap is evaluating mixed integer types. A comparison involving an unsigned operand may convert a signed value to an unsigned type. It is better to check a negative value before a conversion than to hope the usual arithmetic conversions preserve its mathematical meaning.

2.8 Bitwise operations

Bit masks encode independent boolean properties in one integer. Shift counts must be valid for the promoted left operand's width. For portable protocol data, perform shifts in an explicitly selected unsigned type.

```
#include <stdint.h>

enum {
```

```
    FRAME_READY = 1u << 0,
    FRAME_ERROR = 1u << 1,
    FRAME_RETRY = 1u << 2
};

unsigned flags = FRAME_READY | FRAME_RETRY;
bool retry = (flags & FRAME_RETRY) != 0;
flags &= ~FRAME_RETRY;
```

The `|` operator combines selected bits, `&` tests or retains bits, and `~` complements all bits in the promoted type. Using a mask is not the same as testing a boolean expression with `||` or `&&`; those logical operators produce zero or one and may short-circuit.

An important embedded distinction: masking an integer held in RAM is straightforward; read-modify-write of a device register may trigger hardware-specific behavior. Consult the peripheral's reference manual before applying an apparently innocent `register |= bit`.

2.9 Evaluation order is independent of precedence

Multiplication binds more tightly than addition, but precedence tells the compiler how expressions are grouped, not the order of their operands' evaluation. For example, `f() + g()` does not impose an ordering between calls. The programmer can create a sequence explicitly:

```
int left = f();
int right = g();
int result = left + right;
```

Now the calls happen in the order of the statements, though addition still requires a representable result. Avoid constructions that both modify and read a scalar without the necessary sequencing. Compact code is not an excuse for surprising evaluation.

2.10 A checked average

For integers in a constrained environment, accumulating into a wider type can help, but a wider type is not a proof of no overflow. Even a 64-bit sum can overflow if the input is long enough. A fully safe routine must either check each addition, constrain the input count and element ranges, or use a documented alternative algorithm.

A good interface for an average also distinguishes the empty-input case from a real average of zero. A return value alone does not always provide enough information.

Further exercises

2.5 Write a function that sets and clears one bit of a `uint32_t` given a bit position. Reject positions outside 0–31. **2.6** Show an example in which adding two valid `int` values overflows, and implement a guard that avoids evaluating the addition. **2.7** Explain the difference between `a & b == 0` and `(a & b) == 0`. **2.8** Design a loop over `unsigned` values that handles `UINT_MAX` without wrapping forever.

2.11 Representable values before casts

A cast requests a conversion; it does not assert that the mathematical value fits. If an external protocol provides a `uint32_t` length and a function expects an `int` length, test the range *before* conversion. A cast performed first may already have discarded information.

```
#include <stdint.h>
#include <limits.h>
#include <stdbool.h>
```

```
bool length_to_int(uint32_t length, int *out)
{
    if (out == nullptr || (uintmax_t)length > (uintmax_t)INT_MAX)
        return false;
    *out = (int)length;
    return true;
}
```

Here `uintmax_t` gives a common unsigned domain capable of representing both operands; the positive limit `INT_MAX` is safe to convert. The routine assigns its output only on success. This pattern works for sizes, array indexes, and serialized integer fields.

C23 requires two's-complement representation for signed integers, but that does **not** make signed overflow defined. `INT_MAX + 1` is not a portable wraparound operation. The representation model and the rules for evaluating expressions are separate questions.

2.12 Division, remainder, and a dangerous minimum

For integer division, the fractional part is discarded toward zero. The remainder satisfies `(a / b) * b + a % b == a` when the quotient is representable and `b` is nonzero. The expression `INT_MIN / -1` can overflow the signed result on a typical two's-complement `int` and has undefined behavior. Division by zero is undefined for integer arithmetic.

An absolute-value function with signed output cannot represent the absolute value of `INT_MIN` on the usual two's-complement implementation. Instead of writing `abs(INT_MIN)` and assuming success, choose a wider type where available or provide a checked interface.

2.13 Floating-point accuracy is a contract

Floating-point arithmetic has finite precision and a finite exponent range. A result can round differently from the exact real-number result; adding a very small number to a very large one may change nothing at all. Reassociating sums can alter their results:

```
double a = 1.0e20;
double b = -1.0e20;
double c = 3.0;
/* (a+b)+c and a+(b+c) need not be equal. */
```

A numerical algorithm should state its permitted error, expected input scale, handling of infinities and NaNs, and acceptable loss of significance. The rules can also depend on whether the target implements IEC 60559 floating-point facilities. In embedded computation, fixed-point is sometimes chosen not because floating point is categorically wrong, but because reproducibility, speed, or resource bounds favor an integer representation with an explicit scale.

2.14 A precise conversion exercise

Write a checked function accepting a signed `long` and returning a `uint16_t` if and only if the input is between zero and `UINT16_MAX`. The correct order is: reject negative input; compare the nonnegative input in an appropriate common type; then perform the cast. Test `-1`, zero, one, the maximum, and one beyond the maximum wherever that value is representable.

Be careful with `sizeof`: its result is `size_t`, not `int`. Printing it with `%d` is an argument-type mismatch. Use `%zu` for `size_t`. Comparing object sizes and subtracting indexes also requires careful thought: an unsigned subtraction cannot express a negative distance unless the design explicitly uses a signed difference type such as `ptrdiff_t` and checks its range.

Further problems: Derive a safe check for `a * b` with two positive `size_t` values. Show the difference between `(unsigned)-1` and a signed negative one. Explain why shifting a negative signed value is a poor choice for portable bit-manipulation code.

2.15 Promotions are part of the computation

Small integer types are often promoted before arithmetic. Suppose `uint8_t a = 200, b = 100;`. On a common target where `int` represents every value of `uint8_t`, the expression `a + b` is evaluated in `int`, producing 300. Assigning that result back to `uint8_t` then performs a conversion that produces a value modulo 256. These are two different stages.

Do not infer the expression's type solely from the declared type of its operands. When shifts or bitwise complement are involved, promotions can be particularly surprising. A `~` applied to a narrow unsigned value can first promote it to `int`, yielding a value with more significant bits than the programmer expected.

If a protocol needs an eight-bit mask, perform the operation in a known unsigned domain and intentionally narrow the result only after masking:

```
#include <stdint.h>

uint8_t low_byte(uint32_t value)
{
    return (uint8_t)(value & UINT32_C(0xff));
}
```

The mask states exactly which bits belong in the result.

2.16 Comparisons between signed and unsigned values

Consider `int delta = -1;` and `size_t count = 4;`. The expression `delta < count` can convert the negative `int` to an unsigned type, depending on the usual arithmetic conversions. On a typical 64-bit platform, it evaluates in a way a casual mathematical reading does not predict.

A reliable design checks negativity first:

```
bool valid_index(int index, size_t count)
{
    return index >= 0 && (size_t)index < count;
}
```

The `&&` operation evaluates the cast and comparison only after establishing that the signed input is nonnegative. The cast is then value-preserving whenever `size_t` has sufficient range for the nonnegative `int`; implementations with unusual widths deserve a more general conversion policy.

Prefer to keep array positions as `size_t` from the beginning when negative indexes are not meaningful. But do not convert a potentially negative external value to `size_t` before validating it.

2.17 Division edge cases

Integer division by zero is undefined. The signed quotient may also be unrepresentable: on ordinary two's-complement systems, `INT_MIN / -1` is the familiar example. A library routine parsing a user-supplied divisor should test both conditions before evaluating the operation.

```
#include <stdbool.h>
#include <limits.h>

bool divide_int(int a, int b, int *out)
{
    if (out == nullptr || b == 0 ||
```

```
        (a == INT_MIN && b == -1))
        return false;
    *out = a / b;
    return true;
}
```

The output is unchanged on failure. If the function returns a quotient and remainder, compute them only after the same validation. C integer division truncates toward zero; the remainder has the sign rules implied by that division, not the always-nonnegative modulo convention used in some mathematical texts.

2.18 Floating-point comparisons require a scale

A universal test such as `fabs(a - b) < 1e-6` is usually not a universal numeric contract. It may reject two large quantities whose relative difference is tiny, and it may accept two tiny values whose relative difference is enormous.

A numerical application should choose absolute tolerance, relative tolerance, or a combination according to the units and problem:

```
#include <math.h>
#include <stdbool.h>

bool approximately_equal(double a, double b,
                         double absolute, double relative)
{
    if (!isfinite(a) || !isfinite(b) ||
        absolute < 0.0 || relative < 0.0)
        return false;
    double difference = fabs(a - b);
    double scale = fmax(fabs(a), fabs(b));
    return difference <= absolute || difference <= relative * scale;
}
```

This is a teaching fragment with an explicit finite-input policy; even here, intermediate overflow can make `a - b` or `relative * scale` nonfinite for extreme values. That may be acceptable for the application, or it may require a more careful algorithm. A numerical routine is correct relative to a chosen error model, not simply because it uses `double`.

2.19 Derive overflow checks rather than guessing

For two nonnegative `size_t` operands, `a * b` fits when either operand is zero or `a <= SIZE_MAX / b`. Notice why the division is performed only when `b != 0`. The test avoids executing multiplication until representability has been established.

```
#include <stdbool.h>
#include <stddef.h>
#include <stdint.h>

bool multiply_size(size_t a, size_t b, size_t *out)
{
    if (out == nullptr || (b != 0 && a > SIZE_MAX / b))
        return false;
    *out = a * b;
    return true;
}
```

When zero-sized allocations are undesirable, a caller can separately reject zero operands. Arithmetic validity and resource-policy validity are different matters.

Worked questions

What is the type of `sizeof buffer`? Why can a byte count greater than `INT_MAX` not safely be passed as an `int` length without a range check? Why can `INT_MIN` not always be negated? What does `a & b == 0` mean without additional parentheses? For each, answer by applying the language rules rather than by running one compiler and memorizing its output.

2.20 Parsing a signed integer without invoking signed overflow

A decimal parser for a signed type is trickier than one for an unsigned type. The problem is the negative endpoint. On a two's-complement implementation, the negative range contains one more magnitude than the positive range. For a typical 32-bit `long`, `LONG_MIN` is −2,147,483,648 while `LONG_MAX` is 2,147,483,647.

An implementation that accumulates a positive signed magnitude and negates it at the end cannot represent the magnitude of `LONG_MIN` during parsing. Its valid final answer becomes impossible to compute through that intermediate value.

One solution is to accumulate the magnitude in an unsigned type. Parse an optional sign first, then choose the maximum allowed magnitude:

```
const unsigned long limit =
    negative ? (unsigned long)LONG_MAX + 1UL
             : (unsigned long)LONG_MAX;
```

C23's two's-complement requirement makes the negative endpoint's magnitude one larger than the positive maximum. The unsigned type corresponding to `long` can represent that magnitude.

For each decimal digit `d`, the next mathematical magnitude is `10 * magnitude + d`. Check it *before* performing the arithmetic:

```
if (magnitude > (limit - digit) / 10UL)
    return false;
magnitude = magnitude * 10UL + digit;
```

Here every digit is in the range zero through nine, and the limit is far larger than nine for standard `long` ranges. The subtraction and division therefore have valid operands.

At the end, if the input was negative and the magnitude equals `LONG_MAX + 1`, assign `LONG_MIN` directly. For smaller negative magnitudes, cast the unsigned value into a representable signed value and then negate. Never cast the unrepresentable endpoint magnitude into `long` and then hope negation repairs it.

The program `examples/parse_signed.c` implements this design and tests ordinary values, signs, leading zeros, `LONG_MIN`, `LONG_MAX`, malformed strings, and overflow.

2.21 Grammar is independent from arithmetic

The same mathematical value can have multiple textual forms. Is `+00042` valid? Is `42 ` valid? May there be leading whitespace? Are digit separators allowed? Must the parser consume the entire input?

The `parse_signed` example accepts an optional leading plus or minus sign followed by at least one ASCII decimal digit. It rejects spaces before or after the number and rejects all trailing nondigits. The standard function `strtol` intentionally accepts a broader grammar, including leading whitespace, so using it would require checking how that grammar matches the application.

A protocol field containing exactly four ASCII digits may require a still narrower parser that accepts no sign and no variable number of digits. Grammar validation and arithmetic overflow detection are separate responsibilities.

For an externally supplied counted byte span, avoid `strlen`. It is not safe to assume arbitrary data is null-terminated. Instead parse exactly the declared span, test every byte, and return a status explaining whether the grammar and range were valid.

2.22 Convert after proving range

Consider converting a signed `long` into an unsigned 16-bit field. Casting the negative value first may produce a valid-looking unsigned integer that does not represent the user's intended value. Check the sign before the cast; then check the upper limit:

```
#include <stdint.h>
#include <stdbool.h>

bool to_u16(long input, uint16_t *out)
{
    if (out == nullptr || input < 0 ||
        (uintmax_t)input > UINT16_MAX)
        return false;
    *out = (uint16_t)input;
    return true;
}
```

The comparison uses a domain capable of representing the nonnegative input, so the conversion to the destination type occurs only after representability has been established. Notice that the function does not confuse the destination type's maximum with the application's permitted value. A sensor register might have a 16-bit representation while supporting only the numerical range 0–4095; that additional restriction belongs in the API's contract.

2.23 Signed division and remainder

C integer division truncates toward zero. Thus `-7 / 3` yields `-2`, and `-7 % 3` yields `-1`. Some mathematical algorithms require a nonnegative remainder even for negative inputs; implementing them correctly means handling negative operands intentionally rather than assuming C's `%` operator is the mathematical modulo function.

Two special cases must be rejected before signed division: divisor zero, and division of the most negative representable value by -1 when that quotient cannot be represented. In the C23 two's-complement model, `LONG_MIN / -1` overflows its signed result.

A checked division routine can return a boolean success value and leave its quotient output unchanged on failure. If it also returns a remainder, both outputs should be assigned together only after successful validation, so the caller cannot observe a new quotient paired with an old remainder.

2.24 Value, representation, and transport

A number has a mathematical meaning; a C object has an implementation-defined in-memory representation within the language's constraints; a file or network packet has a specified external representation. These are three layers.

For example, the mathematical value 4660 fits in a `uint16_t` if that type is available. Its in-memory octet order depends on the target. A little-endian protocol represents it as octets `0x34` and `0x12`. Decoding those bytes requires applying the protocol's rule, not casting an arbitrary byte pointer to a `uint16_t *`.

That distinction is fundamental for embedded interoperability. Different processors may exchange identical packets while using different internal representations.

2.25 Why tests need all endpoints

A parser that works for 0, 17, and −17 may still fail at `LONG_MIN`. A size calculation that works for small arrays may wrap at `SIZE_MAX`. A shift that works with count one may invoke undefined behavior with count 32.

Construct tests from the type's limits and the function's domain:

- The smallest legal value and one value smaller, when representable.
- The largest legal value and one value larger, when representable.
- The empty or zero-size case.
- Malformed input and null arguments when those are permitted error cases.
- A result that exactly fills available capacity.
- A result that exceeds capacity by one.

Where a mathematical out-of-range value cannot be represented in the function's input type, use an external representation such as a decimal string to construct the test. This is another reason parser tests require both syntax and range cases.

Additional exercises

2.9 Explain why a positive signed accumulator cannot parse `LONG_MIN` on a two's-complement implementation. **2.10** Extend `parse_signed` to work with a pointer and explicit byte length rather than a terminating string. **2.11** Write checked division returning both quotient and remainder and preserving both outputs on failure. **2.12** Define a sensor-register range smaller than `UINT16_MAX` and write a conversion that checks both limits. **2.13** Produce independent decimal-string test cases for the signed maximum and minimum using `snprintf`.

3. Functions and Program Structure

Good C programs put complexity into small, explicit interfaces. Functions make this possible, but they do not hide ownership, sizes, or error conditions unless the programmer makes those matters part of the contract.

3.1 Definitions, declarations, and headers

```
/* statistics.h */
#ifndef STATISTICS_H
#define STATISTICS_H
#include <stddef.h>
double average(const double values[static 1], size_t count);
#endif
```

The `static 1` in an array parameter is a promise that an argument refers to at least one element. It is not a runtime bounds check. For an interface that accepts zero elements, prefer `const double *values` and state whether a null pointer is acceptable when `count == 0`.

```
#include "statistics.h"

double average(const double *values, size_t count)
{
    if (count == 0) return 0.0; /* choose and document policy */
    double total = 0.0;
    for (size_t i = 0; i < count; ++i) total += values[i];
```

```
    return total / (double)count;
}
```

Zero is not always a good error signal. Later we will return status codes and write results through out-parameters.

3.2 Scope and lifetime

A variable defined in a block has block scope. An ordinary local variable generally has automatic storage duration: its lifetime ends when the block invocation ends. A file-scope object has static storage duration. `static` within a function preserves an object's value across calls, but introduces shared state that complicates testing and concurrent execution.

3.3 Recursion

```
unsigned long factorial(unsigned n)
{
    return n <= 1 ? 1UL : (unsigned long)n * factorial(n - 1);
}
```

This mathematical definition is compact, but unsigned overflow makes its result unsuitable for large inputs, and deep recursion consumes stack. A useful recursive function requires both a terminating case and a bounded resource argument. For most production factorial calculations, an iterative implementation with an overflow check is easier to defend.

3.4 Design by contract

A function that converts a decimal string should say whether leading whitespace is accepted, which characters terminate conversion, how overflow is reported, and whether partial parses are accepted. A good interface often has the form

```
bool parse_unsigned(const char *text, unsigned *result);
```

A successful call writes the result. A failed call leaves the output unchanged by convention. We can implement this without guessing whether an error sentinel might also be a valid result.

Exercises. 3.1 Implement `parse_unsigned` with overflow detection using `UINT_MAX`. 3.2 Replace a recursive factorial with an iterative, checked version. 3.3 Create a header/source pair and link it with a separate test file.

3.5 Out-parameters and explicit failure

A function that returns an integer can use every integer value as an ordinary result. That leaves no spare sentinel for failure. An interface can instead return success and write the result through a pointer:

```
#include <stdbool.h>
#include <stddef.h>
#include <limits.h>

bool sum_nonnegative(const unsigned *items, size_t n,
                     unsigned *result)
{
    if (result == nullptr || (n != 0 && items == nullptr))
        return false;

    unsigned total = 0;
    for (size_t i = 0; i < n; ++i) {
```

```
        if (items[i] > UINT_MAX - total)
            return false;
        total += items[i];
    }
    *result = total;
    return true;
}
```

This function makes two promises: failure leaves `*result` unchanged, and success provides the exact sum. The first promise is maintained by assigning to the output only after all checks succeed. Its caller still has an obligation: the pointer must refer to a writable `unsigned` object. A null check cannot detect a dangling pointer.

The result type `bool` expresses a simple yes-or-no outcome. More elaborate interfaces can return an enumeration of errors: invalid argument, overflow, and unsupported operation are often more helpful than one generic false value.

3.6 Header contracts and linkage

Suppose a program has three translation units, `main.c`, `statistics.c`, and `statistics.h`. The header should state the function's public declaration and any types required to call it. The source file defining the function should include the same header; this makes disagreements visible to the compiler.

An identifier with external linkage may be referred to across translation units. A `static` function at file scope instead has internal linkage. Prefer internal linkage for helpers that are not part of the module's interface. It keeps the global symbol namespace small.

Do not put ordinary external object definitions in headers. If each including source file defines the same external object, linking can fail with multiple definitions. Place a declaration such as `extern unsigned counter;` in the header and exactly one definition `unsigned counter;` in a source file.

3.7 Recursion versus iteration

The mathematical definition of a recursive function does not establish that the implementation fits within the machine's resources. Each recursive invocation may need stack space. A function that explores a linked data structure may have recursion depth proportional to its length.

Use recursion when it reflects a bounded structure clearly. Prefer iteration when stack bounds are critical or when the recursive call can be replaced with an ordinary loop. For embedded programs, provide a maximum-depth argument before treating recursive code as suitable for production.

3.8 Exercises

3.4 Design a `parse_temperature` interface distinguishing invalid text from out-of-range input. **3.5** Implement a checked sum for signed `int` elements. **3.6** Split `sum_nonnegative` into a public header, implementation, and executable tests. **3.7** Show how to preserve the output argument's value after each error. **3.8** Explain the difference between linkage and storage duration.

3.9 Choose the shape of an interface

Consider three interfaces for parsing a decimal value:

```
unsigned parse(const char *text);
bool parse(const char *text, unsigned *out);
enum parse_status parse(const char *text, unsigned *out);
```

The first has no unambiguous error channel if every `unsigned` value is valid. The second distinguishes success from failure, and the third can explain why parsing failed. Their usefulness depends on the caller's needs. A library for configuration files might distinguish empty input, invalid character, and range error; a tiny internal helper might only require a boolean result.

An out-parameter contract should explicitly say what happens on failure. A practical convention is *strong failure safety*: validate into local state and assign to `*out` only after success. That avoids forcing callers to guess whether they can reuse a previous result following an error. It does not protect against invalid pointers supplied by callers.

3.10 Function declarations are contracts across files

Imagine `parse.c` defines a function but `main.c` has copied its declaration incorrectly. Separate compilation may not catch a mismatch without a shared header. Include the public header in the implementation file as well as at call sites.

```
/* parse.h */
#ifndef PARSE_H
#define PARSE_H
#include <stdbool.h>
bool parse_unsigned(const char *text, unsigned *out);
#endif
```

The header describes the interface; it does not allocate storage or call the function. A linker can find the definition later when both translation units are linked. Be cautious about external variables in headers: `extern int errors;` is a declaration; `int errors;` can be a tentative definition and should not be duplicated indiscriminately across source files.

`static` at file scope gives helper functions internal linkage. Its purpose is not to make a function execute faster. `inline` also does not compel a compiler to substitute the body at call sites; it is part of the language's definition and linkage rules, not a measured performance guarantee.

3.11 A fully checked factorial

The earlier recursive factorial demonstrates recurrence, not a production overflow policy. A robust iterative version has to guard the multiplication and consider the loop-counter's boundary:

```
#include <limits.h>
#include <stdbool.h>

bool factorial_checked(unsigned n, unsigned long *out)
{
    if (out == nullptr)
        return false;
    unsigned long value = 1;
    for (unsigned i = 2; i <= n; ) {
        if (value > ULONG_MAX / i)
            return false;
        value *= i;
        if (i == n)
            break;
        ++i;
    }
    *out = value;
    return true;
}
```

The explicit final-iteration break prevents an unsigned loop counter from wrapping when `n == UINT_MAX`. In practice the multiplication check will reject very large values long before that, but the loop's correctness should not depend on an unrelated range fact when a simple structure makes termination obvious.

3.12 Functions and test seams

A routine that mixes input reading, parsing, allocation, and printing is harder to test than four smaller operations with explicit contracts. The goal is not an arbitrary number of functions. It is to isolate decisions: one function transforms input bytes; one validates values; one stores them; a boundary function handles the operating environment.

For host-based firmware tests, a function that accepts a byte buffer and a length can be tested without a UART peripheral. A thin UART adapter may read bytes on the device and forward them to the parser. This separation is useful because transport errors and parsing errors have different causes and remedies.

Further problems: Write a header and implementation for a checked decimal parser. Define a status enumeration for invalid syntax versus overflow. Test that failure preserves the output. Explain why a static local variable may complicate recursion and concurrent calls.

3.13 Separate syntax checking from semantic validation

A parser does two jobs: it recognizes a sequence of characters and it interprets the recognized sequence as a value. Combining both stages into an unchecked conversion makes it easy to accidentally accept trailing garbage or overflow.

Consider a decimal parser with this contract: the input must be a nonempty sequence of ASCII digits, its value must fit in `unsigned`, and on failure the output must remain unchanged. A clear algorithm is:

1. Reject a null input or null output pointer.
2. Reject empty text.
3. Start an accumulator at zero.
4. For each byte, reject anything outside `'0'` through `'9'`.
5. Before multiplying by ten, check that the new value is representable.
6. Assign the final accumulator to the output once, after the whole string succeeds.

`examples/parse_unsigned.c` implements this design. It deliberately differs from `strtoul` by rejecting signs and spaces. Neither behavior is inherently better; the application's grammar decides.

For a parser that receives a counted span, reject both a null pointer with nonzero length and any claimed length that is not actually backed by valid storage. The latter is a caller precondition C cannot test generically.

3.14 Return values and out-parameters

Why return a status and write a value through an out-parameter rather than return a structure containing both? Either design can work. In C, a small result structure is often natural:

```
struct parse_result {
    bool ok;
    unsigned value;
};
```

With this interface, the caller does not manage a separate output pointer. The function can return an all-zero result on failure, then fill both members on success. It may be clearer for small values; an out-parameter may be preferable when writing a large object or retaining a long-established ABI.

Whichever design you choose, state the failure contract. A boolean-returning function that partly overwrites the output before reporting failure is much harder for callers to use correctly than one that commits a completed result once.

3.15 File scope, linkage, and modules

C has no built-in class module system. The conventional unit of modularity is a header plus one or more source files. The header exposes declarations; the source file defines those operations and keeps helpers private with `static`.

A source file should include its own public header before defining the corresponding functions. That forces the compiler to compare the declared and defined types. If callers include the same header, interfaces cannot silently diverge without diagnostics.

A declaration is not always a definition. For a shared file-scope object, the header can use `extern unsigned event_count;`, while precisely one source file supplies `unsigned event_count = 0;`. Repeating a non-`extern` external object definition across headers or translation units is not a sound substitute for a module.

3.16 Passing arrays does not pass their sizes

Suppose a function accepts `const int values[10]`. That parameter declaration is adjusted to a pointer type; the compiler does not automatically supply a count of ten and ordinarily cannot check what the caller passed. A more honest general interface accepts `const int *values, size_t count`.

The type `const int *` says the function may not modify the elements *through that pointer*. It does not guarantee ownership, lifetime, or thread safety, and it does not make the underlying object immutable if other permitted aliases exist.

To express a fixed minimum bound, C permits array parameter syntax with `static` in the brackets. Such a declaration is a caller contract, not dynamic bounds enforcement. It should be used only when every valid call really supplies enough elements.

3.17 Recursive functions need a resource argument

Recursion can closely express a tree traversal or mathematical recurrence, but every call consumes some execution resources. A desktop process may have ample stack; a microcontroller task may have only a small fixed stack.

When evaluating recursive code, ask whether the maximum depth is bounded by the input contract. Can a malicious or malformed input create arbitrarily deep nesting? What happens when the recursion reaches its base case? A recursive algorithm with logarithmic depth may be reasonable; a recursive traversal of an unconstrained linked list often deserves an iterative form or a documented maximum depth.

Guided exercises

3.9 Implement the structure-returning parser result and compare its call sites with the out-parameter version. **3.10** Write a self-contained public header and compile it as the first include in a separate test file. **3.11** Replace a recursively implemented byte count with a loop and identify the changed resource bounds. **3.12** Design an API to read a sensor value that can report 'not ready', 'hardware error', and 'valid sample' distinctly.

3.18 Design functions around failure rather than around syntax

A function signature is an engineering contract. It describes argument and result types, but it does not express every precondition or guarantee. A routine that accepts a buffer pointer and a

byte count must still document whether the count can be zero, whether the pointer may be null, whether the buffer is mutable, and what happens if the operation cannot finish.

For example, consider a parser that accepts a counted byte span and produces a 16-bit value. A useful status type could be:

```
enum parse_status {
    PARSE_OK,
    PARSE_EMPTY,
    PARSE_INVALID,
    PARSE_RANGE
};
```

A public interface may then declare:

```
enum parse_status parse_u16(const unsigned char *bytes,
                            size_t count, uint16_t *out);
```

The comments in its header should say that nonzero `count` requires at least that many readable bytes, that null `out` is not permitted, that failure leaves `*out` unchanged, and that accepted input contains only ASCII decimal digits. C cannot generally discover an invalid pointer whose bit pattern happens to be non-null.

Notice what this interface *does not* do: it does not open a file, read from a UART, or display a message. Those tasks belong in adapters that supply counted spans and interpret status codes. Separating them makes the parser reusable on a host and a freestanding target.

3.19 A complete checked operation

The following function illustrates a strong failure postcondition:

```
#include <stdbool.h>
#include <stddef.h>
#include <stdint.h>

bool sum_u32(const uint32_t *numbers, size_t count, uint32_t *out)
{
    if (out == nullptr || (numbers == nullptr && count != 0))
        return false;
    uint32_t sum = 0;
    for (size_t i = 0; i < count; ++i) {
        if (numbers[i] > UINT32_MAX - sum)
            return false;
        sum += numbers[i];
    }
    *out = sum;
    return true;
}
```

The invariant is that `sum` equals the mathematical total of previously visited elements, and `sum <= UINT32_MAX`. The subtraction is valid because `sum` never exceeds the unsigned type's maximum. The check precedes addition; no intermediate sum wraps.

There is a subtle limitation: if `out` points into the `numbers` array, assigning the final result changes one input element, but only after all reads have completed. That is permitted by this synchronous design. A future refactoring that writes the output during iteration would have different aliasing implications. Rather than assuming every pointer is disjoint, document when aliasing matters.

3.20 Make a status stable across versions

A public interface may be used by code compiled at different times. An enumeration can represent useful status categories, but adding a new enumerator does not guarantee that older clients understand it. A caller's switch should have a documented fallback for an unknown result; otherwise a newly introduced failure category may be silently ignored.

The size of an enum and its exact ABI can depend on implementation choices unless the interface specifies an underlying representation supported by the toolchain and ABI. Do not transmit a raw enum object as a wire-protocol field. Serialize a defined numeric code instead.

For a stable binary interface, a structure's field layout and ownership rules are just as important as function names. In an internal application built from one revision of source, rebuilding callers and implementations together simplifies compatibility. A public library with independently deployed clients has stronger requirements.

3.21 Initialization and cleanup symmetry

A resource-managing module might expose functions conceptually named `open`, `read`, and `close`. Its documentation should identify when the handle is valid and who may release it.

An effective pattern initializes a caller-owned structure to an empty state, acquires resources one by one, and guarantees cleanup after partial failure. The corresponding close function is tolerant of the empty state and leaves the object empty after successful release.

Some C APIs return an opaque pointer from an open function and require a matching destroy function. Others use a caller-provided structure. These designs differ in allocation policy and ABI flexibility, but both benefit from explicit ownership transfer and well-defined failure behavior.

Do not assume the fact that a function returns `nullptr` means no cleanup is necessary. An implementation may have acquired and released temporary resources before reporting failure, and those cleanup paths deserve tests.

3.22 Order of evaluation at function calls

When a function has two argument expressions, C does not generally require that the left argument be evaluated before the right. A function's parameters are initialized from the evaluated argument values, but an expression should not depend on one argument modifying state before another reads it.

Avoid:

```
int i = 1;
/* Do not depend on which argument evaluation happens first: */
use_values(i++, i);
```

Instead:

```
int first = i;
++i;
int second = i;
use_values(first, second);
```

The latter makes the sequencing explicit. The function prototype can verify argument types, but a declaration cannot repair unsequenced conflicting accesses in an expression supplied by the caller.

3.23 Choosing between recursion and iteration

A recursive traversal often matches a data structure naturally. But the call stack is a resource. A parser accepting nested syntax from an external source can receive a depth limited only by input

length; an attacker or ordinary malformed file can then exhaust a small task stack.

An iterative parser using an explicit stack can control storage explicitly. For a fixed maximum nesting depth, a bounded array of parser states makes the failure mode predictable: an input exceeding the depth is rejected rather than causing uncontrolled stack growth. The tradeoff is additional state-management code.

A tail-recursive function is not guaranteed to be transformed into a loop by the language. If worst-case stack use matters, write the loop explicitly or rely only on an optimization whose use is verified and required by the deployment toolchain.

3.24 A function review checklist

For each exported function, answer these questions:

1. What input values and storage regions are valid?
2. Which storage is borrowed, retained, allocated, or released?
3. What is written on success? What remains unchanged on failure?
4. Can arithmetic overflow before a later check?
5. Can callbacks or concurrent callers reenter the function?
6. Does the function use a hosted facility unavailable on a freestanding target?
7. How will tests reach its error paths?

A short answer for each is usually more valuable than a long comment repeating what every line does.

Additional exercises

3.13 Implement `sum_u32` and inject an overflowing final element. Verify the caller's original result remains unchanged. **3.14** Define a parser status enumeration and write tests for empty, invalid, overflowing, and valid input. **3.15** Replace a recursive nested-record traversal with an explicit fixed-depth stack. **3.16** Write a public header that compiles independently and explain why its implementation must include the same header. **3.17** Review a resource-acquisition function for partial-failure cleanup.

4. Pointers, Arrays, and Strings

Arrays own contiguous elements; pointers refer to objects or functions. The relationship is close, but the two are not the same thing.

4.1 Address and indirection

```
int value = 12;
int *p = &value;
*p = 13;          /* modifies value */
```

A pointer's validity depends on the referenced object's lifetime, alignment, and access permissions. Dereferencing a null, dangling, or otherwise invalid pointer has undefined behavior. C23 adds the keyword `nullptr` for a null pointer constant; legacy code commonly uses `NULL`.

4.2 Array arguments carry no length

```
#include <stddef.h>

int sum(const int *data, size_t count, long long *out)
{
    if (out == nullptr || (count != 0 && data == nullptr))
```

```
        return -1;
    long long total = 0;
    for (size_t i = 0; i < count; ++i)
        total += data[i]; /* caller must ensure representable result */
    *out = total;
    return 0;
}
```

The signature carries a count because a pointer does not record the bounds of the original array. This implementation still assumes the sum fits in `long long`; a fully checked version needs an overflow test before each addition.

4.3 Strings are terminated byte sequences

A C string is an array of `char` with a terminating zero byte. `strlen` measures the number of bytes before that terminator, not necessarily the number of Unicode characters. `sizeof` measures a type or an object, not the current length of a string.

```
char name[] = "C23";
size_t capacity = sizeof name;  /* 4 including '\0' */
```

UTF-8 represents some characters with multiple bytes. A procedure that uppercases arbitrary UTF-8 by manipulating each `char` is not a correct Unicode case converter.

4.4 Copying bounded text

```
#include <stddef.h>

bool copy_text(char *dst, size_t capacity, const char *src)
{
    if (dst == nullptr || src == nullptr || capacity == 0)
        return false;
    size_t i = 0;
    while (i + 1 < capacity && src[i] != '\0') {
        dst[i] = src[i];
        ++i;
    }
    dst[i] = '\0';
    return src[i] == '\0';
}
```

This function assumes valid source storage and that source and destination do not overlap; it signals truncation rather than silently claiming success. It is not a replacement for a general `memmove`.

4.5 Constness and pointer arithmetic

`const int *p` prohibits modifying an `int` through `p`. `int *const p` prohibits assigning another address to `p`. Pointer arithmetic is defined only within an array object (including the permitted one-past position), subject to the language's constraints. A one-past pointer may be compared or subtracted appropriately but not dereferenced.

Exercises. 4.1 Implement `strlen` using a pointer and compare with an index version. 4.2 Explain the difference between `char *p = "abc";` and `char p[] = "abc";`. 4.3 Implement `reverse` for a mutable byte array of known length. 4.4 Add overlap detection or explicitly reject overlap in a copy API.

4.6 A pointer is not an array descriptor

The expression `sizeof array` is useful inside the scope where `array` is an actual array object. Once an array expression is converted to a pointer for passing to a function, its bound is ordinarily lost.

```
#include <stddef.h>

static size_t count_values(const int *values, size_t count)
{
    size_t found = 0;
    for (size_t i = 0; i < count; ++i)
        if (values[i] != 0) ++found;
    return found;
}

int main(void)
{
    int values[] = {3, 0, 8, 0, 1};
    size_t count = sizeof values / sizeof values[0];
    return count_values(values, count) == 3 ? 0 : 1;
}
```

The caller supplies the element count. The callee accepts that count as a contract; C cannot generally establish that a non-null pointer references as many elements as claimed.

4.7 Two-dimensional arrays

A real two-dimensional array occupies contiguous storage in row-major order:

```
int matrix[2][3] = {
    {1, 2, 3},
    {4, 5, 6}
};
```

The expression `matrix` in most contexts converts to a pointer to an array of three `int`, not an `int **`. This distinction is essential: a pointer-to-pointer usually points to another pointer object, whereas a 2-D array's rows are laid out directly next to each other.

A function that requires a known column count can state it:

```
int matrix_total(size_t rows, const int values[rows][3])
{
    int result = 0; /* contract: sum must fit in int */
    for (size_t r = 0; r < rows; ++r)
        for (size_t c = 0; c < 3; ++c)
            result += values[r][c];
    return result;
}
```

C23 retains variable length array support as specified by the standard, with implementation differences relevant to storage-duration choices. Where fixed dimensions suffice, fixed arrays make capacity easier to reason about.

4.8 Lifetime is as important as address

Never return the address of an automatic local object:

```
int *bad_pointer(void)
{
    int local = 7;
```

```
    return &local; /* invalid after this invocation ends */
}
```

The problem is not that the stack address must immediately change; it is that the object's lifetime has ended. The behavior of later dereferencing that pointer is not defined. Returning a value, writing through a caller-supplied pointer, or returning an allocated object with a documented owner are legitimate alternatives.

4.9 Restrict and aliasing

The qualifier `restrict` describes an access pattern that permits optimizations when the programmer satisfies its association requirements. It is a promise, not a runtime test. Do not add it to a public API merely because it sounds faster. If two pointer arguments can legally refer to overlapping storage, the interface must not assert a restriction that the caller violates.

4.10 Exercises

4.5 Implement an in-place reversal of a `char` array given its exact length. **4.6** Explain why `int **p = matrix;` is incompatible with a two-dimensional `int matrix[2][3]`. **4.7** Write a function to find a value in a slice represented by pointer and length. **4.8** Return both minimum and maximum through checked out-parameters. **4.9** Explain what happens to pointers into a block after successful `realloc`.

4.11 Three meanings of an array name

In many expressions an array expression converts to a pointer to its first element. Important exceptions include operands of `sizeof`, unary `&`, and other contexts with specified non-decay behavior. For a declared array:

```
int items[5] = {};
size_t bytes = sizeof items;
int *element = items;
int (*whole)[5] = &items;
```

`element` points to an `int`; `whole` points to the entire five-element array. Incrementing the former advances by one `int`; incrementing the latter would advance by one whole array. The types convey this difference and permit the compiler to diagnose incompatible pointer uses.

When an array is passed as a function argument, a declaration such as `void process(int values[10]);` still adjusts to a pointer parameter. The `10` in that form does not create a checked ten-element slice. Carry a count unless the function's contract fixes the dimension and callers are required to honor it.

4.12 One past the end is not an element

Pointer arithmetic can produce the one-past position of an array. It is useful as an end marker, but it cannot be dereferenced. For a nonempty array, `begin = &items[0]` and `end = begin + count` can represent a half-open range `[begin, end)`.

Comparing or subtracting arbitrary unrelated object pointers is not the same as comparing array positions. Pointer subtraction is defined for pointers into the same array (including one past), subject to representability in `ptrdiff_t`. When `count` may be enormous, indexes of type `size_t` can be more straightforward than pointer differences.

A null pointer is not an empty array that can be incremented freely. A zero-length buffer interface should explicitly describe whether it accepts `nullptr`, and code must avoid forming `nullptr + 0` merely to make an end pointer.

4.13 Copy, move, and overlap

`memcpy` copies object-representation bytes from one valid region to another but requires nonoverlapping regions. `memmove` supports overlap. Both require sufficiently large regions. Neither can prove that a claimed size is correct.

A safe shift within an array might use:

```
#include <string.h>
/* Preconditions: capacity >= count + 1 and position <= count. */
memmove(&items[position + 1], &items[position],
        (count - position) * sizeof items[0]);
```

Those preconditions matter. Check capacity and `position` before computing an address or multiplication. Also ensure the item type permits moving its object representation in the intended way; plain C value records generally do, whereas pointers inside records may carry ownership semantics not preserved by shallow copying.

4.14 String ownership and views

A string literal has static storage duration; modifying it through a pointer produces undefined behavior. By contrast, `char text[] = "abc";` defines a mutable array whose elements can be changed. The type `const char *` commonly signals that a function borrows text without modifying it, but `const` alone does not guarantee the memory remains alive.

A counted string view, `struct text_view { const char *data; size_t size; };`, can represent slices that are not null-terminated. Such views are convenient for parsers, but passing their pointers to `strlen` or `printf("%s")` is unsafe unless termination is separately guaranteed. A view usually borrows storage and should not outlive its owner.

Further problems: Implement a bounded substring search that never relies on a terminator. Explain the difference between `int (*)[4]` and `int **`. Write tests for a zero-length slice and for the final legal element index.

4.15 Draw the objects before drawing pointers

For a declaration such as

```
int values[4] = {10, 20, 30, 40};
int *p = &values[1];
```

draw four adjacent `int` objects and one separate pointer object. The pointer is not a second element of the array; it stores a value designating the second element. `p + 1` points at the third element. `*(p + 1)` reads that element. `p[1]` is defined in terms of the same indexing relationship.

Now draw `&values`. Its type is pointer to an array of four `int`, whereas `values` ordinarily converts to pointer to its first `int`. These pointers may designate the same starting address in memory, yet their types control pointer arithmetic: incrementing a pointer-to-array advances by the whole array size.

This is why declarations matter even when two pointer values print alike.

4.16 A counted slice

A plain pointer to the first element does not contain a length. A small value type can carry the two pieces together:

```
#include <stddef.h>

struct int_slice {
```

```
    const int *data;
    size_t length;
};
```

The slice usually *borrows* storage. It does not allocate anything, and copying the structure copies the pointer and length rather than copying the elements. The owner must keep the array alive for the duration of every use of the slice.

A safe access function may return a boolean result and use an output pointer:

```
#include <stdbool.h>

bool slice_get(struct int_slice s, size_t index, int *out)
{
    if (out == nullptr || index >= s.length ||
        s.data == nullptr)
        return false;
    *out = s.data[index];
    return true;
}
```

This checks the index against the *declared* slice length. Its precondition still requires `s.data` to point to at least `s.length` readable integers when `s.length` is nonzero. A dangling pointer or dishonest length cannot be repaired by the index check.

The standalone example `examples/slices.c` demonstrates both direct access and the creation of a bounded subslice.

4.17 Derive a subslice without overflow

A subslice starts at `offset` and contains `count` elements. It is tempting to check `offset + count <= length`, but unsigned addition may overflow. Instead, require `offset <= length` and `count <= length - offset`. That subtraction is evaluated only after the first condition holds.

When `count == 0`, you may represent the new slice using a pointer to an existing valid position, including the one-past position of a nonempty array. If the original slice is empty and its data pointer is null, do not form `data + 0`: pointer arithmetic on a null pointer is not a portable way to construct a valid empty slice. An implementation can return a null data pointer for a zero-length result.

4.18 Const qualifies a path of access

Compare these declarations:

```
const int *read_only_elements;  /* elements not modifiable through pointer */
int value = 0;
int *const fixed_address = &value;
```

The first pointer variable may be reassigned, but it cannot be used to assign to the designated `int`. The second pointer variable itself cannot be reassigned, but it can modify the designated `int` if that object is modifiable and the pointer is valid.

A `const`-qualified pointer is not an ownership system. A borrowed `const char *` can become dangling when another part of the program releases its storage. Conversely, a constant object with static storage duration may remain alive for the whole program. Lifetime and permission to modify are separate properties.

4.19 Overlap and copying

Suppose `char letters[] = "abcdef";`. Moving the region `letters[0..3]` one position to the right creates overlapping source and destination regions. `memcpy` is not the right operation; `memmove` is specified to handle such overlap.

Before calling either library function, establish that both source and destination contain the requested number of valid bytes. The function names do not infer buffer capacity. For a type with pointer members, byte-wise copying the enclosing structure still performs a shallow copy; it does not duplicate heap storage referenced by members.

4.20 Pointer lifetime can end without the pointer value changing

A local variable's lifetime ends when the relevant block invocation ends. A heap allocation's lifetime ends when it is freed. Successful `realloc` ends the old allocation's lifetime and may return a different address. In each case an old pointer can retain an address-like bit pattern while becoming unusable for dereferencing the former object.

A good habit is to avoid returning borrowed pointers to transient storage. Instead return a value, let the caller supply output storage, or explicitly transfer ownership of allocated storage under an API contract.

Guided exercises

4.10 Draw a four-element array, a pointer to element two, and a pointer to the whole array; determine the element reached by each valid pointer increment. **4.11** Implement `slice_find` returning an index through an output pointer, distinguishing 'not found' from an invalid input. **4.12** Implement a reverse operation on a mutable slice and test empty and single-element slices. **4.13** Explain why copying a slice structure does not copy its elements. **4.14** Write a subslice function whose arithmetic remains defined at `SIZE_MAX`.

4.21 A worked memory diagram

The array and pointer relationship is easier to understand when we distinguish objects from the values that refer to them. Consider a five-element array and a separate pointer variable:

```
int values[5] = {10, 20, 30, 40, 50};
int *current = &values[2];
```

One possible conceptual picture is:

```
values:   +----+----+----+----+----+
          | 10 | 20 | 30 | 40 | 50 |
          +----+----+----+----+----+
           [0]  [1]  [2]  [3]  [4]
                      ^
current:  +-----------|-----------+
          | pointer to values[2]  |
          +-----------------------+
```

The pointer object `current` has its own storage and type. It contains a value designating `values[2]`, not a copy of the integer 30. The expression `*current = 31;` modifies the array element. The expression `current += 2;` moves the pointer to `values[4]`. One more increment produces a legitimate one-past pointer, but dereferencing that one-past pointer would be undefined.

Nothing in this picture commits C to an address size or to a particular stack layout. The array's five elements are contiguous by language definition; where the implementation stores the automatic objects is a separate matter.

4.22 Decoding difficult declarations

For ordinary declarations, start at the identifier and work outward using the parentheses to determine binding:

```
int *p;                 /* pointer to int */
int a[4];               /* array of four int */
int (*row)[4];          /* pointer to array of four int */
int *function(void);    /* function returning pointer to int */
int (*callback)(int);   /* pointer to function */
```

Compare `int *function(void)` with `int (*callback)(int)`. The first declares a function, because the parentheses following `function` form the parameter list. The second declares a pointer object, because the parentheses group the `*` with `callback` before the function-parameter part is interpreted.

A `typedef` can make a complex interface more readable:

```
typedef bool (*sample_callback)(int value, void *context);
```

It creates a name for the type, not a new distinct type with ownership or lifetime enforcement. Documentation must still tell clients how long the callback and context remain valid and whether invocation can happen asynchronously.

4.23 Arrays in function parameters

The declaration `void normalize(double samples[16]);` is not a request for the language to pass an entire 16-element array by value. The array parameter is adjusted to a pointer to its first element. The number 16 may document intent, but it does not enforce length at runtime.

Use a count when the function should support variable-length input:

```
bool normalize(double *samples, size_t count);
```

This signature also forces a new question: can `samples` be null when `count == 0`? If the answer is yes, do not form an end pointer with `samples + count` before checking the null case. If the answer is no, return an error or state a stronger caller precondition.

For a fixed-size mathematical object, an array pointer can retain the row dimension:

```
void work_on_row(int (*row)[4]);
```

The caller may pass the address of an array of four `int` values. This is not interchangeable with `int **`, which would require a pointer object at its target.

4.24 Building a safe subslice

A pair `(data, length)` describes an array region when the caller's preconditions are met. A subslice adds two numbers, `offset` and `count`, and must ensure the requested interval fits.

The naive condition `offset + count <= length` can overflow, because these quantities are usually `size_t`. Rewrite the condition as two comparisons:

```
if (offset > length || count > length - offset)
    return false;
```

C's `||` operator short-circuits, so the subtraction executes only if `offset <= length`. This is an example of making a proof obligation visible in the expression itself.

The companion `examples/slices.c` demonstrates a full slice, a middle subslice, a failed request with `SIZE_MAX`, and the zero-length case. The function leaves the output view unchanged on failure, a useful promise for callers that maintain a previously valid view.

Why is a view not an owning container? Because copying it does not copy the underlying elements. It borrows the original buffer and cannot safely outlive the object that supplies those elements. When the owner reallocates or frees its storage, the view becomes invalid even if its pointer value is unchanged.

4.25 Multidimensional layout

A declaration `int grid[3][4];` denotes three contiguous arrays, each containing four contiguous integers. The expression `grid[r][c]` selects an element using the row and column. The address calculation is conceptually equivalent to advancing by `r * 4 + c` elements from the beginning, assuming `r < 3` and `c < 4`.

A function accepting that array can be written with a known column dimension:

```
void inspect(size_t rows, const int grid[rows][4]);
```

The parameter is adjusted to a pointer to an array of four `const int` elements. This differs fundamentally from an array of independent row pointers, where each row may live in a different allocation.

Some APIs expose an explicit row stride or pitch because data may be padded, shared with a graphics surface, or stored by a peripheral DMA engine. A stride describes storage layout; it is not always equal to the number of logically meaningful columns. Validate both logical bounds and the physical storage extent before an indexed access.

4.26 String storage and counted text

A string literal includes a trailing zero character and has static storage duration. The statement `const char *name = "C23";` stores a pointer to that literal. The statement `char name[] = "C23";` defines an independently stored mutable array containing four bytes on an eight-bit-byte implementation: `'C'`, `'2'`, `'3'`, and the terminating zero.

A counted text span can represent an input containing embedded zero bytes. Such a span cannot be passed to `strlen` or `printf("%s", ...)` without additional termination guarantees. A fixed-format output like `fwrite(data, 1, length, stdout)` is appropriate for known valid byte storage, and its return value must be checked for short output.

For UTF-8, the span length counts bytes rather than Unicode scalar values or grapheme clusters. Do not assume that truncating a UTF-8 buffer to an arbitrary byte capacity preserves complete characters; a truncation may cut through a multibyte sequence.

4.27 Reasoning about overlap

Suppose a mutable array contains `"abcde"`, and we want to remove its second element. We can move the bytes from the third element onward left by one position, including the terminating zero if the result remains a C string.

```
char text[] = "abcde";
memmove(&text[1], &text[2], sizeof text - 2);
```

Because the regions overlap, `memmove` is required instead of `memcpy`. The code's size calculation is valid for this fixed array, but a general function accepting `char *` must be told the meaningful length and total capacity. The destination remains within the original allocation.

When the array elements are structures that contain owning pointers, moving the record bytes is only a shallow move. The program must ensure it has not accidentally created duplicate owners or left an old record that will be destroyed twice.

Diagnostic questions

Explain the type and validity of `&values[5]` for a five-element array. Explain why `int **` cannot receive `int grid[3][4]` directly. Describe which operation invalidates a slice into a vector after a successful resize. Show a proof that a subslice calculation cannot wrap even when `offset == SIZE_MAX`. Decide whether your interface accepts an empty slice with a null pointer.

5. Structures and Data Representation

Structures associate fields under one type. Enumerations name integral alternatives. Unions allow members to share storage, but they require disciplined tracking of which interpretation is meaningful.

5.1 A value type

```
struct sample {
    unsigned channel;
    double measurement;
    bool valid;
};

struct sample reading = {
    .channel = 2,
    .measurement = 3.75,
    .valid = true
};
```

Designated initializers name members instead of depending on their order. Structure assignment copies member values. Padding bytes may exist between fields; they are not stable serialized data. Do not write a structure directly to a wire protocol or disk format and assume portability across targets.

5.2 Tagged unions

```
enum value_kind { VALUE_INTEGER, VALUE_REAL };

struct value {
    enum value_kind kind;
    union {
        long integer;
        double real;
    } data;
};
```

The tag establishes which member the surrounding program treats as active. An API should expose constructors and accessors rather than relying on callers to maintain an undocumented convention.

5.3 Linked records

```
struct node {
    int key;
    struct node *next;
};
```

A linked list illustrates the difference between a logical collection and its physical representation: nodes may be discontiguous, each allocation may fail, and traversal follows explicit links. In embedded systems, a bounded array can often be simpler and more predictable than a heap-backed list.

5.4 Bit fields and explicit masks

Bit fields can make local state concise, but details of layout and allocation are implementation-defined. For device registers and portable serialized formats, fixed-width unsigned integers with shifts and masks often make data representation clearer. Hardware access adds concerns about `volatile`, synchronization, barriers, and the device's register semantics; `volatile` alone does not make an operation atomic.

Exercises. 5.1 Represent a timestamp with a structure and validate its fields. 5.2 Implement a fixed-capacity pool of list nodes without `malloc`. 5.3 Encode two unsigned 4-bit fields into a byte and decode them.

5.5 Field order, padding, and ABI

A structure establishes a logical association among members, but not a portable wire representation. A compiler may insert padding to meet alignment requirements. The order of named non-bit-field members is preserved in address order, yet `sizeof(struct record)` may exceed the sum of member sizes.

```
struct record {
    unsigned char kind;
    unsigned count;
};
```

There may be padding between `kind` and `count`, or after the final member. Different target architectures or compiler options may create different representations. Avoid `fwrite(&record, sizeof record, 1, stream)` as a long-term file format unless the representation is intentionally fixed for that exact environment.

To serialize an unsigned multi-byte value, select an exact format and explicitly encode bytes. That code is easier to test and more portable than depending on host endianness or struct layout.

5.6 A bounded ring buffer

A ring buffer holds a fixed number of elements and wraps its index back to the beginning. We use a separate count to distinguish an empty buffer from a full one.

```
#include <stdbool.h>
#include <stddef.h>

enum { QUEUE_CAPACITY = 8 };

struct queue {
    unsigned data[QUEUE_CAPACITY];
    size_t head;
    size_t count;
};

bool queue_push(struct queue *q, unsigned value)
{
    if (q == nullptr || q->count == QUEUE_CAPACITY)
        return false;
    size_t tail = (q->head + q->count) % QUEUE_CAPACITY;
```

```
    q->data[tail] = value;
    ++q->count;
    return true;
}

bool queue_pop(struct queue *q, unsigned *out)
{
    if (q == nullptr || out == nullptr || q->count == 0)
        return false;
    *out = q->data[q->head];
    q->head = (q->head + 1) % QUEUE_CAPACITY;
    --q->count;
    return true;
}
```

Initialize the entire queue with `struct queue q = {};`. This implementation is appropriate for sequential calls; it is **not** a concurrent lock-free queue. Sharing it between an interrupt and a thread introduces separate ordering, atomicity, and scheduling requirements.

One invariant states that `count` stays between zero and capacity. Another states that `head` is a valid index. Both remain true after successful and failed operations, assuming a properly initialized queue.

5.7 Opaque structures as a design choice

Sometimes callers need a type's behavior, not its layout. A header may declare `struct handle;` without defining its members. A source file privately defines the structure and exports functions taking pointers to it. This allows implementations to change without recompiling all callers, though allocation and ownership must be carefully arranged.

Opaque types are especially useful for devices, parsers, and protocol sessions. A public, by-value structure is often preferable for tiny value types that need no hidden invariants.

5.8 Exercises

5.4 Trace the queue above across eight pushes, one failed push, three pops, and three more pushes. **5.5** Add a `queue_peek` operation that does not remove an item. **5.6** Test that a failed pop leaves the output unchanged. **5.7** Encode `kind` and `count` in a documented byte order without copying the structure representation. **5.8** Compare the memory requirements of this queue and a linked queue for the same maximum capacity.

5.9 Structure layout as a local implementation detail

A structure value is a collection of logically related members, not a universally portable byte record. A compiler may insert padding before, between, and after members so that each can be accessed at an appropriate alignment. The expression `sizeof(struct sample)` therefore answers a question about this implementation's object layout; it does not define a network packet size.

Use `offsetof` from `<stddef.h>` when you genuinely need a member's offset within a structure. Avoid guessing offsets from the declaration. Embedded systems sometimes depend on a vendor-defined register structure layout, but that dependency should be explicit, reviewed against the compiler ABI, and separated from protocol serialization.

A structure assignment copies the member values, including pointer *values*, not heap allocations reachable through those pointers. If a record contains `char *name`, copying the record makes two references to the same storage. It does not perform a deep copy, extend the pointed-to object's lifetime, or establish a second independent owner.

5.10 Flexible array members

A final structure member can be a flexible array. It has no fixed element count in the structure's definition:

```
#include <stddef.h>
#include <stdlib.h>
#include <stdint.h>

struct message {
    size_t length;
    unsigned char bytes[];
};
```

A caller may allocate extra storage sufficient for the header and a requested number of byte elements. The size computation must be checked against `SIZE_MAX` before the call to `malloc`, and the allocated object must remain alive during access. This representation is useful for variable-length messages in one allocation, but it is inappropriate when a strictly bounded static buffer is preferable.

Flexible array members have restrictions: they belong at the end of an eligible structure, and the containing type cannot be used in every context where an ordinary fixed-size structure can. They also do not turn the member into a bounds-checked array; the actual allocation size still matters.

5.11 Layout-independent serialization

A portable encoder establishes each output byte deliberately. Consider a record with an 8-bit kind and 16-bit little-endian length. Its wire representation can use exactly three octets regardless of padding in the in-memory structure:

```
#include <stdint.h>
#include <stdbool.h>
#include <stddef.h>

bool encode_header(unsigned char *dst, size_t size,
                   uint8_t kind, uint16_t length)
{
    if (dst == nullptr || size < 3) return false;
    dst[0] = kind;
    dst[1] = (unsigned char)length;
    dst[2] = (unsigned char)(length >> 8);
    return true;
}
```

This example assumes an octet-oriented destination and should be used only where the target's byte representation meets that contract. The field width and endianness are design decisions, independent of the host machine.

5.12 Data-structure choice follows requirements

A linked list offers insertion without moving contiguous records, but each node needs storage and a link, and traversal is not random access. A fixed array has locality and predictable overall capacity but may need to shift elements during insertion. A ring buffer is excellent for FIFO traffic but does not support arbitrary deletion efficiently.

For an MCU message queue with a known maximum occupancy, the fixed array or ring buffer may be easier to justify than a heap-backed list. If the queue is shared with an interrupt, however, a nonconcurrent ring implementation remains unsafe without a synchronization design. Capacity, ownership, timing, and concurrency are separate requirements.

Further problems: Implement a deep-copy operation for a record owning a dynamically allocated string. Show how to reject a flexible-array allocation whose size calculation overflows. Write an encoder and decoder for the three-octet header, testing both directions and malformed lengths.

5.13 A structure is a value with invariants

A structure gives related fields one type, but the programmer decides which combinations of fields are meaningful. Imagine a temperature reading containing a numeric value, a unit identifier, and a validity flag. A declaration permits all combinations; the interface should say which represent actual measurements.

One design creates values through a constructor-like function that validates the inputs. An alternative stores an explicit status enumeration instead of a boolean flag so it can represent sensor-not-ready, disconnected, and measured states separately. The point is not to make every structure opaque; it is to choose a representation whose invariants are easy to state and preserve.

Designated initialization helps when a structure evolves because it names the intended members. A missing member is initialized according to the appropriate language rules, which may or may not express a valid application value. A status enumeration whose zero value has no defined meaning can produce a logically invalid initialized object despite being a perfectly valid C object.

5.14 Padding and equality

Two structures can have equal corresponding members while their padding bytes differ. Using `memcmp(&a, &b, sizeof a)` to test semantic equality is therefore not a general solution. Compare the meaningful fields instead:

```
struct measurement { unsigned id; double value; };

bool same_measurement(struct measurement a, struct measurement b)
{
    return a.id == b.id && a.value == b.value;
}
```

The floating comparison here deliberately treats NaNs as unequal; whether this is appropriate depends on the application's data model. If bitwise reproducibility matters, define representation-specific behavior rather than assuming the members' == operators implement it.

For serialization, encode only documented fields in a documented order. Writing `sizeof(struct measurement)` bytes to a file can unintentionally expose padding bytes and platform-specific floating representations.

5.15 Tagged unions and exhaustiveness

A union can store one of several representations in shared storage. It does not automatically record which interpretation is active. A common approach pairs the union with a tag:

```
enum value_kind { VALUE_INT, VALUE_FLOAT };
struct value {
    enum value_kind kind;
    union { int integer; double real; } data;
};
```

A function consuming `struct value` should examine the tag before accessing the corresponding member. C23 does not provide algebraic data types with exhaustive pattern matching; a `switch` over an enum can still have invalid values due to corrupted input or casts. An explicit default error path is useful at trust boundaries.

A tagged union is particularly effective for a parser returning one of several record kinds. It makes illegal reinterpretations easier to detect in code review and separates the logical type tag from the memory representation of the payload.

5.16 Intrusive versus owning lists

A linked-list node can own its payload, or the payload can contain the linkage needed to participate in the list. An intrusive list avoids a separate node allocation but ties the item's lifetime to list membership. An owning list may allocate a node and payload together or separately.

Before writing the first insert function, choose whether the list owns items. Can a caller free an item while it is linked? Can an item belong to two lists? What does removing an element do to its lifetime? These questions determine correctness more than the syntax of `next`.

On microcontrollers, a fixed pool of nodes and explicit occupancy flags may be preferable to unbounded heap allocation. It bounds memory, can eliminate allocation fragmentation, and makes exhaustion an ordinary status that tests can exercise.

5.17 Ring buffer wraparound

The earlier queue has capacity eight, with independent `head` and `count` fields. To find the tail, it computes `(head + count) % capacity`. This works because the queue invariant constrains `head` and `count`, and their sum is small enough to avoid overflowing `size_t`.

Trace a full cycle. Push values 0 through 7; the queue is full. A ninth push fails without modifying state. Pop 0, 1, and 2; now head is 3 and count is 5. Push 8, 9, and 10; the physical array slots reused are 0, 1, and 2. Subsequent pops return 3 through 10 in order.

The same logical sequence can be represented differently in physical storage. Clients should observe only the queue's documented behavior, not assume elements are contiguous in chronological order inside its array.

Exercises

5.9 Write an equality function for a structure containing an integer, a floating value, and an optional identifier; state the NaN policy. **5.10** Add a status tag to a variant record and reject an unsupported tag on decoding. **5.11** Design a fixed-capacity node pool for a list and describe release behavior. **5.12** Trace ring-buffer state after each push and pop, identifying array index and logical position.

5.18 A diagram of logical fields and physical layout

The declaration

```
struct event {
    unsigned char kind;
    unsigned count;
    bool valid;
};
```

establishes three members with those types and in that order. It does not specify an identical packed sequence of their representations across every target.

An implementation might arrange them conceptually as follows:

```
+------+---------+----------------+-------+-----------+
| kind | padding |     count      | valid | padding   |
+------+---------+----------------+-------+-----------+
```

The sizes and amount of padding in this sketch are deliberately unspecified. The compiler may choose a different layout to satisfy each member's alignment requirements. `offsetof(struct event, count)` can report the actual offset for the selected implementation, but using that result does not create a portable external file format.

The structure's *semantic* equality should compare members under a documented policy. Comparing `sizeof(struct event)` raw bytes with `memcmp` can read padding values that need not match even when corresponding members compare equal.

5.19 Representation is not serialization

A common mistake is to write a structure to disk with:

```
fwrite(&event, sizeof event, 1, stream);
```

This can be acceptable for a tightly controlled, versioned, same-ABI internal checkpoint under a deliberately chosen contract, but it is generally unsuitable as a portable protocol or persistent format. The bytes may contain padding, depend on endianness, encode a compiler-specific `bool` representation, and change when the structure definition or ABI changes.

A portable message chooses field widths and byte order separately. For example, an event wire format might define one octet for kind, four little-endian octets for count, and one octet containing 0 or 1 for validity. Six octets are then required regardless of the native structure's `sizeof`, assuming eight-bit octets as required by the format.

A field encoder should write each octet deliberately. It should reject a count outside the format's chosen numerical range if the C source type could represent larger values, and should not read from or write to buffers before checking their declared bounds.

5.20 Padding does not solve ownership

Suppose a structure contains a pointer to allocated text:

```
struct record {
    char *name;
    size_t length;
};
```

Assigning one `struct record` to another copies the pointer and length. It does not copy the characters. If both records now behave as independent owners, destroying each will attempt to free the same allocation.

A correct owning interface must define one of several policies. It can forbid copying, offer an explicit deep-copy operation, or transfer ownership from one record to another while clearing the old owner. C permits expressing each policy, but it does not automatically enforce it.

The same caution applies when records are moved inside a sorted array. Shifting a structure containing a pointer copies that pointer. If the data structure later destroys an overwritten record incorrectly, it can leak or double-release storage. A sorted array of fixed, inline word buffers avoids much of this complexity because each record carries its bytes directly.

5.21 Flexible array members, correctly sized

A flexible array member is a final array member with no declared bound. It permits one allocation to contain a header and a variable-length trailing sequence.

```
struct message {
    size_t count;
    unsigned char payload[];
};
```

An allocation for `count` bytes should use a checked size calculation. The language permits padding differences between `sizeof(struct message)` and offsets reported for its flexible member, so a conservative size computation can be based on `sizeof(struct message)` plus the payload byte count, after checking that the sum fits in `size_t`.

```
if (count > SIZE_MAX - sizeof(struct message))
    return nullptr;
struct message *m = malloc(sizeof(struct message) + count);
```

The snippet requires the appropriate headers and still must check `malloc` failure. The count may also need an application maximum smaller than `SIZE_MAX`. Because `payload` is an array of `unsigned char`, no multiplication is needed for its element size. A flexible array of larger element type would need both checked multiplication and checked addition.

There are important restrictions on structures with flexible array members. They are not a universal substitute for a vector type; the object still cannot grow by simply changing its `count` field. Resizing its allocation would require `realloc` or constructing another object, with the usual pointer invalidation and failure rules.

5.22 A tagged union as a small protocol model

A tagged union stores one of multiple logical value shapes:

```
enum observation_kind { OBS_INTEGER, OBS_REAL };

struct observation {
    enum observation_kind kind;
    union {
        long integer;
        double real;
    } value;
};
```

The tag describes how the program should interpret the union. An input decoder must validate the tag before selecting the member, and it must reject unsupported values. C enums can hold values produced by casts or corrupt external data that do not correspond to any named enumerator.

A useful test constructs both valid variants, then constructs an invalid tag and confirms the consumer rejects it without reading an unintended member. This is a logical correctness test as well as a type-access safety discipline.

5.23 A ring buffer as a state machine

The fixed-capacity ring buffer in Chapter 5 can be described entirely through its state variables `head` and `count`. No pointer allocations are required.

The queue is empty when `count == 0` and full when `count == capacity`. A successful push stores at `(head + count) % capacity` and increments `count`. A successful pop reads `data[head]`, advances head modulo capacity, and decrements count.

To prove memory safety, assume initially that `head < capacity` and `count <= capacity`. Before a push, reject `count == capacity`; the calculated tail is a valid array index. Before a pop, reject `count == 0`; head is a valid index by invariant. Each update preserves the invariants.

The state-machine description does not establish concurrency safety. If an interrupt producer and task consumer modify the same `count` without synchronization, the program can have a data race. Chapter 16 develops a distinct SPSC design with atomic publication positions; the two queues are not interchangeable simply because both have circular arrays.

5.24 Representation tradeoffs for embedded firmware

Choose a representation using four criteria: maximum storage, worst-case operation cost, ownership complexity, and concurrency needs.

A fixed array has predictable allocation and simple indexed access. An intrusive list avoids external node allocations but requires careful membership and lifetime rules. A heap-backed list permits incremental growth but has per-node allocation costs and fragmentation concerns. A fixed pool provides bounded capacity and explicit slot reclamation. A ring buffer is appropriate for FIFO traffic and can be designed for specialized producer-consumer roles.

The best representation is not the one that uses the most advanced C syntax. It is the one whose invariants, resource bounds, and interfaces are easiest to verify for the application.

Additional exercises

5.13 Use `offsetof` to inspect a structure on two compilers without assuming the offsets are universal. **5.14** Write a deep-copy function for a record containing an owned string, including allocation-failure cleanup. **5.15** Derive the checked allocation size for a flexible array member of `uint32_t`. **5.16** Construct an invalid tagged-union value and define a safe error result. **5.17** Prove the ring buffer's `head` and `count` invariants after push and pop.

5.25 A fixed pool for bounded record ownership

An application can know its maximum number of live objects even when it does not know the exact times at which they will be created and released. A sensor interface might allow four outstanding requests. A telemetry task might buffer sixteen fixed-size events. A driver might have a small set of descriptors, each of which is either available or currently assigned to a client.

A fixed pool stores those objects in a statically bounded array and maintains availability metadata. It offers operations resembling allocation and deallocation without repeatedly calling `malloc` or `free`.

The following example uses four integer-valued slots, but the idea generalizes to records:

```
enum { SLOT_CAPACITY = 4 };

struct slot {
    bool occupied;
    bool retired;
    unsigned generation;
    int value;
};

struct pool {
    struct slot slots[SLOT_CAPACITY];
};
```

Every `struct pool` owns its slots. Its storage lasts as long as the containing object. An automatic `struct pool` is destroyed when its lifetime ends; a static one lasts for the relevant program lifetime. Clients must not assume an acquired reference remains meaningful after its owning pool is destroyed.

A pool's maximum storage is fixed and can be inspected with `sizeof(struct pool)` on each target. Padding may make this larger than the sum of the visible field sizes. That matters in a small RAM budget but does not change the pool's logical capacity.

5.26 Why a raw index is not always enough

The simplest handle for a slot pool is an array index. A client receives index one, uses the record at index one, and eventually releases it. But the same index might later be reassigned to a different client.

Consider the sequence:

```
Time    Slot 1               Client A              Client B
----    ------------------   --------------------  ----------------
t0      free                 —                     —
t1      owns value 10        receives index 1      —
t2      released             retains stale 1       —
t3      owns value 99        still has stale 1     receives index 1
t4      owns value 99        reads using index 1   —
```

At time `t4`, checking only that index one is occupied would allow A to read B's new logical object. The memory address is valid; the bug is *logical lifetime*, not necessarily an out-of-bounds access.

A generation-tagged handle adds a number that changes when the slot is released:

```
struct handle {
    size_t index;
    unsigned generation;
};
```

The pool accepts a handle only if its index is valid, the slot is currently occupied, and the supplied generation matches the slot's current generation. A stale handle with an older generation is rejected.

The idea is analogous to versioning a resource identifier. It helps detect a stale reference even when the physical slot has been reused.

5.27 The pool's invariants

A pool operation can be reviewed using a small set of statements:

1. Every slot is either occupied or available. An occupied slot is not considered available to allocation.
2. A retired slot is never allocated again.
3. A successfully acquired handle identifies an occupied slot whose generation equals the handle's generation.
4. A successful release changes the slot's state so the old handle can no longer access it.
5. A failed acquire or release does not accidentally modify an unrelated slot.
6. Slot indexes are checked before array access.

The first allocation searches for an available, non-retired slot. Because `SLOT_CAPACITY` is a fixed small number, a linear scan is simple and takes a bounded number of slot examinations.

The implementation fills the selected slot and publishes the handle only after success. If no slot is available, it returns false and leaves the caller's out-parameter unchanged.

This is a useful *strong failure guarantee*. A caller can keep its previous handle after a failed request without wondering whether the pool partly updated it.

5.28 Stale handles and generation wraparound

Generation counters introduce their own corner case. If the counter is `unsigned`, incrementing it after `UINT_MAX` wraps to zero under C's unsigned arithmetic rules. A very old stale handle whose generation was zero might then match a newly assigned object after enough reuse cycles.

For teaching purposes, the companion `examples/slot_pool.c` avoids that possibility by retiring a slot when its current generation reaches `UINT_MAX`. A retired slot is never reused.

This policy exchanges eventual capacity loss for a simple lifetime argument: generations for a particular reusable slot never repeat. In most short-lived applications, exhaustion of a wide unsigned generation counter may be impractical, but a correctness proof should not rest on the assumption that an event is merely unlikely.

Other systems may choose much wider generation numbers, process-lifetime guarantees, or explicit epoch mechanisms. Their design must account for the length of time old handles can remain in circulation and for what happens after wraparound.

A generation number also cannot prevent every misuse. The teaching handle does not include the identity of its owning pool. If two different pool objects happen to have matching slot indexes and generation values, passing a handle from one pool to the other might be accepted. An API that must reject foreign-pool handles needs an additional identity mechanism or a type/interface boundary that prevents such cross-pool use.

5.29 No heap is not the same as no ownership

Although this pool performs no dynamic allocation, callers still have ownership obligations. One client must know when an acquired logical object can be released. Releasing it twice is invalid. Using the old handle after release is invalid. Destroying the pool while clients hold handles invalidates their logical references.

The storage's location does not eliminate those rules.

A pool also is not automatically thread-safe. Two concurrent clients searching for an available slot may both observe the same slot as free. Correct concurrent allocation requires synchronization or an algorithm specifically designed for multiple participants.

If an interrupt handler accesses the pool, the chosen synchronization mechanism must be appropriate for that context. A general mutex may not be callable from an ISR, and ordinary unsynchronized state updates may constitute data races in threaded C.

5.30 The practical design choices

A four-slot pool that scans at most four entries is easy to analyze. With thousands of entries, an explicit free list or bitmap may locate available slots more efficiently. Those representations introduce their own invariants and corruption hazards, so complexity should follow measured requirements.

A fixed pool is often appealing when:

- maximum simultaneous occupancy is known;
- allocation and release must be predictable;
- object sizes are uniform;
- heap fragmentation is undesirable;
- stale client handles should be detected.

It is less natural when individual items have very different sizes, when the maximum capacity is unknown, or when clients require independently resized objects.

The broader lesson is to choose a data structure around its observable contract: which resources are available, who owns them, when identities can be reused, and how failure is reported.

Worked trace

Starting from an empty four-slot pool, acquire values 10, 20, 30, and 40. The pool is full. A fifth acquire fails and leaves the requested output handle unchanged.

Release the handle for slot one. Its generation increments. Acquire another record with value 99; slot one can be reused, but it now has a different generation. The original handle for slot one fails on lookup, while the new handle succeeds.

The standalone program tests this trace, double release, stale reads, and the retirement path at `UINT_MAX`.

Exercises

5.18 Add an explicit pool identifier to each handle and state how identifiers are kept unique during the pool's lifetime. **5.19** Add a `pool_write` operation that validates generation before updating the value. **5.20** Compare a scan-based pool with a bitmap-based pool for 64 fixed records. **5.21** Explain why permanent retirement avoids a generation wraparound alias but reduces capacity. **5.22** Design the synchronization required when multiple threads may acquire slots concurrently.

6. Input, Output, and Files

C's hosted I/O library is built around streams. A stream may refer to a terminal, ordinary file, or implementation-specific source. Errors should be treated as part of the computation.

6.1 Reading lines safely

```
#include <stdio.h>
#include <string.h>

int main(void)
{
    char line[256];
    while (fgets(line, sizeof line, stdin) != nullptr) {
        size_t n = strlen(line);
        bool complete = n > 0 && line[n - 1] == '\n';
        printf("bytes=%zu complete=%s\n", n, complete ? "yes" : "no");
    }
    if (ferror(stdin)) {
        fputs("input error\n", stderr);
        return 1;
    }
}
```

A full buffer does not imply a complete line. The next `fgets` may read the rest. Embedded applications frequently need a bounded protocol parser rather than a line-oriented parser.

6.2 Formatted I/O

`printf` uses a format string to interpret subsequent arguments. A mismatch between conversion specifications and argument types can yield undefined behavior. Never use untrusted input as the format string:

```
printf("%s", user_text); /* fixed format */
```

For integers of exact width, use `<inttypes.h>` formatting macros. For `size_t`, use `%zu`. Avoid relying on the fact that two types happen to share a representation on one platform.

6.3 Binary I/O

```
unsigned char buffer[1024];
size_t got = fread(buffer, 1, sizeof buffer, stdin);
if (got < sizeof buffer && ferror(stdin)) {
    fputs("read failure\n", stderr);
}
```

A binary protocol needs a documented byte order, field widths, bounds, and error policy. Reading bytes into a native structure is not sufficient. The portable approach decodes fields explicitly.

6.4 Files and cleanup

```
FILE *fp = fopen("output.txt", "w");
if (fp == nullptr) {
    perror("output.txt");
    return 1;
}
if (fputs("record\n", fp) == EOF || fclose(fp) == EOF)
    return 1;
```

The short example needs care: if `fputs` fails, short-circuit evaluation skips `fclose`, leaking the open stream. Real cleanup logic must close the file on every path, then report either the write error or the close error. A successful write call does not guarantee that data is durably stored.

Exercises. 6.1 Write a line counter that identifies I/O failure. 6.2 Read a little-endian 32-bit unsigned integer from four bytes. 6.3 Rewrite the file example so it always closes its stream.

6.5 The difference between text and bytes

A text stream may transform representations, notably line endings on some systems. Binary I/O is suited to exact bytes. If a file is a protocol packet or serialized image, open it with `"rb"` or `"wb"` in hosted implementations and define the byte format explicitly.

A read of `n` bytes is not guaranteed to deliver `n` bytes in one call. A short `fread` result must be interpreted using `feof` and `ferror`, and the caller must decide whether a short record is valid or malformed.

6.6 A bounded line counter

Some tasks need only one pass over input and almost no memory:

```
#include <stdio.h>
#include <stdbool.h>
#include <limits.h>

bool count_newlines(FILE *stream, unsigned long *result)
{
    if (stream == nullptr || result == nullptr)
        return false;

    unsigned long count = 0;
    int ch;
    while ((ch = fgetc(stream)) != EOF) {
        if (ch == '\n') {
            if (count == ULONG_MAX)
                return false;
            ++count;
```

```
        }
    }
    if (ferror(stream))
        return false;
    *result = count;
    return true;
}
```

The function counts newline characters, which may differ from a human notion of lines if the final input has no trailing newline. Documenting that distinction matters more than choosing one rule universally. The caller owns the stream, so this function neither opens nor closes it.

6.7 Reading numeric text

Using `scanf` indiscriminately can make validation difficult: conversion success, partial input, range errors, and leftover text all matter. For untrusted input, reading a bounded line and then using `strtol` or another checked parser often exposes the contract more clearly.

A parser should specify whether whitespace is accepted, whether leading `+` or `-` is legal, whether trailing characters are rejected, and the range of accepted values. A caller should not infer all of those policies from the function name alone.

6.8 File cleanup paths

A file-writing function commonly needs two independent success checks: whether the write succeeded, and whether closing the stream succeeded. Avoid short-circuiting away the close operation:

```
bool ok = fputs("record\n", fp) != EOF;
if (fclose(fp) == EOF)
    ok = false;
```

This makes the cleanup action unconditional once the stream has been opened successfully. It does not guarantee that written data is durable after a power loss, since operating systems and storage devices may buffer writes.

6.9 Exercises

6.4 Define whether a final non-newline-terminated sequence counts as a line, then implement your policy. **6.5** Decode a four-byte length prefix and reject a payload that would exceed a caller-specified capacity. **6.6** Compare a binary stream with a text stream on a platform that transforms line endings. **6.7** Design error reporting that preserves both a write failure and a close failure. **6.8** Explain why a successful `fwrite` is not a promise of power-failure durability.

6.10 A byte-stream copier

Reading and writing streams is not a single operation when short transfers are possible. A robust file copier reads a bounded block, then writes every byte obtained or reports an error. The `fwrite` interface returns the number of complete elements written; when the element size is one byte, that count can be compared directly with the requested byte count.

The repository's `examples/file_copy.c` illustrates this pattern for standard input and output. It checks `ferror(stdin)` when a short read occurs and `fflush(stdout)` before returning success. It still does not guarantee durability against an operating-system crash or power loss. That stronger guarantee is outside the portable C stream contract and may require platform-specific synchronization.

A binary copier preserves byte values; a text-processing program may interpret newline conventions, encodings, and locale. The distinction should be part of the program's specification.

6.11 Why parsing after input is often easier

An unchecked numeric scan mixes syntax recognition with conversion. It can be difficult to determine whether a trailing character remains or a number was only partially consumed. Separating operations makes behavior explicit: read a bounded line; determine whether the entire line matches the required grammar; then perform checked numeric conversion.

For example, a temperature configuration file might permit optional leading whitespace but forbid any trailing nonspace characters. That is a different grammar from a telemetry message consisting of exactly four decimal digits. Neither grammar is universally correct. A well-designed parser names and tests the distinction.

Use `strtol` or `strtoul` when their grammar matches the intended input. Set `errno` to zero before the call, use the end pointer to detect whether any digits were converted and whether unconsumed input remains, and inspect `ERANGE`. Then check that the result fits the application type. A successful library conversion is not proof that all application constraints were met.

6.12 Error reporting preserves the cause

A function returning only zero or one may be sufficient for a toy program. In a larger system, distinguish an open failure, a read error, malformed content, a short write, and a close failure. These conditions have different remedies and should be logged or propagated without losing their meaning.

If a write fails and closing the file also fails, what should the caller see? One design returns the primary failure and logs cleanup failures separately. Another uses a structured result containing both statuses. The choice should be deliberate. Simply overwriting the first error with the last operation's result can hide the original problem.

6.13 Buffering and interleaving

Stream output may be buffered. `fflush` requests transmission of the C library's buffered output to the host environment, not permanent storage. Mixing multiple threads' writes or mixing `stdio` with lower-level operating-system file descriptors can introduce ordering and buffering issues. Portable code should synchronize concurrent stream operations appropriately and should not assume that a line-oriented message is always an indivisible transaction.

6.14 Designing a testable input function

Instead of making a parser call `getchar` itself, define a function accepting `const unsigned char *input` and a `size_t` length. Then a small adapter reads from a stream and passes counted buffers to the parser. Host tests can exercise empty input, malformed records, and boundaries without inventing a terminal or fake file.

Further problems: Specify how a line reader handles a trailing newline, embedded NUL, and a buffer too small for an entire line. Implement a status type distinguishing EOF from error. Explain why `ferror` must be consulted after an I/O operation fails rather than treating every `EOF` result as a clean end of data.

6.15 Streams, records, and short transfers

The C stream library exposes a sequence of bytes or characters; it does not magically preserve an application's record boundaries. Reading 100 bytes from a binary file says nothing about where

one protocol record ends. If the format has a length field, the parser must validate that field and obtain exactly the corresponding number of bytes or explicitly report incomplete input.

A complete fixed-size record reader may loop until it has accumulated the required count, encounters an error, or reaches clean EOF before the record is complete. These outcomes are different. In particular, clean EOF between records is normal, whereas EOF in the middle of a required record often indicates truncation.

For an interactive terminal, line processing introduces other questions. What counts as end-of-line on this host? Is the input a text stream or a binary stream? How should excessively long lines be rejected or drained? Do not use `fgets` as if it guaranteed one whole logical line when its buffer is finite.

6.16 Write operations have partial-success semantics

A `fwrite` call returns the number of complete elements written, and a short result signals failure under its contract. For arbitrary byte counts, choose an element size of one and compare the returned count with the requested bytes. In a broader abstraction over operating-system calls, writes may also return a positive short count and require a loop to finish the request. Do not conflate the contracts of different APIs.

A robust writer distinguishes the amount of data accepted, the error status, and whether retry is meaningful. On nonblocking devices, a temporary inability to write should not necessarily be treated as fatal. Conversely, looping forever on a zero-byte result would violate bounded-progress expectations.

When writing a file, checking `fclose` remains important because some errors can surface after prior buffered operations appeared successful. Even a successful close does not universally mean data is physically durable after sudden power loss.

6.17 Use a parser independent of the transport

A reusable packet decoder can accept a pointer, length, and output destination:

```
bool decode_packet(const unsigned char *bytes,
                   size_t length,
                   struct packet *out);
```

The parser does not call `getchar`, open a file, or read a UART register. Instead, adapters gather bytes from those sources and call the same decoder. A desktop test can supply a deliberately truncated packet, an invalid length, or a checksum mismatch without requiring hardware. Device-level tests can then focus on acquisition and timing.

There is a cost: the adapter must buffer enough bytes for a record, or the parser must be designed as a streaming state machine. The right choice depends on maximum frame size and available RAM.

6.18 Standard output and error output

`stdout` and `stderr` serve different roles. Machine-readable output should be written to `stdout`, while explanatory error messages usually belong on `stderr`. A program that produces a table can then be piped into another tool without contaminating its data with diagnostics.

For command-line tools, establish an exit-status convention: zero for success and nonzero for failure. If there are many error categories, the exact exit values may be less informative than a precise error message, but the contract should still be predictable.

An embedded firmware component may not have any of these streams. It may instead return a structured status to its caller or emit a diagnostic over a separately configured telemetry channel. The error model survives even when the transport differs.

6.19 A defensive byte-count example

Suppose a header says the payload is `length` bytes, and the caller has a `capacity`-byte buffer. Reject an oversized payload before copying. If the total frame needs five extra octets, check `capacity >= 5` and then `length <= capacity - 5`. A direct test of `length + 5 <= capacity` can overflow unsigned arithmetic before the comparison.

This tiny transformation is a recurring pattern in systems programming: convert a potentially overflowing sum into a subtraction after establishing its precondition.

Exercises

6.9 Design a status enumeration for clean EOF, incomplete record, I/O error, and valid record. **6.10** Implement a line reader that rejects overlong input but drains until newline so it can resynchronize. **6.11** Write a host test for a packet decoder that uses no filesystem calls. **6.12** Explain when checking `fflush` and `fclose` matters and what neither operation guarantees about power-loss durability.

6.20 A line is a record, not a read call

A C text stream supplies characters, not guaranteed whole application records. Calling `fgets` with a 16-character buffer does not promise to read an entire logical line: the line may be longer than the buffer or may be split across multiple calls. A robust line-reader API must define how it handles maximum length, separators, final input without newline, and malformed records.

In `examples/line_reader.c`, the grammar is intentionally narrow: at most 16 non-newline, non-NUL bytes form a line. A newline terminates a record and is not copied into the returned text. A final nonempty line with no newline is accepted on clean EOF. Empty input returns an explicit end status rather than an empty line.

Why forbid NUL? The example returns a conventional terminated C string; silently accepting an embedded zero would make later `strlen` or `strcmp` calls interpret only the prefix. Rejecting NUL is one valid policy. A counted binary-record reader could legitimately accept it without treating its result as a C string.

6.21 Define error states before writing the loop

The example uses five result classes:

Status	Meaning	Public output
`LINE_OK`	A complete line was read; includes a final unterminated line	New terminated text and leng
`LINE_END`	Clean EOF before any more record bytes	Unchanged
`LINE_TOO_LONG`	More than 16 ordinary bytes before delimiter	Unchanged
`LINE_INVALID`	Invalid parameters or a forbidden NUL byte	Unchanged
`LINE_IO_ERROR`	Stream reading reported an error	Unchanged

Different libraries can choose a different set of statuses, but their meanings should not depend on what happens to be in the destination after a failed call.

The reader builds a candidate in local storage and publishes it to the caller only on success. This creates a useful invariant: a result other than `LINE_OK` leaves the existing text and reported length untouched. A caller can safely retain its previous valid line after a rejected record.

6.22 Drain malformed input to resynchronize

An overlong record creates a design choice. Returning immediately when the 17th byte arrives would leave the remainder of that record in the stream. A later call might then interpret its suffix as a fresh line, even though it never began at a proper record boundary.

Instead, the reader remembers that the record is invalid and keeps consuming bytes until the next newline or EOF. It ignores their content while retaining the failure reason. The *next* call therefore begins at the start of the next record.

This design guarantees bounded **memory** but not bounded **time** per call: a malicious or broken input stream could provide an arbitrarily long line without a newline, forcing the reader to keep discarding bytes. A real-time service may need a byte-consumption budget, a timeout, a nonblocking incremental interface, or a policy to close the stream.

Memory bounds and execution-time bounds are distinct guarantees. A correct statement of the example's contract should acknowledge both.

6.23 The EOF corner cases

There are at least four interesting endings:

- The stream is empty: the first read sees clean EOF and returns `LINE_END`.
- The last record ends with newline: the reader returns that record; the next call returns `LINE_END`.
- The last record contains text without newline: the reader returns it once and the next call returns `LINE_END`.
- The final record exceeds the limit: the reader returns `LINE_TOO_LONG`, even if the last byte was followed by EOF rather than newline.

Do not confuse `fgetc` returning `EOF` with certainty about *why* it stopped. The result may indicate clean end of file or a stream error. Check `ferror` and handle the reported error rather than treating every sentinel as a normal end.

It is also important to test the exact capacity boundary. If `MAX_LINE` is 16, a line containing 16 ordinary bytes is valid and fits in `char output[17]` with a terminating zero. A line containing 17 ordinary bytes is not valid. Off-by-one mistakes at the terminator are common.

6.24 The test uses a real C stream

A portable hosted C test can create a temporary file with `tmpfile`, write test data, reposition with `fseek`, and then call the line-reader repeatedly. This is valuable because the test exercises the actual `FILE *` interface without depending on a named file path or a user-supplied input file.

The example checks its write count and its repositioning result before reading. It tests the complete record sequence: short valid line, overlong line, valid line after recovery, exactly full line, final line without newline, and clean EOF. It also verifies that the rejected line did not modify either output.

The harness itself must clean up the temporary stream on failures. In a real application, the return value from closing an output stream can reveal delayed write errors. An input-only temporary-stream test still should close its resource rather than leaking it.

6.25 Text and binary streams have different concerns

Text mode may perform implementation-defined translations of line-ending representations. Binary mode is the right abstraction for a protocol defined in exact octet values, subject to the underlying implementation's byte assumptions. A file containing UTF-8 text is still a sequence of octets that needs explicit encoding interpretation; opening it as a C text stream does not magically supply Unicode grapheme semantics.

The book separates three layers:

1. Transport obtains data and reports I/O conditions.
2. Framing identifies complete application records.
3. Parsing interprets the record's content and reports syntax or range errors.

A serial device might deliver an arbitrary number of bytes per interrupt. A network receiver might receive one protocol record in several fragments. Keeping framing and parsing separate prevents code from assuming that the transport's chunk boundary equals a message boundary.

6.26 Move from a blocking function to a state machine

The teaching line reader calls `fgetc` until one record is complete or rejected. That is suitable for small hosted command-line tools. A firmware task that must remain responsive could instead use an incremental interface:

```
enum feed_result {
    FEED_MORE,
    FEED_RECORD,
    FEED_REJECTED
};
```

Each call accepts one byte or a counted fragment and updates a parser-state structure. A result of `FEED_MORE` means that additional input is required; `FEED_RECORD` publishes one complete validated record; `FEED_REJECTED` means a policy-defined invalid or overlong record has been discarded.

The state would track current length and whether the parser is draining an invalid record. The input adapter determines when and how often to call it. Bounded processing per received byte then becomes possible without an unbounded blocking read loop, although the overall protocol still needs timeouts and backpressure policies.

A robust incremental design also needs a reset operation and an explicit rule for stream termination in the middle of a record.

Additional exercises

6.13 Extend the line reader to report whether a rejected record was overlong or contained NUL, without changing output on failure. **6.14** Convert it to an incremental, one-byte-at-a-time state machine. **6.15** State the difference between a bounded buffer and a bounded worst-case execution time. **6.16** Test a valid line exactly at the capacity boundary, and a final valid line without newline. **6.17** Explain why using `strlen` on an arbitrary counted byte buffer is unsafe.

7. Dynamic Memory and Ownership

Every object occupies storage for a particular lifetime. Automatic objects usually live for one block invocation. Static objects live for the execution of the program. Dynamic objects, obtained through allocation functions, persist until explicitly freed.

7.1 Allocation is a request

```
#include <stdlib.h>
#include <stdint.h>
#include <stddef.h>

int *make_array(size_t count)
{
    if (count == 0 || count > SIZE_MAX / sizeof(int))
```

```
        return nullptr;
    return malloc(count * sizeof(int));
}
```

The multiplication is checked before it can overflow. This interface rejects zero-sized requests. Other interfaces may choose different zero-size semantics, but they should document them. New storage from `malloc` has indeterminate contents. Use `calloc` for zero-initialized storage, while still checking the requested dimensions.

7.2 One owner, one release

```
int *samples = make_array(100);
if (samples == nullptr)
    return 1;
/* use samples[0] through samples[99] */
free(samples);
samples = nullptr;
```

The variable `samples` is not the allocated object; it holds an address. After `free`, all aliases that would access the old object are invalid. Resetting one variable does not repair copies of the address.

A useful API states: who allocates, who frees, whether buffers may be shared, which functions can retain pointers, and which operations invalidate earlier pointers. These are conventions implemented by programmers, not runtime guarantees supplied by the language.

7.3 Resize without losing the original

```
void *tmp = realloc(buffer, new_size);
if (tmp == nullptr) {
    /* buffer remains allocated when new_size > 0 */
    return false;
}
buffer = tmp;
```

The example assumes `new_size > 0` and that `buffer` is a valid reallocatable allocation. Assigning `realloc` directly to the only saved pointer risks a leak on failure. Successful reallocation may move storage, invalidating interior pointers.

7.4 Fixed-capacity alternatives

Embedded systems often trade flexible allocation for predictable resource bounds. A fixed-size arena can allocate successive objects from a caller-provided array. It must honor alignment and report exhaustion; an arena reset invalidates all objects in it. Pools that recycle uniform slots provide a different tradeoff. Neither allocator model is automatically thread-safe.

7.5 A resource table

Resource	Acquisition	Release
Heap block	`malloc`, `calloc`	`free`
File stream	`fopen`	`fclose`
Automatic array	enter block	leave block
Static object	program initialization	program termination

Exercises. 7.1 Implement checked dynamic-array growth. 7.2 Write a fixed-capacity arena with explicit alignment policy. 7.3 Identify double-free, leak, and use-after-free paths in a linked-list teardown.

7.6 A checked growable vector

A dynamic array needs three measurements: the pointer to the allocation, the number of initialized elements, and its capacity. The invariant is `length <= capacity`. Growing is exceptional, not required on every insertion.

```
struct vector {
    int *data;
    size_t length;
    size_t capacity;
};
```

A growth policy might double capacity, but the multiplication needs an overflow check. When capacity cannot double, the allocator must either use a smaller sufficient capacity or report failure. The success of `realloc` invalidates pointers into the old allocation even if the returned address happens to equal the old address; callers should obtain fresh pointers from the vector after a change.

The vector's *ownership* can be stated simply: it owns `data`, its caller owns the vector structure, and exactly one destroy operation releases the allocation. Copying the structure by assignment copies its pointer, not its ownership. Two such copies must not both be destroyed.

7.7 An arena with an explicit alignment policy

An arena hands out storage from a fixed block. Its main benefits are bounded overall consumption and cheap bulk reset. Correct alignment must be established for each allocation. A simple arena supporting only objects with alignment no stricter than `max_align_t` can use storage suitably aligned at its base, round offsets upward without overflow, and reject requests that do not fit.

Alignment is part of the contract, not a detail to ignore because a particular CPU tolerates misaligned loads. DMA buffers and cache maintenance impose further target-specific constraints that a generic C arena does not automatically meet.

7.8 Failure-safe initialization

Consider creating an object that owns a buffer and opens a file. If the second step fails, the first resource must be released. Design initialization so that either every required resource is acquired, or every partial acquisition is undone. A cleanup label can be a sensible C technique when a function has many resource-acquisition steps.

```
int status = 1;
void *buffer = malloc(1024);
if (buffer == nullptr) goto done;
FILE *file = fopen("data.bin", "rb");
if (file == nullptr) goto done;
/* use resources */
if (fclose(file) != 0) goto done;
status = 0;
done:
free(buffer);
return status;
```

This sketch requires `<stdio.h>` and `<stdlib.h>`. It closes the file on the successful path, but it is not a general model for cleanup after all possible failures within the use block. For several resources, initialize handles to known values and centralize cleanup that checks each handle.

7.9 Exercises

7.4 Implement a vector with `init`, `append`, and `destroy`. **7.5** Demonstrate how copying a vector structure creates two pointer aliases. **7.6** Design a resettable arena and describe which pointers become invalid after reset. **7.7** Write a function that acquires two resources and releases both on every failure path.

7.10 Distinguish capacity from initialized length

An allocation of space for ten integers does not automatically initialize ten integer values. With `malloc`, reading elements before they are assigned is invalid. With `calloc`, the allocated bytes are zero-initialized; this yields the all-bits-zero representation of every byte, which for integer objects gives the value zero. For pointers and some other types, all-bits-zero bytes are not universally a promise of the semantic null or zero value on every conceivable C implementation.

A vector records both allocated *capacity* and initialized *length*. An append operation writes an element into unused capacity before increasing the length. A destroy operation releases the allocation once and invalidates all borrowed element pointers.

When designing a C API, use distinct names for these concepts. Confusing `length` and `capacity` is a recurring source of out-of-bounds reads and writes.

7.11 Zero-size requests in C23

A historical compatibility trap deserves explicit attention. In C23, calling `realloc(ptr, 0)` has undefined behavior. Avoid using it as a shorthand for freeing a block. Instead, use `free(ptr)` when the desired operation is release, and define an explicit empty-state policy in your container API.

The behavior of `malloc(0)` is not a portable way to obtain a usable object. An API can reject zero-sized element arrays, represent them with a null pointer and zero capacity, or use another documented convention. The key is consistency at every call site.

7.12 Checked growth without losing storage

To grow an array, first choose a new capacity greater than or equal to the required length; then check whether `capacity * sizeof element` fits in `size_t`; then call `realloc` using a temporary pointer. Only on success do you update the stored pointer and capacity.

The example vector in `examples/vector.c` follows this approach for simple `int` elements. Because it uses a doubling policy, a very large capacity may be rejected rather than trying every possible smaller capacity. That is an acceptable documented policy for a teaching example; it is not the only valid design.

No pointer into an allocation may be assumed valid after a successful `realloc`. Recompute interior pointers from the returned base. A failed nonzero-size `realloc` leaves the old allocation intact, which is why assigning the result directly to the only owner is dangerous.

7.13 Alignment and arenas

C object types have alignment requirements. A byte arena can hand out storage for types only when it aligns the returned address appropriately and has enough remaining bytes. Rounding an

offset upward can overflow; check the arithmetic. A generic aligned allocator must consider the actual address of the backing allocation, not just an offset from an arbitrarily aligned base.

If an arena is reset, every allocation it had issued becomes invalid for application purposes, even though the bytes physically remain. The arena is attractive when a whole group of temporary objects has the same lifetime. A free-list pool is better when individual fixed-size objects must be reclaimed independently.

7.14 Failure injection as a learning tool

Ordinary runs rarely exercise allocation failures. For a unit test, wrap `malloc` and `realloc` behind function pointers or an allocator object, then provide a test implementation that fails on a specified request. Confirm that failure neither leaks the old allocation nor changes the vector's logical length or existing element values.

This is a useful general lesson: error paths deserve executable tests, not just reassuring comments.

Further problems: Modify the vector to distinguish allocation failure from arithmetic overflow. Implement `vector_reserve` that leaves all state untouched on failure. State the ownership implications of `struct vector copy = original;`. Write an arena allocation test covering zero bytes, alignment and exhaustion.

7.15 Ownership as an interface contract

C allocation functions do not attach a runtime owner to each block. Ownership is a design convention: one component is responsible for eventually releasing a resource, while other components may temporarily borrow access to it.

A vector may own its element allocation. A slice into that vector borrows the memory. When the vector grows with `realloc`, every previously borrowed element pointer must be considered invalid, even if the new allocation happens to return the same numerical address.

A useful public header should state whether a function borrows its input, retains it after returning, or transfers ownership. For example, `vector_append` can copy an `int` value and retain no caller pointer. An asynchronous send operation that merely borrows a buffer may require the caller to keep that buffer alive until a completion callback. An API that copies the payload on submission has a different resource and timing contract.

These distinctions do not require a new language feature. They require precise documentation and tests.

7.16 Strong and basic failure guarantees

An append operation can promise that a failed allocation leaves the vector exactly as it was: same pointer, length, capacity, and existing elements. That is a strong failure guarantee. A more complex operation might promise only that the structure remains valid and usable, even though its contents have changed; that is a weaker but sometimes acceptable guarantee.

To achieve strong failure behavior, compute requirements locally, allocate into a temporary pointer, and commit state only after success. With `realloc`, a failed nonzero-size request leaves the old object allocated, so the implementation must not overwrite its only saved pointer with a null result.

The choice of a growth factor also affects behavior. Doubling gives amortized efficient append in the ordinary case, but may request more memory than immediately needed and can reject near-maximum capacities. A constrained embedded application may prefer fixed capacity or an arena with known bounds.

7.17 A checked vector implementation

The companion `examples/vector_checked.c` implements a small vector of `int`. It has initialization through an empty aggregate, append with checked capacity arithmetic, retrieval through an out-parameter, and destruction.

Its invariants are:

- `length <= capacity`;
- `data == nullptr` when capacity is zero;
- for positive capacity, `data` is an owned allocation of at least `capacity * sizeof(int)` bytes;
- elements in the half-open range `[0, length)` have been assigned meaningful values.

The append path computes a new capacity, checks both doubling and allocation-byte multiplication, and performs reallocation through a temporary pointer. If allocation fails, its earlier state remains intact. The test checks many appends, retrieval at boundaries, and destruction.

A complete library would also consider whether copying the `struct vector` itself is permitted. A shallow copy creates two apparent owners of the same allocation. For this teaching interface, copying an owning vector by assignment is prohibited by the ownership contract. C cannot enforce that restriction in the type system.

7.18 How to use an arena safely

An arena can provide a bounded amount of dynamic-like storage without making a separate operating-system allocation for each item. But an arena still needs an alignment policy. A request for a `double` object must be aligned suitably for `double`; a byte pointer incremented without regard to alignment is not sufficient.

When resetting the arena, think of every borrowed pointer into it as invalidated by the allocator's application-level lifetime rules. Even if old bytes remain readable on the hardware, the program must not depend on objects whose storage has been repurposed.

A simple arena is not automatically thread-safe. The same considerations apply to a pool of fixed-size objects. Concurrency, memory ordering, and synchronization remain system-level responsibilities.

7.19 Resource cleanup as a state machine

A function acquiring two independent resources can be modeled with states: neither acquired, buffer acquired, buffer and file acquired, and both released. Every failure transition must eventually reach the released state.

One C technique uses a single cleanup block with handles initialized to known safe values. Another uses small helper functions that release partial objects. Neither technique is inherently better in all contexts. The key is that cleanup must run on every exit path that owns resources, including paths triggered by short reads, failures to close a stream, and errors during an intermediate transformation.

Avoid placing release operations inside a boolean expression that short-circuits when an earlier operation fails. Such expressions can unintentionally skip cleanup.

7.20 Testing the allocator's failure paths

Ordinary desktop runs may never fail a small allocation, leaving a large fraction of an owning container's control flow untested. A test allocator can reject the next request predictably. With this seam, verify that failed append does not change length, capacity, pointer, or previously stored values.

Sanitizers can discover use-after-free and certain out-of-bounds errors along executed paths, but they do not prove the ownership design is correct. A written ownership contract plus fault-injection tests is a stronger foundation.

Guided exercises

7.8 Add `reserve` to the vector while preserving the strong failure guarantee. **7.9** Implement an allocator adapter that fails after N successful requests. **7.10** Explain why a shallow copy of the vector must not be destroyed independently. **7.11** Contrast an arena reset with `free` on an individual block. **7.12** Design a fixed-memory message pool for 32 records with allocation, release, and double-release detection.

7.21 A bounded byte arena, step by step

A byte arena offers a useful middle ground between one large static buffer and many individual heap allocations. It manages a fixed byte array and a current offset. A request for `count` bytes either returns the next available region and advances the offset or reports failure without changing state.

The complete example is `examples/arena_bytes.c`. Its essential representation is:

```
enum { ARENA_CAPACITY = 128 };

struct byte_arena {
    unsigned char storage[ARENA_CAPACITY];
    size_t used;
};
```

The invariant is `used <= ARENA_CAPACITY`. The next available byte is `storage[used]` when `used < ARENA_CAPACITY`; `used == ARENA_CAPACITY` describes a full arena.

Imagine that the first request consumes five bytes and the second consumes seven:

```
capacity = 128

+---------+-------------+--------------------------------------+
| 5 bytes |   7 bytes   |            free space                |
+---------+-------------+--------------------------------------+
0         5             12                                     128
                        ^
                        used
```

The next request must fit inside the remaining 116 bytes. The naive condition `used + count <= capacity` can wrap if both values are large unsigned numbers. The safe sequence is to first verify `used <= capacity`, then test `count <= capacity - used`.

This is more than a convenient idiom: it is a short proof that `storage + used` designates a position within or just beyond the array and that the requested range lies within capacity.

A strong-failure allocation function

```
bool arena_take(struct byte_arena *a, size_t count,
                unsigned char **out)
{
    if (a == nullptr || out == nullptr || count == 0 ||
        a->used > ARENA_CAPACITY ||
        count > ARENA_CAPACITY - a->used)
        return false;
```

```
    unsigned char *candidate = a->storage + a->used;
    a->used += count;
    *out = candidate;
    return true;
}
```

This interface explicitly rejects zero-sized requests and leaves the caller's output pointer unchanged on failure. The caller must keep the arena alive. The returned pointer is a borrowed view into arena storage; the caller must not pass it to `free`.

The arena in this example allocates *bytes*, not arbitrarily aligned typed objects. Although the array is perfectly suitable for storing octets, it does not promise all requirements for dynamically creating arbitrary C object types by casting the returned pointer. A general-purpose typed arena needs an alignment policy, suitable backing storage, and careful attention to object access and effective-type rules.

7.22 What reset really means

An arena reset can simply assign `used = 0`. It does not free any separate blocks because the storage belongs to the arena structure. It also does not overwrite the old bytes, which may remain physically present.

Nevertheless, the application's ownership protocol declares every previous allocation invalid after reset. A previous pointer into the array may still numerically point to an accessible byte, but that byte can now belong to a new logical allocation. Reading through the old reference may expose someone else's data or cause a logic error.

This distinction between C object lifetime and *allocator-managed logical lifetime* is important. If the arena is an automatic variable and its enclosing block ends, then the actual C object's lifetime ends as well. If only the arena offset resets, its underlying byte array still exists, but old allocations are no longer valid according to the arena API.

A robust arena user therefore treats reset as an ownership boundary. Reset only after every borrower has finished using the old allocations.

7.23 Alignment is a separate engineering requirement

A processor may require a `double` or 32-bit register value to begin at an address with a particular alignment. Returning the next byte without alignment adjustment may produce an address unsuitable for that type.

There are two parts to a general aligned allocation request: round the current location upward to an address satisfying the requested alignment, then verify that the padding *plus* the requested object size fits. Both arithmetic steps can overflow unless carefully checked.

A safe implementation can either support a fixed set of known alignments or require a documented power-of-two alignment no greater than the backing allocation's guarantee. On freestanding targets, DMA may impose constraints stronger than ordinary C alignment. A buffer can be correctly aligned for its type and still be unusable for an uncached DMA transfer without additional architectural measures.

The byte-arena teaching program avoids claiming those guarantees. That limitation is a feature of its contract, not a defect hidden by a cast.

7.24 Growable vectors versus fixed arenas

A vector owns one resizable contiguous allocation. An arena owns one fixed backing region from which many short-lived byte spans can be borrowed.

A vector is natural when one sequence grows and needs random access. An arena is useful when many temporary objects have a shared lifetime. A fixed pool of uniform records is a third choice when individual objects must be reclaimed without relocating the others.

Compare what happens on exhaustion. A vector may attempt a heap allocation and can fail if memory is unavailable or its required size overflows. An arena simply rejects a request that exceeds its fixed remaining capacity. A pool rejects a request when no free slot remains. Each has a clear failure mode, but none is automatically thread-safe.

For real-time work, do not assume that replacing `malloc` with an arena proves a deadline. You must still account for synchronization, initialization cost, memory contention, and whatever the caller does with returned storage.

7.25 A precise realloc failure guarantee

Suppose a vector has four stored integers and capacity four. A fifth append triggers a resize. Save the pointer, length, capacity, and existing elements before attempting the resize.

If the resizing function returns null for a nonzero request, the original allocation remains allocated. A correct implementation leaves every saved property unchanged. The test in `examples/vector_checked.c` deliberately injects such a failure through a callback, then verifies the strong failure guarantee.

On success, the allocation may be relocated. Even if the new pointer compares numerically equal to the old address, callers must derive new interior references from the returned pointer. A container interface should document that any resize invalidates previously borrowed element pointers.

Two subtle points deserve attention. First, copying the vector structure is a shallow copy, not a transfer of ownership. Second, the library should not expose a pointer into an allocation and then resize it behind the caller's back without a documented invalidation rule.

7.26 Building a cleanup path deliberately

A function that acquires a buffer, opens a file, and transforms data has multiple opportunities for failure. A reliable function records which resources have been acquired and ensures every acquired resource is released on all exits.

An illustrative pattern is:

```
#include <stdio.h>
#include <stdlib.h>

int process_file(const char *path)
{
    int status = 1;
    void *buffer = nullptr;
    FILE *fp = nullptr;

    if (path == nullptr)
        return status;

    buffer = malloc(512);
    if (buffer == nullptr)
        goto cleanup;

    fp = fopen(path, "rb");
    if (fp == nullptr)
        goto cleanup;
```

```
    /* Process input; on error jump to cleanup. */
    status = 0;

cleanup:
    if (fp != nullptr && fclose(fp) == EOF)
        status = 1;
    free(buffer);
    return status;
}
```

This complete function is a *skeleton*: its processing step is omitted. The cleanup logic shows that a failed `fopen` still releases the buffer and a failed `fclose` changes the status. Calling `free(nullptr)` is permitted. A nonzero processing error would be preserved unless code later overwrote it with success, so real implementations should set success only after processing succeeds.

There is no need to avoid `goto` categorically. In C, a clearly labeled cleanup block can be easier to audit than deeply nested conditions, especially when multiple resources have independent acquisition and release steps.

7.27 Ownership exercise with three aliases

Consider:

```
struct vector a = {};
/* append three elements into a */
struct vector b = a;
int *first = a.data;
```

Draw all three objects. The structures `a` and `b` are separate, but both pointers designate the same allocation; `first` also refers to that storage. If one structure calls `free` through its pointer, all aliases become unusable for accessing the former allocation. The other structure must not call `free` again.

A library can avoid such mistakes through opaque owning types, explicit move-like transfer functions, or conventions disallowing structure copying. C does not directly enforce a unique-owner type, so APIs and code review must carry the guarantee.

Additional exercises

7.13 Draw an arena after requests of 5, 7, and 100 bytes; identify the next legal maximum request. **7.14** Explain why `used + count <= capacity` is a weaker validation expression than `count <= capacity - used` after checking `used <= capacity`. **7.15** Add a `remaining` function that never underflows when the arena state is corrupt. **7.16** Modify an arena to support marks and rewinds; define which borrowed views are invalidated. **7.17** Introduce an injected allocation failure in a vector and prove that all prior elements survive. **7.18** Explain why a general-purpose typed arena cannot simply return any byte-aligned pointer to any requested type.

7.28 Pools and arenas solve different lifetime problems

The bounded byte arena earlier in this chapter hands out consecutive spans and reclaims them as a group. Its reset operation makes every previously issued span invalid according to the arena's logical ownership contract. It does not reclaim an individual middle allocation.

A fixed slot pool, introduced in Chapter 5, makes a different choice: each slot can be released independently. An available slot can then be reassigned to another client without relocating the remaining occupied slots.

These differences determine which structure fits a workload:

Property	Byte arena	Fixed slot pool
Backing storage	One fixed byte array	Fixed array of uniform records
Allocation	Advances an offset	Finds an available slot
Individual release	Not normally supported	Supported
Reclamation	Reset whole arena or rewind a mark	Release by validated handle
Address stability	Until reset and reuse	Stable physical slot while occupied
Lifetime identity	Borrowed span plus arena epoch	Index and optional generation
Exhaustion	Insufficient remaining bytes	No available non-retired slot
Typed objects	Requires alignment/access design	Records already have declared types
Concurrency	Requires a synchronization design	Requires a synchronization design

The last row matters. Avoid claiming that a static array is intrinsically thread-safe. Storage policy and synchronization policy are independent.

7.29 A stale handle is a lifetime bug even with valid memory

Suppose client A holds a pointer or index to a record in a fixed pool. A releases the record. Later client B acquires the same physical slot for a different logical object.

A's old pointer could still refer to an address inside the pool array. Dereferencing it may be within the raw bounds of that array, yet accessing B's logical record violates the application ownership model.

Generation-tagged handles help detect this error. A handle contains an index and a generation number. Release invalidates the handle by changing the generation before reuse. A lookup verifies both values.

This does not make the handle a cryptographically secure capability. An arbitrary caller can construct a matching pair of numbers; two distinct pools can reuse the same index and generation; a bug can corrupt the pool metadata. The scheme protects against common accidental stale-reference mistakes under its documented trust and lifetime assumptions.

7.30 Preventing generation reuse

A generation counter that wraps can eventually recreate the value of an old handle. For an unsigned counter, wrapping is *defined* arithmetic—but that defined wraparound may cause a logical identity error.

The `slot_pool.c` example retires a slot when its generation reaches `UINT_MAX`. The slot is no longer eligible for allocation, so no generation value is reused for that physical slot during the pool's lifetime.

This gives a simple identity argument at a cost: repeated acquire/release cycles can eventually reduce the pool's available capacity. A real system might choose a wider generation field and a carefully reasoned lifetime bound, but it should document why generation reuse cannot affect active clients.

The design also needs to distinguish free from retired. A retired slot is not occupied by a client, but the allocator must never select it again.

7.31 Object lifetime and logical lifetime

There are two useful meanings of 'still alive':

- The C object underlying a pool's slot continues to exist because the containing array object is alive.

- The *logical resource allocation* associated with a particular handle exists only between successful acquire and release operations.

After release, the physical slot may remain a valid C object; the logical record identified by the released handle is no longer accessible under the pool API.

This distinction is common in firmware. A descriptor entry, DMA buffer, task-control record, or message slot may remain permanently allocated in RAM while its ownership changes from one subsystem to another.

A correct API should define transitions between logical owners. Access through an outdated owner is wrong even when a simple address sanitizer cannot detect a physical out-of-bounds access.

7.32 Recycling pointers versus recycling identities

Returning raw pointers from a fixed pool can be convenient, but a pointer alone may not distinguish an old allocation from a newly acquired object in the same slot. A generation-tagged handle deliberately requires clients to revalidate identity through the pool before each operation.

There is a cost to this indirection: an index check, an occupancy check, and a generation comparison. For a small control-plane operation these checks may be cheap relative to the value of reliable stale-handle detection.

In a time-critical hot loop, a system may instead acquire a validated record and use a borrowed pointer for a short, clearly delimited interval. The owner must guarantee that the slot will not be released or reassigned during that interval. Such a guarantee can come from task confinement, locking, or another explicitly designed protocol.

7.33 Failure behavior at saturation

A real-time system usually needs a policy for full resource pools. The pool example returns false immediately. A larger system could queue a request, block with a timeout, drop an old event, or reject incoming work.

Those choices have very different timing and data-loss properties. The most predictable choice for a small synchronous pool is often immediate rejection, with the caller deciding what to do.

Never silently overwrite an occupied record simply because a new request arrives. Such behavior can invalidate another client's handle without an agreed ownership transition and can convert a recoverable capacity error into corrupted application state.

Worked exercises

7.19 Design a driver descriptor pool with eight slots and a rule that no handle from one pool may be used with another. **7.20** Decide whether a full pool should reject, wait, or evict for a safety-related telemetry application. **7.21** Prove that generation retirement prevents reuse of a retired slot's handle value. **7.22** Explain why a pointer into a pool may remain physically in bounds after its logical allocation has ended. **7.23** Extend `slot_pool.c` with a validated update function and tests for stale handles.

8. The Preprocessor and Separate Compilation

The C preprocessor transforms translation units before compilation. It helps share declarations and configure programs, but its token-level substitutions must not be mistaken for typed functions.

8.1 Headers describe interfaces

```
/* counter.h */
#ifndef COUNTER_H
#define COUNTER_H

#include <stddef.h>
struct counter { size_t value; };
void counter_increment(struct counter *c);

#endif
```

```
/* counter.c */
#include "counter.h"
void counter_increment(struct counter *c)
{
    if (c != nullptr) ++c->value;
}
```

The header is included by both the definition and callers, allowing the compiler to check their consistency. The include guard prevents repeated definitions within a translation unit.

8.2 Prefer typed operations

```
#define SQUARE(x) ((x) * (x))
```

This macro evaluates its argument twice; `SQUARE(i++)` produces undefined behavior due to unsequenced modifications. A function avoids repeated evaluation:

```
static inline int square(int x) { return x * x; }
```

The function has one parameter evaluation, though signed multiplication can still overflow. `inline` is not a promise that machine code will contain no call.

8.3 Conditional compilation

```
#if defined(DEVICE_TARGET)
    /* device-specific interface */
#else
    /* host implementation */
#endif
```

Keep conditional code small and localized. C23 adds `#elifdef` and `#elifndef` as clearer forms for common conditional cases. Feature-test macros should verify actual support rather than assume the entire standard library is implemented because a compiler accepts `-std=c23`.

8.4 Compile and link

For two files:

```
cc -std=c23 -Wall -Wextra -Wpedantic -c counter.c
cc -std=c23 -Wall -Wextra -Wpedantic -c main.c
cc counter.o main.o -o program
```

The compilation stage separately translates source files; the linker resolves symbols and constructs the executable. An undefined reference usually means the program declared a symbol but no linked object provided its definition.

Exercises. 8.1 Write a minimal `Makefile` for the two-source example. 8.2 Replace a parameterized macro with an inline function. 8.3 Explain the distinction among declaration, tentative definition, and external definition.

8.5 What belongs in an interface?

A header exposes names and types necessary for clients. A private source file keeps the implementation's helper functions, internal objects, and algorithms. Public headers should be self-contained: including one in an otherwise empty translation unit should not require secret prior includes.

A useful test compiles a short file containing only the header and a minimal `main`. This catches hidden dependencies that may otherwise be masked by the include order in a larger build.

8.6 Macros and evaluation hazards

Function-like macros are token substitution, not calls. This matters for evaluation count, operator binding, statement boundaries, and type checking. Compare:

```
#define MIN_BAD(a, b) ((a) < (b) ? (a) : (b))
static inline int min_int(int a, int b)
{
    return a < b ? a : b;
}
```

The macro may evaluate one argument twice, while the function evaluates each call argument once. Even the function does not specify which argument expression is evaluated first; do not write code that depends on call-argument evaluation order.

8.7 Build modes and reproducibility

A program's observed behavior depends on source code, implementation, target, optimization settings, language standard, and linked libraries. Keep warning flags in normal development builds. Use optimization during testing as well as debugging, since undefined behavior may appear differently at different optimization levels.

The C standard does not dictate a build system. A small Makefile can teach separate compilation effectively. Larger projects benefit from explicit dependency tracking, toolchain files, and reproducible CI environments.

8.8 Defensive configuration

Prefer feature tests to compiler-name checks. Detecting a compiler vendor does not prove that a header is available or that a function links with the intended C library. For a C23 library feature, compile and link a minimal program using it before enabling the code path. Document fallbacks for older supported targets.

8.9 Exercises

8.4 Write a header-only compile probe. **8.5** Show why a macro with arguments containing ++ can change program behavior. **8.6** Build a two-file project with a separate `-c` step and explain a linker error. **8.7** Define what toolchain details must be recorded to reproduce a firmware build.

8.10 A source file is not an isolated program

Suppose one source file includes a header declaring `int calculate(int);`, and another defines `double calculate(double)`. Each source can appear locally plausible, but the program violates

the intended cross-file interface. Include the same header in the implementation file, build with warnings enabled, and avoid hand-copied prototypes.

At link time, the toolchain resolves external references. A linker error about an undefined reference can mean the implementation object was not linked, the identifier has internal linkage, or the symbol spelling does not match. Distinguish such linkage errors from a compiler diagnosis about a malformed declaration.

8.11 Build a tiny library deliberately

A three-file project can contain `counter.h`, `counter.c`, and `main.c`. Compile the implementation and main program into separate object files, then link them. In a Makefile, express dependencies so a changed header causes both translation units to rebuild. A project that rebuilds only the implementation after changing a public structure definition may link stale object code with an incompatible layout.

For larger projects, generated dependency files or a build system such as CMake can automate this bookkeeping, but understanding the underlying dependency graph remains valuable.

8.12 Host and target are different builds

An embedded program may compile once for a desktop host and once for an Arm microcontroller. The compiler, target ABI, linker script, startup code, libraries, and permissible memory layout can be different. A program's `sizeof(long)`, pointer representation, object alignment and supported atomics must not be inferred from the host build.

Keep portable algorithms in files that both environments can compile. Isolate peripheral drivers and target-specific register definitions behind narrow interfaces. This makes host-side unit tests useful without pretending they replace hardware validation.

8.13 Preprocessor conditions versus feature probes

A macro such as `__STDC_VERSION__` can tell you which language revision an implementation claims for a translation mode. It cannot prove that optional facilities or the desired target C library are available. To depend on `<stdckdint.h>`, compile and link a small test that actually includes the header and calls a checked arithmetic operation. To depend on `#embed`, compile an actual embedding directive using the chosen toolchain.

The distinction matters because a C front end and its runtime library are often delivered separately. Firmware toolchains may ship specialized or partial libraries.

8.14 Keep generated output reproducible

The manuscript's PDF is generated from Markdown. A reliable release records the source commit, document toolchain versions, fonts, and build command. The page count and layout can change when these vary. Committing a PDF without a verified successful build may give readers a stale document that does not match the latest chapters.

Use a CI workflow to build, inspect, and publish the PDF after source changes, with an explicit failure whenever required tools or referenced files are missing. An automation configuration in a repository is a plan for a build, not evidence that the build actually passed.

Further problems: Write Makefile rules for two object files sharing a header. Explain the meaning of `-c`. Design a two-compiler CI matrix and identify how to distinguish compiler-support failure from a programming error.

8.15 A real three-file C module

A header becomes useful when more than one translation unit agrees to an interface. The companion example under `examples/modules/` consists of a header, an implementation, and a separate client program. It computes a checksum over a bounded byte span. The calculation is intentionally elementary; the lesson is how declarations, definitions, and build dependencies cooperate.

The header contains the declaration:

```
#ifndef BOOK_CHECKSUM_H
#define BOOK_CHECKSUM_H

#include <stdbool.h>
#include <stddef.h>

bool book_xor_checksum(const unsigned char *bytes, size_t count,
                       unsigned char *out);

#endif
```

The implementation includes that same header, rather than independently restating the function's type. It also places any helper functions under internal linkage with `static`. The client includes the header and calls the function. The build compiles both source files and links their object code.

Why insist that the implementation include its own header? Suppose a developer changes the parameter from `size_t` to `unsigned` in only the implementation. Without the header, different translation units could compile against incompatible assumptions. With a common header, the compiler can diagnose disagreement where the function is defined.

8.16 Rebuild dependencies

A Makefile must know more than the final executable name. If `checksum.h` changes, both the implementation that defines the operation and the client that calls it may need recompilation. For a small demonstration, an explicit recipe that depends on all three files is sufficient:

```
modules_demo: examples/modules/main.c examples/modules/checksum.c examples/modules/checksum.h
    $(CC) $(CFLAGS) examples/modules/main.c examples/modules/checksum.c -o $@
```

For larger projects, compile to separate object files and generate header dependencies automatically. Otherwise a stale object file may be linked against an updated structure layout or changed interface.

This particular checksum is not a cryptographic digest. It is a teaching example for module boundaries and input-length contracts. Using it for message authentication would be a security design error.

8.17 Header hygiene and include order

A well-designed public header should compile when included first in an otherwise minimal translation unit. It should include the standard headers needed by its own declarations instead of relying on the client to include them beforehand.

Include guards prevent repeated textual inclusion within one translation unit, not multiple external definitions across separate source files. Define ordinary global storage in one implementation file; put only an `extern` declaration in the header if clients truly need to refer to it.

An API can often avoid exporting global state at all. Pass a context structure or opaque handle through function arguments so tests can create independent instances without accidental interference.

8.18 Libraries are composed, not merely named

A static library usually packages compiled object files into an archive. A shared library adds platform-specific dynamic-linking rules and lifetime constraints. The C language itself specifies translation and linkage concepts but does not require a particular operating-system library format.

An embedded firmware image may link application code, an RTOS, a runtime library, startup code and a linker script in one final step. The C code can be portable while its final link process is highly target-specific.

Exercises

8.8 Compile the module example with separate `-c` steps. **8.9** Remove the implementation file from the link command and explain the resulting diagnostic. **8.10** Delete `<stddef.h>` from the header and determine why standalone inclusion now fails. **8.11** Add a unit-test client using the same header but a different `main`.

8.19 Follow the module through the compiler

The repository's `examples/modules/` directory contains three files: `checksum.h`, `checksum.c`, and `main.c`. They illustrate a complete but small source-level library.

The header declares an operation that computes a byte-wise XOR of a counted input:

```
#ifndef BOOK_CHECKSUM_H
#define BOOK_CHECKSUM_H

#include <stdbool.h>
#include <stddef.h>

bool book_xor_checksum(const unsigned char *bytes, size_t count,
                       unsigned char *out);

#endif
```

The implementation includes this header. That makes it impossible for its function definition to silently disagree with the declaration without the compiler having the opportunity to diagnose the mismatch.

A caller also includes the same header. The compiler can then type-check the function call, even when it has not yet seen the implementation's source file.

The entire demonstration is built as a single executable in the book's main Makefile. However, its two source files can also be compiled separately to object files:

```
cc -std=c23 -Wall -Wextra -Wpedantic -c examples/modules/checksum.c -o checksum.o
cc -std=c23 -Wall -Wextra -Wpedantic -c examples/modules/main.c -o module_main.o
cc checksum.o module_main.o -o modules_demo
./modules_demo
```

These commands are run from the `modernc_book` directory and assume a compiler named `cc` that understands C23. The final link command is what combines the two compiled translation units into a runnable program.

8.20 A header is a contract, not a shared implementation file

A public header normally provides declarations and the types needed by callers. Ordinary externally linked function definitions belong in implementation files. If an externally linked definition is pasted into a header included by many source files, the linker may encounter multiple definitions.

There are legitimate header-defined functions, including appropriately declared `static inline` functions and implementation-specific patterns. Those require understanding the function's linkage and emission rules. The straightforward starting design is one externally visible definition in one source file, with declarations visible through a self-contained header.

The include guard stops repeated textual inclusion within one translation unit. It does not resolve multiple definitions across different translation units. These are separate concerns.

An implementation file should generally include its own public header first. If the function signature changes accidentally, a compile-time discrepancy is detected close to the source of the mistake.

8.21 The difference between syntax errors and link errors

Imagine the caller includes the header and invokes `book_xor_checksum`, but the object file containing the function definition is omitted from the link. Each source file may compile perfectly well on its own. The final link step then fails to resolve the function's external symbol.

That is not a missing declaration; the declaration was present. It is a missing definition in the linked program.

Conversely, if the caller omits the header and uses an undeclared function, modern C requires a diagnostic for the invalid declaration situation. A later linker success cannot repair the source-level type contract.

Distinguishing these cases makes build failures easier to diagnose:

Symptom	Likely stage	First investigation
Unknown type `size_t`	Compile	Header's own dependencies
Conflicting function types	Compile	Declaration versus definition
Undefined reference to a known function	Link	Missing object or library
Multiple definition of a symbol	Link	Repeated external definitions
Program runs but output is wrong	Execution	Algorithm and input contracts
Works on host but not on MCU	Target integration	Runtime, ABI, drivers and linker script

These are categories, not a complete error taxonomy; compilers and linkers use different diagnostic messages.

8.22 Automate dependencies rather than relying on luck

Suppose `checksum.h` changes but a build system recompiles only `checksum.c`. The old `main.o` may have been compiled using a different interface assumption. A successful link is not always enough to establish source-level agreement if objects come from incompatible revisions.

Makefile dependencies must therefore include included headers as well as source files. On many GCC-compatible toolchains, options such as `-MMD -MP` can generate dependency files listing included non-system headers. The exact flag support should be checked on the chosen compiler.

The book's small Makefile can list the three module files explicitly, avoiding a need for generated dependencies in the teaching example. Larger projects should move to reliable dependency generation or a build system that handles it.

A clean build is useful for validation, but it should not be the only way to avoid stale object files in daily work.

8.23 Host and target have different responsibilities

The C language defines source and translation semantics, but it does not choose a particular microcontroller linker script, reset vector, RTOS runtime, memory map, or device driver.

An embedded build can involve:

- a cross-compiler targeting a different instruction set;
- startup code that initializes memory before calling application entry points;
- a linker script locating code and data in flash and RAM;
- a chosen implementation of the C library, sometimes with reduced functionality;
- hardware-specific interrupt and peripheral definitions;
- a flashing and debugging mechanism.

A C module can remain portable even while the final firmware image is highly platform-specific. Isolate target details in separate adapters and keep the module's public contract independent of register addresses where possible.

A hosted `FILE *` test does not prove that the firmware image has a filesystem, and a hosted `<threads.h>` test does not prove that the MCU toolchain supplies C threads. Test those interfaces separately.

8.24 Feature tests must include the library

A compiler accepting `-std=c23` does not guarantee that its selected C library provides every new C23 header. A feature probe should include the intended header, use a representative interface, compile, link, and when meaningful, run the result.

For `<stdckdint.h>`, a probe calls `ckd_add` on fitting and overflowing operands. For `<stdbit.h>`, it calls the desired bit utility. For `#embed`, a probe includes a small known binary resource and verifies its compiled contents. Those are different checks because some features are language-front-end functionality and others depend on the library or build resource handling.

Do not silently downgrade a test to C17 and still report that it verified C23 support. Record the exact dialect, compiler, library, target, and observed result.

8.25 Reproducible does not mean identical machine code

Two conforming compilers may generate different instructions from the same C program while preserving its required observable behavior. Different optimization levels, ABIs, standard libraries and linker configurations also affect the resulting executable.

A reproducible software workflow should identify its source revision, toolchain and options so an observed build can be repeated. Bit-for-bit reproducible binaries require additional controls, such as deterministic archives, build paths, timestamps and toolchain versions.

For this book, the immediate goal is narrower but important: each example should compile and execute under the specified C23-capable CI compilers, and the PDF should be generated from the intended manuscript revision.

8.26 A teaching exercise in failure classification

Deliberately introduce one fault at a time into the checksum module:

1. Remove `<stdbool.h>` from the header and compile a client that includes it first.
2. Change the function's output pointer type in the implementation only.
3. Omit the implementation object when linking the client.
4. Change the algorithm to XOR one byte fewer than required.
5. Change the source's accepted null-pointer contract but leave its tests unchanged.

For each, record whether compilation fails, linking fails, execution fails a test, or a contract mismatch remains hidden by incomplete tests.

This exercise emphasizes the limits of each stage. Compilers catch some interface errors. Linkers resolve definitions. Runtime tests check selected behaviors. Code review and specifications establish the requirements against which those behaviors should be judged.

Further exercises

8.12 Compile the module into separate object files and link them by hand. **8.13** Make a header self-contained by including its required standard type definitions. **8.14** Add a generated header-dependency file for a two-source project. **8.15** Create separate host and embedded build configurations without changing the checksum algorithm. **8.16** Document the difference between a missing function declaration and a missing function definition.

9. The C23 Language

C23 is the 2024 revision of the ISO C standard, commonly called C23. It modernizes familiar C without turning it into C++. Its features are best adopted where they improve clarity or correctness; portable projects must still verify compiler and standard-library implementation support.

9.1 Booleans and null pointers

```
bool valid = true;
int *ptr = nullptr;
if (!valid || ptr == nullptr) {
    /* handle absence */
}
```

`bool`, `true`, and `false` are language keywords in C23. `nullptr` has type `nullptr_t` and denotes a null pointer constant without relying on a macro such as `NULL`.

9.2 Type inference with auto and typeof

```
int count = 3;
auto copy = count;            /* copy has type int */
typeof(count) another = 4;   /* another has type int */
```

The C23 `auto` form is a type inference declaration, not the C++ rules in their entirety. `typeof` provides a type based on another expression or type name. C23 also supplies `typeof_unqual`, discarding qualifiers and atomicity from the resulting type in its specified way. These mechanisms can reduce duplicated types but should not obscure API contracts.

9.3 Constant expressions

```
constexpr int channels = 4;
int measurements[channels];
```

C23 introduces `constexpr` for certain objects initialized by constant expressions. This is not C++ compile-time function evaluation. Keep the distinction between an expression computable at translation time and a runtime `const` object.

9.4 Width-aware integers

Where supported, `_BitInt(N)` and `unsigned _BitInt(N)` provide bit-precise integer types. Their availability and limits depend on implementation support. They can be valuable for numerical

kernels and hardware-oriented code, but ordinary public interfaces may prefer portable fixed-width types when widths are already covered by `<stdint.h>`.

9.5 Binary literals and digit separators

```
unsigned mask = 0b1010'0101;
```

Binary literals communicate bit patterns; the single-quote separators make long constants easier to read. A leading `0b` is now standardized rather than a vendor extension.

9.6 Attributes

```
[[nodiscard]] int save_record(void);
[[maybe_unused]] static int debug_setting = 0;
```

Standard attribute syntax provides optional diagnostics and intent. `[[nodiscard]]` encourages checking a result; the exact diagnostic is implementation-dependent. Attributes do not magically enforce a full error-handling policy. C23 also specifies `[[deprecated]]`, `[[fallthrough]]`, and `[[noreturn]]` as well as forms of `[[unsequenced]]` and `[[reproducible]]`.

9.7 New preprocessor capabilities

C23 adds `#embed` for incorporating binary data at translation time and `#warning` for diagnostics. `__VA_OPT__` permits more reliable conditional comma handling in variadic macros. Implementation support for `#embed`, in particular, requires explicit checking.

9.8 Checked arithmetic and bit utilities

C23 specifies `<stdckdint.h>` with `ckd_add`, `ckd_sub`, and `ckd_mul` for checked integer arithmetic, and `<stdbit.h>` for unsigned integer bit operations. These facilities may be missing in otherwise C23-capable runtime libraries. A project that depends on them should run configure-time feature tests. Do not assume a working `-std=c23` flag implies complete header support.

9.9 A migration policy

First build with maximum practical warnings. Remove implicit declarations and invalid conversions; make allocation-size checks explicit; audit dangerous signed arithmetic; identify any reliance on extensions. Introduce `nullptr`, boolean keywords, binary literals, designated/empty initializers, and attributes only when supported by the actual deployment toolchains. A robust migration preserves behavior before improving style.

Exercises. 9.1 Replace a platform-specific `__typeof__` declaration with standardized spelling where available. 9.2 Use `[[nodiscard]]` on a status-returning function and observe diagnostics. 9.3 Write a feature-test program for `<stdckdint.h>`. 9.4 Explain why `constexpr` is not simply another spelling for `const`.

9.10 Generic selection

C's `_Generic` chooses an expression based on a controlling expression's type. It supports type-directed interfaces while remaining a compile-time selection mechanism. Generic selection is not dynamic dispatch.

```
#define magnitude(x) _Generic((x), \
    float: fabsf, \
    double: fabs, \
    long double: fabsl \
)(x)
```

This example requires <math.h>. It illustrates a common pattern: select an appropriately typed function, then call it. The supported type list is intentionally explicit; passing an unsupported type is a constraint violation unless a default association is provided. _Generic predates C23, but remains important when designing a modern C interface.

9.11 Safer use of typeof

When an expression's type is cumbersome, typeof can keep an interface aligned with the expression itself. It does **not** turn C into a language with automatic lifetime management. A pointer inferred from another pointer has the same potential lifetime hazards.

Use inferred types when duplication creates maintenance risk, but write explicit types when they communicate important width, ownership, or protocol constraints.

9.12 Checked arithmetic in C23

When available, <stdckdint.h> supports checked integer calculations that report overflow through a boolean result rather than leaving the programmer to reconstruct each type's boundary conditions.

```
#include <stdckdint.h>
int result;
if (ckd_add(&result, a, b)) {
    /* the mathematical result is not representable in result's type */
} else {
    /* result contains the sum */
}
```

The above is a C23-library-dependent fragment, not a guarantee that every compiler installing a C23 front end also ships this header. For targets without it, a tested manual check or compiler-specific checked-arithmetic primitive may be appropriate.

9.13 Selected portability cautions

Not all useful C23 additions are equivalent from a portability standpoint. Binary literals and attributes may be available before other features. #embed support can vary independently from the core language. Bit-precise integers may have target-specific maximum widths. Headers such as <stdbit.h> may be absent from older C libraries.

Keep a portability table stating which implementation combinations were actually tested. The standard defines what a conforming implementation must do; deployment engineering determines which implementation is in hand.

9.14 Exercises

9.5 Build a _Generic wrapper that chooses the correct absolute-value function for three floating types. **9.6** Write a small compile probe for nullptr, typeof, and constexpr. **9.7** Compare manual overflow checks with ckd_add when available. **9.8** Explain why a compiler's version string alone cannot establish support for <stdbit.h>.

9.15 What actually changes when moving from C17

C23 is not merely a new spelling of old constructs. It removes traditional, unprototyped function declarations and old-style function definitions. In C23, void work() declares a function taking no arguments; older C dialects treated an empty parameter list differently outside a function definition. This is especially important in legacy headers.

C23 also fixes signed integer representation to two's complement. It does **not** define signed overflow. The two statements are compatible: an implementation has a required representation, while the result of evaluating an overflowing signed arithmetic expression remains undefined.

The macro `__STDC_VERSION__` is defined as `202311L` in C23 mode for conforming implementations. Checking this can select a language-level path, but it cannot certify that a particular runtime library or optional capability is present.

9.16 The boundary between auto and typeof

C23 introduces type inference for declarations with `auto` and standardizes `typeof` and `typeof_unqual`. These facilities eliminate some redundant type spellings. They do not provide templates, classes, overload resolution, or C++-style deduction rules.

A library's public signature should normally state the types important to its binary contract: `uint32_t`, `size_t`, or a named structure. Locally, inferred types can be useful where the type is unambiguous from the initializer and repeating it would be error-prone.

C23 `constexpr` applies to certain constant objects under specific restrictions; it is not general C++ constant evaluation. For example, C23 does not make an arbitrary user-written function eligible for execution during translation merely because it is called from a constant initializer. A `constexpr` object should not be used to promise behavior that the actual compiler has not implemented.

9.17 Modernized enumerations and attributes

C23 adds facilities for enumerations with explicitly selected underlying integer types. They can make intent clear when enum values appear in compact data structures or interfaces. Still, an enum object in a wire format must be serialized as an explicitly chosen integer representation; the in-memory layout should not be transmitted wholesale.

Attributes such as `[[nodiscard]]` help advertise contracts. They improve diagnostics but are not enforcement mechanisms: a caller may still discard a return value and a conforming implementation's diagnostic behavior depends on the attribute rules. Code review and tests remain necessary.

9.18 Checked arithmetic belongs at boundaries

When the implementation supplies `<stdckdint.h>`, `ckd_add`, `ckd_sub`, and `ckd_mul` can help test whether the mathematical result is representable in the result type. Use them when dealing with byte counts, sizes and input-derived values. Do not confuse this check with validating the semantic limits of a protocol. A packet length may fit in a `size_t` while still exceeding the application's maximum frame length.

Some deployments provide a C23 compiler without the corresponding C23 library headers. The right adaptation is a feature check at configuration time, or a small, tested fallback module with exactly the same contract. A bare compiler-version comparison is less reliable than verifying the required operation.

9.19 C23 migration example

Suppose a legacy API relied on `malloc(count * width)`. Migrating to C23 is a chance to fix the arithmetic contract before introducing new syntax:

```
#include <stdint.h>
#include <stddef.h>
#include <stdlib.h>
#include <stdbool.h>
```

```
bool allocate_bytes(size_t count, size_t width, void **out)
{
    if (out == nullptr || count == 0 || width == 0 ||
        count > SIZE_MAX / width)
        return false;
    void *memory = malloc(count * width);
    if (memory == nullptr)
        return false;
    *out = memory;
    return true;
}
```

The routine refuses zero-sized requests and leaves the output unchanged on failure. It does not assert that all targets offer a usable heap. In freestanding firmware, a caller-provided arena may be preferable.

9.20 A testable feature inventory

Record support separately for language syntax, library headers, implementation-defined properties, and target environment. The first category includes `nullptr`, `typeof`, and `constexpr`. The second includes `<stdckdint.h>` and `<stdbit.h>`. The third includes integer widths and alignment. The fourth includes available heap, streams, threads, and atomics.

Further problems: Port an old K&R-style declaration to a C23 prototype. Test whether a compiler supports `#embed` independently from `-std=c23`. Show that a C23 `constexpr` object is not a user-defined compile-time function. Design a feature probe for a C23 standard-library header.

9.21 A concrete subset of C23

The following complete program deliberately selects language features that are supported by the book's intended GCC 14 and Clang 18 test matrix. It does not exercise the newer library headers or `#embed`, which must be checked independently.

```
#include <stdio.h>

struct point { int x, y; };

int main(void)
{
    auto flags = 0b1010'0101u;
    typeof(flags) another = 0b0000'0011u;
    struct point origin = {};
    int *missing = nullptr;
    bool good = (flags & another) == 1u;

    static_assert(sizeof(flags) == sizeof(unsigned));
    if (!good || origin.x != 0 || origin.y != 0 ||
        missing != nullptr)
        return 1;
    puts("C23 essentials: OK");
}
```

The integer literal is explicitly unsigned. The digit separator is a single quote between digits. The `auto` declaration infers the type from its initializer, and `typeof(flags)` names that type again. Empty initialization initializes both structure members to zero. `nullptr` provides a null pointer constant. The `static_assert` condition is evaluated at translation time.

The repository keeps a corresponding standalone program in `examples/c23_basics.c`. When teaching this listing, compare what the compiler knows at translation time with what the program checks at runtime.

9.22 Why not use every C23 feature in the first example?

A language revision consists of many individually specified changes. A compiler may support the new keywords but lack the new embedding directive. Its system C library may lack `<stdckdint.h>` or `<stdbit.h>`. A microcontroller toolchain may implement only a subset of hosted interfaces.

Choosing a well-supported subset for the introductory example makes the core lesson reproducible. Dedicated feature probes are the right place to demonstrate `constexpr`, bit-precise integers, the newer enumeration facilities, and the new library interfaces across toolchains.

9.23 Initialization is not byte filling

`struct point origin = {};` uses language-level initialization, not a raw `memset`. C's initialization rules set members to their appropriate empty-initialization values. This distinction matters for structures containing pointers or floating members, because an all-bits-zero representation is not universally promised to be the semantic null-pointer or floating zero representation.

For an ordinary aggregate, correct language initialization is usually both clearer and more portable than relying on representation bytes.

Further problems: Rewrite the example without `auto` and `typeof`. Identify which expressions are evaluated at translation time and which during execution. Build the same file under a C17 dialect and record the diagnostics instead of assuming all failures have the same cause.

9.24 A language revision is a collection of independent features

Using `-std=c23` selects a language mode, but the compiler and standard library are not always delivered or updated together. A cross-compiler may recognize `nullptr` and `typeof` while its libc lacks `<stdckdint.h>`. The correct deployment procedure tests the exact construct or library interface that the project uses.

Classify features before adopting them. Some are purely syntactic improvements, such as binary literals and more convenient forms of initialization. Some change type expressiveness, such as bit-precise integers. Others require new translation support, such as `#embed`. Still others require implementation of additional library headers.

That taxonomy prevents a migration effort from becoming a blind replacement of every old idiom with a new-looking construct.

9.25 C23 integer literals and readable masks

Binary literals represent bit patterns compactly:

```
unsigned control = 0b1010'0001u;
```

The single quotes separate digit groups without affecting value. This is useful when a device control register is documented by bit positions. It does not solve the hardware semantics: whether a register is read-only, write-one-to-clear, or requires reserved bits depends on the device manual.

Clang 18 in some configurations diagnoses the standardized C23 binary-literal syntax with a GNU-extension warning. Our CI retains `-Werror` but suppresses that one diagnostic for Clang 18. A warning emitted by a compiler version is not necessarily evidence that the source violates the language standard.

9.26 nullptr is a value, not an owner

C23's `nullptr` is a null pointer constant with type `nullptr_t`. It makes pointer intent clearer than writing integer zero in a pointer context. But replacing `NULL` with `nullptr` does nothing to prevent a non-null dangling pointer, a wrongly sized array, or a double free.

The following check rejects absence, not invalid lifetime:

```
if (buffer == nullptr)
    return false;
```

A buffer that was freed but whose old pointer has not been reset may still compare non-null. The API must prevent access after the owner's release, not hope that an address test will discover it.

9.27 Type inference should not hide the contract

C23's `auto` can infer a local type from its initializer. This can avoid spelling a complex expression type twice, and `typeof` can express a related declaration. However, the public interface of a protocol decoder should not hide that its input is `const unsigned char *` with a `size_t` length or that a field is a fixed-width integer.

At public boundaries, explicit types communicate ABI and range requirements. Inside a function, type inference is helpful when the initializer makes the meaning obvious. Use it as a maintenance aid, not a substitute for reviewing conversions.

9.28 Checked integer arithmetic in a C23 library

The `ckd_add`, `ckd_sub`, and `ckd_mul` interfaces from `<stdckdint.h>` express checked operations with a result pointer. They are especially useful for sizes and accumulated counts.

A feature probe should include the header, call a chosen macro with values that fit and values that overflow, and link the resulting program using the actual target C library. If the header is missing, a separately tested implementation of equivalent arithmetic checks may be reasonable. A compiler version alone is insufficient evidence.

The application may need a limit stricter than the type's representable range. A frame size of 65,000 bytes can be representable in `size_t` yet impossible to buffer in a 2 KB MCU arena. Checked arithmetic and application resource checks solve different problems.

9.29 Choosing what to modernize

A conservative migration can proceed in three phases. First, make the old program correct under its current contracts: fix implicit assumptions, mismatched prototypes, unchecked sizes and invalid conversions. Next, select the C23 language mode and confirm equivalent behavior under tests. Finally, adopt newer features where they improve readability or reliability.

Do not rewrite a working fixed-size state machine solely to exhibit `_BitInt` or `typeof`. A textbook should show what the feature contributes and what its portability cost is. For firmware, support across required cross-toolchains is often more important than showcasing every new keyword.

Guided exercises

9.9 Create a compilation probe for `constexpr` and compare its syntax with `const`. **9.10** Implement a feature check for `<stdckdint.h>` without assuming its existence. **9.11** Explain what `nullptr` does not protect against. **9.12** Convert a simple C17 program to C23 mode while preserving the behavior of every input test. **9.13** Document the actual warning produced by two compilers for a binary literal and distinguish a diagnostic from a normative rule.

9.30 One standard, several implementation layers

C23 is the language revision published as ISO/IEC 9899:2024. The same phrase 'supports C23' can describe very different states of implementation. A compiler front end may accept new syntax while the selected C library is missing a corresponding header. A hosted implementation may offer features unavailable in a freestanding MCU runtime. A particular back end may support a bit-precise type only up to a limited width.

For a reliable project, record support in separate columns:

Question	Example of evidence
Does the language front end recognize the feature?	Compile a minimal translation unit in C23 mode
Does the required library facility exist?	Include the header, use its operation, then link
Does the target support its representation?	Build and execute target-specific range and size tests
Is the feature permitted in the deployment context?	Review the implementation and RTOS/ISR documentat

The macro `__STDC_VERSION__` has the value `202311L` in a conforming C23 translation mode. It communicates the claimed language revision; it is not a substitute for the additional checks above.

The GCC and Clang projects publish changing C-standard support matrices. Use those project pages as starting points, then verify the exact compiler, library, target and feature in the release being deployed.

9.31 constexpr is an object feature, not C++ constexpr functions

C23 introduces `constexpr` for certain objects with constant initializers. It does not provide C++'s general facility for executing arbitrary user-written functions during translation.

An illustrative C23 declaration is:

```
constexpr int samples_per_frame = 160;
static_assert(samples_per_frame > 0);
```

The initializer must meet the feature's translation-time requirements, and the object cannot be modified afterward. C23 `constexpr` has restrictions, including exclusions for atomic and volatile objects and for most non-null pointer values.

Do not assume that every compiler supporting `-std=c23` implements `constexpr` objects in the selected release. Compile this minimal example separately before relying on it. If a firmware target uses an older compiler, an enumeration constant or a suitably scoped macro may meet an application requirement without forcing a toolchain migration.

9.32 Bit-precise integers have specific ranges

The `_BitInt(N)` family makes it possible to request an integer type with a chosen value width, within an implementation-supported range. For a signed nine-bit value, the mathematical range is −256 through 255. An unsigned nine-bit value can represent 0 through 511.

```
_BitInt(9) signed_sample = -200;
unsigned _BitInt(9) channel_code = 500;
```

These types are especially interesting for tightly specified bit-oriented computations and representations. Yet a bit-precise C type does not, by itself, dictate a portable nine-bit wire format or guarantee compact storage without padding on every target. External packets still need explicit packing and extraction.

Implementation limits, type promotions, ABI conventions and library-format support deserve attention. Do not pass an arbitrary bit-precise value to `printf` using a format intended for `int` without first establishing a safe conversion and format contract.

9.33 Checked arithmetic from the C23 library

The header `<stdckdint.h>` provides `ckd_add`, `ckd_sub`, and `ckd_mul`. They test whether an integer calculation is representable in the destination type and report an overflow condition.

A compile-and-run feature probe can be written as a standalone file:

```
#include <limits.h>
#include <stdckdint.h>

int main(void)
{
    int result = 0;
    if (ckd_add(&result, 10, 20) || result != 30)
        return 1;
    if (!ckd_add(&result, INT_MAX, 1))
        return 2;
    return 0;
}
```

A nonzero result or failure to compile or link means the application cannot rely on this facility as tested. On overflow, do not assume the result object preserves an earlier value; if your API promises unchanged output on failure, compute into a temporary and assign only after the checked operation succeeds.

A more restrictive application limit remains independent. A count can fit in `size_t` but exceed the capacity of an arena, message queue, or peripheral packet buffer.

9.34 Bit utilities are not rotate operators

C23's `<stdbit.h>` provides operations for tasks such as counting set bits, counting leading or trailing zeros, querying bit width, and computing bit-floor or bit-ceil values. It does not provide a C interface named `std::rotl` or `std::rotr`, which belong to C++.

If an implementation provides the new header, a small feature probe that calls `stdc_count_ones` on an unsigned integer can establish the facility needed by an application. The portable rotation functions in Chapter 14 use ordinary unsigned shifts and explicitly avoid shifting by the type width.

Do not assume the binary representation of a C object equals the byte order of a file or external wire format. Bit operations compute on integer values; serialization independently defines where each octet goes.

9.35 Embedding a resource into a program

C23's `#embed` directive is designed to incorporate the bytes of a resource at translation time. A possible use is storing a small static lookup table or model constant inside firmware:

```
static const unsigned char resource[] = {
#embed "coefficients.bin"
};
```

This is a feature illustration, not a guaranteed portable build for every compiler used by this book. A real build must supply the resource file, arrange an appropriate search path, test the compiler's support, and validate the embedded format and length.

It also must consider the target linker script and memory placement. The C source alone does not promise that a large resource will reside in flash rather than consuming precious RAM. A generated C array may provide a fallback on an older toolchain.

Embedding bytes also does not authenticate the resource. The build pipeline must establish where the file came from and whether its contents match the expected model or calibration data.

9.36 Attributes assist review but do not replace contracts

C23 attribute syntax can communicate useful information to tools. For example, `[[nodiscard]]` can document that a result should not normally be discarded. That may produce a compiler warning if the caller ignores the result, but it does not make incorrect error handling impossible.

A declaration such as

```
[[nodiscard]] bool write_record(const unsigned char *data,
                                size_t length);
```

still needs a contract: which lengths are valid, when null is permitted, what failure means, and whether any bytes might have been written before returning false. Attributes improve communication; they do not settle resource ownership or partial side effects.

9.37 Treat migration as a tested engineering change

For legacy source, first correct implicit assumptions: unprototyped declarations, unsafe conversions, unvalidated sizes, and ownership confusion. Build and run the old program in its supported mode, recording results. Next select C23 mode and repair genuine standard incompatibilities. Finally adopt new features where they clarify a design and are supported by required toolchains.

Changing language mode and rewriting every algorithm in the same commit makes regressions difficult to attribute. A staged migration retains meaningful test evidence.

The book's CI deliberately uses a conservative tested feature subset for standalone examples. More advanced features such as `constexpr`, `_BitInt`, and `#embed` should receive their own compatibility probes before being made dependencies of the full build.

Further exercises

9.14 Write and run a `constexpr` translation probe, recording the exact compiler version. **9.15** Determine the valid ranges of signed and unsigned nine-bit integers without guessing their storage size. **9.16** Build the `ckd_add` probe and distinguish a missing header from a failed arithmetic test. **9.17** Construct a `#embed` example using a known binary file and verify the number of generated octets. **9.18** Design a fallback when the target C library lacks `<stdbit.h>`. **9.19** Audit an exported function for invalid C23 assumptions and document its preconditions.

10. Reliable Systems Programming

The language gives precise rules for certain operations and deliberately leaves other matters to implementations or gives no meaning to invalid programs. Reliability begins with knowing the boundary.

10.1 Undefined, unspecified, implementation-defined

An **undefined** operation has no required behavior: indexing past an array or signed overflow are examples. **Implementation-defined** behavior obliges an implementation to document a choice, such as properties of `char`. **Unspecified** behavior allows a choice among permitted outcomes without requiring that choice to be documented.

Never make the program's correctness depend on one observed result of undefined behavior, even if a particular build seems stable.

10.2 Check sizes before multiplying

```
#include <stdint.h>
#include <stdlib.h>
```

```c
#include <stddef.h>

bool allocate_samples(size_t count, int **out)
{
    if (out == nullptr) return false;
    if (count == 0 || count > SIZE_MAX / sizeof(int)) return false;
    int *p = malloc(count * sizeof *p);
    if (p == nullptr) return false;
    *out = p;
    return true;
}
```

The caller must release `*out` after success. The function leaves `*out` unchanged on failure. For C23, `bool` is a keyword. For maximum compatibility with older dialects, include `<stdbool.h>`.

10.3 The limits of volatile

`volatile` is useful for accesses with externally observable behavior, including certain memory-mapped I/O, when used according to implementation and hardware rules. It is not a synchronization primitive and does not prevent data races. Atomics from `<stdatomic.h>` provide operations with explicit memory-order semantics, when supported by the target's implementation.

10.4 Host tests and device tests

Test numerical functions on a host where instrumentation is available. Enable sanitizers such as AddressSanitizer and UndefinedBehaviorSanitizer where toolchains support them. Unit tests do not replace target validation for memory maps, interrupts, scheduling latency, peripheral behavior, or floating-point differences.

10.5 Protocol parsing

A parser accepting a length and a byte buffer is easier to audit than one that searches indefinitely for a terminator. Distinguish malformed input, insufficient data, overflow, and unsupported versions. Parse into temporary state and publish a complete result only after validation.

10.6 Review checklist

Ask of each interface: Are sizes explicit? Are output objects initialized on success only? Is ownership documented? Are integer conversions checked? Are assumptions about endianness and alignment written down? Can cleanup happen on every error path? Is concurrent access synchronized?

Exercises. 10.1 Enumerate five undefined-behavior risks in a string decoder. 10.2 Design a bounded ring buffer and define its full and empty states. 10.3 Explain why `volatile int flag` is not generally a safe cross-thread signal.

10.7 Establish invariants before optimizing

An invariant is a property required to remain true across operations. A ring buffer may maintain `count <= capacity`; a vector may maintain `length <= capacity`; a parser may maintain `position <= input_length`. The easiest time to discover a broken invariant is at the operation that first violates it.

Checks should precede the risky operation. Checking a pointer after dereferencing it, or checking a sum after signed overflow, is too late. An optimizer may assume undefined behavior never occurs and transform subsequent code accordingly.

10.8 Bound both time and storage

An embedded function should be evaluated not only for logical correctness but also for its worst-case time and memory needs. A loop over a fixed-size buffer has an explicit bound. A dynamically grown table or a scan to an unbounded terminator may not.

Real-time behavior adds scheduling, interrupt latency, contention, and priority inversion. None can be solved merely by choosing C23 syntax. The language supports expressing the operations; the system architecture determines when they can run.

10.9 Concurrency and atomics

Sharing a non-atomic object between threads without appropriate synchronization can create a data race, resulting in undefined behavior. `volatile` does not repair this. Atomics in `<stdatomic.h>` provide carefully specified atomic operations; mutexes and other primitives may be preferable where a critical section covers several related objects.

Memory-order choices require a proof about which values must be visible to which threads. Start with sequentially consistent operations unless the design has a demonstrated reason to use weaker ordering. Interrupt-to-thread communication also depends on the implementation's interrupt model and target hardware.

10.10 Errors are part of the API

A function's result contract should specify what happens when it receives a null pointer, an invalid length, an unrecognized packet, insufficient output storage, or a resource failure. The failure behavior may need to preserve caller state. Document that guarantee and test it.

The absence of a crash does not prove memory safety. A parser can mis-handle a length while still writing inside allocated storage, and a numerical routine can return a plausible but incorrect value after an invalid conversion.

10.11 Exercises

10.4 Write invariants for a fixed-capacity queue and construct tests preserving them. **10.5** Make a parser reject overflow before computing `offset + length`. **10.6** Describe what synchronization an interrupt-driven producer and thread consumer would require. **10.7** Identify whether each of four sample failures is undefined, unspecified, or implementation-defined behavior.

10.12 A bounded packet decoder

A practical parser should never trust an unvalidated input length. The companion program `examples/packet_decode.c` reads a miniature binary packet with magic byte `0xA5`, version byte `1`, a two-byte little-endian payload length, at most 64 payload bytes, and a one-byte XOR checksum. This deliberately simple checksum demonstrates framing but does **not** provide cryptographic integrity or meaningful protection against intentional tampering.

The parser first checks the minimum header and checksum size. Next it validates the magic and version. It constructs the declared payload length from bytes, without casting the input buffer to a structure. Before accessing the payload, it verifies the declared length against both the fixed maximum and the available input bytes. Only after validating the complete packet and checksum does it write the decoded result into the caller's object.

One subtle design choice is to build a local candidate object and assign it to the caller only on success. This gives failure an important contract: the output remains unchanged. The implementation uses the subtraction `size - HEADER_SIZE - 1` only after proving the total size is at least `HEADER_SIZE + 1`, so the subtraction cannot underflow.

The program also exercises an invalid checksum and truncated input. It was compiled and executed locally using GCC 14 in C2x dialect, with `-Wall -Wextra -Wpedantic -Werror` and AddressSanitizer and UndefinedBehaviorSanitizer enabled. This verifies that executable under the tested configuration, not every possible input or toolchain.

Exercises. 10.8 Add a failure reason enum without changing the output-on-failure guarantee. 10.9 Generate exhaustive byte-sequence inputs for packets of length zero through four. 10.10 Replace the XOR checksum with a documented CRC and explain why even a CRC is not a cryptographic authentication code.

10.13 Preconditions, invariants, and postconditions

A robust C function can be described by three kinds of statements. Preconditions specify what the caller must supply; invariants describe properties that remain true during execution; postconditions describe what success and failure mean.

For a byte decoder, a precondition may be that any non-null buffer pointer refers to at least the supplied length of readable storage. An invariant may be that the parser index never exceeds the input length. The successful postcondition is a fully decoded object; the failure postcondition is that the output object remains unchanged.

These distinctions matter because not every invalid pointer can be detected with a null check. A caller that lies about an object's bounds violates the contract. The function can still protect itself against internally computed out-of-range indexes by validating lengths before access.

10.14 A two-stage parser

An effective pattern is *validate, then commit*. Use temporary state to decode a candidate; inspect the entire record; only then write the result.

```
struct header candidate = {};
/* Parse candidate from a counted byte buffer. */
if (!valid_magic || !valid_length || !valid_checksum)
    return false;
*out = candidate;
return true;
```

This is a fragment, not a standalone example. The important point is that errors do not leave a partially updated public result. Chapter 10's packet example uses a bounded payload and this publication policy.

10.15 Safety is not security

Memory safety protects against invalid storage access. A parser can be memory-safe yet accept maliciously large, inconsistent, or semantically forbidden requests. A one-byte XOR checksum may detect some accidental errors but is not authentication; an attacker can usually alter data and restore a valid XOR.

For an external protocol, consider resource exhaustion, replay, malformed framing, version downgrades, and authorization separately from buffer bounds. The task determines whether cryptographic integrity, rate limits, and stronger protocol validation are required.

10.16 Bounded execution requires a system model

Worst-case running time is more than an algorithmic big-O expression. In a real-time task, include interrupt preemption, cache behavior, memory contention, blocking calls, and other scheduling delays. A loop of exactly 128 iterations is bounded in iteration count; it is not automatically guaranteed to meet a microsecond deadline under every scheduler and cache state.

Avoid heap allocation and unbounded blocking on a critical path unless their timing has been proven acceptable. For many embedded applications, fixed-size data structures, preallocation, and explicit timeout behavior simplify the argument.

10.17 Reviews use independent evidence

Different validation methods find different errors. Compiler warnings detect suspicious constructs. Static analysis explores possible paths. Sanitizers catch classes of runtime faults when the relevant paths execute. Property tests explore behavioral invariants. Hardware-in-the-loop tests reveal device-specific assumptions.

No single passing test suite proves correctness, particularly for concurrent programs. For a critical algorithm, pair executable tests with a short written proof sketch: define ranges, explain why every memory access is within them, and explain why every loop terminates.

10.18 A fault-injection exercise

Build a streaming decoder with an injected byte source that can return a short read, an I/O error, or clean EOF. Give each outcome a distinct status and assert that the caller never sees half-decoded output. Repeat with a deliberately full output buffer and with a record whose claimed payload length is larger than available storage.

Further problems: Construct a threat model for the toy packet protocol. Explain how a full buffer and an overflowed integer differ. Write a reviewer checklist for atomic queue shutdown and object lifetime.

10.19 Defensive decoding: from bytes to a validated object

A packet decoder is a good place to practice several C contracts at once. A caller supplies a byte buffer and its length. The decoder must establish a valid format before touching output state.

For a toy protocol, define the frame as a one-byte magic number, one-byte version, two-byte little-endian payload length, payload bytes, and a checksum octet. The decoder first confirms that at least the fixed header and checksum are present. Only then does it read the header fields. Next it verifies the payload length against both its application-defined maximum and the remaining input capacity. Finally, it computes the checksum and copies the completed candidate into the caller's output.

The order matters. If code reads `bytes[3]` before checking `length >= 4`, a later bounds check cannot undo the invalid read. If it calculates `payload_length + 5` without checking representability, a large unsigned value could wrap. If it writes the public result field-by-field and later discovers an invalid checksum, the caller sees partially updated state.

A strong decoder therefore has a simple publication point: all validation succeeds, and only then is `*out` assigned.

10.20 Define three failure classes

Not all failures have the same meaning. A useful API might distinguish malformed input, incomplete input, and exhausted output capacity. Malformed input violates the format even if more bytes were supplied. Incomplete input may be a valid prefix that needs more data. Insufficient output capacity means the input might be valid, but the caller did not provide enough destination storage.

An RTOS message receiver can react differently: wait for the rest of an incomplete frame, discard malformed input and resynchronize, or apply backpressure when output storage is full. Collapsing all three into a boolean failure can make the transport logic awkward.

The simplest teaching program can return `bool`, but a production interface should choose the status representation based on its callers.

10.21 Error recovery is a protocol decision

After rejecting a malformed frame, should the receiver scan for the next magic byte? Doing so may recover quickly, but payload bytes could themselves contain that marker. A more robust framing design might escape reserved bytes, use a fixed header with a validated length and checksum, or rely on an out-of-band boundary.

The algorithm's *resynchronization* rule belongs in the protocol specification. Otherwise two devices may interpret the same corrupt stream differently despite both having memory-safe parsers.

10.22 Worst-case resource accounting

To evaluate firmware use, compute worst-case storage and work. A parser with a maximum 64-byte payload, a four-byte header, and a checksum can bound its input record storage. The actual C output structure may be larger due to padding, so assess `sizeof` on the target when reserving a pool of output records.

The per-record checksum loop has at most a fixed number of iterations. That proves an iteration bound, not a deadline: execution time also depends on scheduling, memory placement, compiler settings, and hardware. A task that blocks on a mutex or output stream may take longer than its local loop suggests.

Exercises

10.11 Implement a status enumeration with valid, incomplete, malformed and no-capacity cases. **10.12** Design a resynchronization policy when a payload contains the magic byte. **10.13** Calculate an upper bound on parser storage independent of input stream length. **10.14** Test that all error paths preserve the output structure's meaningful members.

10.23 A bounded binary record format

A type-length-value (TLV) record stores each field as a type identifier, a length, and a sequence of value bytes. This pattern appears in binary protocols, embedded device configuration, and extensible message formats. The exact field widths and constraints must be specified; the letters 'TLV' alone do not define a universal byte layout.

For our teaching protocol, each field has:

- One octet identifying the tag.
- One octet containing an unsigned byte count.
- Exactly that many following payload octets.

A complete message is a concatenation of zero or more fields. The application imposes two further limits: at most four fields and at most eight payload octets per field. Repeated tags are permitted, and a zero-length field is valid.

Those limits bound memory independent of the input's advertised lengths. The parser rejects a message that exceeds them rather than allocating more storage or truncating a field.

The complete implementation is `examples/tlv_decode.c`.

10.24 Parsing is a sequence of narrowing proofs

A byte array supplied by the caller does not have the same type as the application record. The decoder must establish that every external field fits the chosen internal representation.

Suppose `cursor` is the number of input bytes already consumed and `size` is the supplied input length. Maintain the invariant:

```
0 <= cursor <= size
0 <= decoded_field_count <= 4
```

Before reading a two-octet field header, test `size - cursor >= 2`. That subtraction is safe because the cursor invariant already holds.

Read tag and length only after this test. Advance cursor by two, then check that the length is no greater than eight and that `length <= size - cursor`. Only after these checks is it valid to copy the payload bytes from the source.

This approach avoids forming potentially overflowing sums such as `cursor + header_size + length` before checking bounds. It also avoids reading the length octet from a truncated one-byte header.

10.25 A strong failure guarantee

A decoder that fills its caller's result structure incrementally can leave partially decoded data on a malformed input. That is not always forbidden, but it complicates callers that want a previous valid configuration to survive a failed update.

The TLV decoder instead begins with a local empty candidate. It appends each validated field to that candidate and assigns the whole result to `*out` only after consuming the complete input successfully.

```
struct tlv_message candidate = {};
/* Validate every field and fill candidate. */
*out = candidate;    /* Exactly one publication point. */
return true;
```

The program intentionally retains a simple `bool` result: malformed input, excessive field count, and oversized field values all return false. For a production protocol, an enumeration distinguishing those failure causes may be more useful.

On failure, no meaningful member of the output result is modified. The implementation tests this guarantee after a valid two-field message by attempting several malformed messages and checking that the earlier result is retained.

10.26 What zero-length input means

The grammar permits an empty message containing zero TLV fields. Consequently, the decoder accepts `bytes == nullptr` when `size == 0` and produces an empty result. It rejects a null input pointer when the supplied size is nonzero.

This explicit convention avoids a common C pointer mistake: attempting to construct an end pointer by writing `nullptr + 0`. The decoder uses indexes and never performs pointer arithmetic on a null input.

An application might instead require at least one field and reject an empty message. That is a valid alternative, but the rule belongs in its specification rather than being implied by an implementation accident.

10.27 Truncation and extra bytes

Consider these byte sequences:

```
01 03 0A 0B 0C   Valid field: tag 1, three bytes 0A 0B 0C
01 04 0A 0B 0C   Invalid: claims four bytes, supplies three
```

```
01               Invalid: incomplete field header
01 00 02 00      Valid: two empty fields, tags 1 and 2
```

The decoder does not accept the valid prefix of a malformed message and ignore trailing or missing bytes. Its loop must consume exactly the supplied length in complete TLV fields.

The tests also supply a field whose claimed length exceeds eight and a message containing five empty fields. Both are rejected by application limits even though the one-byte length value itself is representable.

Memory safety is necessary but not sufficient for protocol correctness. A decoder could remain within every buffer while silently ignoring an unsupported tag or interpreting an unknown field incorrectly. A real application often adds tag validation, mandatory-field checks, versioning, and field-order rules.

10.28 Unknown fields and forward compatibility

A flexible protocol may allow unknown tag values to be skipped using their lengths. This can help newer senders communicate with older receivers. But skipping a field is safe only after its claimed length has been validated against the available input bytes.

Different tags may also have different allowed lengths. A timestamp might require exactly four octets, while a device name might be a variable-length text field. A general decoder can first validate the outer TLV shape and then run a second semantic validation stage for each recognized tag.

Do not assume unknown fields are harmless. A security- or safety-critical field marked mandatory might need to cause rejection when the receiver cannot interpret it. Protocol evolution requires an explicit compatibility policy.

10.29 Storage and timing bounds

The example's result structure reserves four field slots, each containing up to eight value octets plus tag and length metadata. Its maximum storage is known from the C type, although padding means `sizeof(struct tlv_message)` need not equal a simple sum of wire-format octets.

The decoder's loop can accept at most four fields before reaching its success condition. It rejects an attempted fifth field. Its payload copies are limited to eight octets each, establishing a straightforward bound on local work for valid and rejected records.

This is a useful design for small embedded messages: resource exhaustion becomes an ordinary error value rather than unexpected heap allocation or unbounded parsing.

However, if the surrounding transport waits indefinitely for a complete message, the application's end-to-end latency is not bounded merely because the parser is. Transport and parser timing must be assessed separately.

10.30 From parser to authenticated command

The educational TLV example intentionally has no checksum, authentication, anti-replay protection, or authorization. It must not be treated as a secure command protocol simply because it performs strict bounds checking.

A production command receiver should distinguish the integrity of transmitted data, the identity or authorization of the sender, and the semantic validity of the command. A CRC may detect accidental corruption; cryptographic authentication may be required when malicious modification is in scope. Bounds checks cannot provide those properties.

The separation is valuable architecturally. A framing layer obtains a complete message, an integrity layer verifies the appropriate check, the TLV decoder constructs a typed candidate, and the application verifies allowed operations before committing state.

Additional exercises

10.15 Modify the TLV decoder to return distinct statuses for truncated header, truncated payload, excessive field count, and oversized value. **10.16** Specify mandatory tags and reject duplicate instances of those tags. **10.17** Add an unknown-field skipping policy while preserving all bounds checks. **10.18** Build an independent expected-value test for two TLV fields rather than relying on an encoder written with the same algorithm. **10.19** Explain why a bounded parser is not automatically a secure or real-time protocol implementation.

10.31 Framing one byte at a time

The complete TLV decoder earlier in the chapter assumes that its caller already has a complete candidate message. Real serial links often provide only a few bytes at a time. A UART receiver might produce one octet per interrupt; a DMA operation might deposit a block; a network API may return a fragment that ends in the middle of a field.

A streaming frame parser makes progress incrementally. It keeps a small state object between calls and processes one received octet at a time. It never needs to allocate a buffer proportional to an untrusted advertised length.

For an educational frame format, define these octets:

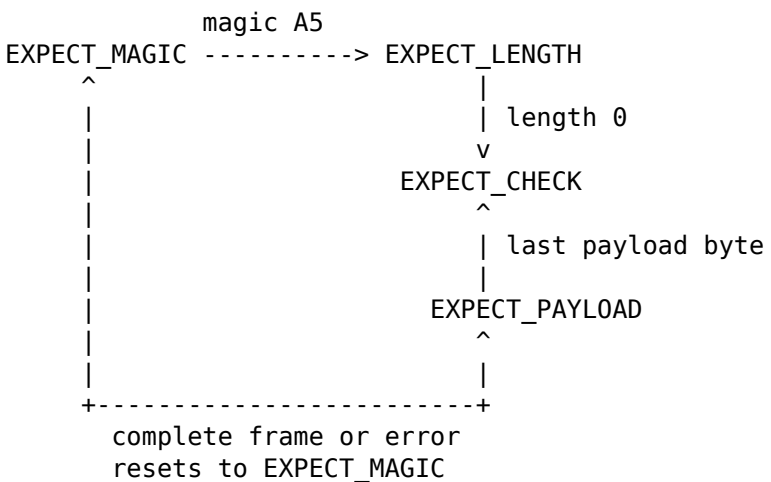
```
+--------+--------+----------------------+----------+
| magic  | length | payload (0..8 octets) | checksum |
+--------+--------+----------------------+----------+
  0xA5    0..8     exactly length bytes    XOR below
```

The checksum is the bitwise XOR of the length octet and all payload octets. The magic octet is not included. The format is intentionally simple enough to hand-calculate expected frames. It is not cryptographically secure, nor is its XOR checksum a robust choice for all error-detection environments.

The implementation appears in `examples/frame_stream.c`. It requires eight-bit octets and makes that assumption explicit at compile time.

10.32 Four parser states

The parser recognizes a frame using four states:



The actual transitions are:

- `EXPECT_MAGIC`: accept only the framing marker and move to length; other octets are rejected.
- `EXPECT_LENGTH`: reject a value above eight; otherwise store the expected payload count and choose `EXPECT_CHECK` for zero-length payload or `EXPECT_PAYLOAD` for a nonempty payload.
- `EXPECT_PAYLOAD`: append exactly the announced number of octets, updating the running XOR.
- `EXPECT_CHECK`: compare the next octet with the computed XOR and either publish one complete frame or reject it.

This explicit state design is useful for RTOS tasks and event-driven firmware. Each call performs a bounded amount of work, independent of how long the input stream runs.

It does not guarantee that the task receives input at a bounded interval; scheduling, interrupt latency, and transport backpressure are separate requirements.

10.33 Invariants of the parser state

A helpful proof begins by identifying what each state makes true.

In `EXPECT_MAGIC`, no partial frame remains. The parser can accept a new framing marker.

In `EXPECT_LENGTH`, the magic octet has been accepted but the length has not yet been read.

In `EXPECT_PAYLOAD`, the declared length is between one and eight, and `received < length` before appending the next octet. Since the destination array has eight elements, the access `candidate.bytes[received]` is within bounds.

In `EXPECT_CHECK`, the announced payload length has been fully received, including the zero-length special case. The running checksum corresponds to the length and every received payload octet.

Every rejected frame resets the parser to its initial state. On success, the parser assigns its completed candidate to the output and then resets. That single publication point ensures callers never see a partly decoded public message.

In the example, the output is a `struct message` containing a count and an eight-element array. The parser owns a separate candidate structure and only copies it once validation has succeeded.

10.34 Bounded work per byte

The parser performs at most a few comparisons, one array store in the payload state, and an XOR operation for each input byte. It never calls `malloc`, loops over an untrusted count, or copies a variable-length region in one invocation.

Its memory requirement is constant: a parser state object and an output message supplied by the caller.

This establishes a useful algorithmic property: processing any one octet requires a bounded number of C operations. It does **not** establish a precise worst-case execution time on every MCU. Compiler optimization, interrupt preemption, cache behavior, memory wait states, and synchronization can affect actual latency.

When moving the parser to an RTOS, a good design separates the interrupt-level byte acquisition from higher-level frame interpretation. An ISR may deposit octets into a bounded queue; an ordinary task can then call `frame_feed`. Queue overflow and input dropping must have a documented policy.

10.35 An explicit three-result interface

The feed function returns one of:

Result	Meaning	Public message
`FEED_MORE`	A possible frame is incomplete	Unchanged
`FEED_COMPLETE`	A complete valid frame has been received	Updated
`FEED_REJECTED`	Invalid marker, length, checksum, or API state	Unchanged

The caller can process completed messages without needing to inspect a partial buffer. It can ignore or count rejected octets according to its transport policy.

An error return does not necessarily mean an exceptional software fault. In a noisy communications link, malformed input is normal data that a decoder must handle safely.

The parser treats `FEED_MORE` as a legitimate result. A caller must not interpret it as a failure and reset the parser after every input fragment, or multibyte frames could never complete.

10.36 Checksum validation and publication

A frame with payload `0A 14 1E` and length three has a checksum obtained by XORing `03 0A 14 1E`:

```
A5 03 0A 14 1E 03
|  |  |  |  |  +-- XOR checksum
|  |  +--+--+----- three payload octets
|  +-------------- length 3
+----------------- magic
```

The checksum happens to be `03` for this example. That coincidence is not part of the general format; different payloads produce different XOR results.

A failed checksum must not modify the caller's previously completed message. The parser resets its internal candidate and consumes the failing checksum octet.

The code checks this property by setting a recognizable output length, sending a corrupted frame, and verifying that the old output remains intact.

10.37 The subtle problem of resynchronization

The educational parser uses the simple policy **consume the rejected octet and reset**. It does not reconsider that octet as a possible magic marker.

This has a consequence. If the bad checksum octet equals `A5` and simultaneously would have been the start of a new frame, that potential new frame start is lost. Likewise, a magic octet inside a payload is treated as payload, not as a new frame boundary.

Neither behavior is necessarily a bug; it follows from the specified framing policy. However, a robust protocol must decide how receivers recover from corruption and where true message boundaries can be discovered.

A stronger design might use byte stuffing or escaping, a delimiter that cannot appear unescaped inside a payload, an authenticated record layer, or a scanning strategy with carefully tested recovery rules.

Searching blindly for the next `A5` is not automatically correct, because payload data may naturally contain that octet. The format, not just the parser code, determines what resynchronization can guarantee.

10.38 Timeout, truncation, and reset

An input source can stop providing bytes midway through a frame. For a parser at its initial state, clean stream termination means no incomplete frame remains. For a parser in any other state, final termination means the last frame is truncated.

The caller can query whether the parser is at a clean boundary and, on truncation, record an incomplete-input error. In an ongoing UART session, a temporary pause may not justify resetting the parser immediately. A higher-level timeout policy determines how long partial frames are retained.

The example exposes a reset operation for explicitly abandoning a partial frame. Reset is an ownership and protocol action: any bytes previously accumulated are discarded and cannot be recovered.

For applications requiring deadlines, a timestamp of the most recent received byte or an expected completion deadline can accompany the parser state. Timekeeping itself should remain outside the pure byte-decoding operation so host tests can supply deterministic timestamps.

10.39 How to test the state machine

Use independently calculated packets rather than relying exclusively on an encoder written by the same developer. The test suite covers:

- One valid frame containing three payload octets.
- A valid zero-length payload.
- An invalid advertised length above eight.
- A bad checksum that must not publish a result.
- A valid prefix that ends before all payload bytes arrive.
- Explicit reset after abandoning a truncated frame.

Additional recommended tests include every payload length from zero to eight, all possible single-byte corruptions of representative packets, and splitting a valid frame at each transport boundary.

A property-based test can generate valid frames and feed them one octet at a time. The resulting message must match the generated payload. A separate set of manually constructed vectors remains important because both a generator and decoder can share a format mistake.

10.40 Choosing a protocol interface

A whole-record parser is often simpler when the input transport supplies bounded complete messages. An incremental state machine is useful when records arrive in fragments or the application has a small per-invocation work budget.

Neither approach universally replaces the other. A stream parser often delegates completed frames to a whole-record semantic decoder. For example, `frame_feed` could identify and validate a small payload, then a TLV decoder could interpret fields inside that payload.

Each layer would have a distinct contract:

1. Transport supplies octets and reports delivery errors.
2. Framing recognizes one complete message.
3. Content decoding validates field representation and lengths.
4. Application logic checks the meaning, authorization, and allowed state changes.

That separation makes it easier to test algorithms on a desktop and replace only the transport adapter when moving to embedded hardware.

Exercises

10.20 Construct the byte sequence for an empty frame and explain every state transition. **10.21** Prove that the payload store can never write beyond index seven. **10.22** Design a separate timeout policy for an incomplete frame. **10.23** Explain what can happen when a failing checksum octet is also a magic marker. **10.24** Extend the parser with error categories for invalid marker, invalid length, and checksum failure. **10.25** Describe how an RTOS task would feed bytes acquired through a UART interrupt-driven ring buffer.

11. An Extended Example: A Word Index

A word counter is a small program that demonstrates streaming input, fixed-capacity state, tokenization, searching, and error reporting. We deliberately avoid dynamic allocation so the memory bound is explicit.

11.1 Requirements

Read standard input one byte at a time. Treat ASCII letters and digits as part of words, fold uppercase ASCII to lowercase, store up to 1024 distinct words of at most 31 bytes, and print each word with its occurrence count. Reject excess distinct words rather than corrupting the table.

This is an ASCII-oriented utility, not a general Unicode word segmenter. Its fixed-size table and linear search make worst-case costs visible.

11.2 Implementation

```
#include <stdio.h>
#include <string.h>
#include <stdbool.h>
#include <stddef.h>

enum { MAX_WORDS = 1024, MAX_LEN = 31 };

struct entry {
    char word[MAX_LEN + 1];
    unsigned long count;
};

static struct entry entries[MAX_WORDS];
static size_t used;

static bool letter_or_digit(int c)
{
    return (c >= 'a' && c <= 'z') ||
           (c >= 'A' && c <= 'Z') ||
           (c >= '0' && c <= '9');
}

static char lower_ascii(int c)
{
    if (c >= 'A' && c <= 'Z') c += 'a' - 'A';
    return (char)c;
}

static bool add_word(const char *word)
```

```
{
    for (size_t i = 0; i < used; ++i) {
        if (strcmp(entries[i].word, word) == 0) {
            if (entries[i].count == ~0UL) return false;
            ++entries[i].count;
            return true;
        }
    }
    if (used == MAX_WORDS) return false;
    strcpy(entries[used].word, word); /* bounded by tokenizer */
    entries[used].count = 1;
    ++used;
    return true;
}

int main(void)
{
    char word[MAX_LEN + 1];
    size_t len = 0;
    int ch;
    bool too_long = false;
    while ((ch = getchar()) != EOF) {
        if (letter_or_digit(ch)) {
            if (len < MAX_LEN)
                word[len++] = lower_ascii(ch);
            else
                too_long = true;
        } else if (len != 0 || too_long) {
            if (too_long) {
                fputs("word exceeds limit\n", stderr);
                return 1;
            }
            word[len] = '\0';
            if (!add_word(word)) {
                fputs("word table full or counter overflow\n", stderr);
                return 1;
            }
            len = 0;
        }
    }
    if (ferror(stdin)) {
        fputs("input error\n", stderr);
        return 1;
    }
    if (too_long) {
        fputs("word exceeds limit\n", stderr);
        return 1;
    }
    if (len != 0) {
        word[len] = '\0';
        if (!add_word(word)) return 1;
    }
    for (size_t i = 0; i < used; ++i)
        printf("%-31s %lu\n", entries[i].word, entries[i].count);
    return ferror(stdout) ? 1 : 0;
}
```

The total table storage is fixed. The program checks every insertion, rejects too-long words, and refuses counter wraparound. The final output is in order of first appearance, not lexicographic order.

11.3 Complexity and extensions

Each word may be compared against up to 1024 existing entries, so the lookup time is linear in the number of distinct words. A hash table can improve average lookup, but then needs collision handling, a load-factor policy, and carefully bounded worst-case behavior. Sorting entries before printing provides alphabetical output without changing the tokenizer.

Exercises. 11.1 Sort the final table alphabetically. 11.2 Replace linear lookup with open addressing. 11.3 Count overflowed word lengths rather than aborting. 11.4 Implement Unicode-aware tokenization using a deliberately selected library, documenting its normalization rules.

11.4 From linear search to a sorted index

The first implementation searches entries in order of first appearance. If there are *u* distinct words, one lookup may require *u* comparisons. We can instead keep the table sorted and use binary search to locate an existing word or the place to insert a new one.

For a sorted array with bounds `lo` and `hi`, the active range is half-open: `[lo, hi)`. At each iteration, compute `mid = lo + (hi - lo) / 2`; this avoids the potential overflow of `(lo + hi) / 2`. Compare the word at `mid` with the query. If it sorts earlier, the search continues after `mid`; otherwise it continues at or before `mid`. The range narrows on every iteration.

When `lo == hi`, the search ends. The same position is the insertion point for a new word. Shifting entries one position to the right preserves sort order. Lookup costs logarithmic comparisons, but insertion still takes linear time in the number of entries. This is a useful reminder that improving one operation does not automatically improve the entire algorithm.

11.5 A table with explicit invariants

The sorted-array representation has three important invariants: (1) `used` never exceeds `MAX_WORDS`, (2) entries in `[0, used)` are sorted lexicographically, and (3) every stored word contains at most `MAX_LEN` ASCII bytes followed by a zero terminator. Before incrementing a count, test against `ULONG_MAX`. Before inserting, confirm unused capacity. These checks protect both memory bounds and numerical meaning.

The companion complete program, `examples/word_index.c`, demonstrates sorted indexing, ASCII normalization, fixed storage, and failure reporting. It deliberately treats words exceeding the maximum length as an error instead of silently truncating them into potentially ambiguous keys.

11.6 Complexity, measurement, and design

For a text with *n* input bytes and *u* unique words, lexical scanning is linear in *n*. Binary lookup is *O(log u)* string comparisons per completed word; moving records for insertion is *O(u)* in the worst case. Since each string is at most 31 bytes, comparison work has a fixed bound. The static table capacity also provides a fixed memory bound.

A hash table could give expected near-constant lookup, with tradeoffs for collision policy, load factor, hashing cost, and predictability. A balanced tree could avoid moving array entries but would require more bookkeeping. Neither is categorically superior: requirements determine the data structure.

11.7 Additional exercises

11.5 Prove that the binary-search interval strictly shrinks on every iteration. **11.6** Add a configurable maximum word count, without introducing dynamic allocation. **11.7** Implement a frequency-first ordering at print time, preserving alphabetical order on ties. **11.8** Test the empty input, a final word without newline, repeated separators, a full table, and an overlong word. **11.9** Benchmark insertion workloads with vocabulary ordered forward, reverse, and randomly.

11.8 Designing the input grammar

The index accepts ASCII letters and decimal digits and treats everything else as a separator. That simple grammar has two important consequences: punctuation splits words, and any byte outside the specified ASCII set is not recognized as part of a word. A UTF-8 text file may therefore produce surprising tokens unless the program is explicitly extended to decode Unicode.

The rule for overlong words is equally important. Truncating a word after 31 bytes is not merely losing display information; two different long words can collapse into the same key. Rejecting overlong tokens prevents an incorrect count from looking authoritative.

The program also chooses to count only complete words and to accept a final word that has no trailing separator. That behavior is tested separately because streaming programs often get end-of-file boundaries wrong.

11.9 Proving the sorted insertion

The maintained invariant is that `entries[0]` through `entries[used-1]` are sorted and distinct. Binary search finds the first position whose word is not less than the input. There are two cases:

- If an entry at that position compares equal, increment its count after checking for overflow.
- Otherwise, if capacity permits, move the suffix one element right, copy the new word into the gap, set its count to one, and increment `used`.

The move proceeds from the right edge toward the insertion point. Moving in the opposite direction with ordinary assignments could overwrite entries before they were copied. The special bound `used < MAX_WORDS` ensures one free slot exists before any shift.

The search interval `[lo, hi)` shrinks because each iteration either increases `lo` beyond the midpoint or lowers `hi` to the midpoint. Since its endpoints are integers in a finite range, the loop terminates. The midpoint calculation `lo + (hi-lo)/2` avoids adding two potentially large bounds.

11.10 Complexity with meaningful constants

Let `w` be the number of word occurrences, `u` the number of distinct words, and `L` the maximum key length. Tokenization scans input bytes once. A binary search performs at most logarithmically many comparisons in `u`, and each comparison can inspect up to `L` bytes. A new entry may require copying `O(u)` records. Thus workloads dominated by repeated words may be substantially cheaper than workloads dominated by inserting new distinct words.

For a fixed table of 1024 entries and 31-byte keys, the resource ceiling is explicit. That bounded design may matter more in firmware than the asymptotic advantage of a dynamic hash table. A hash implementation is a useful improvement exercise, not a universally better choice.

11.11 Test cases that reveal design errors

Use independent expected outputs for: no input; one word; a final word without newline; repeated separators; uppercase and lowercase variants; words exactly at the maximum length; a word one byte too long; the maximum distinct-word count; and an attempted extra distinct word.

A property test can also compare the sum of all printed counts with the number of successfully recognized word occurrences. This is not a complete proof, since two errors could cancel, but it detects many accidental drops and duplicates.

11.12 Extending the implementation responsibly

An open-addressed hash table needs a defined hash function, collision resolution, load limit, and behavior when no free slot exists. Returning failure is better than silently overwriting an unrelated entry. If the application must preserve sorted output, sort the final records or maintain a separate ordering; hashing by itself does not provide alphabetical order.

A Unicode-aware version is a different product. It must define normalization and word-boundary rules, decide how case folding works, and bound its memory for multibyte text. Simply applying `tolower` to each UTF-8 byte is not enough.

Further problems: Provide a proof that the insertion loop cannot access index `MAX_WORDS`. Write a test generating 1024 distinct terms. Compare sorted-array and hash-table lookup under high repetition. Design a stable frequency ordering for ties.

11.13 Two implementations and two output contracts

This chapter intentionally presents an initial *linear-search teaching implementation* before developing the sorted-index design. The first complete listing stores new words in the order in which it first encounters them. Its final loop prints entries in that insertion order, using a padded word column.

The standalone executable `examples/word_index.c`, in contrast, implements the **sorted-index version**. It uses binary search to locate existing words or an insertion position and shifts records to preserve lexicographic ordering. It emits unpadded `word count` lines and is the version exercised by `tests/test_word_index.sh`.

These are related algorithms, not byte-for-byte copies of the same program. Their important common contracts are bounded storage, case folding for the stated ASCII-compatible environment, count-overflow rejection, and failure rather than silent truncation. Their output ordering and formatting are deliberately different.

For example, the input `pear apple` causes the initial insertion-order implementation to print `pear` before `apple`, while the sorted implementation prints `apple` before `pear`. That is an observable behavioral distinction. A user-facing application or regression test should state which output contract is required.

When migrating from the first implementation to the second, preserve the tokenizer's accepted grammar and error behavior, then change the indexing algorithm and explicitly update the expected ordering. The regression suite tests the second program, not the first chapter listing. A future manuscript-wide listing audit should compile and test both independently.

11.14 A search algorithm with a proof obligation

Binary search is brief enough to fit on a few lines, but it should be understood through its invariant. At the start of each iteration, the insertion point lies in the half-open interval `[lo, hi)`. Every entry strictly before `lo` compares less than the requested word. Every entry at or after `hi` compares greater than or equal to it.

Set `mid = lo + (hi - lo) / 2`. If the middle entry compares less than the request, the insertion position must follow `mid`, so set `lo = mid + 1`. Otherwise, it is at or before `mid`, so set `hi = mid`. When `lo == hi`, exactly one insertion position remains.

The implementation can then determine whether the requested word already exists at that position. If so, it increments a counter only after verifying that increment is representable. If not, it checks table capacity before shifting existing records.

The proof is useful when changing the data structure. One off-by-one mistake can produce a program that appears correct for a small vocabulary but corrupts storage when inserting at the beginning or end.

11.15 The cost of sorted insertion

Binary search reduces the number of comparisons needed to locate a word, but maintaining a sorted array costs more when inserting new records. If there are already `u` distinct words and the new one belongs at the front, the program shifts `u` entries. The worst-case cost of inserting `u` distinct keys is therefore quadratic in `u`, even though each lookup performs only logarithmically many comparisons.

That is not automatically unacceptable. With a hard limit of 1,024 short keys, the worst-case memory usage is small and predictable. A microcontroller may value a fixed allocation and simple failure policy more than average-case insertion speed. For a large text corpus with millions of terms, a hash table or tree is likely more appropriate.

Measure the actual workload: repeated tokens benefit from quick lookup and do not shift records, while a sorted stream of unique tokens repeatedly inserts near one end. Big-O notation tells us the growth trend, but a real resource bound must still account for maximum keys and fixed string lengths.

11.16 Error handling belongs inside the index

Rejecting a long token rather than truncating it prevents two different words from becoming the same key. Rejecting an eleventh-hundredth distinct key rather than writing past the table prevents memory corruption. Rejecting counter overflow prevents the program from silently reporting a smaller count than the truth.

After any rejection, decide whether the program should stop, skip the offending token and continue, or expose a detailed status. The teaching implementation stops at the first error. A long-running service might recover, but only with a documented strategy that avoids partially committing a word.

The internal operation `add_word` returns success or failure; a larger program could use an error enumeration to distinguish table-full from count-overflow. That would improve diagnostics without changing the sorted-array algorithm.

11.17 Independently test the expected output

A decoder and encoder written with the same wrong byte order can agree with each other. Likewise, a word index and an expected-result generator copied from the same algorithm can share the same defect. For a handful of text samples, write the expected alphabetical output manually.

Then add generated larger cases with invariants: total counts must equal accepted word occurrences, entries must be strictly sorted and unique, and no count may be zero. For a stable random test, record the seed so failures can be reproduced.

Exercises

11.10 Trace the search interval for inserting a word at the beginning, middle, and end of a five-element table. **11.11** Estimate the maximum number of shifts for inserting 1,024 distinct words in reverse order. **11.12** Design an error enumeration that distinguishes a long word, full table, input

failure, and counter overflow. **11.13** Write an independent expected output for a file containing repeated words with differing capitalization.

11.18 Trace a complete input

The word-index program is small enough to reason about from its first input byte to its final output. Consider the stream:

```
Pear apple pear BANANA apple
```

The tokenizer recognizes letters and decimal digits and treats the intervening space as a separator. Under its intended ASCII-compatible execution encoding, it converts the uppercase letters to lowercase before updating the index.

The resulting logical sequence of words is:

```
pear, apple, pear, banana, apple
```

The index maintains entries in sorted order, so its final output is:

```
apple 2
banana 1
pear 2
```

This output is independently determined by the input, rather than constructed by running the index algorithm again. That distinction is useful in testing: the expected result must not rely on the implementation being correct.

11.19 A trace of sorted insertion

Start with an empty table and insert `pear`: the search interval is `[0,0)`, so the insertion position is zero.

Insert `apple`: the search finds position zero because `apple` sorts before `pear`. The program shifts the existing record right and inserts `apple` at the first slot.

Insert `pear` again: the search locates the existing entry at position one and increments its count rather than inserting a duplicate.

Insert `banana`: the new word belongs between `apple` and `pear`, so the program moves `pear` right by one slot. The table now contains three distinct words.

Finally, inserting `apple` again increments the first entry's count. No shifting occurs for repeated words.

At each step, the invariant is that exactly the first `used` entries contain initialized words, those words are strictly increasing according to `strcmp`, and all counts are positive.

The implementation can therefore print the table directly without a final sorting pass.

11.20 Why the midpoint is written carefully

The binary search uses the half-open interval `[lo, hi)`. Its midpoint is:

```
size_t mid = lo + (hi - lo) / 2;
```

Before this expression, the algorithm maintains `lo <= hi <= used`. Therefore `hi - lo` is defined without unsigned underflow, and adding at most half that difference to `lo` yields a result between `lo` and `hi`.

The superficially simpler expression `(lo + hi) / 2` may overflow a sufficiently narrow unsigned type even when both endpoints are valid indexes. In this particular fixed-capacity example, `used`

never exceeds 1,024 and such overflow is not realistic on its supported hosts. The subtraction-based midpoint is nevertheless a clear reusable pattern.

When the middle entry is lexically less than the requested word, the insertion position must follow it; the search assigns `lo = mid + 1`. Otherwise it assigns `hi = mid`. The interval shrinks until the endpoints meet.

The final position identifies either an existing word or the location where a new word should be inserted.

11.21 Moving entries without losing one

Suppose there are `used` initialized entries and a new word belongs at index `lo`. Before insertion, confirm that `used < MAX_WORDS`. The program moves entries from right to left:

```
for (size_t j = used; j > lo; --j)
    entries[j] = entries[j - 1];
```

On the first iteration, the destination is `entries[used]`, which is available because the capacity check established that one extra slot exists. The source is `entries[used - 1]`, the final initialized entry.

Moving right to left is necessary. Moving left to right would overwrite an entry before its old contents had been copied to the next slot.

When `lo == used`, the loop executes zero times and the word is appended. When `lo == 0`, every existing entry shifts one position. This creates a linear worst-case cost for insertion.

Each `struct entry` includes its word bytes directly, so structure assignment copies the complete inline representation. There is no owning pointer that would require a deep-copy operation. This is a deliberate simplification of the data structure.

11.22 Fixed resource bounds

The example defines:

```
enum { MAX_WORDS = 1024, MAX_LEN = 31 };
```

Each entry stores a word buffer of 32 `char` elements and an `unsigned long` count. Thus the table uses the storage required by 1,024 such structures, including any implementation-selected padding or alignment. Its exact RAM footprint is `sizeof entries` on the selected target.

The tokenizer uses a single 32-element buffer plus a length and an overlong-word flag. No storage grows with the input file size.

The application therefore has predictable memory consumption, but input processing time depends on the number of bytes and words. A new distinct word near the beginning may shift many entries; repeated words require a binary search and count increment. Once 1,024 distinct words exist, a new distinct word is rejected.

A production application might decide to evict old entries, use a fixed-size hash table, or retain approximate counts. Those choices change the observable output contract. The teaching program instead reports exhaustion explicitly.

11.23 What 'ASCII' means in this program

The source uses numerical range tests such as `c >= 'A' && c <= 'Z'`, and it converts uppercase letters by adding the difference between the lowercase and uppercase initial letter codes.

This assumes an execution character encoding with contiguous alphabetic ranges and uniform uppercase/lowercase offsets, as is true on common ASCII-compatible systems. The C language

guarantees that decimal digits are consecutively encoded, but does not require every alphabetic character range to be contiguous on every conforming implementation.

The program's intended scope is ASCII-oriented text processing on a compatible hosted target. It should not be described as a universally portable Unicode tokenizer.

For a locale-sensitive C application, character classification functions in `<ctype.h>` may be appropriate, with their argument-domain restrictions respected. For UTF-8, byte-range classification alone does not identify Unicode letters. A Unicode tokenizer would need decoding and higher-level character-property decisions before indexing.

Explicitly stating this limitation is more accurate than claiming that a small ASCII example automatically supports every text encoding.

11.24 The three end-of-input cases

The tokenizer recognizes a word when a separator is encountered after word bytes. But a stream can end immediately after the final word without a separator.

The program therefore has a final step after the input loop: if the current word length is positive, terminate the buffer and insert that last word. Without this step, a file ending in an ordinary letter or digit would silently omit its final token.

There is another case: an overlong word reaches EOF without a separator. The program must still reject it. It does not become legal merely because a delimiter never appeared.

Finally, `getchar` returning `EOF` does not alone establish clean end of input. The program checks `ferror(stdin)` before reporting success. These cases illustrate why stream termination must be part of the algorithm, not an afterthought.

11.25 Improve the regression tests

The original test feeds a small mixed-case sentence, checks sorted output and counts, verifies empty input, and expects a word longer than 31 characters to be rejected.

A stronger boundary suite should also test:

- Exactly 31 valid word bytes.
- A 32-byte word rejected without truncated insertion.
- A final valid word with no trailing newline.
- Multiple separators in a row.
- A repeated word at the beginning and end of the stream.
- Output order when inserting distinct words in reverse alphabetical order.

The program exposes failure through a nonzero exit status and a diagnostic written to `stderr`. Tests must inspect the exit status, not infer success from the presence of output alone.

Further exercises

11.14 Produce an insertion trace for five words in reverse alphabetical order and count the total number of shifted entries. **11.15** Derive the exact worst-case number of shifts for inserting 1,024 distinct words at the front of the table. **11.16** Add boundary tests for exactly 31 and 32 word bytes. **11.17** Explain what would need to change before the tokenizer could correctly process UTF-8 letters outside ASCII. **11.18** Replace the fixed inline word buffers with dynamically owned strings, then describe the additional ownership and failure cases created by that design.

12. Exercises and Further Reading

The best way to learn C is to predict a program's behavior before compiling it, then explain any discrepancy. Exercises should include ordinary cases, boundaries, and failure cases.

12.1 A progression of projects

Project A — Bounded line reader. Implement a function that fills a caller-supplied buffer, reports bytes stored, distinguishes newline from EOF, and signals truncation. Write tests for zero-capacity buffers, one-character buffers, long lines, embedded zero bytes, and input errors.

Project B — Integer parser. Parse unsigned decimal text into `unsigned` without `atoi`. Reject empty input and nondigits; detect overflow before multiplying by ten or adding the next digit. Show why checking after an arithmetic overflow is too late.

Project C — A fixed pool. Implement allocation and release of equal-sized records from a fixed array. Reject invalid and repeated releases. Describe whether the pool can be used concurrently, and if so how it is synchronized.

Project D — Binary message reader. Decode a header with a version byte, 16-bit little-endian payload length, and checksum. Check all bounds before decoding. Test malformed lengths and truncated buffers.

Project E — Word index. Improve Chapter 11 with a hash table, sorting, and a test driver. Benchmark it using increasing vocabulary sizes and measure memory usage.

12.2 How to study source code

Read short functions and annotate their contracts. What objects must exist at entry? What inputs are valid? Which values may overflow? What state changes after success? What state is preserved after failure? Which implementation-defined assumptions are present?

Compiler warnings are evidence to inspect, not text to suppress reflexively. Sanitizers find bugs only on executed paths; code review and carefully constructed tests remain essential.

12.3 References

- ISO/IEC 9899:2024, *Programming languages — C* (normative specification).
- cppreference, C language reference, especially its C23 features and library pages.
- GCC C standards implementation status and command-line dialect options.
- Clang C language status and diagnostics documentation.
- The ISO C Secure Coding Rules and CERT C guidance, for additional defensive-programming perspectives.

12.4 A note on editions

This is an independent original textbook. Classic C teaching remains valuable for its economy of examples, but 2026 readers also need accurate discussion of defined behavior, lifetime, ownership, toolchain variation, safer interfaces, and C23's additions. Future manuscript revisions should add tested solutions, further real-world worked examples, portable build checks, and an index before a complete book-length edition is declared.

12.1 Exercises as specifications

An exercise is complete only when it states what the program accepts, what it returns, and how failure is reported. Consider a decimal parser: what does it do with an empty string, leading

spaces, a plus sign, invalid characters, and a number larger than the destination type? The interesting part of the problem is often deciding these details before coding.

12.5 Testing boundaries

For a bounded collection with capacity N, test insertion at sizes zero, one, N−1, and N, then one additional insertion that must fail without damaging existing records. For signed arithmetic, test the largest and smallest representable values, opposite signs, and zero. For a byte parser, test all shorter-than-required inputs.

12.6 Integration project: a streaming index

Read Chapter 11's sorted word index and complete the following experiments:

1. Validate a file containing zero words and a file ending with an unterminated word.
2. Check that capitalization is folded only for ASCII letters and that digits remain in the word.
3. Generate 1,024 unique words and verify the table fills without overrun.
4. Attempt one more unique word and confirm the operation fails.
5. Compare counts against an independent reference implementation for randomly generated ASCII input.
6. Measure runtime for sorted and reverse-sorted unique vocabularies to investigate insertion cost.
7. Replace the sorted array with a fixed-capacity open-addressed hash table; state the collision policy, load factor, and failure semantics.
8. Describe how a UTF-8 version would need a more sophisticated definition of words, code points, and normalization.

12.7 Integration project: a bounded vector

Start from `examples/vector.c`. Add `reserve`, `clear`, and `get` operations with contracts. Make failed growth preserve the old allocation and data. Test failed allocation by injecting an allocator that rejects selected requests. Discuss why assigning one vector object to another is not a deep copy.

12.8 Reading C with a reviewer

For every function, identify which variables represent length and capacity, which arithmetic might overflow, which objects are borrowed versus owned, whether outputs change on failure, which operations may allocate, and whether concurrent calls can safely share state. A program may be short and still require serious reasoning.

12.9 Challenge: a portable protocol reader

Design a packet decoder accepting `const unsigned char *bytes` and `size_t length`. A packet contains a magic byte, a version, a two-byte little-endian payload length, payload bytes, and an eight-bit checksum. Reject incomplete packets before reading fields; reject lengths inconsistent with the buffer; do not write the output structure until the checksum passes. Provide cases for exact minimum length, maximum supported payload, bad version, invalid checksum, and length overflow.

12.10 Challenge: concurrency

Design a producer-consumer queue for one interrupt-side producer and one thread-side consumer. State why Chapter 5's plain queue is not automatically safe under concurrent access. Choose either an RTOS synchronization primitive with appropriate interrupt semantics or a target-validated

atomic design. Document memory-order assumptions and why the chosen operations are legal in both execution contexts.

12.11 Further reading in practice

After solving a problem, read the relevant normative language clause or trusted reference and compare it with the implementation's documentation. A standards-conforming program is not merely one that a compiler accepts; conformance involves the behavior required by the language and library specification.

12.12 How to write convincing solutions

A solution should include more than a working program. State assumptions, list invariants, justify any bounds arithmetic, and show representative tests. A design that accepts invalid input only because one particular compiler tolerated it is not a complete solution.

A useful submission format has five short parts: (1) contract, (2) algorithm, (3) implementation, (4) test table, and (5) limitations. In a small exercise this may fit on one page. In a hardware-related problem, also identify dependencies on the toolchain and target.

12.13 Progressive memory exercises

Start with a fixed array of integers. Add a function returning its count of nonzero elements. Next write a version accepting a pointer and a length. Then add dynamic allocation with a checked size calculation. Finally add a failure-injecting allocator and verify that unsuccessful growth preserves existing state.

These stages teach the same abstraction with increasing responsibility. They also show how a simple algorithm can remain unchanged when its storage policy changes.

12.14 Progressive text exercises

First write an ASCII decimal parser that rejects every non-digit. Next permit leading spaces while still requiring at least one digit. Then add an optional sign and specify the behavior for signed minimum and maximum values. Finally switch from a terminating string to a counted byte span.

For each stage, list the accepted grammar explicitly. Tests should include zero, negative values when allowed, empty input, trailing junk, maximum representable values, and one value beyond the chosen range.

12.15 A bounded data pipeline

Combine three earlier projects into an application: read a binary message, validate its header and checksum, then append a decoded value to a fixed-size queue. Your contract must explain what happens when the queue is full. Does the program reject the message, drop the oldest value, or report backpressure? Each is a meaningful policy choice.

The test should inject short packets, malformed lengths, checksum failures, and queue saturation, and verify that rejected packets do not alter queue contents.

12.16 A portability investigation

Compile a set of probes that report `CHAR_BIT`, `sizeof(int)`, `sizeof(void *)`, the availability of `uint32_t`, and support for selected C23 features. Do not treat the output of one workstation as universal. Repeat on a second compiler, a different architecture, or a freestanding target where possible.

Compare the results with the preconditions stated in the book's byte-oriented and dynamic-memory examples. This exercise joins language rules to engineering practice.

12.17 A review rubric

Assess work in four dimensions, not one. **Correctness**: do the algorithms meet their contracts? **Safety**: are arithmetic and storage bounds checked? **Clarity**: can another engineer understand the code and its assumptions? **Verification**: do the tests challenge boundaries rather than merely demonstrating one successful input?

A concise, correct implementation with explicit limits is preferable to a larger one that hides failures in complicated abstractions.

Further problems: Review a peer's parser without running it, then compare your findings with compiler and sanitizer output. Build a small executable that verifies a public header compiles independently. Explain which claims in a test report are observed and which remain assumptions.

12.18 Guided project: a bounded arena

A fixed byte arena is a compact way to practice both arithmetic and ownership. The repository's `examples/arena_bytes.c` contains a reference implementation. Before reading it, write your own design specification.

Your arena contains an array of 128 `unsigned char` elements and a `size_t used` offset. It must accept a request for a nonzero number of bytes if the request fits in the unallocated suffix. A successful request returns a pointer to that region and advances the offset. A failed request must leave the offset and the caller's output pointer unchanged.

Work through three stages.

Stage A: capacity proof. Write the invariant `used <= capacity`. Show that after checking it, the expression `capacity - used` is defined and nonnegative in the mathematical sense. Check `count <= capacity - used` before adding count to the offset.

Stage B: behavior. Allocate five bytes, then seven bytes. Confirm that the second returned pointer lies immediately after the first region. Request more than the remaining capacity and ensure failure changes no state. Fill the arena exactly and verify that the next one-byte request fails.

Stage C: reuse. Reset the arena offset. Confirm a fresh request starts at the first byte. Explain why every previously borrowed region must now be treated as invalid under the arena's logical lifetime contract.

This project allocates byte spans, not arbitrary typed objects. To extend it into a general-purpose typed arena, investigate alignment, backing-storage guarantees, object access rules and target DMA constraints before introducing casts.

12.19 Guided project: signed decimal input

Write a parser for `long` with an optional sign followed by one or more ASCII decimal digits. Reject spaces and trailing garbage. Failure must leave the output unchanged.

The challenge is `LONG_MIN`. On a two's-complement implementation it has a magnitude one greater than `LONG_MAX`. If you accumulate a positive signed magnitude, the parser cannot reach the most negative legal result without first trying to represent an illegal intermediate value.

Accumulate an unsigned magnitude with a sign-dependent limit. Before each multiplication by ten, verify that the new value will fit. When the final input is the exact negative endpoint, assign `LONG_MIN` directly. Use `snprintf` to produce boundary strings on the actual implementation, rather than hardcoding an assumed `long` width.

A good test sequence includes empty input, a bare plus or minus sign, zero, negative zero, leading zeros, ordinary positive and negative numbers, `LONG_MAX`, `LONG_MIN`, and strings whose magnitudes exceed both endpoints.

Compare the final solution with `examples/parse_signed.c`. Then adapt the interface to accept a counted byte buffer so it does not require null termination.

12.20 Guided project: portable rotation

Implement a left and right 32-bit rotation. Define rotation counts modulo 32; this means counts 0, 32, and 64 leave the value unchanged.

Your implementation must never evaluate a shift whose count is 32. Handle the normalized zero case separately. Prove that for all other counts, both shift quantities are between one and 31 inclusive.

Test against independent hexadecimal constants. Then test the inverse property that rotating left and right by the same count restores the original value. This property is useful but insufficient by itself; two incorrect functions could still cancel one another.

Finally, explain why a target CPU's presence or absence of a rotate instruction does not change the C source's required semantics. Optional assembly belongs behind a target-specific interface and must be justified by measurement.

12.21 Guided project: ownership and aliases

Draw the memory objects involved in a vector whose structure contains pointer, length and capacity. Add three integers through its append operation, then create a second structure by copying the first.

Identify which fields have independent storage and which pointer values designate the same heap allocation. Does modifying an element through one structure affect the other? What happens if the first vector is destroyed? Can the second now be destroyed safely?

Next, create a borrowed pointer to the first element and append enough values to trigger a successful reallocation. Explain why that old pointer is invalid even if the allocator returns an address with the same numerical value.

Finally, add an injected allocation failure. Confirm that a correctly implemented append leaves every existing element and ownership field unchanged. This is an example of a strong failure guarantee, not an attempt to make shallow copies safe.

12.22 Review a protocol without compiling it

Given a packet consisting of a magic octet, version, two length octets, payload, and checksum, write a code-review checklist before implementing the parser. At what point may each byte be read? How is the length decoded? When can a subtraction underflow? What is the maximum payload? What happens if extra trailing data exists? When is output state published?

Now implement the decoder, write manual golden test vectors, and run it with sanitizer instrumentation if available. Keep the input-buffer contract explicit: a non-null pointer and a numeric length do not by themselves prove that storage of that size exists.

This exercise combines contracts from Chapters 4, 7, 10 and 14. Its outcome should include an implementation, a test matrix and a short reasoning argument.

12.23 Writing up the result

For each project, include a short report: the precise accepted input domain, key invariants, failure semantics, tests, compiler mode, and known limitations. Differentiate ordinary input errors from caller contract violations and from implementation-specific assumptions.

A particularly useful final section is *what the tests do not prove*. A host byte-arena test does not establish general typed-object alignment. A sequential queue test does not prove concurrent safety. A passing frame-decoder test does not authenticate a message. A clean sanitizer run cannot prove that unexecuted paths are safe.

That level of precision distinguishes a professional systems-programming exercise from a demonstration that happened to run once.

12.24 Capstone: a bounded telemetry receiver

This project combines several of the book's smaller components into one coherent embedded-style design. The goal is to receive short binary frames, validate their content, and retain a bounded number of decoded messages for a consumer.

The receiver has three layers:

```
incoming octets -> frame parser -> fixed slot pool -> application consumer
```

The first layer supplies octets. The second recognizes complete, valid frames. The third stores completed messages until the application consumes them. Each layer has a different contract, and no layer needs to allocate memory dynamically.

A finished implementation should make all failure paths visible: malformed input, an incomplete frame, exhausted message slots, a stale handle, and shutdown while a message is still outstanding.

Part 1: define the external format

Use the same compact frame as Chapter 10: magic A5, a length octet between zero and eight, the declared number of payload octets, and an XOR checksum covering the length and payload.

Create a table containing at least these independent test frames:

Meaning	Hexadecimal frame
Empty message	A5 00 00
One octet 05	A5 01 05 04
Three octets 0A 14 1E	A5 03 0A 14 1E 03
Invalid length	A5 09
Incorrect checksum	A5 01 05 00

For the last two entries, a complete application test must check that no new message is published. The invalid-length example represents a rejected prefix; it does not need a valid checksum because length validation fails first.

Explain why successful encoding and decoding in a loop is not enough to verify the format. The independent table protects against two implementations that share the same mistake.

Part 2: identify the state owners

Create one frame parser and a pool containing four fixed message records. Make a single receiver task the owner of the parser. The receiver calls the feed function for each incoming octet.

When a complete message appears, the receiver attempts to acquire a free pool slot. If acquisition succeeds, it copies the completed payload into the slot and publishes a handle to the application

consumer. If the pool is full, the receiver must report or count a dropped message according to the chosen policy.

The application consumer eventually retrieves a message using the validated handle and releases its slot. After release, the old handle is no longer valid.

This arrangement separates the lifetime of a temporary candidate message within the parser from the lifetime of a completed message retained by the pool.

Part 3: choose a queue and saturation policy

A real producer and consumer often run at different rates. A pool alone stores objects but does not define the order in which completed-message handles are delivered.

Add a fixed-capacity queue of handles. When the receiver decodes a message successfully, it acquires a slot, stores the message, and enqueues the handle. The consumer dequeues a handle, processes the corresponding message, and releases the slot.

Specify what happens if either the pool or queue is full. Possible choices include rejecting the newest message, dropping the oldest queued message, or applying backpressure to the transport. Dropping an old message requires care: remove its handle from the queue and release its pool slot exactly once.

Do not silently overwrite a live slot. A bounded memory design is useful only when saturation has predictable semantics.

Part 4: distinguish host tests from device behavior

On a desktop, a test can supply input bytes from a fixed array and run receiver and consumer operations sequentially. Verify the exact decoded data, queue order, handle generations, full conditions, and stale-handle rejections.

That test establishes the algorithm's expected behavior for the chosen sequence. It does not demonstrate that concurrent producer and consumer access is safe.

If the project uses an RTOS and UART interrupts, select or implement a queue whose operations are supported in the relevant execution contexts. Establish how the ISR hands data to a receiver task and how that task signals the consumer. Avoid having the ISR call an arbitrary C function that may block or use unsupported library facilities.

Part 5: bound resources

Document the maximum storage required by:

- the frame parser's candidate payload;
- all four message-pool records;
- a fixed queue of message handles;
- any transport byte buffer;
- the stacks of the participating tasks.

These are separate allocations or objects, and their sizes may include alignment padding. Confirm their actual sizes on the target toolchain using its build output or suitable instrumentation.

Next, bound local operations. One call to the frame parser processes one octet; a pool acquire examines at most four slots; a queue enqueue performs a fixed amount of index work under its stated conditions.

This gives useful operation bounds but not a complete worst-case deadline analysis. Real-time completion also depends on preemption, queue contention, memory access times, and the scheduling of the consumer.

Part 6: design shutdown

Shutting down the receiver requires a policy for incomplete frames and pending messages. One possible sequence is to stop new input, allow the receiver to finish processing already queued octets, signal the consumer, drain or deliberately discard the remaining messages, join or stop all participating tasks, and only then reclaim storage.

If a queued handle is discarded, release its pool slot. If a consumer still holds a handle, the pool cannot be reset or destroyed until that use is finished under the selected ownership contract.

Shutdown bugs often arise because ordinary tests exercise steady-state operation but never release resources while work remains.

Required test report

Submit a short report describing the accepted frame grammar, all data-structure invariants, capacity and drop policy, lifetime of every borrowed pointer and handle, executable tests, target-specific assumptions, and remaining unverified properties.

A particularly useful test combines several failures: begin a partial frame, abandon it on timeout, fill the pool with valid messages, reject an additional completed frame, release one slot, then accept a new message into that slot. Confirm that an old handle does not refer to the new record.

A clear architecture with documented limits is more important than using the most complicated available synchronization technique.

12.25 A short review rubric for the capstone

A high-quality submission should demonstrate correctness at four levels:

Representation. Each binary field is decoded according to the stated wire format, with no dependence on native structure padding or endianness.

State. Parser transitions, pool occupancy, handle generations, and queue indexes preserve their written invariants.

Ownership. Every acquired slot is released once, stale handles are rejected, and shutdown does not destroy storage that active clients still use.

Execution. Tests cover valid boundaries, truncation, corruption, saturation, and recovery. Concurrency and hardware-related claims are qualified by appropriate target-specific evidence.

The goal is not to build a production-ready communications stack in one exercise. It is to practice decomposing a small real system into contracts that can be reasoned about and tested independently.

13. Testing, Diagnostics, and Portability

A C program can compile successfully and still be wrong. Compiler diagnostics report only what a tool can establish from the program and its selected options. Tests sample behavior for chosen inputs. Static analysis may identify suspicious paths. Sanitizers instrument executable operations. None of these alone proves that a C program is correct, but used together they provide an effective development discipline.

This chapter develops a concrete workflow for the small programs in this book.

13.1 Start with the contract

Before building a test, state what success means. Suppose a function computes the sum of a sequence of unsigned integers. Its contract might say:

- The input is a pointer to at least `count` elements, or a null pointer if `count` is zero.
- The result pointer must address writable storage.
- On success, the output is the exact mathematical sum.
- On failure, the output is unchanged.
- Overflow is an error rather than a request for modular arithmetic.

These statements determine useful tests. They also identify cases that a test **cannot** legitimately exercise: passing a dangling pointer or claiming a buffer has more elements than it actually has violates the caller's contract. Running such a case and observing a crash does not constitute a useful portable API result.

Tests should therefore separate *valid edge cases* from intentional demonstrations of undefined behavior.

13.2 Organize tests by dimensions

For a bounded operation, look for boundaries in several dimensions:

Dimension	Representative tests
Count	zero, one, capacity minus one, capacity, capacity plus one
Value	zero, ordinary value, largest representable value
Text	empty, valid, invalid digit, too long, no trailing newline
Storage	sufficient capacity, insufficient capacity, allocation failure
State	empty collection, full collection, wraparound, repeated use
I/O	clean EOF, short record, read error, write error

A useful test does not merely show that a function returns the expected value. It checks all promised postconditions. If failure must preserve an output, initialize it to a recognizable value before invoking the function, then verify it remains unchanged.

13.3 Assertions for tests, not input validation

The `assert` macro in `<assert.h>` checks a condition when assertions are enabled. Failed assertions terminate the program with a diagnostic. Defining `NDEBUG` can disable them.

```
#include <assert.h>
#include <stddef.h>

static int first(const int *values, size_t count)
{
    assert(values != nullptr && count != 0);
    return values[0];
}
```

This is appropriate only if callers are required to satisfy the precondition and an assertion failure is an acceptable development-time response. It is not a substitute for checking user-supplied input in a production interface. A release build may omit the assertion altogether, after which a bad call still attempts the invalid access.

Never place an operation with required side effects only inside `assert`; disabling assertions would also remove that operation.

13.4 A test harness without external libraries

For a small project, a conventional `main` that returns nonzero on failure is enough:

```c
#include <stdbool.h>
#include <stdio.h>

static bool is_even(unsigned n)
{
    return n % 2U == 0;
}

int main(void)
{
    if (!is_even(0) || !is_even(8) || is_even(7))
        return 1;
    puts("tests passed");
    return 0;
}
```

The build system can run the executable and use its exit status to determine success. Larger suites should report the particular case that failed, but a simple test program can be both readable and valuable.

A command such as `printf 'a b c' | ./word_count` tests a whole program through its standard input and output. Such integration tests complement direct function tests, which are better at reaching internal boundary cases.

13.5 Warnings and language modes

A practical first compiler invocation is:

```
gcc -std=c23 -Wall -Wextra -Wpedantic -Wconversion -Wshadow \
    -Werror -O2 example.c -o example
```

The exact useful flags vary by compiler, codebase, and target. In particular, `-Wconversion` can be noisy in legitimate numeric work, and `-Werror` can cause builds to fail when the compiler introduces new diagnostics. The correct response is to review warnings and make an explicit project policy, not to silence them indiscriminately.

Compile in the actual language mode. A compiler accepting one C23 keyword does not necessarily implement every feature or library facility. When a project supports more than one compiler, build with both. Different diagnostics often reveal different classes of mistakes.

13.6 Sanitizers

On supported host targets, AddressSanitizer can detect many out-of-bounds accesses, use-after-free errors, and other memory defects. UndefinedBehaviorSanitizer can detect selected undefined operations, such as certain signed overflows or invalid shifts. They do not cover every form of undefined behavior, and their effectiveness depends on executing the erroneous path.

A useful build on a supported GCC or Clang environment is:

```
cc -std=c23 -O1 -g -fno-omit-frame-pointer \
   -fsanitize=address,undefined example.c -o example_san
./example_san
```

Sanitizer-enabled executables are typically host-oriented. Their instrumentation, runtime overhead, and dependencies can make them unsuitable for microcontrollers. Test algorithms on a host, then separately verify target-specific assumptions.

Do not interpret one clean sanitizer run as proof of a fully safe program. Test case selection remains decisive.

13.7 Unit tests and property tests

A unit test checks a specific expected value. A *property test* checks a relationship that should hold for many inputs.

For example, consider an in-place reversal of an array. If the operation is correct, reversing twice should restore the original sequence. The property holds for empty arrays, one-element arrays, and arrays with duplicates. A test driver can generate random arrays and check this relationship.

Similarly, if a serializer and parser implement the same format, decoding an encoded valid value should produce an equivalent value. This does not prove either routine matches an external specification: they could share the same mistake. Include independent golden test vectors alongside round-trip tests.

13.8 Fuzzing a parser

A fuzzer supplies many automatically generated byte sequences and observes whether the program crashes, violates a memory safety check, hangs, or triggers an assertion. A parser interface like

```
bool decode_packet(const unsigned char *bytes,
                   size_t length,
                   struct packet *result);
```

is well suited to fuzz testing because its input is a bounded byte sequence. The harness should supply valid storage of the declared size, initialize output state, invoke the parser, and verify that failure preserves the output.

Fuzzing discovers bugs; it does not establish absence of bugs. Use corpus inputs for known valid packets and edge cases, and preserve any newly discovered failing inputs as deterministic regression tests.

13.9 Compare optimization modes

Undefined behavior can cause surprising differences between `-O0` and `-O2`. A compiler is permitted to optimize under the assumption that no undefined operation occurs. It may therefore eliminate checks that seem defensive if the source has already performed an invalid operation.

When debugging a difference between builds, investigate lifetime, aliasing, integer overflow, uninitialized objects, sequencing, and incorrect function declarations. Do not rely on a particular assembly listing as the definition of C semantics.

13.10 Run the actual target

A host test cannot substitute for hardware validation when code depends on interrupt timing, cache coherency, memory-mapped registers, DMA, alignment restrictions, an RTOS, or the implementation's floating-point behavior.

The most effective architecture separates portable algorithms from target-specific adapters. For a firmware project, a packet decoder can be tested exhaustively on the host while the UART driver is validated on the real board. Integration tests must then establish that the adapter passes the right byte sequences and lengths to the decoder.

13.11 Document verified claims

A verification report should record the compiler and version, C library if relevant, command line, target architecture, test suite, and result. State the difference between *committed test code* and *tests actually executed*. If a test was not run, do not write that it passed.

For this manuscript, `make check` invokes a shell-based regression suite, while a GitHub Actions workflow can provide repeatable hosted checks. Both should be run on the precise committed revision before a stable release is declared.

13.12 Exercises

13.1 Add a test for unsigned-sum overflow that verifies the output argument remains unchanged.

13.2 Write a reversibility property test for a bounded array reversal.

13.3 Create a fuzzer harness that invokes the packet decoder on random byte sequences and checks output preservation.

13.4 Compare warning diagnostics on the same example under GCC and Clang. Explain differences without suppressing them globally.

13.5 State which properties of a ring buffer can be established with single-threaded tests and which need concurrency-specific reasoning.

13.6 Design a verification matrix for a program that must run on an x86-64 development host and an ARM Cortex-M microcontroller.

13.13 Make tests deterministic

A regression test should produce the same expected result from the same inputs under documented conditions. Tests based on timing thresholds, random seeds without recording them, or unspecified evaluation order are difficult to reproduce. When randomness is useful, record the seed and provide a way to replay the failing case exactly.

For the book's word index, generate a fixed set of vocabulary and expected counts. For the bounded queue, push and pop a known sequence through wraparound. For the byte decoder, use independent golden vectors defined by the protocol rather than relying only on round trips.

13.14 Test failures, not just successes

Many defects hide on error paths. Test the allocation path that fails, the input that is one byte too short, the array that is exactly full, and the output that must not change when conversion fails.

A useful method is *fault injection*. A test allocator can fail on the third request; a fake byte source can return an I/O error after a chosen number of bytes; a fake output sink can accept only a prefix of a write. The production algorithm should treat these failures as ordinary results.

When an API promises strong failure safety, initialize the output with an unusual value and verify it remains unchanged after every rejection. Be careful not to compare structure padding bytes with `memcmp` and interpret differences as changes in member values.

13.15 Static analysis and sanitizers complement each other

A static analyzer inspects source or intermediate representation without necessarily running every path. It may detect impossible conditions, potential null dereferences, and unchecked return values. False positives exist, and the analyzer's assumptions must be understood.

AddressSanitizer can catch many invalid memory accesses when they execute. UndefinedBehaviorSanitizer can catch selected invalid shifts, arithmetic, and other errors. Neither usually establishes absence of data races; ThreadSanitizer is a different instrument and has platform-specific limitations. Sanitizers also change timing and memory layout, so embedded hardware behavior still requires separate validation.

For numerical kernels, compiler warnings and differential tests with independent reference implementations may reveal issues that memory sanitizers cannot.

13.16 Differential testing

When implementing a custom parser, compare its accepted strings and decoded values with a second trusted implementation **after matching their grammars**. If one parser allows leading whitespace and the other forbids it, disagreement may be by design rather than a bug.

For a numeric function, compare against a reference with higher precision or a well-understood library routine over representative ranges. For a protocol codec, compare byte-level outputs against test vectors written from the format specification.

A test that encodes and decodes using two functions sharing the same mistaken endianness can pass while violating the external protocol. Independent expectations are essential.

13.17 CI as reproducible evidence

A GitHub Actions job should report its actual compiler and version, the selected dialect, and the command that ran. Separate compilation, unit tests, sanitizer builds and document generation into distinguishable steps so a failed PDF toolchain is not mistaken for failing C code.

A CI configuration file is not itself proof of a passing build. Evidence requires inspecting an actual completed workflow run for the exact commit. If the repository cannot expose run status through the available integration, say that verification is pending.

13.18 Choose a release criterion

A reasonable first reading edition can require all examples to compile under two supported compilers, all deterministic tests to pass, all Markdown links to resolve, and the generated PDF to include every chapter and appendix. A publishable technical book needs more: technical review, a coherent index, verified page layout, exercise solutions, and clear statements of compiler-specific features.

Further problems: Design a differential test for a little-endian decoder using independent literal expected values. Explain why a clean sanitizer run cannot prove every array index is valid. Add a fault-injectable allocation layer to a dynamic vector and report which postconditions remain true.

13.19 A test is a contract experiment

Testing becomes more systematic when each case is derived from a documented precondition, invariant, or postcondition. For a vector, the invariants include `length <= capacity` and valid ownership of allocated storage. For a packet parser, failure must not modify the public result. For an SPSC queue, the producer and consumer must observe values in FIFO order under the single-producer/single-consumer protocol.

A test should state what it establishes. A successful sequential queue test demonstrates ordering and wraparound for the tested operations. It does not establish freedom from data races if producer and consumer execute simultaneously. A clean AddressSanitizer run shows that the tested execution did not trigger any instrumented errors; it does not prove all paths safe.

13.20 Use fault injection to reach unlikely branches

If a container allocates only a few dozen bytes, ordinary development runs rarely experience allocation failure. A design that can accept a resizing callback allows a test to deliberately return `nullptr` and verify strong failure safety. The book's `vector_checked.c` follows this pattern.

An analogous technique works for output interfaces. The complete `callback_sink.c` example supplies a fake sink whose write operation accepts at most two bytes per call. A `write_all` wrapper must retry the unwritten suffix until the whole buffer has been accepted. A second fake mode returns no progress, and the wrapper must report failure rather than loop forever.

These are not theatrical failures. Embedded UART queues can fill, filesystem writes can fail, network links can stall, and bounded buffers can run out of capacity. Making errors injectable produces reproducible tests for behavior that otherwise appears only in the field.

13.21 Test the smallest complete interface

A function-level test can verify a single boundary, but a project also needs tests for interfaces between modules. The module example in Chapter 8 exercises compilation through a header, implementation, and client. The word-index test exercises the complete command-line program through standard input and output. The PDF build tests documentation generation separately from C compilation.

Keeping these levels distinct helps diagnose failures. If compilation fails because of a C23 syntax diagnostic, rewriting a parser algorithm is unlikely to help. If a program compiles but returns the wrong table, investigate semantics and test data. If the PDF builder fails while C tests pass, investigate the document toolchain.

13.22 Measure coverage without worshipping a number

Line and branch coverage can reveal unexecuted paths, especially error-handling branches. They do not say whether a test checks the right result. A test might run the overflow check but ignore the output state afterward. A branch can be covered without verifying that it preserves a container's ownership contract.

For high-risk functions, review a small table of input classes: minimum size, maximum size, invalid value, missing data, exhausted resource, and valid boundary. Include a human explanation of why each case matters.

13.23 Reproducible CI

A useful CI run records the exact commit, compiler versions, flags, and exit status. Different compilers discover different issues, but changing to a newer toolchain can also change diagnostics. When the compiler misclassifies valid C23 syntax as an extension, handle that specific diagnostic rather than globally disabling `-Werror`.

For publication, add a second gate: the generated PDF must contain every chapter and appendix, have a readable table of contents, and be built from the same source revision being published. A source repository containing an older PDF should not be treated as proof that the latest manuscript has been typeset.

Exercises

13.7 Write failure-injection cases for an allocator and a short-writing output sink. **13.8** List what a passing host test cannot establish about MCU interrupt timing. **13.9** Add a CI check that separately reports compiler-test and PDF-build failures. **13.10** Design independent golden vectors for a three-byte header codec.

13.24 A repeatable verification session

The book's standalone examples are intended to be compiled using the actual C23 language mode, not merely whichever dialect a host compiler selects by default. A reproducible local session should begin with an explicit compiler version check and a clean build:

```
gcc --version
make clean
make CC=gcc CFLAGS='-std=c23 -Wall -Wextra -Wpedantic -Werror -O2' check
```

On a system where an older compiler does not recognize `-std=c23`, install or select a compatible compiler rather than quietly switching to an earlier language dialect and assuming the results prove C23 compatibility.

The GitHub Actions matrix in this repository uses GCC 14 and Clang 18. It preserves warnings as errors, with a targeted workaround for Clang 18's diagnostic that classified standardized C23 binary literals as a GNU extension. This does not justify suppressing all pedantic warnings.

13.25 Verify the errors that matter

For a signed parser, ordinary numbers are necessary tests but not the interesting boundary. Verify `LONG_MIN`, `LONG_MAX`, bare signs, whitespace policy, trailing garbage, and an out-of-range decimal string. Initialize the output to a sentinel and ensure failed calls leave it unchanged.

For a byte arena, fill its remaining storage exactly, attempt one byte more, and check that neither the allocation cursor nor the output pointer changes on failure. Test reset, while respecting the rule that borrowed regions from before reset are no longer valid under the arena API.

For bit rotations, use literal expected values and count zero, 31, and 32. The tests are valuable because an invalid shift count may seem harmless on a particular CPU while still invoking undefined C behavior.

13.26 Distinguish instrumentation from proof

Host sanitizers can reveal many use-after-free errors, out-of-bounds accesses, and undefined operations when the erroneous path executes. They do not establish all possible behavior by merely completing a smoke test. Property testing can broaden input coverage, but an inverse-operation property may be shared by two mutually incorrect implementations.

Combine three forms of evidence: independent expected values for carefully chosen cases, invariants derived from the language and algorithm, and repeated executable tests across compiler configurations. Use target hardware tests for properties not represented by the host execution environment.

A good verification report names the source commit, compiler versions, flags, test inputs, observed outcomes, and remaining unverified assumptions. The presence of a test file does not mean a test ran; similarly, a PDF workflow file does not mean the latest reading edition was published.

Additional exercises

13.11 Construct a CI matrix separating compiler tests, sanitizer tests, and PDF generation. **13.12** Write a failure-injection test that makes a vector resize fail and checks all saved invariants. **13.13** Produce an independent expected-value table for 32-bit rotations. **13.14** Write a one-page review explaining which properties of a fixed byte arena are proven by bounds checks and which depend on caller-provided storage and lifetime.

13.27 Model-based testing for a fixed pool

A fixed pool is small enough that a separate test model can describe its intended behavior without sharing its implementation. For four slots, the model can record which logical resources are live and which handles are stale. Generate a sequence of acquire, read, release and invalid-handle operations. After every step, compare the implementation's result with the model's expected result.

This is stronger than one happy-path test, because operations can be interleaved in many orders. The model should not simply copy the pool's actual loop or generation-update code; an independent model helps detect a mistake shared by several implementation functions.

A minimal regression sequence exercises all of these: fill all slots; reject one extra acquire; release one slot; reject its old handle; reacquire the slot with a different generation; reject double release; verify that every still-live handle reads its original value.

Then simulate generation exhaustion by initializing one test-only slot's generation at UINT_MAX, acquiring and releasing it, and checking permanent retirement. This branch would be practically impossible to exercise by waiting through billions of ordinary allocation cycles.

13.28 State-transition coverage for a streaming parser

Line coverage can report that each parser state was visited, yet miss important transitions. For a framed byte protocol, tests should cover every state and each type of outgoing transition.

A practical transition table includes:

Starting state	Byte class	Expected result
Expect magic	Correct marker	More input required
Expect magic	Other byte	Rejected
Expect length	Zero	Expect checksum
Expect length	Valid positive length	Expect payload
Expect length	Above maximum	Rejected
Expect payload	Ordinary payload byte	More input required
Expect payload	Final payload byte	Expect checksum
Expect checksum	Correct byte	Completed message
Expect checksum	Wrong byte	Rejected

After every rejected candidate, assert that the previously published result remains unchanged. After every successful frame, assert that parser state has returned to its initial boundary. Test both a zero-length payload and a maximum-length payload.

A byte-at-a-time parser is especially amenable to exhaustive exploration over short inputs, but the full 256-byte input alphabet grows exponentially with sequence length. Partition bytes by behavior and combine that model with selected exhaustive small cases and generated longer streams.

13.29 Independent golden vectors

An encoder and decoder can agree on an incorrect interpretation of a protocol. If both accidentally omit the length octet from a checksum, an encode/decode round trip may still pass.

For at least a few tests, calculate the expected wire representation independently from the format specification. In the example frame, a length of three and payload 0A 14 1E yields XOR checksum 03, so the complete frame is A5 03 0A 14 1E 03. The test should compare the entire decoded payload and count, not just the fact that the parser returned success.

A second independent vector should test an empty payload. Then deliberately change one byte without updating the checksum and confirm that rejection preserves the previous result.

The same principle applies to numerical routines. An inverse property is useful, but two wrong functions may be inverses of one another. Combine properties with independently established outputs.

13.30 Fuzzing and its limits

Fuzzing supplies many generated or mutated inputs to a function, looking for crashes, sanitizer reports, assertion failures, hangs, or unexpected statuses. A bounded parser is a natural candidate: its input is a byte sequence with a known maximum meaningful record length.

Build a fuzz harness that initializes a new parser for each input, feeds the bytes, and checks invariants after each transition. The harness should never treat malformed input as a reason to abort the process. Instead, malformed input should produce a documented rejection status.

Instrumented fuzzing can be especially valuable for off-by-one errors and incomplete-input transitions. However, absence of a crash does not show that the parser implements the intended wire format. Include assertions for output publication, state reset, resource limits, and independence from input chunk boundaries where those properties are part of the API.

Fuzzing also cannot by itself establish ISR timing or hardware DMA coherency. Those properties require target-specific analysis.

13.31 A compact evidence hierarchy

Different evidence supports different claims:

- A successful compilation establishes that the source was accepted by the selected compiler and flags.
- A passing deterministic test establishes observed behavior for specified inputs.
- A property or randomized test broadens the explored input space.
- A sanitizer run can expose certain runtime defects along executed paths.
- A written invariant argument explains why a property should hold across every permitted iteration.
- Hardware measurement characterizes the real target's timing and environmental behavior.

A technical book should label these distinctions accurately. A repository with green CI is a good teaching artifact, but not proof that every illustrative fragment is production-ready on every embedded toolchain.

Further exercises

13.15 Create a model-based randomized test for the four-slot pool and record the random seed needed to reproduce failures. **13.16** Write a transition coverage table for all four states of the frame parser. **13.17** Build two independent golden vectors and a corrupted-vector test. **13.18** Design a fuzz harness that never interprets malformed external input as a programming assertion failure. **13.19** Prepare a verification report distinguishing host observations from target-hardware claims.

14. Bits, Bytes, and Portable Data

C gives programmers direct control over integer representations and bytes. That power is especially useful for embedded devices, persistent files, and network protocols. But an integer's mathematical value is distinct from its in-memory representation, and a portable program must not confuse the two.

14.1 Storage units

An object occupies one or more bytes. In C, a byte is the size of a `char` and contains `CHAR_BIT` bits; while eight-bit bytes are overwhelmingly common, the language does not require `CHAR_BIT == 8`. A format specified in *octets* requires eight-bit units. A portable implementation must check that its target can represent and address the format as intended.

```
#include <limits.h>
#include <stdio.h>

int main(void)
{
```

```
    printf("bits per byte: %d\n", CHAR_BIT);
}
```

Exact-width integer types such as `uint32_t` exist when the implementation offers a type of exactly the specified width without padding bits. Fixed-width names clarify the protocol's mathematical range, but they do not choose byte order.

14.2 Masks, shifts, and flags

Use unsigned values for bit-level work. A left shift with a count greater than or equal to the promoted operand's width has undefined behavior, as does a negative shift count. The shift operators have their own integer-promotion rules; explicitly selecting a wide enough unsigned operand avoids accidental signed arithmetic.

```
#include <stdint.h>

static uint32_t extract_byte(uint32_t value, unsigned position)
{
    return (value >> (position * 8u)) & UINT32_C(0xff);
}
```

This helper requires the precondition `position < 4`. Without it, the shift count can be invalid. A production interface can check the position and return success through a separate status indicator.

A register read-modify-write operation may not have the semantics of an ordinary RAM variable. For instance, writing a one may clear a device status bit. Apply shifts and masks according to the hardware reference manual, not just the C expression's arithmetic meaning.

14.3 Endianness is part of the format

A little-endian 32-bit unsigned field stores its least significant octet first. A big-endian field stores the most significant octet first. For portable decoding, use shifts and masks on bytes rather than reinterpreting an input buffer as a pointer to `uint32_t`.

```
#include <stdint.h>
#include <stddef.h>
#include <stdbool.h>

bool decode_u32_be(const unsigned char *src, size_t n, uint32_t *out)
{
    if (src == nullptr || out == nullptr || n < 4)
        return false;
    uint32_t v = ((uint32_t)src[0] << 24)
               | ((uint32_t)src[1] << 16)
               | ((uint32_t)src[2] << 8)
               | (uint32_t)src[3];
    *out = v;
    return true;
}
```

This assumes eight-bit input bytes as specified by the format. If the environment permits wider bytes, a protocol layer should explicitly define how octets are obtained. The output is assigned only after the bounds check.

14.4 Representation and alignment

An address in a byte stream may not meet the alignment required by an integer type. Converting that address and dereferencing a misaligned integer pointer may be undefined. Even on processors that allow unaligned loads, compiler assumptions or peripheral restrictions may differ. Byte-wise decoding avoids this issue.

Structure padding and bit-field layout may differ across implementations. Avoid transmitting raw structures as protocol records. Instead, define field lengths, byte order, permitted ranges, and checksums. Encode fields explicitly and test them against documented vectors.

14.5 The meaning of volatile

A memory-mapped device register may require volatile-qualified access so that the compiler performs observable reads and writes. But `volatile` does not specify the hardware bus transaction width, cache behavior, atomicity, memory ordering across cores, or synchronization with interrupts. Those require target-specific facilities and device documentation.

When two threads share ordinary application state, prefer atomics or appropriate synchronization rather than a volatile flag. The language's concurrency rules and a hardware peripheral's rules are related but not interchangeable.

14.6 A complete byte codec

See `examples/byte_codec.c`, which encodes and decodes a 32-bit value in big-endian octet order and verifies the result. It checks `CHAR_BIT == 8` at translation time and uses unsigned shifts. Its public-style functions take explicit output capacity and return `false` when storage is insufficient.

The tests exercise zero, a mixed-byte value, and the maximum 32-bit value. An actual protocol should also test malformed lengths, endianness against independent known vectors, and behavior when the caller supplies invalid storage.

14.7 Exercises

14.1 Write `decode_u16_le` with an explicit length and output pointer, preserving the output on failure.

14.2 Explain why `*(const uint32_t *)buffer` is not a portable replacement for a byte decoder.

14.3 Implement a 32-bit rotate-left operation without shifting by the type's width. Test shift counts zero, one, 31, and 32 after defining whether counts are reduced modulo 32.

14.4 Write a packet encoder with a documented maximum length, a version field, and a checksum. Prove that it never writes past its capacity.

14.5 Compare a host byte codec with the actual peripheral register interface of a chosen microcontroller. Identify which assumptions belong to C and which belong to the vendor's device specification.

14.8 Fixed-width data is not fixed layout

Declaring `uint32_t word;` specifies a 32-bit unsigned integer type if the implementation provides it. It does not say which byte of its object representation occupies the lowest address. Nor does it authorize unaligned access to an arbitrary buffer address after casting it to `uint32_t *`.

A format that defines 32-bit fields in network byte order requires explicit big-endian encoding and decoding, regardless of whether the host processor happens to use little-endian loads. Keeping representation conversion in small functions makes protocol code easy to test on multiple architectures.

14.9 Shifts need a domain proof

The expression `1 << position` has a signed left operand. For sufficiently large `position`, the result may not fit, or the shift count may equal or exceed the promoted width. Using an unsigned fixed-width operand addresses the signedness issue but not invalid counts.

```
#include <stdbool.h>
#include <stdint.h>

bool make_u32_mask(unsigned position, uint32_t *out)
{
    if (out == nullptr || position >= 32)
        return false;
    *out = UINT32_C(1) << position;
    return true;
}
```

This example rejects positions outside the 32-bit domain before performing the shift. It does not assume a particular memory representation of `uint32_t`.

14.10 Check the complete frame size

Suppose an output frame contains a four-byte header, a payload, and a one-byte checksum. If the caller gives a buffer of capacity `capacity`, it is tempting to test `capacity >= payload_length + 5`. But that sum may overflow `size_t`.

A safer shape is:

```
if (capacity < 5 || payload_length > capacity - 5)
    return false;
```

Only after this check is it safe to index all fields in a correctly allocated buffer of the declared capacity. This is an arithmetic *and* storage contract: the caller must actually provide a valid region that large.

14.11 MMIO is not just a pointer cast

In embedded programming, a peripheral register may be accessed through a volatile-qualified lvalue or vendor-provided accessor. The C language does not itself specify the address map, the side effects of reading, or whether writing a one clears a flag. The device manual and compiler/ABI documentation remain authoritative for such accesses.

A compound assignment like `REG |= MASK` may compile into a read followed by a write. It may be unsuitable for a register whose read has side effects, whose reserved bits must retain particular values, or whose updates must be atomic with respect to an interrupt. Some peripherals offer dedicated set/reset registers; use the documented mechanism.

14.12 Field tests beat raw dumps

For a byte codec, test zero, all bits set, alternating-bit patterns, and a distinct-byte value such as `0x12345678`. The distinct-byte case is especially useful for finding reversed byte order. Then test output buffers that are too small, null pointers when prohibited, and any nonstandard target assumptions.

Further problems: Write an encoder for a 24-bit unsigned field stored in three octets. Show that each shift count is valid. Explain why an ordinary struct copy and a wire serialization are different operations. Analyze a hypothetical write-one-to-clear status register.

14.13 A wire protocol is a specification, not a structure

Suppose a device transmits a frame with this byte-level layout:

Offset	Width in octets	Meaning
0	1	Magic value `0xA5`
1	1	Version `0x01`
2	2	Unsigned payload length, little-endian
4	0–64	Payload
4 + length	1	XOR of all preceding octets

This is an intentionally simple teaching format. The XOR field detects some accidental errors; it does not authenticate a sender and is not adequate against deliberate modification.

The format does not mention `struct`, padding, alignment, or host byte order. Those are properties of the implementation, not the external message. A portable encoder explicitly writes each field as bytes. A portable decoder reads each field only after establishing that the corresponding bytes exist.

14.14 Encode without partial publication

An encoder can reject an invalid request before writing any output: reject a null destination, insufficient capacity, a payload length beyond 64, and a nonzero payload length with no valid payload storage. Then compute exactly how many bytes are needed, using arithmetic already bounded by the format's maximum.

If failure must leave the destination unchanged, construct the complete frame in a local fixed-size temporary array and copy it only after validation. That costs one extra bounded copy but gives callers a simple postcondition. For larger frames, an alternative API may write incrementally and report partial output; either policy is acceptable if explicit.

The companion `examples/frame_codec.c` demonstrates a strong-failure interface for both encoding and decoding. It uses `memcpy` only after checking capacities. Its local buffer size is determined by the maximum defined frame size rather than by untrusted input.

14.15 Decode in a safe order

For decoding, perform these checks in sequence:

1. Confirm that the input pointer and output pointer satisfy the API and that at least five octets exist.
2. Check magic and version fields.
3. Decode the two-byte length from individual octets.
4. Ensure length is at most 64 and the input contains exactly the declared frame length.
5. Verify the checksum.
6. Copy payload bytes into a local candidate and commit the result.

If the function subtracts a fixed overhead from the input length, it must first establish that the input length is at least that overhead. This prevents unsigned underflow. Because the maximum payload is small, the final length calculation can be proven to fit in `size_t` on supported targets.

14.16 Endianness and alignment are independent

A machine can allow unaligned access and still have a byte order different from the protocol. Conversely, a target may use the protocol's byte order but require alignment for native multi-byte loads. Casting the input pointer to a multi-byte integer pointer therefore makes more than one unsafe assumption.

The explicit decoder `(uint16_t)bytes[2] | ((uint16_t)bytes[3] << 8)` communicates the wire format and avoids reading a misaligned native integer. Since `uint16_t` might undergo integer promotions, the operation remains carefully bounded by the known eight-bit octet values.

14.17 Testing message boundaries

Test a zero-length payload, a payload at maximum length, one byte less than a complete frame, an extra trailing byte, a corrupted checksum, and an unsupported version. Confirm that failures leave the output object unchanged. Also test against a manually constructed frame rather than relying exclusively on an encoder/decoder round trip.

Exercises

14.6 Change the checksum from XOR to a documented CRC and provide independent golden test vectors. **14.7** Add an error enumeration for malformed, incomplete and oversized frames. **14.8** Design a streaming decoder using bounded state and explain how it handles resynchronization after corrupted input.

14.18 Rotation is not the same as shifting

A shift discards bits at one end of an integer and fills positions at the other end according to the operation's rules. A rotation instead moves discarded bits around to the opposite end. Many checksums, hash functions, cryptographic constructions, and bit-manipulation algorithms use rotation.

For a 32-bit unsigned integer, rotating the hexadecimal pattern `0x12345678` left by four bits yields `0x23456781`. The top hexadecimal digit wraps around to the lowest position.

A mathematical rotate-left can be expressed as the union of two shifted regions:

[\operatorname{rotl}_{32}(x,n)

(x \ll n);|;(x \gg (32-n)),]

for (1 \le n \le 31). But that range condition is essential. In C, shifting a 32-bit value by 32 or more is undefined behavior. Using an unsigned left operand avoids signed-left-shift problems, but it does not make an out-of-range count legal.

The fully safe implementation handles the zero count separately:

```
#include <stdint.h>

uint32_t rotate_left_32(uint32_t x, unsigned count)
{
    unsigned n = count % 32u;
    if (n == 0)
        return x;
    return (x << n) | (x >> (32u - n));
}
```

The count-reduction rule is part of the function's contract: values 0, 32, 64, and so on all mean an unchanged result. Once the zero case is excluded, both shift counts are between 1 and 31. The proof is short, and it should appear beside the code rather than remaining an unstated assumption.

14.19 Rotate-right and inverse properties

The corresponding rotate-right operation is:

```c
uint32_t rotate_right_32(uint32_t x, unsigned count)
{
    unsigned n = count % 32u;
    if (n == 0)
        return x;
    return (x >> n) | (x << (32u - n));
}
```

An important property is that applying a left rotation and then a right rotation by the same count restores the original value. That property is useful for automatically generating tests over numerous count values.

The standalone `examples/rotate_bits.c` verifies zero and full-width counts, checks manually calculated constants, and tests the inverse property across counts 0 through 63.

Round-trip tests should accompany, not replace, independent expected values. Two incorrect operations could be inverses of each other and still fail to implement the required rotation convention. The example therefore includes explicit hexadecimal expected results.

14.20 Why direct machine-code tricks do not define C

A processor may have a single rotate instruction, and an optimizing compiler may recognize this unsigned expression and select it. That is a performance matter, not a language guarantee. The source must be valid whether or not the target has a rotate instruction.

Handwritten assembly can be appropriate in a carefully scoped architecture-specific module after measurement shows a real need. It should not become a substitute for specifying the mathematical operation.

The C23 `<stdbit.h>` header provides several bit-counting and bit-width facilities, but not the C++ `std::rotl` / `std::rotr` interface. Use the portable C helper or a documented implementation extension when appropriate.

14.21 Device registers are not ordinary bit containers

A local `uint32_t flags` variable can be safely updated with masking operations when the arithmetic is valid and access is appropriately synchronized. A memory-mapped peripheral register may behave differently. Reading a status register may acknowledge an interrupt. Writing a one may clear a bit. Some registers require reserved positions to be written as zero, while others require preserving particular bits.

Therefore `REG |= MASK` is not automatically safe. It generally implies a read, logical OR, and write, which can have surprising hardware side effects and may not be atomic relative to interrupts. Prefer the peripheral's documented set/reset mechanism when available, and isolate register semantics in target-specific driver code.

The C `volatile` qualifier can be relevant to observable device accesses, but it does not guarantee bus ordering, cache coherency, or safe multi-thread synchronization. Those concerns belong to the target's memory model and the device documentation.

14.22 Proving an encoder's byte range

Consider a protocol with a fixed five-octet overhead and at most 64 payload octets. The total frame size is at most 69 octets, so the arithmetic `overhead + payload` fits on every target supporting an array that large. The important checks are then whether the input payload meets the application maximum and whether the destination capacity is sufficient.

For a *general* protocol with unbounded payload length, `payload + overhead` might overflow `size_t` before comparison. The safe test is:

```
if (capacity < overhead ||
    payload_length > capacity - overhead)
    return false;
```

This separates a simple mathematical property from an important C-specific evaluation rule. It also shows why the order of tests matters: subtraction is permitted only after `capacity >= overhead`.

The packet-codec example in this book constructs a candidate frame before copying it to the caller's output. That gives the encoder a strong failure guarantee: an invalid request leaves the destination bytes unchanged. For large frames, this extra buffer may be too expensive, and the API might instead document partial output. Neither approach is universally right; their contracts and resource costs differ.

14.23 Independent bit-level tests

A bit-manipulation test should include:

- Zero and all-one values.
- A pattern with distinct nibbles such as `0x12345678`.
- Rotation by zero, one, 31, 32, and 33.
- Values with only the highest bit set or only the lowest bit set.
- Round-trip properties over many counts.

For a serializer, also use literal expected byte sequences. For example, `0x12345678` encoded as big-endian octets must yield `12 34 56 78`, while little-endian octets must yield `78 56 34 12`. These independent vectors distinguish errors that a self-consistent encoder and decoder would not reveal.

Additional exercises

14.9 Prove why each shift in the rotate implementation is in the range 1–31. **14.10** Implement a checked bit-setting function that rejects positions outside the selected unsigned type's width. **14.11** Explain which rules come from C and which come from a vendor peripheral manual when manipulating a status register. **14.12** Construct an independent expected-value table for left rotation of `0x80000001` by 0, 1, 4, and 31 bits.

15. Expressions, Objects, and the Abstract Machine

A C program is not defined by the machine instructions observed in one build. It is defined by the language's abstract machine: a model of evaluation, objects, values, and observable behavior. A compiler is free to transform an implementation as long as it preserves the behavior required of a conforming program.

This principle explains both C's performance and its hazards. A short expression can have a precise meaning, an implementation-dependent meaning, or no defined meaning at all.

15.1 Values and objects

An *object* is a region of data storage whose contents can represent values. An *expression* can produce a value, designate an object, or have side effects. Consider:

```
int number = 7;
int *address = &number;
*address = 9;
```

The pointer contains a value that designates `number`. The indirection expression `*address` is an lvalue; the assignment modifies the designated object. The validity of the operation depends on `number` being alive, accessible, and represented by a suitable pointer.

Not every expression is a modifiable lvalue. A constant expression, a nonmodifiable qualified lvalue, and an array expression cannot simply be treated as destinations for assignment.

15.2 Storage duration and scope answer different questions

Scope describes where an identifier may be used in source code. *Storage duration* describes the lifetime of storage associated with an object. A local `static` object has block scope but static storage duration:

```
unsigned next_identifier(void)
{
    static unsigned value = 0;
    return ++value;
}
```

The object persists across calls. The function has a new problem: unsigned wraparound eventually repeats an identifier. It is also not inherently safe for unsynchronized concurrent calls. A compact interface does not guarantee suitable semantics.

An automatic object, by contrast, usually exists only for the relevant invocation. Returning its address does not extend its lifetime.

15.3 Sequenced and unsequenced evaluations

Consider the expression

```
int result = f() + g();
```

The language does not require the call to `f` to finish before the call to `g`, or vice versa. It does require both results before their addition can produce the final result. If the functions only compute independent values, that freedom usually does not matter.

If both calls modify the same non-atomic scalar object, their unsequenced side effects can make the expression undefined. Write separate statements when an ordering is required:

```
int left = f();
int right = g();
int result = left + right;
```

The full expressions are sequenced in source order. Still check that the addition itself is representable for signed types.

The operators `&&`, `||`, and comma (when used as an operator) introduce sequencing properties that differ from ordinary arithmetic operators. A function-call argument list is not an ordering guarantee.

15.4 Integer overflow and conversions

For an unsigned integer type, arithmetic is performed modulo one more than its maximum representable value. Signed arithmetic overflow has undefined behavior. These are not implementation details that can safely be ignored because a processor uses two's complement.

```
#include <limits.h>
#include <stdbool.h>

bool increment_int(int value, int *out)
{
```

```
    if (out == nullptr || value == INT_MAX)
        return false;
    *out = value + 1;
    return true;
}
```

The guard precedes the addition. A post-addition check would already be too late.

Conversion introduces another boundary. Narrowing an integer to a smaller unsigned type follows modular rules. Converting an out-of-range value to a signed integer type has implementation-defined behavior or can raise an implementation-defined signal. Prefer range checks before narrowing, particularly when parsing files or messages.

15.5 Effective type and aliasing

Object access rules restrict which lvalue types may be used to read or write stored objects. A programmer cannot generally take the address of a `float`, cast it to `int *`, dereference it, and expect portable reinterpretation. Doing so can violate alignment and effective-type rules as well as make code sensitive to representation.

For object-representation inspection, an appropriately sized array of `unsigned char` and `memcpy` are usually clearer. Representation inspection does not imply that every bit pattern is a valid value of every destination type.

```
#include <string.h>
#include <stdint.h>

uint32_t bits_of_float(float x)
{
    uint32_t bits = 0;
    /* Requires equal sizes, verified for the selected target. */
    if (sizeof bits == sizeof x)
        memcpy(&bits, &x, sizeof bits);
    return bits;
}
```

This example exposes representation bytes without violating aliasing rules, but the resulting integer's meaning depends on the target's floating-point representation and byte order. It is **not** a portable way to serialize a floating value.

15.6 Order, lifetime, and optimizers

A valid optimizer may assume that undefined behavior does not occur. Consequently, source checks appearing after invalid operations may be removed. The correct style is to validate the condition *before* executing the operation it protects.

For array access, verify the index against a known count before evaluation of `array[index]`. For allocation, verify multiplication before computing required bytes. For pointer use, ensure the object has not been freed or otherwise had its lifetime ended.

15.7 A small checked interval

Suppose a program writes `count` bytes beginning at `offset` into a buffer of `capacity` bytes. The tempting check `offset + count <= capacity` can itself overflow when the quantities use an unsigned type. Rewrite the requirement into conditions that can be checked safely:

```
#include <stdbool.h>
#include <stddef.h>
```

```
bool fits(size_t offset, size_t count, size_t capacity)
{
    return offset <= capacity && count <= capacity - offset;
}
```

The && operator short-circuits. Therefore `capacity - offset` is evaluated only when `offset <= capacity`. This both avoids underflow and expresses the intended invariant.

A complete executable version with boundary tests appears in `examples/checked_interval.c`.

15.8 Exercises

15.1 Explain why `i = i++ + 1` does not provide a portable way to increment a scalar.

15.2 Give an example of block scope combined with static storage duration. Describe the consequences for recursive calls.

15.3 Implement a checked multiplication of two `size_t` values, preserving its output argument on failure.

15.4 Design a test matrix for `fits` including zero capacity, exact fits, large offsets, and `SIZE_MAX`.

15.5 Explain why reading an integer directly from an arbitrary byte-buffer address can violate alignment or effective-type rules.

15.6 Distinguish a language's unspecified evaluation order from an implementation-defined integer conversion.

15.9 The optimizer reasons from the language rules

Compilers are not required to preserve an apparent intention that depends on undefined behavior. Consider:

```
int *p = /* supplied by caller */;
int value = *p;
if (p == nullptr)
    return -1;
```

The program already dereferences `p` before testing it. A compiler may optimize under the assumption that the dereference had defined behavior; moving the test after the access has made it ineffective. The correct ordering is to check the pointer first and then dereference.

Likewise, `int sum = a + b; if (sum < a) ...` does not provide a general overflow test for signed integers, since overflowing addition already has undefined behavior. Establish representability before evaluating the addition.

15.10 Object representation versus value

Every object has an object representation consisting of bytes. The interpretation of those bytes depends on the type. Some object representations can contain padding or trap representations for some types or implementations. Reading arbitrary bytes as if they were a fully initialized object of any chosen type is not generally safe.

The character types have special significance for inspecting object representations. Copying bytes using `memcpy` is commonly preferable to violating aliasing rules through an incompatible pointer cast. But reproducing the bytes of a pointer or floating-point value on another machine does not guarantee the same value there.

15.11 Sequence and indirection

An expression such as `array[i++] = array[i]` deserves suspicion: side effects and value computations may be unsequenced in a way that violates the rules. A clearer version splits responsibilities:

```
size_t index = i;
int value = array[index];
array[index] = value;
++i;
```

This example is intentionally trivial, but each step now has an explicit order. The larger lesson is not to ban compact expressions, but to separate operations when sequencing is a necessary part of correctness.

15.12 Bounds and pointer provenance

A pointer is more than a numeric address for the purposes of the C abstract machine. Pointer arithmetic and dereference are constrained by the object and its lifetime. Converting an arbitrary integer to a pointer and dereferencing it is not portable ISO C, even on targets where hardware maps devices at fixed numeric addresses.

Freestanding systems often provide implementation-specific extensions for memory-mapped I/O and linker-defined memory locations. Such code should be isolated and documented rather than presented as a portable language feature.

15.13 A reasoning template for code reviews

For each expression ask: Which types participate? What conversions occur? Which operations can overflow or have invalid operands? What is the evaluation ordering? Which object is accessed? Is its lifetime active, and is the access permitted through this lvalue type?

This short set of questions is particularly effective for parsing, arithmetic kernels, and pointer-heavy utilities.

Further problems: Rewrite an overflow-after-addition test into a valid precondition check. Identify the violated rule in a dangling-pointer return. Explain why `memcpy` may safely move representation bytes even though a cast-and-dereference technique is not generally interchangeable.

15.14 Draw the memory model before optimizing

A useful sketch separates objects and pointers. For a vector, draw one structure containing a pointer, a length, and a capacity. Draw a separate heap block containing the elements. A shallow assignment of the vector copies its three fields into another structure, creating another pointer to the same block. It does not allocate a second block.

Now suppose one vector calls `realloc`. The old block's lifetime ends on success. Any pointer borrowed from the old block is invalid; the second shallow-copied owner also has an invalid pointer. Destroying both structures can produce invalid releases. The simplest fix is not a clever cast or compiler option but an ownership rule forbidding such copies.

15.15 Distinguish value equality from byte equality

For an ordinary `struct`, two values may have equal members but different padding bytes. Comparing their raw representations with `memcmp` is not a general semantic equality test. For floating values, even member-wise == needs an application policy concerning NaNs and signed zeros.

Serialization is a further step: it needs a stable external format independent of structure padding, compiler ABI, and machine byte order. This is why Chapter 14 constructs protocol fields byte-by-byte instead of writing the in-memory structure directly.

15.16 Expression transformations need preconditions

A mathematically valid rearrangement is not always valid for finite-width C arithmetic. The expression `offset + count <= capacity` looks equivalent to `count <= capacity - offset`, but only the latter can be evaluated safely after first checking `offset <= capacity` when all quantities have type `size_t`.

Similarly, `a + b` can overflow for signed integers even when a later comparison attempts to detect it. A valid transformation must preserve the language-defined behavior for all permitted inputs, not merely produce the same result on small examples.

15.17 Observable behavior and the optimizer

Compilers may reorder and eliminate operations when permitted by the language's abstract-machine rules. The fact that a debugger shows a particular sequence of machine instructions does not make that sequence the semantics of the source program.

For concurrent code, atomics and synchronization express ordering requirements across threads. `volatile` affects certain observable accesses but is not a general mechanism for sharing ordinary variables safely. For device registers, additional target-specific rules determine bus transactions and barriers.

Exercises

15.7 Draw the heap and stack objects involved in `vector_append`, then redraw them after successful growth. **15.8** Identify when pointer aliases become invalid. **15.9** Compare two structures semantically without reading padding bytes. **15.10** Transform a potentially overflowing length sum into a checked subtraction formulation.

15.18 A pointer can be in bounds yet logically stale

An out-of-bounds pointer is an obvious hazard. A subtler problem occurs when the pointer still designates storage inside a valid array but the application has reassigned that storage to another logical owner.

Consider a fixed array of driver descriptors. A client acquires descriptor slot two, releases it, and later another client acquires the same slot. The original pointer might still address the slot's C object. A bounds checker may have no reason to complain. But the old client no longer owns that logical descriptor.

The generation-tagged pool from Chapter 5 separates physical slot identity from logical allocation identity. Instead of trusting the raw address, an operation verifies the handle's slot index and generation against current pool metadata. The example is a reminder that a language-level memory safety argument and an application-level ownership argument are related but not identical.

An object's *C lifetime* can remain active while its application-level role changes. Conversely, an allocated object's actual lifetime ends when its storage is released. The two cases must not be conflated: accessing an object after `free` is a language-level lifetime violation, whereas accessing a reused pool slot through an obsolete handle can be a logical protocol violation even though the array itself remains alive.

15.19 Object representations and the character types

C allows an object's representation to be inspected as bytes through character-type access. This is useful for checksums, diagnostics and controlled serialization, but raw object bytes are not automatically a stable wire format.

Imagine a structure containing a `bool`, a 32-bit integer, and a pointer. Its representation can depend on padding, pointer format, alignment, and the target ABI. Writing its bytes to disk may also leak padding content or meaningless pointer addresses.

For a portable external record, specify the exact fields and encode them explicitly. For example, store a 32-bit unsigned field into four eight-bit octets by shifting and masking unsigned values. Use `memcpy` for copying known valid object representations between properly sized regions, not as a substitute for format design.

The distinction is particularly important when testing. A `memcmp` over two structures can report different padding bytes even when every meaningful member has equal value. Tests should compare the actual contract, not incidental representation.

15.20 Effective type and aliasing

A compiler reasons about which lvalue types may be used to access an object. Reinterpreting a `float` object by casting its address to an incompatible integer pointer and dereferencing it is not a general portable way to inspect floating-point bits.

A representation-oriented operation can copy bytes into another object using `memcpy`, subject to both objects being valid and sufficiently large. For example, if an application needs the implementation's raw representation of a 32-bit floating type, it can copy those bytes into an array of `unsigned char`, then inspect the bytes. It must still avoid claiming the resulting sequence is the IEEE-754 format unless the target actually provides that representation.

A compiler's optimizer may use the aliasing rules to transform memory operations. The fact that an invalid cast produces an expected value at optimization level zero is not evidence that the program is defined under optimization.

15.21 `restrict` expresses a promise, not a check

A restricted pointer can help an implementation optimize accesses when the programmer guarantees the required non-aliasing relationships for the relevant scope.

Consider:

```
void add_arrays(size_t count,
                double *restrict result,
                const double *restrict left,
                const double *restrict right)
{
    for (size_t i = 0; i < count; ++i)
        result[i] = left[i] + right[i];
}
```

This is a fragment requiring `<stddef.h>` for `size_t`. Its caller must provide valid arrays with the necessary restricted-access relationships. It would be wrong to present `restrict` as a mechanism that checks whether the three pointers overlap. It does not.

The exact C restrict association rules deserve careful reading before adding the qualifier to an existing public interface. Optimizing a small loop should not come at the cost of silently making common aliasing call patterns invalid.

If an application must support in-place operations or partially overlapping buffers, design and test those behaviors explicitly instead of placing `restrict` on every pointer parameter.

15.22 Provenance and hardware addresses

A microcontroller may map device registers to fixed numeric addresses. The C language does not independently promise that converting a numeric constant into an arbitrary pointer and derefer-

encing it is portable across implementations.

Embedded toolchains, vendor headers and the target ABI provide additional contracts for such operations. A well-structured driver isolates these assumptions and documents the hardware register layout, access width, and side effects.

Ordinary C synchronization also does not necessarily control a device's memory ordering or DMA coherency. Appropriate barriers, cache maintenance and peripheral-specific read/write sequences must come from the target's documentation.

Separating the portable algorithm from the hardware adaptation keeps the language rules easier to reason about and permits meaningful host-side tests.

15.23 The optimizer cannot repair undefined arithmetic

A particularly common anti-pattern performs an invalid operation and then tries to detect it:

```
int result = a + b;
if (result < a) { /* assumed overflow */ }
```

For signed integers, if the addition overflows, its behavior is already undefined. A compiler is permitted to reason under the assumption that valid programs do not execute undefined signed overflow and may transform code in ways that defeat the intended post hoc test.

Instead, establish the result's representability before evaluating the addition or use an available checked arithmetic facility with the correct contract.

The same ordering principle applies to pointers: check the input range before computing a potentially out-of-bounds address, and check divisor conditions before dividing. A later condition cannot undo an invalid earlier operation.

15.24 A review sequence for low-level C

For a suspicious expression, review in this order:

1. **Objects:** Which actual objects exist, and are their lifetimes active?
2. **Types:** What are the declared types, integer promotions, and conversions?
3. **Ranges:** Are every index, size, shift count, and arithmetic result valid?
4. **Aliasing:** Is the access permitted through this lvalue type, and do any restrict promises apply?
5. **Sequencing:** Can two evaluations modify or read the same state without the required sequencing?
6. **Concurrency:** Are potentially conflicting shared accesses synchronized?
7. **Environment:** Does the operation rely on implementation-defined ABI or hardware behavior?

This checklist is more reusable than memorizing a collection of undefined-behavior anecdotes. It directs attention to the conditions under which each expression is supposed to be correct.

Exercises

15.11 Explain why an obsolete pool handle is invalid under the API contract even if its index remains inside the array. **15.12** Replace a type-punning pointer cast with a representation copy and list the representation assumptions that remain. **15.13** Determine whether an array-addition routine may support `result == left` when its parameters are restricted. **15.14** Rewrite an overflow-after-addition test so no signed overflow is executed. **15.15** Design a driver adapter that isolates integer-to-pointer conversions and MMIO ordering from portable parsing code.

16. Atomics, Threads, and Communication

Two threads executing the same valid sequential code do not necessarily produce a valid concurrent program. In C, unsynchronized conflicting accesses to a non-atomic object can form a data race and yield undefined behavior. The standard atomic library gives us a way to coordinate access, but it does not eliminate the need to design ownership and progress.

16.1 A race on a counter

Suppose two threads each execute `++counter` one thousand times. The operation is a read, computation, and write; it is not necessarily one indivisible action. Without suitable synchronization, concurrent increments of an ordinary `int` are a data race. A result of two thousand is not guaranteed, and the program has undefined behavior rather than merely a risk of losing a few increments.

An atomic object makes individual accesses atomic:

```
#include <stdatomic.h>

atomic_uint counter = 0;

/* Called by either worker: */
void count_event(void)
{
    atomic_fetch_add_explicit(&counter, 1u, memory_order_relaxed);
}
```

Relaxed ordering suffices when all that matters is a single independent count, not the visibility of some other data. It does not order ordinary memory operations in a way that can publish a complex object.

16.2 Publication with release and acquire

Imagine a producer fills a payload and then signals that it is ready. The consumer must not read the payload before the producer's writes have been made visible through synchronization.

```
#include <stdatomic.h>
#include <stdbool.h>

static int payload;
static atomic_bool ready = false;

void publish(int value)
{
    payload = value;
    atomic_store_explicit(&ready, true, memory_order_release);
}

bool try_consume(int *out)
{
    if (out == nullptr)
        return false;
    if (!atomic_load_explicit(&ready, memory_order_acquire))
        return false;
    *out = payload;
    return true;
}
```

This is a **one-shot publication example**. It is valid only under a protocol in which one producer publishes once, after which `payload` is never modified, and consumers do not access it before an acquire load observes `true`. Reusing the flag for repeated publications without an additional state transition and synchronization would be incorrect.

A release store synchronizes with an acquire load that reads from it, establishing a happens-before relationship for earlier writes. The atomic flag does not itself turn every access to `payload` into an atomic access.

16.3 A bounded single-producer/single-consumer queue

For repeated communication we need a real protocol. A single-producer/single-consumer (SPSC) ring buffer can keep separate atomic positions. Only the producer modifies the write position, and only the consumer modifies the read position.

Reserve one slot to distinguish full from empty. With capacity eight, the queue can hold at most seven items. The producer writes a non-atomic data element, then publishes its advanced write position with release semantics. The consumer acquires the write position before reading the element. The consumer releases its advanced read position after finishing; the producer acquires it before reusing the slot.

The example in `examples/spsc_queue.c` implements precisely this one-producer/one-consumer algorithm. It is **not** a multiple-producer or multiple-consumer queue. It also does not directly establish that its operations are safe to invoke from an interrupt handler on every implementation: that is a separate target-specific question involving lock-free guarantees, runtime support, and interrupt semantics.

C23 initialization note. Older atomic examples may use `ATOMIC_VAR_INIT`, but that macro was removed in C23. The standalone SPSC example uses ordinary `0` initialization for its atomic index members before either participant starts. Once concurrent operation begins, the participants access those positions only through atomic operations with the documented ordering.

16.4 Why index wraparound is not numeric overflow

An SPSC ring normally advances an index with `(index + 1) % CAPACITY`. If index values are always less than capacity, and capacity is suitably small, the addition stays in range. The modulo computes a position within the array. It should not be confused with allowing arbitrary integer overflow.

An invariant for the chosen queue is `0 <= read_index < CAPACITY` and `0 <= write_index < CAPACITY`. Equal positions signify empty; advancing the writer to the reader's current position would signify full and must be rejected.

16.5 Lifetime and shutdown

A queue's memory must remain alive until both participants stop accessing it. Atomic operations do not rescue a buffer freed while a worker still runs. A threaded application therefore needs a termination protocol and a join or equivalent completion mechanism.

This often has greater practical importance than micro-optimizing a memory order. A formally correct queue implementation is still unsafe if its containing object is destroyed too early.

16.6 Mutexes remain valuable

Lock-free algorithms are not necessarily faster, simpler, or more predictable. A mutex protects a group of related operations and is often the right design for moderate contention. In an RTOS, priority inheritance and interrupt-safe API restrictions matter; do not assume that a host C mutex can be used inside an interrupt handler.

Atomicity, mutual exclusion, wait-freedom, and real-time boundedness are different properties. Assess them independently.

16.7 C23 and implementation support

The C atomic facilities are specified by the language and standard library, but target availability and lock-free properties depend on the implementation. Check `atomic_is_lock_free` when relevant and verify the supported interrupt and threading model on the actual hardware. Freestanding environments may provide only a subset of hosted facilities.

On common hosted implementations, compiling threaded examples may require additional compiler and linker options. Keep those options separate from programs that do not require threading.

16.8 Exercises

16.1 Explain why `volatile unsigned counter` does not fix a data race.

16.2 Draw the release/acquire relationships for one successful SPSC enqueue and dequeue.

16.3 Extend the SPSC example with a clear shutdown protocol. Explain object lifetime requirements.

16.4 Determine whether a capacity-eight ring that leaves one slot empty can hold eight elements. Provide tests for full and empty transitions.

16.5 Explain why changing two producer threads to call the same SPSC enqueue operation breaks the algorithm.

16.6 Compare an SPSC queue with a mutex-protected queue for latency, correctness, maintainability, and portability.

16.9 Identify the exact shared objects

Concurrency reasoning starts by listing which objects are accessed by more than one participant. In the SPSC queue, the producer publishes `tail` and observes `head`; the consumer publishes `head` and observes `tail`. Both positions are atomic. Data slots are ordinary objects, but the publication protocol ensures the producer and consumer do not access the same occupied slot without the appropriate ordering.

Merely declaring the indices atomic is not sufficient if the buffer is accessed outside that protocol. Every operation must respect the rule that only the producer enqueues, only the consumer dequeues, and the queue object remains alive until both stop.

16.10 Understanding the release/acquire pair

A producer writes a slot, then uses a release store to publish the new tail. A consumer that reads the published tail with an acquire load can safely observe the preceding slot write. After consuming the element, the consumer releases its new head position. The producer uses an acquire load of head before deciding that a slot can be reused.

The two directions solve distinct problems: publication of newly written data and reclamation of space after consumption. Using only one direction would not establish all of the required cross-thread ordering in the intended design.

16.11 Sequential tests do not demonstrate concurrency correctness

A single-threaded test that pushes seven items, observes full, pops three, and wraps around is useful for verifying index arithmetic and FIFO ordering. It does not establish absence of a data race during simultaneous operations.

A concurrency test can run one producer and one consumer with a known sequence and verify that every item is transferred exactly once and in order. A stress test with variable delays may expose mistakes, but even millions of successful transfers cannot prove the memory-order argument. Reasoning about the C memory model remains necessary.

16.12 Interrupts are outside a generic hosted-thread proof

An interrupt handler is not automatically equivalent to a C standard thread. The compiler, runtime, architecture, and RTOS specify which atomic operations are safe in that context, whether they are lock-free, and how memory barriers interact with devices.

An atomic operation implemented through a library helper or lock may be unsuitable in an interrupt. Even lock-free machine instructions can require cache coherency or architectural barriers beyond what a generic application-level description conveys. Verify the target before advertising an SPSC queue as interrupt-safe.

16.13 Lifetime and cancellation

Stopping a producer is not the same as ensuring it has stopped. A queue owner must establish that neither participant will access the queue before releasing its memory. A shutdown protocol might signal cancellation, wake blocked consumers, join worker threads, and only then destroy the queue.

The choice between draining and discarding pending items should be explicit. Either can be correct depending on the application, but implicit destruction while work remains creates lifetime hazards.

Further problems: Create a two-thread FIFO stress test recording item sequence numbers. Explain the effect of replacing a release store with a relaxed store. Sketch a shutdown protocol for a queue shared between a worker thread and an RTOS task.

16.14 Memory ordering is a protocol, not a performance switch

The SPSC queue works because it establishes two publication relationships. The producer writes an ordinary slot and then publishes an updated tail through a release operation. The consumer acquires that tail before reading the slot. In the opposite direction, the consumer finishes reading the slot and releases its updated head; the producer acquires head before reusing that storage.

It is tempting to label every atomic access `memory_order_relaxed` for speed. But relaxed atomics guarantee the atomicity and modification order of the atomic object, not the required visibility of ordinary slot writes to another thread. The release/acquire relationships are part of this algorithm's proof.

Conversely, using `memory_order_seq_cst` everywhere does not magically repair a design that lets two producers concurrently modify a slot intended for one producer. Memory ordering and permitted roles are separate requirements.

16.15 Read the single-producer guarantee literally

An SPSC queue has one producer and one consumer *at a time under the documented protocol*. Calling `push` from a second producer can make both read the same tail, write the same slot, and publish conflicting updates. Atomic tail storage does not turn the remaining sequence into an atomic transaction.

A multiple-producer queue usually needs additional synchronization or a different algorithm. If a single interrupt is the producer and one task is the consumer, you must still verify that the compiler, target architecture and RTOS support the chosen atomic operations in that interrupt context.

A callback that sometimes runs on the producer thread and sometimes runs in an interrupt is not automatically a single producer under a safe schedule. Identify the actual execution contexts.

16.16 Progress guarantees are different

A nonblocking operation returns without sleeping for a mutex, but the terms *lock-free* and *wait-free* have precise algorithmic meanings. Lock-free generally concerns system-wide progress; wait-free concerns per-operation completion in a bounded number of steps. Neither necessarily means low latency under every cache, interrupt or scheduling condition.

The small SPSC queue has bounded local index work, but whether its atomic primitives are implemented without locks depends on the target. On a small MCU, a supposedly atomic operation may be lowered to a helper routine that is inappropriate in an interrupt handler.

Measure real target execution and verify the atomic implementation before making a real-time guarantee.

16.17 Shutdown and ownership

Memory ordering does not establish object lifetime by itself. Imagine a queue owner frees the queue while a consumer is about to load its head. Even though the member is atomic, the storage is gone and the access is invalid.

A shutdown protocol can announce cancellation, stop the producer, let the consumer drain or abandon remaining items by policy, wait for both to complete, and only then reclaim storage. A good API states who owns the queue object and which operation establishes that no in-flight access remains.

For embedded systems, task deletion and ISR unregistering can each require target-specific synchronization. Test the shutdown path, not only steady-state traffic.

Exercises

16.7 Draw the two release/acquire edges on an eight-slot queue. **16.8** Explain why a second producer is unsafe despite atomic index variables. **16.9** Compare mutex-based and atomic-based designs for a real-time telemetry queue. **16.10** Describe a shutdown sequence that makes destruction safe.

16.18 A complete two-thread atomic counter

The repository's `examples/thread_counter.c` creates two hosted C threads through the facilities in `<threads.h>`. Each worker performs 20,000 increments on one shared `atomic_uint`. The main thread joins both workers and verifies that the final result is 40,000.

The worker's essential operation is:

```
atomic_fetch_add_explicit(&ctx->count, 1u, memory_order_relaxed);
```

This read-modify-write operation is atomic. Two workers cannot accidentally overwrite one another's increments through the classic load/increment/store race on an ordinary integer. The example deliberately chooses a total that fits even in the standard minimum range of `unsigned int`.

Why is relaxed ordering sufficient? The program cares only about the counter's value. It does not use that counter to publish a separate ordinary data structure. The atomic object has an atomic modification order, and successful thread joins establish that the workers have completed before the main thread checks the result.

If a producer filled an ordinary object and then incremented the counter to announce that the object was ready, the design would be different. Relaxed operations on the counter alone would not establish the necessary ordering of writes to the accompanying non-atomic object.

16.19 Partial thread creation is a resource problem

Suppose creating the first worker succeeds and creating the second fails. The first worker may still be executing and using a pointer to the caller's stack-local context.

The correct cleanup principle is to remember how many workers started and join them all before allowing that context's lifetime to end. Returning immediately after the second creation error could leave the first thread with a dangling pointer.

This is the same ownership discipline used for partially initialized files, allocations, and device resources: acquire incrementally, track what succeeded, and clean up what was acquired on every failure path.

The example checks each creation and join result, although a failed join in a production thread framework may require a more comprehensive recovery policy. A passing test of ordinary creation and completion does not prove all exceptional thread-system conditions are recoverable.

16.20 One-shot release/acquire publication

An atomic counter is not the same as publishing a payload. Consider a shared ordinary `int payload` and an atomic readiness flag. The intended one-shot protocol is:

```
Producer                              Consumer
--------                              --------
payload = value;
store ready=true (release)            load ready (acquire)
                                      if true, read payload
```

When the consumer's acquire load reads from the producer's release store, the release/acquire synchronization establishes that the preceding payload write happens-before the later read.

This explanation depends on a **one-shot** lifetime rule: the producer writes payload once and does not overwrite it after publication. Reusing the flag and publishing a second value without coordinating with the consumer is not automatically safe. A later write could race with the earlier consumer's read.

Using `volatile` instead of an atomic flag would not establish this general thread synchronization guarantee.

16.21 Two distinct directions in an SPSC ring

In the book's single-producer/single-consumer queue, the producer writes a slot and release-stores the updated tail. The consumer acquire-loads the tail before reading the newly available slot.

The opposite direction matters too: after the consumer finishes reading a slot, it release-stores an advanced head; the producer acquire-loads head before deciding that the old slot is safe to reuse. One direction publishes data; the other publishes reclaimed capacity.

The queue assumes exactly one producer and one consumer under this protocol. Adding a second producer is not a harmless increase in workload. Two producers may observe the same tail, write the same slot, and publish conflicting states. Atomic positions alone do not turn a sequence of operations into a multiple-producer transaction.

16.22 Different tests answer different questions

A sequential queue test verifies index arithmetic, FIFO behavior, full/empty conditions, and wraparound for chosen operations. A concurrent producer-consumer test may expose interleaving failures. The two-thread counter demonstrates that the test environment supports concurrent atomic read-modify-write operations.

None of those tests proves all possible memory-ordering arguments for an unrelated lock-free algorithm. Correctness also requires reviewing which participant owns each slot, which objects are shared, what establishes happens-before relationships, and when storage may be destroyed.

A host ThreadSanitizer run can detect certain data races on executed paths, but its instrumentation changes timing and may not model MCU interrupt or DMA semantics.

16.23 What the C thread library does not promise for a microcontroller

The hosted threads facility is not universally provided by freestanding C implementations. A desktop build may need a compiler/linker flag such as `-pthread`; an embedded project instead uses RTOS task facilities or target-specific interfaces.

The C memory model provides a useful conceptual foundation, but whether a particular atomic operation is lock-free or safe inside an interrupt handler is an implementation-specific matter. An atomic operation lowered to a runtime library helper may be inappropriate in an ISR. DMA buffers also introduce coherency requirements outside the ordinary two-thread example.

Do not equate lock-free with deadline-safe. System-wide progress is not the same as a proven maximum completion time for every caller. A real-time design must account for preemption, cache, scheduling, blocking and priority behavior.

Additional exercises

16.11 Explain why `memory_order_relaxed` is sufficient for a standalone atomic count but not for publication of an ordinary payload. **16.12** Draw all resources after one thread creation succeeds and the second fails. **16.13** Label the two release/acquire handoff directions of an SPSC queue. **16.14** Describe the additional requirements for making the queue safely usable by an MCU interrupt handler. **16.15** Distinguish lock-free, wait-free, and bounded real-time execution.

17. Function Pointers, Callbacks, and Interfaces

An ordinary function call names an operation directly. A callback allows a caller to supply the operation itself. This is useful for sorting, traversing data, handling events, and separating reusable code from application policy.

17.1 A function pointer

Consider a function that compares two integers:

```
static int compare_int(int a, int b)
{
    return (a > b) - (a < b);
}
```

The subtraction produces -1, 0, or 1 without calculating `a - b`, which could overflow. A pointer to such a function can be declared as:

```
int (*comparison)(int, int) = compare_int;
int result = comparison(4, 7);
```

The parentheses around `*comparison` are essential. Without them, the declaration would describe a function returning a pointer rather than a pointer to a function.

17.2 A callback with context

Practical interfaces often accept both a callback and an opaque context pointer. The context lets the caller maintain state without resorting to global variables:

```
#include <stddef.h>
#include <stdbool.h>

typedef bool (*visit_fn)(int value, void *context);

bool visit_all(const int *items, size_t count,
               visit_fn visitor, void *context)
{
    if (visitor == nullptr || (count != 0 && items == nullptr))
        return false;
    for (size_t i = 0; i < count; ++i)
        if (!visitor(items[i], context))
            return false;
    return true;
}
```

The visitor returns `true` to continue and `false` to stop. In this design `false` from the traversal means either an invalid argument or an early stop; a richer status type could distinguish them.

The `void *` context is opaque to the traversal function. The caller is responsible for supplying a pointer to the appropriate object and keeping it alive for the duration of the call. Converting an object pointer to `void *` and back preserves its intended use when the underlying type and lifetime are correct.

17.3 A stateful visitor

```
struct accumulator { long long total; };

static bool accumulate(int value, void *context)
{
    struct accumulator *state = context;
    state->total += value; /* assume the sum is representable */
    return true;
}
```

A callback can change the caller's state without changing the traversal algorithm. In production, this example should either check overflow or document a bound on the count and input values.

Callbacks are usually synchronous in simple C library interfaces, but not every callback API is synchronous. An asynchronous system must state whether it retains the context pointer, which thread invokes the callback, and how cancellation and destruction interact.

17.4 Sorting and the comparison contract

The standard `qsort` function takes a comparator that accepts pointers to two objects and returns a negative, zero, or positive result. A comparator must be consistent; it must not produce contradictory orderings for the same values.

```
static int order_ints(const void *left, const void *right)
{
```

```
    int a = *(const int *)left;
    int b = *(const int *)right;
    return (a > b) - (a < b);
}
```

The casts are appropriate when the comparator is called by `qsort` on an array of `int`. A comparator should not simply return `a - b` because signed subtraction may overflow. The function cannot assume that its arguments refer to arbitrary or mismatched types.

17.5 Function pointers are not object pointers

The language distinguishes pointers to functions from pointers to objects. Do not assume a function pointer can be stored portably in `void *`. Some operating systems provide dynamic-loading interfaces with platform-specific conventions, but those conventions are not general ISO C guarantees.

Likewise, do not manufacture a function pointer from an integer address and call it without platform-specific guarantees. Embedded startup code and vector tables sometimes rely on implementation-defined representations; such code belongs behind a documented target interface.

17.6 Interfaces as boundaries

A driver interface might offer methods to read, write, and close a device:

```
#include <stddef.h>
#include <stdbool.h>

struct byte_sink {
    bool (*write)(void *context, const unsigned char *data, size_t size);
    void *context;
};
```

An application can then accept a `struct byte_sink` instead of knowing whether output goes to UART, a file, or a memory buffer. This style permits host tests using a fake sink. Yet it carries obligations: the context must remain valid, operations need failure semantics, and concurrent access must be synchronized if shared.

17.7 Costs and limits

Callbacks add indirection and may limit compiler inlining. That cost is often negligible compared with I/O and other operations, but it can matter in a small real-time inner loop. Measure rather than assume. A callback-based interface also needs careful error handling so that failures are not silently ignored.

17.8 Exercises

17.1 Implement a visitor that counts elements greater than a threshold supplied in its context.

17.2 Use `qsort` to order signed integers. Test `INT_MIN` and `INT_MAX` to show why a subtracting comparator is unsafe.

17.3 Design a byte sink whose write function may accept fewer bytes than requested, and define a complete-write wrapper.

17.4 Explain why casting a function pointer to `void *` is not a portable storage strategy.

17.5 Show how a synchronous callback's context lifetime differs from an asynchronous callback's lifetime.

17.9 Make callback contracts explicit

A callback interface has two contracts rather than one. The host function promises when and how it invokes callbacks. The callback promises which values and side effects it supports. A traversal API should specify whether it visits elements in order, whether it may stop early, and whether callbacks are allowed to modify the traversed collection.

Reentrancy is another decision. If a callback invokes the traversal function again, will the operation remain correct? A traversal that uses only local state may be naturally reentrant; one that uses a static cursor may not. If callbacks can be invoked concurrently, the context object also needs an appropriate synchronization policy.

17.10 Context lifetime and asynchronous operations

A synchronous visitor that returns before its caller exits can safely borrow a stack-local context if it does not retain the pointer. An asynchronous driver that stores the callback and context for a later event cannot use a context that has already gone out of scope.

```
struct counter_state { unsigned count; };

void register_callback(void (*callback)(void *), void *context);
```

This declaration alone says nothing about whether the callback is invoked immediately, on another thread, or after the registration function returns. A real API must specify that behavior. It must also provide a safe way to unregister and determine when in-flight callbacks have stopped before the context is destroyed.

17.11 Design a complete-write adapter

An abstract byte sink can report the number of bytes actually accepted rather than simply success or failure. A wrapper then loops until the entire record is written or the sink reports no progress or error. Such an adapter can unify a UART, a buffered memory sink, and a file stream while preserving their differing behavior at the boundary.

The wrapper must use subtraction to calculate remaining bytes, handle a zero-byte result without looping forever, and enforce the caller's storage contract. If the sink is nonblocking, `would block` may deserve a distinct status rather than being treated as fatal.

17.12 qsort and ordering

A comparator used with `qsort` must induce a consistent ordering. For integers, use relational comparisons rather than subtraction to avoid overflow. For structures, compare fields in a documented priority order. For floating values, decide how NaNs should be treated; ordinary relational comparisons do not automatically provide a total order over every floating value.

Sorting also changes element positions. Any pointer to an element's current location may no longer designate the same logical record after sorting. The ownership of the array itself does not change, but references to records by index may need to be updated.

17.13 Interfaces and performance

Function-pointer indirection can inhibit inlining and complicate static call-graph analysis, but sometimes it improves modularity enough to outweigh its cost. In firmware, an interface through function pointers may be suitable for the device layer while a hot DSP inner loop uses direct calls.

Measure on target hardware when performance matters. A desktop microbenchmark may not predict instruction cache, branch prediction, or linker optimizations on a Cortex-M core.

Further problems: Write a reentrant traversal that avoids global state. Define unregister semantics for an asynchronous callback API. Implement an ordered comparator for records containing a timestamp and a unique identifier.

17.14 A practical byte-sink interface

A callback can separate a useful algorithm from the device performing its I/O. Define a writer that receives a context pointer, source bytes, and a count, and returns how many bytes it accepted:

```
#include <stddef.h>

typedef size_t (*sink_write_fn)(void *context,
                                const unsigned char *data,
                                size_t count);

struct byte_sink {
    sink_write_fn write;
    void *context;
};
```

The interface contract must forbid returning more bytes than requested. A zero result means no progress under this simplified synchronous protocol, so a complete-write wrapper reports failure rather than spinning indefinitely.

17.15 Handling partial writes

A wrapper starts at offset zero. It calls the sink with `data + offset` and `count - offset`, then increases the offset by the number of bytes accepted. Since the call occurs only when `offset < count`, these operations are within a valid caller-supplied byte buffer. If the sink returns zero or a value larger than the remaining count, the wrapper reports failure.

The loop's progress proof is straightforward: on every successful iteration, offset strictly increases and never exceeds count. The loop therefore terminates after a finite number of accepted writes, even if the sink accepts just one byte at a time.

This design is particularly relevant to a UART driver or nonblocking transport. A real asynchronous system may need a 'would block' status and a state object retained across callbacks. That is a different protocol from the simple synchronous function here.

17.16 Test using a fake device

The companion `examples/callback_sink.c` uses a fake writer that can accept only two bytes per call. The complete-write wrapper must repeatedly call it to deliver a longer message. The test then switches the fake into a mode that accepts zero bytes, verifying that the wrapper fails without looping forever.

A fake device is useful because it isolates the algorithm from real hardware and permits exact, repeatable failure behavior. Hardware validation remains necessary for UART timing, DMA, interrupt interaction, and other target-specific concerns.

17.17 Lifetime and reentrancy

The sink's context pointer is borrowed during each synchronous call. If the sink stores that pointer and uses it after returning, a new lifetime contract is required. Likewise, a sink used simultaneously by multiple threads may require a mutex or atomics. The callback pointer itself does not provide thread safety.

A reusable interface should specify whether a callback may call back into the same interface recursively. Such reentrancy can be safe for a purely local algorithm but dangerous for a driver with shared mutable state.

Exercises

17.6 Modify the sink to report 'would block' separately from a permanent error. **17.7** Write a wrapper that resumes a partially sent message on a later call. **17.8** Design a byte sink with a maximum record length and checked capacity. **17.9** Explain what the caller must guarantee about the lifetime of its context object.

17.18 Sorting records with more than one key

The simplest `qsort` example compares two integers. Real software often sorts records according to a hierarchy of keys. A telemetry log might order events by priority, then source identifier, then the order in which they arrived.

The companion `examples/sort_records.c` defines records with three fields:

```
struct record {
    unsigned priority;
    char name[16];
    size_t arrival;
};
```

Its contract is that `name` contains a valid, zero-terminated byte string within its array. The sort orders priority numerically from low to high, then names according to `strcmp`, then arrival number numerically.

The last field is useful because the standard `qsort` interface does not promise a stable sort. When two elements compare equal, their relative order need not remain the same as it was originally. Including a unique, order-preserving arrival field can produce the desired deterministic result through the comparator itself.

A deterministic tie-breaker does not make the underlying algorithm a stable sorting algorithm. It defines a different total order that distinguishes records the earlier keys regarded as equal.

17.19 A comparator is a mathematical relation

The comparator receives two `const void *` values pointing at objects in the array. It converts them to pointers to the actual element type and returns a negative result, zero, or a positive result according to the selected ordering.

A correct comparison should be consistent. In particular, if it says `a` precedes `b`, it must not also say that `b` precedes `a`. Equal values should compare equal, and comparisons should not depend on changing global state.

The record comparator first checks priority. When that key differs, the remaining keys are irrelevant. Only when the priorities match does it compare names; only when names also match does it compare arrival order. This is called *lexicographic ordering* by keys.

The function need not calculate numerical differences to express the result. For integers, a useful three-way comparison is:

```
return (a > b) - (a < b);
```

Each relational result is zero or one, so their difference is between minus one and one. It cannot overflow, regardless of how widely separated the original integer values are.

By contrast, returning `a - b` for signed `int` keys can overflow when one key is `INT_MIN` and the other is `INT_MAX`. The comparator would then invoke undefined behavior in precisely the unusual cases an ordering routine should handle reliably.

17.20 Why the callback has no context argument

The standard `qsort` comparator type takes only two object pointers. It does not have a general-purpose user-context parameter. This differs from the byte-sink callbacks earlier in this chapter, which carry a context pointer alongside the function pointer.

Suppose an application wants to sort based on a configuration object, such as ascending or descending priority chosen by a user. One portable option is to place the relevant ordering key inside each record or precompute it before sorting. Another is to write a small specialized sort operating on the application's own context.

Some platforms provide `qsort_r`-style extensions with a context parameter, but signatures and availability differ among systems. Such an extension must not be presented as if it were part of portable ISO C.

Global configuration used by a comparator can introduce reentrancy and concurrency problems. Two simultaneous sorts might require different ordering settings. Avoid using a mutable global merely to simulate a missing callback context unless the program explicitly controls access to it.

17.21 The type of `void *` does not carry ownership

A callback accepting `void *context` can be flexible, but the type does not tell the callback whether it may modify, retain, or release the pointed-to object. Those rules require a separate contract.

In the byte-sink example, the context points to a fake output device that remains alive for the entire synchronous call. The callback may inspect or modify its state, but it does not own the structure or free it. If the sink stored the context pointer to use later, the caller would need to keep the object alive until that asynchronous operation finished.

Casting a `void *` to the wrong object pointer type does not produce a dynamically checked conversion. A common pattern is to use a dedicated callback type alongside clear documentation that identifies the expected context structure.

A C module can further reduce mistakes by hiding its context representation behind an opaque structure pointer and exposing initialization and operation functions.

17.22 Return values for partial progress

Consider a callback that accepts a requested byte count and returns how many bytes it consumed. The caller may need to retry with the unwritten suffix. A complete-write adapter can maintain the invariant `offset <= count` and call the sink with `data + offset` and `count - offset`.

If the callback returns zero while bytes remain, blindly looping would cause a busy wait with no forward progress. If it returns more bytes than requested, the caller must reject the invalid result before adding it to the offset.

This simple interface deliberately conflates permanent failure and temporary inability to accept data, treating both as failure when no bytes are accepted. An asynchronous or nonblocking writer generally benefits from a richer status:

```
enum send_status {
    SEND_PROGRESS,
    SEND_WOULD_BLOCK,
    SEND_CLOSED,
    SEND_ERROR
};
```

Such an API should return the accepted byte count separately or inside a small result structure. A `WOULD_BLOCK` result tells the caller to arrange another attempt when the device becomes ready, rather than spin continuously.

17.23 Building a testable device adapter

A callback with a context pointer is a powerful test seam. The production context may contain a UART handle, while a host test substitutes an in-memory sink. The shared algorithm need not contain conditional compilation for every device.

A fake sink can accept at most two bytes per call, record them into a fixed local array, and count callbacks. When a five-byte message is sent through the complete-write adapter, the expected call count is three.

Next configure the fake to accept zero bytes. The adapter should fail immediately without modifying the accepted-byte count. A final test can supply an invalid callback or a null byte pointer with a positive count.

These tests exercise error contracts independently from UART drivers, DMA, or interrupt scheduling. They make the algorithm more reusable without claiming to test the target hardware.

17.24 When a function pointer is the right abstraction

A callback is useful when one component needs to invoke an operation whose implementation varies. Sorting, event handlers, output sinks, memory allocators, and parser visitors are examples.

A function pointer is not necessarily the best choice for every tiny operation. If the set of behaviors is fixed at compile time, ordinary functions and static selection can be simpler and easier for a compiler to optimize. For a flexible library boundary, callbacks can reduce dependencies and permit tests to inject failures precisely.

Performance claims should be measured on the real target. On some compilers a known callback may be inlined after optimization; on others an indirect call may impose measurable cost or complicate worst-case execution analysis. Correct lifetime and error semantics remain necessary in either case.

17.25 A complete comparator test

A strong comparator test should construct records whose keys exercise each tie-break level. The companion example includes priorities zero and three, names such as `alpha`, `gamma`, and `zeta`, and distinct arrival numbers.

After `qsort`, verify the exact expected arrival sequence. Then compare every adjacent pair and confirm that the comparator considers the earlier element strictly less than the next. Testing only a single pair of records is insufficient to demonstrate multi-key ordering.

A more comprehensive test can generate records and check antisymmetry and transitivity over triples, but these properties must be implemented carefully. A test oracle that repeats the exact comparator logic could share its defects, so hand-computed examples remain valuable.

Further exercises

17.10 Extend the record comparator with a timestamp field and state the complete key priority. **17.11** Explain why a unique arrival field can make the final output deterministic without `qsort` becoming a stable algorithm. **17.12** Design a context-aware sorting interface that uses only ISO C and does not rely on mutable globals. **17.13** Add a fake sink mode that returns `WOULD_BLOCK` and explain how the caller resumes a partial write. **17.14** Prove that a three-way comparison built from relational operators cannot overflow as an `int`.

18. Text, Encodings, and Robust Parsers

Text is often treated as the simplest form of input, but a text interface combines several independent conventions: how bytes are encoded, what counts as a character, where a line ends, and which text represents a valid value. C gives programmers byte-oriented primitives; a program must choose and document its interpretation of those bytes.

18.1 The byte is not the character

A C string is a sequence of `char` values terminated by a zero byte. The library function `strlen` counts units before the terminator. For UTF-8 text, that count is not necessarily the number of Unicode scalar values or user-perceived characters.

UTF-8 encodes a code point in one to four octets. Continuation octets do not stand alone. A decoder must reject overlong sequences, surrogate code points, values beyond U+10FFFF, and incomplete sequences. A robust decoder also must not read past the caller's buffer when the leading octet suggests more input than remains.

18.2 Parse lengths explicitly

An untrusted message need not have a terminating zero byte. A parser can accept a pointer and a length, so no operation depends on searching outside the supplied region.

```
#include <stddef.h>
#include <stdbool.h>

bool ascii_digit(unsigned char c)
{
    return c >= '0' && c <= '9';
}
```

This function classifies one byte without invoking locale-sensitive character classification. For applications that require whitespace recognition under a locale, `isspace` is useful, but its argument must be `EOF` or a value representable as `unsigned char`.

18.3 Decimal parsing as a checked transition

A decimal parser maintains an accumulated number. Each incoming digit requests a transition from `value` to `value * 10 + digit`. For an unsigned maximum `MAX`, that transition is valid only when

[value \leq \left\lfloor\frac{MAX-digit}{10}\right\rfloor.]

The inequality must be tested *before* multiplication. A checked parser can therefore reject overflow without ever evaluating the overflowing expression.

A complete implementation appears in `examples/parse_unsigned.c`. Its contract is intentionally narrow: only nonempty ASCII decimal digits are allowed, and failure leaves the output unchanged. This is different from `strtoul`, which has defined syntax for optional whitespace, signs, and bases and uses `errno` for range errors.

18.4 A bounded UTF-8 decoder

The companion program `examples/utf8_decode.c` decodes a single Unicode scalar value from a counted input buffer. It reports how many octets were consumed on success and does not change either output on failure.

For example, the sequence `E2 82 AC` represents U+20AC (the euro sign). But `C0 80` must be rejected: it is an overlong representation of U+0000. The byte sequences `ED A0 80` and `F4 90 80` must also be rejected because they denote disallowed scalar values.

The example's purpose is decoding, not Unicode normalization, segmentation, case folding, or character rendering. Those operations require additional specifications and, in real-world applications, frequently a carefully selected text library.

18.5 Failure and incomplete input are distinct

A streaming decoder may encounter a prefix of a potentially valid sequence, such as the first two octets of a three-octet UTF-8 character. A one-shot parser can report failure and ask its caller to provide more bytes. A streaming parser often benefits from three results: valid, invalid, and incomplete.

That distinction prevents a network receiver from prematurely rejecting input merely because it arrived in pieces. It also prevents an actual malformed sequence from being treated as indefinitely incomplete.

18.6 The language's execution character set

C programs have implementation-defined character representations for source and execution environments, within the language's requirements. The ordinary literal `'A'` does not universally imply the integer 65. An ASCII-oriented parser must state its ASCII requirement explicitly.

The examples in this chapter use octet values for UTF-8 and the digit range for decimal input. Such assumptions are appropriate when decoding formats defined in ASCII or UTF-8, but they are not claims about every C execution character set.

18.7 A secure parsing checklist

Before parsing an untrusted input, identify its maximum length, byte encoding, permitted grammar, numeric ranges, truncation policy, and error reporting. Validate a complete candidate before writing durable output state. Avoid relying on null terminators unless the interface guarantees them.

Tests should include zero-length input, all valid boundary values, incomplete prefixes, extra trailing bytes, and malformed encodings. Pair handwritten test vectors with fuzzing, and preserve discovered failures as regression tests.

18.8 Exercises

18.1 Implement a parser for unsigned hexadecimal digits with an explicit buffer length and checked overflow.

18.2 Extend the UTF-8 decoder to distinguish incomplete input from malformed input.

18.3 Explain why byte length, Unicode scalar count, and displayed character count can differ.

18.4 Add fuzz tests checking that malformed input never changes the decoder's outputs.

18.5 Design a streaming line protocol that accepts CRLF and LF line endings while bounding the line length.

18.6 Compare a locale-sensitive classification function with a format-specific ASCII byte test.

18.9 UTF-8 validity is more than continuation bytes

A four-byte UTF-8 sequence is not valid merely because its final three bytes begin with `10` in binary. The decoded value must also lie between U+10000 and U+10FFFF. A three-byte sequence must not encode a value below U+0800 or any surrogate in U+D800–U+DFFF. A two-byte sequence must not encode an ASCII value.

These rules reject overlong encodings and invalid Unicode scalar values. A robust decoder should validate both the structure of each sequence and the numerical range of its result before publishing it.

18.10 Character count versus display width

UTF-8 code points are not synonymous with visual characters. A base letter followed by a combining accent can use two scalar values but appear as one grapheme cluster. Some visible characters use multiple code points joined by special sequences. Consequently, a user-interface operation such as cursor movement or truncation by displayed character count needs Unicode segmentation rules beyond a basic decoder.

A low-level C parser can be correct while knowing nothing about grapheme clusters. Its contract should say that it counts bytes or scalar values, not vaguely claim to count characters.

18.11 Incremental decoding as a state machine

A streaming decoder can retain a partial code point, a count of expected continuation bytes, and enough information to reject an invalid range. Each input byte updates that state. The result can be one of three conditions: a scalar value completed, more bytes needed, or malformed input.

On an I/O boundary, incomplete is not necessarily invalid. If the stream ends while the decoder still expects bytes, the incomplete sequence becomes a final input error according to the protocol's policy. This mirrors the word-counting state machine from Chapter 1, but with stricter validity requirements.

18.12 Parsing machine-oriented text

Protocol text is not necessarily locale-dependent natural language. A command such as `SET 1234` may be explicitly ASCII with no whitespace variation; a JSON document is UTF-8 with its own grammar and escape sequences. Use a parser matching the format rather than applying broad locale-sensitive rules indiscriminately.

A counted span makes boundaries explicit. For an ASCII command field, scanning exactly its declared number of bytes avoids relying on an unrelated terminator. Reject embedded NUL bytes if the grammar forbids them, even though an ordinary C string function might stop at the first one and ignore trailing bytes.

18.13 Validate before normalizing

Converting malformed byte sequences to lowercase or replacing bytes before validating the input can introduce ambiguity. For format-specific ASCII tokens, validate each byte first, then normalize the allowed subset. For general Unicode text, decoding and normalization need a documented Unicode library and version.

Further problems: Extend the decoder to report incomplete separately from invalid. Produce golden test vectors for one-, two-, three-, and four-byte values. Design a text protocol that rejects embedded NUL and trailing garbage rather than silently ignoring it.

18.14 Separate three kinds of text boundaries

A byte stream has transport boundaries, an encoding has code-unit and code-point boundaries, and a user interface has grapheme or displayed-character boundaries. These are not interchangeable.

A UTF-8 character may be split across two `fread` calls or UART DMA transfers. The decoder must retain the incomplete prefix or the adapter must buffer sufficient bytes. Even after a valid scalar value has been decoded, two adjacent values may combine into a single user-perceived character. A simple MCU command parser often needs only the first two concepts, while a text editor needs all three.

A counted input buffer should always be processed according to its supplied length. A zero byte inside the buffer is valid as a byte value; whether it is allowed by the *protocol* is a separate grammar decision. Using `strlen` on arbitrary counted data can stop early at an embedded zero or read past the allocation if there is no terminator.

18.15 Build the UTF-8 validity proof

For one-byte sequences, the leading byte lies in 00–7F. Valid two-byte sequences begin with C2–DF; C0 and C1 are excluded because they would produce overlong encodings. Valid three-byte sequences begin with E0–EF but need additional restrictions on the second byte or, equivalently, a final decoded-scalar range check. Four-byte sequences begin with F0–F4 and likewise require a result no greater than U+10FFFF.

Every continuation byte must have its top two bits `10`. After combining all payload bits, reject decoded values below the shortest possible value for that sequence length. Reject surrogate code points U+D800–U+DFFF. These tests, in that order or an equivalent validated order, establish that the input represents a Unicode scalar value.

The standalone `utf8_decode.c` handles one code point and reports its consumed length on success. A complete streaming implementation needs an additional status to distinguish a truncated but potentially valid prefix from a definitely malformed sequence.

18.16 Avoid accidental locale dependencies

C's `<ctype.h>` functions operate on character values according to their documented rules and may be locale-sensitive. For a protocol defined in ASCII, the explicit range check `c >= '0' && c <= '9'` may make the grammar clearer, subject to the execution character representation. For raw UTF-8 bytes, interpreting every byte through `tolower` is incorrect Unicode processing.

A production system should choose its textual contract at its boundary: ASCII protocol tokens, UTF-8 source text, or locale-sensitive human input. Once chosen, the parser should not silently switch conventions halfway through processing.

18.17 Bounded streaming text

A command reader with a maximum length of 64 bytes needs both an accumulation buffer and a policy for overflow. If a command exceeds the limit, immediately terminating the connection may be appropriate for a strict protocol. Alternatively, it can discard subsequent bytes until a line delimiter and then resume with the next record.

Do not accept the first 64 bytes as a complete valid command: truncation can change semantics. A long dangerous command might share a harmless-looking prefix with a valid shorter command. Rejecting the entire overlong record preserves the intended grammar.

A streaming decoder also needs an EOF policy: an incomplete multibyte sequence at the end of a stream is not the same thing as clean EOF between complete records.

Exercises

18.7 List valid and invalid UTF-8 leading-byte ranges. **18.8** Build a state machine that distinguishes 'need more bytes' from 'invalid encoding'. **18.9** Explain why NUL may be a valid byte while still being forbidden inside a chosen text protocol. **18.10** Specify the overflow and resynchronization behavior of a bounded line reader.

18.18 A streaming decoder that remembers incomplete input

The one-shot decoder earlier in this chapter receives a buffer containing all bytes for one candidate scalar value. Actual input often arrives in arbitrary fragments. A UART interrupt might deliver one byte, while a socket read might deliver several complete characters followed by part of another. The parser must not equate a transport read boundary with a character boundary.

A streaming decoder can retain just three pieces of state:

- **Accumulator:** the bits of the scalar value built so far.
- **Minimum:** the smallest legal scalar value for the selected sequence length.
- **Remaining:** the number of continuation octets still required.

For ordinary ASCII (00–7F), decoding completes with one byte and no persistent state. For a multibyte leading octet, the decoder records an initial bit field and waits for its continuation octets. Each continuation contributes six more bits.

The implementation in `examples/utf8_stream.c` expresses this using a small state structure:

```
struct utf8_decoder {
    uint32_t value;
    uint32_t minimum;
    unsigned remaining;
};
```

The decoder does not allocate memory and its state size does not grow with the input stream. That is important for constrained embedded devices and untrusted network input.

18.19 Three results are more useful than two

A one-byte-at-a-time decoder needs to distinguish:

Result	Meaning	Output scalar
`UTF8_MORE`	The prefix may still become a valid scalar, but more octets are required	Unchanged
`UTF8_COMPLETE`	One valid Unicode scalar has been decoded	Updated
`UTF8_INVALID`	The input violates the encoding rules, or API arguments are invalid	Unchanged

A boolean result would conflate a valid but incomplete prefix with malformed input. That distinction matters in streaming I/O: receiving the leading octet E2 by itself is not malformed; receiving E2 followed by an invalid continuation is.

The implementation publishes `*out` only when a complete scalar has passed every validation. On any failure or incomplete prefix, the caller's earlier output remains unchanged.

That strong failure rule allows a caller to retain its last successfully decoded scalar without inventing a separate validity flag for the output storage.

18.20 Validate the leading octet first

UTF-8's leading-octet ranges determine the expected length:

```
00..7F     ASCII scalar, complete in one octet
C2..DF     start of a two-octet sequence
E0..EF     start of a three-octet sequence
F0..F4     start of a four-octet sequence
other      invalid as a leading octet
```

C0 and C1 are not valid starting octets because they could produce overlong representations of ASCII values. The bytes 80–BF are continuation octets, not standalone leading octets. F5–FF are outside the permitted leading range for Unicode scalar values.

After choosing a sequence length, the decoder records a minimum allowed scalar value: 0x80 for two octets, 0x800 for three, and 0x10000 for four. These lower limits later reject overlong encodings without requiring dozens of special leading/second-octet cases.

18.21 Every continuation byte has a shape

A continuation octet begins with binary bits `10`. The test

```
if ((byte & 0xc0) != 0x80)
    return UTF8_INVALID;
```

rejects any byte that is not a continuation when one is required. A correct decoder also resets its partial state on an invalid continuation according to a clearly documented recovery policy.

After validation, the low six bits are incorporated:

```
value = (value << 6) | (byte & 0x3f);
```

This operation is performed on an unsigned accumulator. At most four UTF-8 octets are accepted for one scalar; the accumulated value is consequently bounded well below the maximum of `uint32_t`. The shift count is the fixed value six, valid for the type.

The decoder decrements `remaining`. If more continuation bytes are required, it returns `UTF8_MORE`. When the remaining count reaches zero, it performs final scalar-range validation.

18.22 A complete sequence is not necessarily a valid scalar

Even correctly shaped continuation octets can assemble an invalid scalar. A final validity check rejects:

1. Values below the minimum for their sequence length (overlong encodings).
2. Surrogate code points U+D800 through U+DFFF.
3. Values greater than U+10FFFF.

For example, E0 80 80 has three valid-looking octets but decodes to zero, which must be represented using the one-octet form 00. ED A0 80 decodes to a surrogate. F4 90 80 80 exceeds the maximum Unicode scalar.

The accompanying tests include all three cases. They also verify that a failure does not overwrite the previously decoded scalar.

Checking only the structural shape of octets is therefore insufficient.

18.23 End of stream is a separate event

A decoder may be sitting in a valid partial state when its input source reaches clean EOF. The byte prefix was not necessarily malformed when first encountered, but the now-finished stream does not contain a complete scalar.

A small finishing operation can ask whether `remaining == 0`. If it is nonzero, the complete input was truncated. A higher-level parser should report that condition according to its data format: an incomplete UTF-8 text file, a malformed protocol field, or an input requiring retransmission.

Do not silently discard the final incomplete prefix and report clean success, because truncation can change what a document or command means.

An incremental network protocol may keep the partial state between reads and call the finish check only when it has received an actual end-of-message indication. A temporary lack of available bytes is not automatically EOF.

18.24 Recovery is not universal

The example resets decoder state after a malformed sequence and allows the next supplied byte to begin a new scalar. That is a simple teaching policy. A replacement-character decoder may instead emit U+FFFD and consume a precisely defined maximal malformed subsequence. A strict protocol decoder may reject the entire surrounding record and refuse any recovery.

These are different interfaces. A caller must not assume that a particular number of bytes have been consumed on failure unless the decoder explicitly reports that fact.

Unicode normalization is also outside this example. Decoding two valid scalar values does not imply they form two independently displayed characters. Applications requiring canonical normalization or grapheme segmentation need additional Unicode-aware processing.

18.25 A complete test matrix

A good streaming test must challenge transitions, not only full strings:

Input	Expected behavior
ASCII `A`	Complete after one byte
E2 82 AC	Two `MORE` responses, then U+20AC
F0 9F 99 82	Three `MORE` responses, then U+1F642
E2 at temporary input boundary	`MORE`, not malformed
E2 at final EOF	Incomplete final input
E0 80 80	Reject overlong sequence
ED A0 80	Reject surrogate
F4 90 80 80	Reject value above U+10FFFF
E2 followed by `X`	Reject invalid continuation, reset state

An important additional property is that a decoder's result should not depend on the way the same valid byte stream is divided into transport chunks. One byte at a time and multiple chunks should produce identical Unicode scalar values.

The provided example has no heap allocation and a constant-size state object. This makes its memory requirements predictable, but does not by itself prove bounded application latency if the surrounding code waits on I/O.

18.26 From scalar decoding to a full text pipeline

A production text-processing system may have the following pipeline:

```
byte source -> framing -> UTF-8 decoding -> grammar parsing -> application
```

Each stage has different errors. The source can report read failure. Framing can reject an overlong record. UTF-8 decoding can reject malformed sequences. Grammar parsing can reject a command using an unsupported keyword. The application can reject a valid command that violates authorization or resource constraints.

Do not merge all these into a generic 'bad character' result when the caller needs to react differently. Conversely, a small application may deliberately map several low-level errors to one external response to avoid exposing unnecessary implementation detail.

Keeping the layers separate lets the same UTF-8 decoder operate on file input, a network stream, and data buffered by an RTOS task without depending on those transport mechanisms.

Further exercises

18.11 Adapt the decoder to accept counted chunks while preserving partial state between calls. **18.12** Design a result type that distinguishes invalid input from final truncated input. **18.13** Specify a policy for recovering from malformed UTF-8 using U+FFFD. **18.14** Verify that valid decoding results are independent of transport chunk boundaries. **18.15** Explain why even a valid UTF-8 scalar decoder cannot compute user-perceived character count without additional Unicode rules.

19. Compile-Time Programming and the Preprocessor

The C preprocessor is a separate, token-oriented facility that runs before ordinary translation. It can select platform-specific code, define constants, construct repetitive expressions, and include headers. Its power makes restraint important: a macro is not a typed function, and conditional compilation should not obscure the program's logic.

19.1 Include files and translation units

A header normally contains declarations and interface-related definitions. A source file includes that header and supplies the corresponding external definitions. Inclusion is conceptually textual: the preprocessor inserts the header's tokens into the translation unit.

```
#ifndef BUFFER_H
#define BUFFER_H
#include <stddef.h>
struct buffer_view { const unsigned char *data; size_t size; };
#endif
```

The guard prevents repeated processing of the same interface within one translation unit. It does not make two incompatible versions of an interface agree or prevent multiple external definitions across translation units.

19.2 Object-like macros

An object-like macro can name a manifest constant:

```
#define BUFFER_CAPACITY 256
```

For a numeric quantity with an appropriate integer type, an enumeration constant or C23 `constexpr` object can be clearer, subject to the required constant-expression context and toolchain support. A macro has no type of its own; it substitutes preprocessing tokens that are interpreted later.

19.3 Function-like macros and double evaluation

```
#define MAXIMUM(a, b) ((a) > (b) ? (a) : (b))
```

This macro can evaluate a chosen argument twice. An invocation involving `i++` is dangerous even if its parentheses are correct. Replacing a macro with a `static inline` function usually improves

type checking and evaluation behavior, but an inline function still has to handle overflow and other ordinary C hazards.

19.4 Stringizing and token pasting

The # operator can stringify a macro argument, and ## can paste preprocessing tokens. These mechanisms are helpful for diagnostics and repetitive declarations but can create opaque interfaces.

```
#define STRINGIZE(x) #x
#define CONCAT(a, b) a ## b
```

For example, `STRINGIZE(hello)` produces the string literal `"hello"`. Token-pasting does not create arbitrary source text; its result must be a valid preprocessing token. Expand-once helper macros may be necessary when callers pass macro arguments whose expanded values should be concatenated.

19.5 Conditional compilation

```
#if defined(PLATFORM_DEVICE)
    /* device adaptation */
#elif defined(PLATFORM_HOST)
    /* host adaptation */
#else
    #error "Choose a supported platform"
#endif
```

C23 also provides `#elifdef` and `#elifndef`. Use feature probes where possible: checking a vendor name is not proof that a particular header or library function is available. Keep platform branches behind narrow adapter interfaces so most application code can be compiled identically everywhere.

19.6 C23 embedding and diagnostics

The C23 `#embed` directive is designed to incorporate resource data into a translation unit. It is particularly interesting for fixed firmware resources such as calibration tables or static model parameters. Actual support differs among compilers, so a production project needs a feature probe and a fallback generation step.

C23 also standardizes `#warning`. Diagnostics emitted this way can be useful when a configuration is supported but discouraged. Unlike `#error`, a warning does not necessarily stop translation.

19.7 Static assertions

Use `static_assert` to make fundamental assumptions visible at translation time:

```
#include <limits.h>
static_assert(CHAR_BIT == 8, "octet storage required");
```

The constraint is appropriate for an octet-specific wire format; it would be unnecessarily restrictive for a general-purpose algorithm that does not depend on byte width.

19.8 Generic selection

The `_Generic` selection mechanism works on types rather than token spelling. It can choose an appropriate typed operation:

```c
#include <math.h>
#define abs_floating(x) _Generic((x), \
    float: fabsf, double: fabs, long double: fabsl)(x)
```

The selected function takes the value once, avoiding a common multiple-evaluation problem of expression macros. This example requires a supported floating type. A more complete public API should decide how to diagnose unsupported types and document conversion policy.

19.9 An example of compile-time checks

The companion program `examples/compile_time.c` combines `static_assert`, stringizing, token pasting, a C23 empty initializer, and a small function. Its checks are deliberately modest: their purpose is to show that preprocessing and language-level type checking cooperate but are not identical mechanisms.

19.10 Exercises

19.1 Explain why a function-like macro can evaluate an argument more than once.

19.2 Implement two-level stringizing that expands a macro before turning its value into a string.

19.3 Write a configuration header that rejects simultaneous host and device target selections.

19.4 Write a compile probe for `#embed` and document a fallback for a toolchain that lacks it.

19.5 Define the differences among `#if`, `static_assert`, and a runtime `if` statement.

19.6 Use a `_Generic` selection to choose one of two distinct, typed integer-printing functions.

19.11 Preprocessing is not type checking

The preprocessor expands tokens before the compiler evaluates ordinary C types and expressions. This means a macro can construct an expression that looks plausible but has surprising side effects. A typed function receives evaluated arguments once for each parameter, but the order of evaluation among arguments is still not specified merely by the call.

Prefer `enum` constants, appropriately typed `constexpr` objects where supported, and `static inline` functions for tasks that do not require token substitution. Reserve macros for conditional compilation, stringizing, token-pasting, or genuinely generic constructs that cannot be expressed as ordinary C functions.

19.12 Expand macros deliberately

Two-level macros are needed when a macro argument should be expanded before being stringized:

```c
#define STR_DIRECT(x) #x
#define STR_EXPANDED(x) STR_DIRECT(x)
#define MAJOR_VERSION 3

const char *version = STR_EXPANDED(MAJOR_VERSION);
```

`STR_DIRECT(MAJOR_VERSION)` yields the identifier spelling, while `STR_EXPANDED(MAJOR_VERSION)` yields `"3"`. Macro expansion rules are precise but easy to misread. Prefer explicit, small helper macros with a test illustrating their purpose.

19.13 Conditional compilation and dead branches

A `#if` directive removes or retains preprocessing tokens before the compiler type-checks ordinary statements. A runtime `if` statement, even if its condition is a constant expression, still belongs to the translated C program and is subject to language constraints.

Use conditional compilation for platform-specific includes, unsupported instructions, or truly unavailable APIs. When both implementations are valid C on the current target, a normal function call or runtime branch may be easier to maintain.

19.14 Using embed responsibly

C23's `#embed` is valuable for incorporating binary resources into source initialization. A resource might contain firmware calibration constants or a model binary. Treat the bytes as untrusted data from a build-input perspective: validate their length, version, and format before interpreting them, and record their provenance.

Embedding data does not necessarily guarantee where the linker places it, whether it resides in ROM or RAM, or whether a given MCU can fetch it efficiently. Such placement is governed by target toolchain and linker configuration. Provide a generated C array fallback for compilers without the directive.

19.15 Static assertions expose assumptions early

A compile-time assertion can check that an integer width, structure-related condition, or application constant meets a documented requirement. It cannot check a runtime packet's length or prove that an allocation will succeed.

C23 accepts `static_assert` directly as a keyword. To detect a feature's availability, compile a tiny representative translation unit with the intended compiler flags rather than assuming support from a marketing claim about C23 compliance.

19.16 Build diagnostics as a teaching aid

The book's `compile_time.c` demonstrates simple macro expansion and C23 initialization. Add variations that intentionally fail `static_assert` in a scratch file and observe the compiler's message. Then compare that translation-time diagnostic to a runtime check that reports an error without preventing compilation.

Further problems: Explain the difference between `STR_DIRECT(VERSION)` and `STR_EXPANDED(VERSION)`. Implement a portable fallback for an embedded byte array. Design a compile-time condition selecting one of two platform adapters and fail explicitly when neither is supported.

19.17 A preprocessor macro is not a generic function

A function-like macro may evaluate an argument more than once, and even a macro wrapped in parentheses does not become type-safe. The familiar `MAX(a, b)` substitution illustrates the danger when a caller passes `index++`. The expression chosen by the conditional operator may be evaluated again.

A `static inline` function for a specific type avoids duplicate evaluation of its arguments. The caller's argument expressions are still evaluated in an order that C does not necessarily prescribe, so do not write calls that depend on sequencing between them.

A `_Generic` selection can choose among several typed functions at translation time, allowing a type-directed wrapper without converting every value to one common type. Be careful to specify supported types and a clear policy for unsupported arguments.

19.18 Use static assertions for real assumptions

A portable byte decoder may not need to assume an eight-bit byte if it operates on abstract values in the standard character domain, but an octet-specific wire codec does need an explicit policy

about eight-bit units. A compile-time assertion can make that dependency visible to future maintainers.

Do not scatter assertions about irrelevant architecture properties throughout a general library. A function operating on `size_t` lengths and opaque pointers may be perfectly valid on targets with different pointer widths, whereas a hardware register interface can have genuine address and alignment requirements.

19.19 Configuration is part of the reproducible build

A macro such as `DEVICE_TARGET` changes which tokens are translated. Therefore it can change program behavior just as surely as a source edit. Record the selected configuration, compiler version, flags, and target alongside test results.

For embedded code, a host adaptation may use a fake device sink, while the MCU build uses a UART driver. The public interface should remain the same; conditional compilation should select the adapter, not duplicate the entire application algorithm.

19.20 Keep the generated book reproducible

The book's PDF is also a build artifact. A manuscript change should trigger generation from the current source and a check that required headings appear in the output. A page count should be measured from the PDF itself, not inferred from the Markdown word count.

The build is only one acceptance gate. A PDF may technically compile but have broken line wrapping, duplicated section numbering, or code examples that run off the page. The final editorial pass requires inspecting rendered pages and checking the technical content independently.

Exercises

19.7 Write a two-level stringizing macro and demonstrate its expansion with a numeric macro. **19.8** Compare a macro and a typed inline function when passed an increment expression. **19.9** Design a compile-time configuration that selects a host or target byte sink. **19.10** Explain why a successful document build is not proof that every C listing compiles.

19.21 Generic selection is type-directed dispatch

C offers a limited but useful compile-time mechanism for choosing an expression based on the type of another expression. The `_Generic` selection form can select between named functions:

```
static int maximum_int(int a, int b)
{
    return a > b ? a : b;
}

static long maximum_long(long a, long b)
{
    return a > b ? a : b;
}

#define max_same_type(a, b) \
    _Generic((a), int: maximum_int, \
                  long: maximum_long)((a), (b))
```

This macro is intended for two arguments of the same supported type. If the first is `int`, the generic selection chooses `maximum_int`; if it is `long`, it chooses `maximum_long`. An unsupported first-argument type produces a constraint diagnostic because no selection matches and no default association is supplied.

The selected result is a function designator; the arguments are then supplied to that function as in a normal call. The controlling expression is not evaluated at runtime merely for the selection. This matters for side effects.

For example:

```
int count = 0;
int answer = max_same_type(++count, 7);
```

The increment in the controlling expression does **not** happen once during generic selection and again during function invocation. It occurs as part of the actual function argument evaluation, so `count` becomes one, not two.

This contrasts with a traditional function-like macro that copies argument text into both sides of a comparison and may evaluate the same argument repeatedly.

19.22 Generic selection still needs a contract

It is tempting to call a type-directed macro fully type-safe. But a selection based only on the first argument does not force the second argument to have that same type.

A caller might write `max_same_type(1, 2L)`; the `int` function would be selected, and the second argument would be converted to `int` according to the call's function prototype. If its value is out of range for `int`, the conversion may not produce the intended result.

A production helper may need a stronger interface, more specific functions, or a carefully designed second type check. Alternatively, it can state explicitly that callers must provide equal supported types and rely on compiler conversion warnings and code review.

The central rule is that `_Generic` selects an expression based on type. It does not invent a new language-level overloaded function system with the same constraints as C++ templates.

19.23 What the generic associations mean

A selection can include more types and a `default` association. It must not contain multiple compatible type associations in a way prohibited by the language's constraints.

The selection's controlling expression is used for type matching, not for a dynamic switch on its current numerical value. Thus `_Generic(x, int: foo, long: bar)` chooses according to the type of `x`, even if `x` happens to hold a value representable in both types.

For an array or qualified expression, the controlling-expression type rules deserve particular care because the language's ordinary conversions and type compatibility can affect matching. If the distinction between two closely related types is essential, compile and test the exact declaration under the targeted compiler mode rather than reasoning from spelling alone.

A small feature probe can use `static_assert` and a constant generic selection to document an expected match in the build:

```
static_assert(_Generic((int)0, int: 1, default: 0));
```

That statement checks a translation-time property. It does not execute the selected program function.

19.24 Why use real functions behind the macro

The generic macro can select functions defined in ordinary C. Those functions have normal parameter types, normal prototypes, and ordinary evaluation semantics. A developer can set a breakpoint inside one, test it directly, or reuse it without the generic dispatch.

This reduces the risks of macros that implement their entire algorithm through textual substitution. It also gives the compiler a familiar opportunity to inline or optimize a small typed function when appropriate.

The design is not a universal replacement for callbacks. Generic selection occurs at translation time based on the known type, while a callback is a runtime-selected operation. Use the former when supported types are known in source and the latter when behavior must be supplied dynamically by a caller or test.

For example, a host test that substitutes a failing allocator needs a callback or equivalent runtime seam; a math wrapper choosing between float and double functions may benefit from generic selection.

19.25 A small validation example

The standalone `examples/generic_select.c` contains typed maximum functions and tests for both `int` and `long`. It includes `INT_MIN\), INT_MAX), LONG_MIN\) and LONG_MAX)` comparisons without subtracting their values.

The test also increments an integer in a macro argument and confirms that the operation occurs once. This directly checks the pedagogical claim that the macro does not duplicate argument evaluation.

The test does not establish that mixed-type inputs are safe. The macro's documented caller contract requires matching supported types, so the example illustrates one constrained application of `_Generic`, not every possible selection design.

19.26 Macro expansions can mislead a debugger

The preprocessor operates before ordinary compilation of expressions. An object-like or function-like macro expands to tokens, not a separately callable runtime function. Error messages and debugger stepping may therefore point into the macro expansion or into a user call site rather than behave like a normal function call.

When a macro becomes complicated, first ask whether a `static inline` function would express the same logic. If the macro's real purpose is compile-time selection, keep the macro small and delegate work to typed functions.

A second source of confusion is precedence. A simple macro `#define SQUARE(x) x*x` misbehaves when passed `1 + 2`, because substitution does not introduce the parentheses a function call would. Parenthesizing parameters and the result improves such macros, but it does not solve double evaluation when the argument appears twice.

19.27 Compile-time contracts versus runtime checks

Several different mechanisms detect different kinds of errors:

Mechanism	When it applies	What it can establish
`static_assert`	Translation	A constant expression meets a required property
`_Generic`	Translation	Selects a matching association by expression type
`#if` and macros	Preprocessing	Chooses tokens using preprocessing conditions
Function prototype	Translation	Declared argument/return type relationships
Ordinary `if`	Execution	Checks runtime state and input
Tests and sanitizers	Execution	Observe behavior on exercised paths

No one mechanism subsumes the others. A compile-time size assertion cannot guarantee a runtime packet has sufficient available bytes. A runtime null pointer check cannot prove that a pointer

refers to a still-alive object. A generic selection cannot verify an allocator's logical ownership rules.

Use each mechanism for the property it is capable of expressing.

19.28 Readability wins over maximum metaprogramming

C's compile-time facilities can eliminate certain kinds of repetition, but they can also obscure meaning when used excessively. An ordinary named function is often the best interface. A small `static_assert` is useful for a real target assumption. A few lines of generic selection can simplify a family of typed operations.

Do not build a complex macro-based type system merely to avoid writing a small number of clear functions. In particular, firmware intended for multiple compilers and static-analysis tools may benefit from conservative, explicit interfaces that every tool can understand.

The best criterion is whether the change makes the program easier to audit, test, and maintain under the actual supported toolchains.

Further exercises

19.11 Add an `unsigned` maximum function and extend the generic selection to choose it. **19.12** Create a compile-time diagnostic for an unsupported controlling-expression type. **19.13** Investigate how a mixed `int` and `long` call is converted and design an interface that rejects it. **19.14** Replace a double-evaluating macro with a typed inline function. **19.15** Compare the role of `_Generic` with a runtime callback using the same operation on input data.

Appendix A. A Working Reference to C23

This appendix is a navigational guide, not a replacement for ISO/IEC 9899:2024. Implementations differ in optional features and in how far their C libraries have caught up.

A.1 Translation model

A source file is preprocessed and translated as part of a *translation unit*. Header contents are incorporated by inclusion; macro substitution and conditional inclusion precede ordinary compilation. External symbols are resolved when separately compiled translation units are linked. String literals and static objects are initialized according to the rules for their storage duration.

A.2 Fundamental types

- `void` describes absence of a value.
- `bool` is C23's boolean type. `true` and `false` are keywords.
- Character types include `char`, `signed char`, and `unsigned char`. Plain `char` is a distinct type, with implementation-defined signedness.
- Standard signed integer types are `signed char`, `short`, `int`, `long`, and `long long`; each has an unsigned counterpart.
- Floating types include `float`, `double`, and `long double`, with additional optional IEC-related facilities.
- A pointer type refers to an object or function; it does not automatically encode the size or lifetime of the referenced storage.
- Array types contain sequences of elements. Structures contain named members; unions share storage; enumerations provide named integer constants.
- C23 introduces bit-precise integer types written `_BitInt(N)` and `unsigned _BitInt(N)`, subject to implementation limits.

The sizes of fundamental types are not universally fixed. Test the actual target's limits using `<limits.h>`, `<stdint.h>`, and `<float.h>`.

A.3 Expressions at a glance

From highest binding strength to lowest, a practical *simplified* precedence guide:

Level	Operators or forms	Association
Postfix	`[] () . ->` postfix `++ --`	left to right
Unary	prefix `++ --`, `& * + - ! ~`, `sizeof`, `alignof`	right to left
Cast	`(type)`	right to left
Multiplicative	`* / %`	left to right
Additive	`+ -`	left to right
Shift	`<< >>`	left to right
Relational	`< <= > >=`	left to right
Equality	`== !=`	left to right
Bitwise AND	`&`	left to right
Bitwise XOR	`^`	left to right
Bitwise OR	‘	‘
Logical AND	&&	left to right
Logical OR	‘	
Conditional	`?:`	right to left (special grammar rules)
Assignment	‘= += -= *= /= %= «= »= &= ^=	=‘
Comma	,	left to right

Precedence does not imply evaluation order. Thus `f() + g()` does not specify which call happens first. The distinction matters whenever expressions change shared state.

A.4 Declarations

Read a declaration outward from the name, respecting parentheses:

```
int *p;                  /* pointer to int */
int a[8];                /* array of eight int */
int (*fp)(double);       /* pointer to function returning int */
int *make_value(void);   /* function returning pointer to int */
```

A prototype specifies parameter types. In C23, an empty function-parameter list declares a function with no parameters; `(void)` remains explicit and clear. Avoid relying on earlier C conventions for nonprototype declarations.

A.5 Storage, linkage, and qualifiers

Automatic storage normally begins on block entry and ends when execution leaves the block. Static storage lasts for the program execution. Thread storage is associated with a thread where supported. Allocated storage lasts from successful allocation to deallocation.

At file scope, `static` can give an identifier internal linkage; `extern` can declare an externally linked identifier. `const` restricts modification through a qualified lvalue; `volatile` affects observable accesses but does not provide atomic synchronization; `restrict` is a promise about pointer-based accesses and must only be used when its aliasing conditions hold.

A.6 Control statements

`if` chooses a branch; `switch` dispatches among case labels. `for`, `while`, and `do` iterate. `break` exits a loop or switch; `continue` advances to the next loop iteration; `return` exits a function. `goto` transfers control to a label in the same function, but must obey restrictions on entering variably modified scopes and similar rules.

A.7 Behavior categories

- **Defined:** the standard requires specified results.
- **Implementation-defined:** the implementation chooses and documents a permitted result.
- **Unspecified:** the implementation may choose among permitted results.
- **Undefined:** no requirements are imposed; programs must not depend on an observed behavior.

A program may compile without diagnostics and still execute undefined behavior.

A.8 Selected C23 additions

C23 standardizes binary integer literals, digit separators, empty initialization, `nullptr`, `typeof`/`typeof_unqual`, `auto` type inference, `constexpr` objects, bit-precise integers, attributes, `#embed`, `#elifdef`, and `#elifndef`. The library expands with, among other things, checked integer operations and bit utilities. Do not confuse adoption of C23 syntax in a compiler with complete deployment availability of every library feature.

A.9 A disciplined compatibility check

Compile tiny feature probes using the real target compiler and link against the real target C library. Record compiler version, C library version, standard mode, target architecture, and the precise feature probed. Treat a feature as available only after the probe compiles, links, and where possible executes.

A.10 A compact declaration guide

Declarations become much easier to read when they are decomposed into base type, identifier, and type constructors. The following are different contracts:

```
int *p;                 /* pointer to an int */
int values[4];          /* four int elements */
int (*row)[4];          /* pointer to an array of four int */
int *factory(void);     /* function returning pointer to int */
int (*callback)(int);   /* pointer to function taking int */
```

The parentheses decide whether * applies to a named object or to the result of a function declarator. Prefer typedef names for complicated function-pointer interfaces once the meaning is understood; typedef does not create a new distinct type, but it can improve readability.

A.11 Conversions checklist

- **Integer promotions:** narrow integer types are often promoted to `int` or `unsigned int` before arithmetic.
- **Usual arithmetic conversions:** operands in a binary arithmetic operation may be converted to a common type.
- **Assignment conversion:** the right-hand value is converted to the destination type if permitted.
- **Pointer conversions:** object pointers may convert to and from `void *` in the specified circumstances, but pointer arithmetic and access rules still apply.

- **Array-to-pointer conversion:** most array expressions convert to pointers to their first elements, with specified exceptions.

For each conversion, ask whether the mathematical value survives and whether the destination type supports the intended operation. A syntactically valid cast is not proof of a meaningful conversion.

A.12 Alignment and lifetime at a glance

Every complete object type has an alignment requirement. Automatic storage, static storage, thread storage, and allocated storage differ in lifetime rules. A dynamically allocated region can generally hold suitably aligned ordinary objects, but special over-aligned objects may require dedicated allocation facilities or implementation-specific support.

A pointer to a released allocation is no longer a valid way to access the former object. Resetting one alias to `nullptr` does not repair other aliases. A borrowed pointer is safe only while the owner continues to supply a live object.

A.13 C23 compatibility notes

C23's `bool`, `true`, and `false` are keywords. `nullptr` is a null pointer constant with its own type, `nullptr_t`. `static_assert`, `alignof`, and `alignas` also have modern language spellings. A new program need not imitate the macro-based spellings used historically for compatibility.

C23 has no old-style, unprototyped function declaration. Prefer `f(void)` when zero parameters are intended, even though `f()` also means zero parameters in C23. Doing so keeps intent clear when readers encounter older dialects.

The language's bit-precise integer facility, floating-point options, and standard-library additions are not uniformly available in every cross compiler. Test the exact feature on the target, record the version and mode, and provide a fallback when necessary.

A.14 Selected undefined operations

The following are examples to recognize, not experiments whose observed results define the language:

- Out-of-bounds array accesses and invalid pointer dereference.
- Signed integer overflow and integer division by zero.
- Using an automatic object's address after its lifetime ends.
- Violating effective-type or alignment access restrictions.
- Accessing a shared non-atomic object in a data race.
- Using `realloc` with a size argument of zero in C23.
- Performing an invalid shift or violating sequencing rules.

Some implementation-defined behaviors, by contrast, are perfectly legitimate once the implementation documents them and the program accepts that platform dependency.

A.15 References for the careful reader

The normative authority is ISO/IEC 9899:2024. For accessible navigation and implementation details, consult the C language pages at cppreference, GCC's C standards support page, and Clang's C status page. Those references help locate a topic but do not replace the exact standard text when a subtle conformance question matters.

Useful online links:

- https://en.cppreference.com/w/c/23
- https://en.cppreference.com/w/c/language

- https://gcc.gnu.org/projects/c-status.html
- https://clang.llvm.org/c_status.html

Appendix B. The Standard Library by Task

This is a field guide to the most commonly needed hosted C facilities, not a complete list of the standard's clauses.

Need	Header	Key facilities
Input, output, files	`<stdio.h>`	`FILE`, `fopen`, `fclose`, `fread`, `fwrite`, `fgets`, `printf`
Byte and string operations	`<string.h>`	`memcpy`, `memmove`, `memcmp`, `strlen`, `strcmp`
Allocation, conversion	`<stdlib.h>`	`malloc`, `calloc`, `realloc`, `free`, `strtol`, `qsort`
Integer limits	`<limits.h>`	`INT_MAX`, `CHAR_BIT`, etc.
Exact-width integer types	`<stdint.h>`	`uint32_t`, `SIZE_MAX` and related types/macros
Integer formatting	`<inttypes.h>`	`PRIu32`, `SCNx64` etc.
Type and offset	`<stddef.h>`	`size_t`, `ptrdiff_t`, `offsetof`
Character classification	`<ctype.h>`	`isdigit`, `isspace`, `tolower`
Error indications	`<errno.h>`	`errno`, `ERANGE`
Mathematics	`<math.h>`	`sqrt`, `sin`, `fabs`, `isfinite`
Assertions	`<assert.h>`	`assert`
Variadic functions	`<stdarg.h>`	`va_list`, `va_start`, `va_arg`, `va_end`
Atomics	`<stdatomic.h>`	atomic types and operations
Threads	`<threads.h>`	`thrd_t`, mutexes, condition variables where supported
Checked integer arithmetic	`<stdckdint.h>`	`ckd_add`, `ckd_sub`, `ckd_mul` (C23)
Bit utilities	`<stdbit.h>`	leading/trailing bit counts, population count, bit width, and bit

B.1 Length is not capacity

`strlen` finds a terminating zero byte. It does not accept a buffer size, so calling it on a nonterminated array is unsafe. Functions like `snprintf` take output capacity, and their return values should be checked to detect truncation.

```
char buffer[16];
int needed = snprintf(buffer, sizeof buffer, "%u", 12345U);
if (needed < 0 || (size_t)needed >= sizeof buffer) {
    /* formatting error or truncation */
}
```

B.2 Integer conversion

`strtol` is usually preferable to `atoi` when handling untrusted text. Set `errno = 0` before conversion, inspect the end pointer, and check `ERANGE`. Even after a successful `long` result, verify the destination type's range.

B.3 Byte copy versus overlapping copy

`memcpy` requires source and destination regions not to overlap. `memmove` permits overlap, but both require valid regions of at least the requested size. Neither performs object-bound checks for you.

B.4 Character conversion

The `<ctype.h>` functions accept `EOF` or a value representable as `unsigned char`. Passing a negative signed `char` directly is generally invalid. For byte-oriented classification, cast:

```
unsigned char byte = (unsigned char)some_char;
if (isdigit(byte)) { /* ... */ }
```

The character functions can be locale-sensitive. They are not general Unicode text processing primitives.

B.5 Optionality and freestanding environments

A freestanding embedded system may implement only a subset of the hosted library. Standard features may also depend on toolchain, linker, and C library combinations. Document whether examples require a hosted environment, and test all target-specific functions directly.

B.6 A checked numeric conversion

A typical `strtol` call is more careful than merely checking whether the return value is zero:

```
#include <errno.h>
#include <limits.h>
#include <stdbool.h>
#include <stdlib.h>

bool parse_positive_int(const char *text, int *out)
{
    if (text == nullptr || out == nullptr)
        return false;
    errno = 0;
    char *end;
    long value = strtol(text, &end, 10);
    if (end == text || *end != '\0' || errno == ERANGE ||
        value < 0 || value > INT_MAX)
        return false;
    *out = (int)value;
    return true;
}
```

This accepts leading whitespace and an optional plus sign because `strtol` does. Despite its name, it also accepts zero; rename it `parse_nonnegative_int` if the name is meant to describe the precise grammar. It rejects trailing text. The output is unchanged on failure.

B.7 Formatted-output contracts

With `printf`, each conversion specification must match the argument's promoted type. For `size_t`, use `%zu`; for a `uint32_t` whose formatting differs across platforms, use the macros supplied by `<inttypes.h>`.

A format string obtained from outside the program can be dangerous: passing it as the first argument may introduce unexpected conversion directives. To print user-controlled text, use a fixed format such as `printf("%s", input)` when a valid terminated string is guaranteed; use a counted-byte write operation for untrusted spans without terminators.

For `snprintf`, a nonnegative return value is the number of characters that would have been written, excluding the terminating zero. If that number is at least the supplied capacity, output was truncated. A negative result signals an encoding or formatting error under the interface's rules.

B.8 Searching and sorting

`qsort` arranges objects in an array using a comparator. The comparator must interpret the correct object type and provide a consistent ordering. For integers, the result of subtracting the two

values is not always representable; use relational comparisons.

`bsearch` assumes the array is sorted consistently with the supplied comparator. It does not validate this precondition. If records with equal keys have distinct associated data, the API contract must say whether any matching record is acceptable.

B.9 Allocation and zeroed storage

`malloc` allocates uninitialized storage; `calloc` writes zero bytes to its allocated region; `realloc` may resize and move an allocation. Check all requested-size calculations before allocation. In C23, `realloc(p, 0)` is undefined and must not be used as a portable deallocation idiom.

All-bits-zero is not guaranteed to represent a null pointer or floating-point zero on every implementation. Ordinary initializer syntax such as `struct value v = {};` sets members according to the language's initialization rules, which is a different concept from a raw byte memset.

B.10 C23 bit utilities, precisely

The new `<stdbit.h>` header supplies operations including `stdc_leading_zeros`, `stdc_trailing_zeros`, `stdc_count_ones`, `stdc_has_single_bit`, `stdc_bit_width`, `stdc_bit_floor`, and `stdc_bit_ceil`, plus related variants and byte-order macros.

It does **not** provide the C++ `std::rotl` and `std::rotr` interface. If you need rotations in C, write a properly bounded unsigned helper or use a documented implementation extension.

The associated header `<stdckdint.h>` supplies checked integer arithmetic macros. In a practical project, test whether both headers exist and link successfully with the intended C library before treating them as dependencies.

B.11 Platform limitations

C's hosted library includes facilities that may be absent or restricted in freestanding firmware. A microcontroller environment may use retargeted `printf`, a specialized allocator, or no filesystem at all. In such a system, keep the byte-level transformation algorithms separate from device transport and I/O.

Threads and atomics also depend on implementation support. Never assume that a function which works on a desktop is safe to call inside an interrupt handler.

B.12 Error policy examples

When a library function signals failure with a sentinel, immediately consult the documented mechanism for distinguishing failure from a normal result. For `fread`, inspect `ferror` and `feof` following a short read. For `strtol`, inspect `errno` and the end pointer. For `malloc`, check the returned pointer while remembering that zero-size requests have special semantics.

Practice: Create a small table containing every library function used in your program, its return-value meaning, and your application's chosen behavior on failure.

B.13 Selecting string functions by their contracts

The C library contains several functions for operations that look similar but have importantly different preconditions.

`strlen` counts bytes in a terminated C string, excluding its final zero character. It has no separate maximum-length argument. Before calling it, the program must already know that the supplied pointer designates a valid sequence containing a terminator within accessible storage.

An arbitrary network packet or counted file buffer does not satisfy that contract merely because it contains characters. If it has no terminating zero, a call to `strlen` can read beyond the buffer. Use the supplied length for counted data.

`strcmp` compares terminated strings according to the unsigned-character values of their bytes. It is appropriate for keys stored in arrays with guaranteed termination. It is not a Unicode collation operation and does not implement locale-aware natural-language sorting.

`memcpy` copies a specified number of bytes between valid, nonoverlapping regions. `memmove` is designed for overlapping regions. Both require that the requested region fits within accessible storage. Neither checks that a caller's numerical size is truthful.

These distinctions recur throughout the book: a function name does not infer the intended storage extent, and a pointer does not carry a verified length.

B.14 Why a bounded copy name may mislead

The traditional `strncpy` function is sometimes used as though it meant “copy a safely terminated string.” Its actual behavior is more complicated. If the source has at least as many characters as the specified count, the destination may not receive a terminating zero. If the source is shorter, the function fills the remainder of the specified region with zero bytes.

That contract can be useful for certain fixed-width fields, but it is not a general replacement for deliberate validation and copying. If the application expects a terminated string, establish the destination capacity, choose a maximum accepted length, and ensure termination explicitly.

For untrusted inputs, a clear counted-span API is often preferable. The parser can reject a length exceeding its documented maximum instead of silently truncating distinct keys into the same stored representation.

B.15 Check formatted output carefully

The `snprintf` family permits bounded output into a supplied buffer. A typical return value represents the number of characters that would have been written, excluding the terminating zero, when no encoding error occurs. That means an apparently successful positive result can still indicate truncation.

For an ordinary nonnegative return value `n` and destination capacity `capacity`, the formatted result fits without truncation only when `(size_t)n < capacity`. The check must follow a test that `n >= 0`, so a negative error result is not converted into a huge unsigned value.

Consider:

```
char buffer[32];
int n = snprintf(buffer, sizeof buffer, "%u", value);
if (n < 0 || (size_t)n >= sizeof buffer) {
    /* Handle encoding failure or truncation. */
}
```

This fragment assumes a valid `unsigned` value and the required `<stdio.h>` declaration. The caller can choose to reject truncation, retry with a larger buffer, or use a known fixed-size representation. What matters is that the behavior is intentional.

In a freestanding environment, the availability and completeness of formatted I/O depends on the supplied C library. Some embedded libraries intentionally omit portions of floating-point formatting to reduce code size.

B.16 Input conversion with `strtol`

`strtol` is valuable when its grammar matches the application's requirements. It can handle a leading sign and may skip leading whitespace; it also provides an end pointer so callers can see how much input was consumed.

A robust conversion should:

1. Verify that the input is a valid terminated C string.
2. Set `errno = 0` before conversion.
3. Call `strtol` with an explicit base when appropriate.
4. Check whether the end pointer equals the start, indicating no conversion.
5. Check `errno == ERANGE` for range errors.
6. Reject unexpected trailing characters according to the application's grammar.

The parser in Chapter 2 deliberately implements a stricter grammar—optional sign followed immediately by decimal digits, with no whitespace. That is why replacing it with a bare `strtol` call would change observable behavior unless the caller added equivalent validation.

Never assume that converting a valid `long` value to a narrower destination is safe. Apply the destination's representability test separately.

B.17 Sorting is not stable by contract

`qsort` accepts a base pointer, element count, element size, and comparator. It sorts according to that comparator but does not promise to retain the original order of elements that compare equal.

A comparator must interpret pointers as the array's actual element type and provide a consistent ordering. A comparator that subtracts signed integer keys can overflow, and one that depends on mutable global state can be inconsistent across calls.

Chapter 17's `sort_records.c` example combines priority, name, and unique arrival order. The arrival field is a final tie-breaker that produces the application's desired deterministic order without assuming the sorting algorithm is stable.

A comparator may examine pointer members inside a record only if their targets have the required lifetimes and represent valid strings or other objects throughout the sort.

B.18 Functions involving external resources

Hosted file streams and allocated memory are external resources with lifetimes. A program that obtains them should establish where cleanup occurs, including when an intermediate operation fails.

For example, `fread` can return fewer elements than requested because of end of file or an error. Inspect `feof` and `ferror` as appropriate. `fwrite` returning fewer elements than requested indicates failure under its contract. A close can report an error after buffered output operations appeared to succeed.

A reliable program must also check counts before using the bytes as a complete record. Returning a shorter record from a read operation is not the same thing as parsing a complete valid packet.

The bounded line reader in Chapter 6 demonstrates the distinction between clean EOF, complete input, malformed input, and too-long records. Its resource model is hosted and does not imply that a microcontroller environment supplies a filesystem.

B.19 A task-oriented checklist

Before choosing any standard-library function, identify:

- Whether the input is terminated text or counted bytes.

- Whether the source and destination storage may overlap.
- Whether the operation can allocate, block, or otherwise consume external resources.
- How it reports errors and whether any output may already have changed.
- What type conversions occur in its arguments and return value.
- Whether the target C library implements the required facility in the selected language mode.

The standard library is most useful when a program deliberately satisfies these contracts. It cannot infer a valid object lifetime, repair a dishonest buffer length, or supply a hardware feature that the deployment platform lacks.

Reference exercise

Take one use each of `strlen`, `memcpy`, `snprintf`, `strtol`, and `qsort` in a program you have written. For each, state its actual preconditions, the relevant representation or ownership assumptions, and exactly how failure or truncation is detected. Then decide whether its use matches the intended application behavior.

Appendix C. Selected Worked Exercises

These solutions illustrate contracts and tests rather than prescribe one clever implementation.

C.1 Absolute difference without unsigned underflow

```
unsigned difference(unsigned a, unsigned b)
{
    return a >= b ? a - b : b - a;
}
```

Both subtractions occur only in nonnegative order. Test `(0, 0)`, `(1, 0)`, `(0, UINT_MAX)`, and both operand orders.

C.2 Checked parsing of an unsigned decimal

```
#include <stdbool.h>
#include <limits.h>

bool parse_unsigned(const char *text, unsigned *out)
{
    if (text == nullptr || out == nullptr || *text == '\0')
        return false;

    unsigned value = 0;
    for (const unsigned char *p = (const unsigned char *)text;
         *p != '\0'; ++p) {
        if (*p < '0' || *p > '9') return false;
        unsigned digit = (unsigned)(*p - '0');
        if (value > (UINT_MAX - digit) / 10U) return false;
        value = value * 10U + digit;
    }
    *out = value;
    return true;
}
```

The key inequality is evaluated *before* a potentially overflowing multiplication. The output is unchanged on failure. This version accepts no leading sign or whitespace, by design.

C.3 Iterative factorial with an overflow check

```
#include <stdbool.h>
#include <limits.h>

bool factorial(unsigned n, unsigned long *out)
{
    if (out == nullptr) return false;
    unsigned long result = 1;
    unsigned i = 2;
    while (i <= n) {
        if (result > ULONG_MAX / i) return false;
        result *= i;
        if (i == n) break;
        ++i;
    }
    *out = result;
    return true;
}
```

The explicit final-iteration break also avoids unsigned loop-counter wraparound if `n == UINT_MAX`. Checking multiplication alone is not the whole correctness proof.

C.4 Little-endian byte decoding

```
#include <stdint.h>
#include <stddef.h>
#include <stdbool.h>

bool read_u32_le(const unsigned char *bytes, size_t count, uint32_t *out)
{
    if (bytes == nullptr || out == nullptr || count < 4) return false;
    uint32_t result = (uint32_t)bytes[0]
                    | ((uint32_t)bytes[1] << 8)
                    | ((uint32_t)bytes[2] << 16)
                    | ((uint32_t)bytes[3] << 24);
    *out = result;
    return true;
}
```

The cast happens before shifting, so promotions do not make the top byte shift a signed overflow. Endianness is described by the wire format, not assumed from host memory.

C.5 Cleanup on every path

```
#include <stdio.h>
#include <stdbool.h>

bool write_record(const char *path)
{
    FILE *fp = fopen(path, "w");
    if (fp == nullptr) return false;
    bool ok = fputs("record\n", fp) != EOF;
    if (fclose(fp) == EOF) ok = false;
    return ok;
}
```

Notice that `fclose` runs even after `fputs` fails. The function reports stream-level success, not guaranteed durable storage on physical media.

C.6 A safe interval check

An interval starting at `offset` and containing `count` bytes fits within `capacity` when the start is valid and the count fits the remaining space:

```
#include <stddef.h>
#include <stdbool.h>

bool fits(size_t offset, size_t count, size_t capacity)
{
    return offset <= capacity && count <= capacity - offset;
}
```

The short-circuit operator prevents unsigned subtraction from being evaluated if `offset` already exceeds capacity. Test zero capacity, exact fit, one-byte overrun, and `SIZE_MAX` boundaries.

C.7 Checked allocation dimensions

```
#include <stdint.h>
#include <stddef.h>
#include <stdlib.h>
#include <stdbool.h>

bool allocate_ints(size_t count, int **out)
{
    if (out == nullptr || count == 0 ||
        count > SIZE_MAX / sizeof(int))
        return false;
    int *p = malloc(count * sizeof *p);
    if (p == nullptr)
        return false;
    *out = p;
    return true;
}
```

The function rejects zero-sized requests and leaves the output unchanged on failure. The caller owns the returned allocation and must call `free` once. The result is not initialized; assign to elements before reading them.

C.8 A consistent qsort comparator

```
static int compare_ints(const void *left, const void *right)
{
    int a = *(const int *)left;
    int b = *(const int *)right;
    return (a > b) - (a < b);
}
```

Returning `a - b` could overflow for extreme signed values, invalidating the intended ordering. Use the comparator with `qsort` only when both pointers refer to valid `int` elements supplied by that call.

C.9 A non-modifying failure test

Suppose a parser promises not to modify its output on failure. Before invoking it on an invalid string, set `unsigned result = 0x5a5aU;`. Verify the parser returns false *and* that the result remains `0x5a5aU`. Repeat with empty input, trailing junk, and an out-of-range number. A failed result alone does not establish the postcondition.

C.10 A boundary test plan for a ring buffer

Initial state	Operation	Required behavior
Empty	Pop	Report false; do not change output
Empty	Push	Report true; count increases
One element	Pop	Return first item; become empty
Full	Push	Report false; preserve existing items
Partially consumed	Wraparound pushes	FIFO order is preserved
Returned to empty	Pop	Report false again

Sequential tests establish logical behavior, not thread safety. The atomic SPSC queue in Chapter 16 needs additional concurrent reasoning and testing.

C.11 How to check an exercise solution

For each solution, write down preconditions, postconditions on both success and failure, possible arithmetic overflow, and ownership effects. Then test exact boundaries. If your implementation uses a library function, read the library contract rather than assuming it matches the function's name.

Solutions are examples of reasoning, not necessarily the only acceptable design. An alternative is valid when it meets the same explicit contract.

C.12 A counted slice and its ownership contract

A counted slice has a data pointer and a length. It does not automatically own the referenced array. The caller must keep storage alive for any operation reading the slice.

```
#include <stddef.h>
#include <stdbool.h>

struct int_slice {
    const int *data;
    size_t length;
};

bool slice_get(struct int_slice s, size_t index, int *out)
{
    if (out == nullptr || s.data == nullptr || index >= s.length)
        return false;
    *out = s.data[index];
    return true;
}
```

On failure, the function leaves `*out` unchanged. It requires that a nonempty slice identify at least `length` accessible elements; a misleading length is a caller contract violation, not something a simple C function can discover from the pointer.

A subslice also needs to avoid unsigned overflow. Its valid-range test should be `offset <= original.length && count <= original.length - offset`, not `offset + count <= original.length`. With an empty original whose data pointer is null, return a null pointer for an empty result rather than attempting arithmetic on a null pointer.

The complete program in `examples/slices.c` exercises these rules, including `SIZE_MAX` and one-past positions.

C.13 Failure-injected vector growth

The vector in `examples/vector_checked.c` owns its element allocation. An append operation uses a function pointer to a resizing function so a test can force reallocation to fail at a chosen point.

```
typedef void *(*resize_fn)(void *, size_t);

static void *deny_resize(void *old, size_t bytes)
{
    (void)old;
    (void)bytes;
    return nullptr;
}
```

After filling the vector's first allocation, save the pointer, length, capacity and last element. Invoke append with `deny_resize`. A correct strong-failure implementation returns false without changing any saved property.

The second test repeats the append with a real `realloc` implementation; it then verifies the new value, the count, and safe destruction. Crucially, it does not use the old saved pointer after successful reallocation, because that pointer may be invalid even when its bit pattern looks unchanged.

This injection technique can be extended to `malloc` and to fake I/O sources. It is more reliable than hoping the operating system happens to run out of memory during a small test.

C.14 Explain a failing test

A failed assertion is a starting point for diagnosis. Before modifying source, identify whether the program violated its own contract, the test made an invalid assumption, or the compiler did not implement the chosen language/library facility.

For example, a compiler warning that classifies a standardized C23 binary literal as a vendor extension does not automatically mean that the source is nonconforming. The response may be to use a more recent compiler or to suppress that one incorrect diagnostic while retaining other warnings. By contrast, using a macro removed in C23 is a source compatibility problem and should be corrected in the example itself.

A good verification record contains the failing command, exact diagnostic, cause, smallest corrective change, and the result of running tests on the corrected version. Avoid claiming an entire book has been verified when only a single executable has been compiled.

C.15 A bounded frame encoder and decoder

A frame format should be described in bytes before it is represented by C types. The Chapter 14 example uses a magic octet, version octet, two little-endian length octets, zero to 64 payload octets, and an XOR checksum.

The encoder's preconditions include a writable destination of the declared capacity and a readable payload of the declared length. It rejects payloads longer than 64 and destinations smaller than `5 + length`. Because the maximum length is already fixed at 64, the total frame size can be proven

representable on every target supported by the example. It constructs the complete frame in a local candidate buffer and then copies it to the caller's destination; this preserves the destination on validation failure.

The decoder checks fixed header size before reading fields, validates format identifiers, verifies that the declared payload matches the complete input length, and checks the checksum. Only after every check passes does it publish its local candidate result.

Testing: Start with a manually computed empty frame, then test a full 64-byte payload. Corrupt one payload byte without updating the checksum. Truncate a frame by one byte. Supply an extra byte. Confirm the output remains unchanged after rejection. The complete program is `examples/frame_codec.c`.

An XOR checksum offers limited accidental error detection, not sender authentication. A stronger checksum such as a CRC addresses some error patterns but is also not cryptographic authentication.

C.16 A partial-write callback

The byte sink in Chapter 17 returns the number of bytes accepted in a call. A complete-write wrapper must never advance by more than the remaining count, and must stop when the sink reports no progress.

Let `offset` be bytes successfully accepted and `count` the original request. Maintain the invariant `offset <= count`. On each loop iteration, ask the sink for `count - offset` bytes starting at `data + offset`. Reject a returned value of zero or a value greater than requested. Otherwise add the positive count to `offset`.

Every successful iteration strictly increases `offset`, proving termination for a bounded input. This proof says nothing about how long a callback blocks; an RTOS deadline requires additional assumptions about the device operation.

The program `examples/callback_sink.c` uses a fake writer accepting at most two bytes per call. A five-byte message therefore requires three calls. A second mode always returns zero so the wrapper can demonstrate failure without looping indefinitely.

An asynchronous writer needs a richer design: a 'would block' result may be normal, and the wrapper may have to retain state across calls. Never use the synchronous implementation as an automatic substitute for an interrupt-safe or nonblocking API.

C.17 Separate compilation as a testable contract

The module example consists of `examples/modules/checksum.h`, `checksum.c`, and `main.c`. The header declares a function taking a byte pointer, count, and output pointer. The implementation includes that header so an incompatible definition is diagnosed at compile time. The client also includes the same header and verifies that an invalid argument leaves the result unchanged.

The checksum is intentionally XOR, which is simple enough that the program's structure is more instructive than the calculation. The example should not be mistaken for a cryptographic hash.

A useful extra test is to compile `main.c` without linking `checksum.c`; the expected undefined-reference error illustrates the linker's job. Then compile `checksum.c` by itself with `-c`, compile `main.c` similarly, and link the object files.

A public header should include its own dependencies. It must not require clients to include `<stddef.h>` or `<stdbool.h>` before including it.

C.18 What an example demonstrates—and what it does not

A passing byte-sink test demonstrates that the wrapper handles its tested partial-write and no-progress cases. It does not prove that a real UART driver handles DMA coherency or interrupt priority correctly. A passing frame-decoder test demonstrates behavior on its supplied packets, not exhaustive validation of every possible byte sequence. A passing module example demonstrates correct linkage under the chosen build command, not platform independence of every possible implementation.

For each exercise solution, separate observed test results from the assumptions and reasoning needed to use it safely in another environment. This distinction is essential in production embedded systems.

C.19 Review checklist for the three complete examples

When reviewing `frame_codec.c`, verify that no length or checksum byte is read before its bounds check. Compare the decoded frame length with the precise number of supplied bytes, not merely with buffer capacity. Confirm that the encoder's temporary frame fits the fixed maximum and that a failed encode never writes into the caller's destination. The XOR checksum is not a security guarantee.

For `callback_sink.c`, prove that `offset` starts at zero, never exceeds the requested count, and increases on every successful loop iteration. A zero-byte callback result must terminate with failure rather than create a busy loop. The callback's promise not to return more than requested is checked; its promise to read only the provided source span remains part of the callback contract.

For the checksum module, compile the header in isolation and verify that it declares every type it needs. Link a test client with the implementation file, then intentionally omit the implementation and observe the expected linker failure. These exercises distinguish a language error, interface error, runtime test failure, and missing link dependency.

The important lesson is that tests, code review, and explicitly written invariants reinforce each other. A short, compile-tested demonstration can still require careful reasoning before its logic is reused in embedded production code.

C.20 The exact boundary of a signed decimal parser

The most difficult valid input for a signed-decimal parser is often the negative endpoint, `LONG_MIN`. On a two's-complement implementation its magnitude is one greater than `LONG_MAX`, which means a positive signed accumulator cannot hold that magnitude.

The worked solution in `examples/parse_signed.c` instead accumulates an `unsigned long` magnitude. It selects the maximum permitted magnitude based on the sign: `LONG_MAX` for a nonnegative result and `(unsigned long)LONG_MAX + 1UL` for a negative result.

Before accepting each digit, it checks:

```
if (magnitude > (limit - digit) / 10UL)
    return false;
```

This inequality is derived from the requirement that `10 * magnitude + digit <= limit`. The subtraction and division occur before the multiplication. The parser rejects an input once it would exceed the permitted mathematical range, so the accumulator never overflows.

At the final step, an exact negative endpoint is assigned as `LONG_MIN` rather than converted from an out-of-range positive signed value. For all smaller magnitudes, conversion to `long` is representable and a unary minus can safely be applied.

The grammar is also part of the contract: optional sign, at least one digit, no whitespace, no trailing characters. `+` and `-` by themselves are invalid. The output remains unchanged on failure.

Boundary tests: Use `snprintf` to create exact strings for `LONG_MIN` and `LONG_MAX` on the test machine. Add ordinary signed values, leading zeros, sign-only inputs, trailing garbage and much-too-long decimal strings. Confirm the output sentinel is not modified for any rejected string.

C.21 A byte arena proof

Let the arena capacity be (C), its current used count be (u), and the requested count be (n). The invariant is (0 \le u \le C). A request for (n>0) bytes is accepted only when (n \le C-u).

The expression `C - u` is safe because the function first checks `u <= C`. It follows that (u+n \le C), so the updated offset is valid and the returned interval fits in the backing array.

```
if (arena->used > ARENA_CAPACITY ||
    count > ARENA_CAPACITY - arena->used)
    return false;
```

The routine computes the returned pointer only after the checks. A failed request leaves both the `used` field and the out-parameter unchanged.

A reset changes `used` back to zero. It does not destroy the underlying byte array, but the allocator's logical contract makes all old borrowed regions invalid for subsequent use. Those bytes may be overwritten by new allocations. This distinction should be explicit in the interface documentation.

The example is a **byte** arena. It does not promise the alignment or object-access rules required for arbitrary typed allocations.

C.22 Rotations without undefined shifts

For a 32-bit unsigned value, a left rotation by a count (n) in the range 1–31 consists of `(x << n) | (x >> (32 - n))`. The count zero must be handled separately, since shifting a 32-bit operand by 32 is invalid.

Normalize any supplied count with `n = count % 32u`. If zero, return the original value. Otherwise both shift counts are in the interval 1–31, satisfying the language's shift-count requirement.

The companion `examples/rotate_bits.c` tests the following independent constants:

Input	Rotation	Expected value
`0x12345678`	left 0	`0x12345678`
`0x12345678`	left 4	`0x23456781`
`0x12345678`	right 4	`0x81234567`
`0x12345678`	left 32	`0x12345678`

It also checks that rotating left and then right by equal counts restores the original value for 64 count choices. That round-trip property is valuable but should not replace the independent constants, because two incorrect inverse functions could agree with each other.

C.23 Equality and padding

Suppose two `struct event` values contain equal member values. Can `memcmp(&a, &b, sizeof a) == 0` be relied upon to express semantic equality? No. Structure padding bytes are not generally meaningful members, and their stored representations need not be identical. Floating-point members and pointers can add representation-specific concerns as well.

Define equality by comparing the meaningful fields under an explicit application policy. For floating fields, decide whether NaNs and signed zero need special treatment. For a string pointer, decide whether equality means the same pointer identity or identical string contents.

A wire format requires an additional design step: explicitly encode field widths, order and validity values. A raw structure copy is not a general serialization scheme.

C.24 The ownership trace

Consider an owning vector `a`, a shallow copy `b = a`, and a borrowed pointer `p = a.data`. Initially all three pointer values refer to one allocation.

If `a` grows successfully through `realloc`, the old allocation's lifetime ends, even if the returned pointer value appears unchanged. The borrowed pointer `p` is invalid for accessing the old allocation. The pointer inside `b` is likewise invalid. Destroying `b` would attempt to release storage it no longer owns correctly.

If the vector instead fails to grow, the old allocation remains valid and the strong failure guarantee leaves `a` unchanged. But the copy `b` is still a second apparent owner, so destroying both will remain unsafe. Avoiding shallow ownership copies is a separate rule from handling failed allocation.

A correct design may use an opaque owning type, an explicit deep-copy function or a transfer operation that leaves the source in an empty state. C allows these conventions but does not enforce them automatically.

C.25 A reviewer’s test matrix

Operation	Successful boundary	Rejection boundary	Preserve on failure?
Signed parser	`LONG_MIN`, `LONG_MAX`	too many digits	Yes, output unchanged
Byte arena	exactly remaining bytes	one byte too many	Yes, offset and pointer uncha
32-bit rotate	counts 0, 31, 32	all counts permitted via modulo	Not applicable
Counted subslice	ending exactly at length	offset greater than length	Yes, output view unchanged
Vector append	growth succeeds	injected resize failure	Yes, all prior elements
Frame decoder	complete valid frame	truncated or corrupted frame	Yes, output record unchange

A passing test for every listed row provides useful evidence. It does not establish arbitrary pointer validity, interrupt safety, or correctness for all possible inputs. Those conclusions require the API contracts and independent reasoning developed in the chapters.

C.26 An overlong line does not become two records

Consider a maximum of 16 ordinary bytes per line. A line containing 17 bytes is invalid. If a reader returns immediately after observing its 17th byte, remaining bytes from that same logical line may still be in the stream. The next call could incorrectly treat those bytes as the beginning of another record.

A bounded line reader should define a recovery policy. The example in `examples/line_reader.c` continues reading and discarding bytes until newline or clean EOF when it encounters an overlong or malformed line. That makes the next call begin at a genuine record boundary.

It builds a candidate line in a fixed local array and commits the text and reported length only on `LINE_OK`. Any other status leaves both caller outputs unchanged.

The test deliberately arranges the sequence:

```
"short\\n"
"12345678901234567\\n"  (17 bytes: reject)
"recovered\\n"
"1234567890123456\\n"   (16 bytes: accept)
"final"                   (unterminated EOF: accept once)
```

The test must confirm that the failed record did not modify the previously returned line and that `recovered` is read next. The exact-capacity record should be accepted, with a terminating zero written in the extra slot. After `final`, the following call returns clean end of input.

There is a nonobvious tradeoff. Memory consumption is bounded by the candidate buffer, but a read call that drains an arbitrarily long invalid record has no finite input-independent bound on execution time. An event-loop or real-time design should use incremental input with a work budget rather than a single blocking drain operation.

A NUL byte is rejected here because the interface returns a C string. A counted binary record API could accept that byte under a different contract.

C.27 Relaxed atomic increments and join

The `thread_counter.c` example creates two workers, each performing 20,000 atomic increments. The mathematical result is 40,000, which fits the minimum standard range of unsigned int.

Why is `memory_order_relaxed` enough? The counter is the only shared data whose value matters. Every increment is atomic, and the program does not use the counter as a signal that unrelated non-atomic payload bytes are ready.

Thread joins ensure that the program reaches its final count check after workers have completed. The context must remain alive until all started threads are joined. If creating the second worker fails, the first still needs to be joined before returning.

What the test does not establish: It does not show that relaxed memory ordering suffices to publish ordinary data, that an atomic operation is interrupt-safe on every MCU, or that the program can meet a real-time scheduling deadline. Those are different claims with different evidence requirements.

C.28 A one-shot publication proof

A producer writes an ordinary payload and release-stores a ready flag. A consumer acquire-loads the flag and reads the payload only after observing true. If the acquire reads from the release operation, the producer's preceding writes happen-before the consumer's subsequent reads.

The proof assumes one publication and no subsequent writes to the payload while the consumer reads it. Resetting the ready flag and overwriting the payload without a separate handoff can create a data race. Atomicity of the flag is not a general lock on every object around it.

A repeating producer-consumer queue requires a protocol for both data publication and slot reclamation. This is why the SPSC ring example has release/acquire relationships for both head and tail.

C.29 C23 feature probes as executable specifications

A compiler flag selects a dialect, but it is not an exhaustive compatibility test. A feature probe should compile a minimal translation unit that actually uses the feature.

For `constexpr`, a scalar constant object initialized by a permitted constant expression can be verified through `static_assert`. For `_BitInt`, use the exact requested width and check the value range intended by the algorithm. For `#embed`, include a known small binary file and verify that the array contains the expected bytes and length.

For `<stdckdint.h>`, the test must also include and link the header's functionality:

```
#include <limits.h>
#include <stdckdint.h>

int main(void)
```

```
{
    int result;
    if (ckd_add(&result, 10, 20) || result != 30)
        return 1;
    return !ckd_add(&result, INT_MAX, 1);
}
```

On overflow, the test checks the boolean indication rather than making assumptions about the prior contents of the result. A wrapper API promising strong failure safety should use a temporary result.

A failed feature probe may indicate unsupported syntax, a missing library header, an unresolved library function, or a runtime mismatch. Report which step failed. Do not quietly downgrade the language mode while claiming C23 verification.

C.30 What a complete review report looks like

For one example, write a review under six headings:

Contract: accepted input, output and failure behavior.

Invariants: statements preserved through every iteration, including bounds and ownership.

Resource use: maximum storage, allocation behavior, work per call, and whether it can block.

Portability: exact language assumptions, required headers, compiler version and target features.

Verification: independent expected values, error injection, executed compiler flags and CI run.

Limitations: untested interleavings, unsupported platforms, and properties requiring hardware or normative review.

This report can be shorter than the program. Its value is making the conditions of correct use explicit.

C.31 An incremental UTF-8 decoder

The central challenge in decoding UTF-8 from a stream is that a transport boundary may divide one encoded scalar value. A decoder that reads only complete strings can treat missing continuation bytes as failure, but a stateful decoder needs to return *incomplete* without discarding valid prefix information.

The `utf8_stream.c` example maintains three fields: a partial unsigned scalar accumulator, the minimum valid scalar value for the chosen sequence length, and the number of continuation octets still needed.

Its transition table is:

Current state	Incoming byte	New state or result
Ready	00–7F	Complete ASCII scalar
Ready	C2–DF	Store five payload bits; expect one continuation
Ready	E0–EF	Store four payload bits; expect two
Ready	F0–F4	Store three payload bits; expect three
Ready	Other	Invalid
Waiting	80–BF	Append six bits; decrement remaining
Waiting	Other	Invalid; reset
Final continuation	Result in permitted scalar range	Complete; publish value and reset
Final continuation	Overlong, surrogate or too large	Invalid; reset

The implementation does not commit `*out` while waiting or on malformed input. This is the same strong failure-postcondition pattern used throughout the book: accumulate into local or private state and publish an answer only when the complete candidate has passed validation.

When the producer reports that no more input will ever arrive, a state with `remaining > 0` means truncated UTF-8 input. It is not necessarily an error when the source is merely waiting for its next packet or DMA buffer.

Independent test vectors: E2 82 AC must produce U+20AC; F0 9F 99 82 must produce U+1F642; E0 80 80 must be rejected as overlong; ED A0 80 must be rejected as surrogate; F4 90 80 80 must be rejected as greater than U+10FFFF.

A useful property test splits the *same valid octet sequence* into different input chunks and compares the completed scalar sequence. The result should depend on the bytes, not on how they arrived.

What the example does not do

It does not normalize Unicode, identify grapheme clusters, apply case folding, or choose replacement-character behavior for malformed input. It deliberately rejects malformed sequences and resets state, which is a reasonable strict-parser policy but not a universal text-rendering policy.

It is also not a length-counted entire-text decoder: it emits one scalar at a time. An application must decide whether to append, count, compare, or otherwise process each completed scalar.

C.32 A bounded TLV decoder

For the TLV decoder, let `size` be the count of input octets and `cursor` the count already consumed. The critical invariant is `cursor <= size`.

Before reading the tag and length, require at least two unconsumed bytes:

```
if (size - cursor < 2)
    return false;
```

The subtraction is defined under the invariant and cannot underflow. After reading two header octets, advance cursor by two; that remains within the input because the check already proved two octets were available.

Then validate the announced payload length against both limits:

```
if (length > TLV_MAX_VALUE || length > size - cursor)
    return false;
```

The first comparison is an application storage limit; the second is an input-completeness check. They are not interchangeable. A length can be perfectly valid in the protocol but unavailable in the truncated buffer, or fully present in a huge input but larger than the application permits.

Once both checks pass, `memcpy` receives a valid, bounded source span and a fixed-capacity destination field. The candidate's field count increases only after the complete field has been validated.

Why commit once?

The decoder builds a local `struct tlv_message candidate = {};` and copies it into `*out` only after it reaches the exact end of the input. This prevents partial configuration updates. For example, a message containing one valid field followed by a truncated second field must not overwrite a previously valid configuration with the first field alone.

The sample tests a valid two-field record, incomplete header, incomplete payload, declared value length above the limit, and excessive field count. Each rejection must leave the previous output values intact.

Resource reasoning

There are at most four fields and each contains at most eight value octets. That bounds the work of a successful decode and bounds the candidate's storage. Input bytes beyond the fifth field header cause rejection without copying an unbounded amount of data.

The TLV message may represent an empty field sequence. The sample contract accepts a null source pointer when and only when the supplied input length is zero. This avoids constructing any pointer through arithmetic on null.

Semantics beyond TLV shape

A structurally valid TLV stream may still be an invalid command: it might contain repeated mandatory tags, unsupported tag codes, or semantically inconsistent fields. A real protocol should validate those requirements separately, before committing application state.

A TLV parser does not authenticate the sender. Nor does it guarantee freedom from replay or malicious changes. Those are independent protocol and security concerns.

C.33 A comparison of decoding contracts

Contract	Line reader	UTF-8 stream	TLV decoder
Input unit	Hosted stream byte	One octet	Counted octet buffer
Output unit	Terminated line	One Unicode scalar	Structured message
Incomplete input	Waits or EOF handling	`UTF8_MORE`	Rejected as incomplete
Capacity limit	16 bytes per line	Constant decoder state	Four fields, eight octets each
Failure output	Unchanged	Unchanged scalar	Unchanged record
Recovery	Drain to delimiter	Reset on malformed sequence	Reject entire record
Environment	Hosted streams	Any octet source	Counted buffer
Security guarantee	None	Encoding validity only	Bounds and structural validity

The table shows why a general-purpose parser interface is not one-size-fits-all. A blocking line reader owns the task of obtaining bytes. A one-byte decoder depends on the caller to provide the next byte. A whole-message TLV decoder requires that a framing component has already supplied the complete candidate record.

Each design can be correct when its preconditions and error semantics match the surrounding application.

C.34 Turning an exercise into a reviewable result

For both examples, write a small verification report with these headings:

Input grammar. State exactly which byte sequences are legal.

Invariants. Identify the bounds and state relationships maintained after every step.

Failure behavior. State which outputs are unchanged, which parser state is reset, and which conditions require additional bytes.

Limits. Give the maximum allocated storage and maximum local parsing work, distinguishing those from transport waiting time.

Evidence. Provide independent test vectors, the compiler and language mode, and the resulting test run.

Unverified properties. Identify deployment-specific assumptions, unsupported Unicode operations, authentication, concurrency, and any untested code paths.

A reliable solution is one whose contracts can be reviewed without relying on lucky runtime behavior.

C.35 Fixed pool allocation with generation-tagged handles

The fixed pool in examples/slot_pool.c manages four integer-valued records without a heap. Each slot has an occupancy flag, a retirement flag, a generation counter, and a value. A handle contains a slot index and the generation observed when the slot was allocated.

The central invariant is that a handle is valid only if its index is within the array, the slot is occupied, and the generations match. Checking just the index is insufficient: a freed slot may have been assigned to a different client.

Acquire and release

An acquire operation scans at most four slots and chooses the first slot that is neither occupied nor retired. It writes the value, marks the slot occupied, and publishes the newly formed handle only after success. If no slot is available, the caller's output handle remains unchanged.

On release, the function first verifies the index, occupancy, and generation. It marks the slot unoccupied and increments its generation. An old handle now fails validation even if its original physical slot is reused for another value. A second release of the same handle is also rejected.

The implementation retires any slot whose generation has reached UINT_MAX. Ordinary unsigned arithmetic would wrap to zero, which could eventually recreate a previously issued handle identity. Retirement prevents that reuse at the cost of reducing the available pool capacity after extremely many cycles.

A worked trace

Event	Slot 1 generation	Occupancy	Old handle valid?
Initially	0	Free	No handle
Acquire value 20	0	Occupied	Yes, handle (1,0)
Release	1	Free	No
Acquire value 99	1	Occupied	No; new handle is (1,1)
Release again	2	Free	Neither old handle is valid

The implementation tests a full pool, failed acquire, successful release, stale read, double release, reuse at a new generation, and retirement at the maximum generation.

Limits of generation handles

A generation number is not authentication. A client can manufacture an index and number that happen to match. The teaching handle also lacks an owning-pool identifier; a handle from another pool may coincide with a valid pair. If clients are untrusted or handles cross subsystem boundaries, a stronger identity and permission scheme is necessary.

The pool is not thread-safe. Concurrent acquire operations can select the same available slot unless appropriate synchronization or task confinement prevents interleaving. The lack of dynamic allocation does not remove the need for ownership discipline.

C.36 Worked byte-at-a-time frame parser

The parser in examples/frame_stream.c accepts frames with this layout:

```
A5 | length | zero to eight payload octets | checksum
```

The checksum is XOR of length and every payload octet. It is intended only as a teaching error-detection mechanism, not authentication.

The state machine has four phases: wait for magic, read length, collect payload, and check checksum. Every call consumes one input octet and performs a bounded amount of work.

A complete message is published only when the checksum succeeds. While a frame is incomplete, the parser updates its private candidate and returns FEED_MORE without changing the caller's previously completed message. Malformed input resets the parser and returns FEED_REJECTED, again without publishing partial output.

Independently calculated vector

A length-three payload containing hexadecimal octets 0A, 14, and 1E has checksum 03, because 03 XOR 0A XOR 14 XOR 1E equals 03. The full frame is therefore:

```
A5 03 0A 14 1E 03
```

The first five feed calls should return FEED_MORE; the sixth returns FEED_COMPLETE and publishes three payload octets. An empty frame is A5 00 00 and should also complete.

Bounds proof

While collecting payload, the declared length is in the interval one through eight. Before storing the next byte, the received count is smaller than the declared length. Therefore the destination index is less than eight and lies within the fixed array.

The parser accepts no length greater than eight. No loop runs over an untrusted length inside one feed operation. This supports a bounded per-byte work argument, though the actual MCU execution time still depends on the compiler, memory system, scheduling, and interrupts.

Truncation and resynchronization

A prefix such as A5 02 07 is incomplete. The parser can retain its private state between input fragments, or the caller can reset it after a definite EOF or timeout.

A rejected checksum is consumed even if that byte coincides with the magic marker A5. The simple recovery policy can therefore miss the start of a possible following frame. The external protocol must explicitly define how resynchronization works; scanning blindly for the next magic octet is not always correct when that value may appear in payload data.

C.37 The distinction between state and published results

The two examples illustrate different failure guarantees. A failed pool acquire preserves both pool availability and the caller's output handle. A stale release preserves all pool state. A frame feed returning FEED_MORE changes its internal parser state intentionally while preserving the public output. A rejected frame resets its internal state while still preserving the public output.

Consequently the phrase ”no change on failure” is incomplete unless it specifies the object and state whose contents are unchanged.

For a large system, write down this distinction for every API: which state is private, which is externally visible, and exactly when a completed result is published.

C.38 Concurrency and shutdown review

These examples are deterministic when one owner controls each mutable object. A real embedded application must also decide who feeds the parser, who owns the pool, and what happens when a producer outruns a consumer.

A common architecture places incoming UART bytes into a bounded ISR-safe queue and processes them in a normal RTOS task. The task can own the frame parser, while clients request pool operations through a controlled interface. This avoids mixing task and interrupt contexts indiscriminately.

The queue itself still needs a suitable overflow policy, its atomic operations must be supported on the target, and shutdown must ensure no execution context continues using released state.

Extended exercises

C.1 Add an owning-pool identity to handles and document how identity changes when a pool is reinitialized. **C.2** Add detailed parser statuses for invalid length, checksum failure, and incomplete input. **C.3** Test every valid payload length from zero to eight with independent expected frames. **C.4** Design the reset and recovery action after a simulated input-queue overflow. **C.5** Write a review explaining which properties are tested, which follow from invariants, and which require real hardware verification.

C.40 A comparator that cannot overflow

Suppose an application stores events with priority, name, and arrival order. It wishes to sort by ascending priority, then alphabetically by name, then by increasing arrival number.

The tested `examples/sort_records.c` defines a comparator for `qsort`. Its integer comparisons follow this pattern:

```
return (a > b) - (a < b);
```

For any two integer values, each relational expression yields either zero or one. Their difference therefore lies in the interval -1 through 1. The expression cannot overflow as an `int`.

Do not use `return a - b;` as a general signed-integer comparator. If `a` is `INT_MIN` and `b` is `INT_MAX`, the mathematical subtraction cannot fit in an `int`. Its undefined signed overflow makes that comparator invalid even though it may seem to work in ordinary cases.

The multi-key comparator returns as soon as it finds a differing key. For equal priorities it calls `strcmp` on the two stored names; only if those names compare equal does it compare the arrival numbers.

This establishes a lexicographic order. A test constructs an explicit expected arrival sequence, sorts the records, and compares each output position against that independent expectation. It also confirms that each adjacent pair is in ascending comparator order.

Stability versus a tie-breaking key

ISO C's `qsort` does not promise that records comparing equal retain their original positions. Including the unique original arrival number as a final key distinguishes those records. Their relative positions then follow the comparator's specified order.

This does **not** make the sorting algorithm stable. It defines a total ordering that already contains the desired arrival-order behavior.

If arrival values are not unique, the comparator can still report equality for distinct records. Whether that is acceptable depends on the application's output contract.

The comparator's pointer types

The `qsort` function passes two pointers to array elements as `const void *`. The callback converts them into `const struct record *` because the array being sorted contains records of that type.

A callback that interprets them as the wrong object type may violate the language's access rules or produce meaningless comparisons. C does not dynamically attach a type tag to a `void *`; the caller and callback must agree through the API contract.

The `strcmp` call also assumes both fixed-size name arrays contain terminating zero bytes. The teaching initializer satisfies that condition. A parser that populated names from untrusted packets would need to validate or supply the terminator before sorting.

Extension

Add a fourth key, `timestamp`, and write the complete sort specification in prose before implementing it. Test equal priorities, equal names, equal timestamps, and different arrival values. A reviewer should be able to derive the expected array order from the stated contract independently of the code.

C.41 A type-selected operation that evaluates once

The companion `examples/generic_select.c` uses `_Generic` to choose between ordinary typed functions:

```
static int maximum_int(int a, int b)
{
    return a > b ? a : b;
}

static long maximum_long(long a, long b)
{
    return a > b ? a : b;
}

#define max_same_type(a, b) \
    _Generic((a), int: maximum_int, \
                  long: maximum_long)((a), (b))
```

The controlling expression `(a)` determines which function is chosen by type. It is not executed a second time as part of that selection. The arguments to the chosen function are evaluated as for an ordinary function call.

A crucial test is:

```
int side_effect = 0;
int result = max_same_type(++side_effect, 7);
```

The expected `side_effect` after the call is one, and `result` is seven. This contrasts with the dangerous textual macro `#define MAX(a,b) ((a)>(b)?(a):(b))`, which can evaluate the chosen argument more than once.

The standard generic selection syntax was introduced before C23, but remains useful in modern C code. The standalone program is built under C23 alongside the other examples.

The unstated conversion trap

The macro chooses its function using the type of `a` alone. If a caller passes an `int` as `a` and a `long` as `b`, the selection chooses the `int` function. The second argument is then converted to `int` through the function prototype.

Such conversion may discard information when the `long` value lies outside the range of `int`. The macro is therefore documented for arguments of matching supported types.

This is an important lesson in API design: compile-time type selection is not automatic enforcement of every relationship between two arguments.

Additional type support

Extend the example with an unsigned function:

```
static unsigned maximum_unsigned(unsigned a, unsigned b)
{
    return a > b ? a : b;
}
```

Add an association for `unsigned` and test values near `UINT_MAX`. Then try using a `double` first argument and observe the diagnostic for an unsupported association.

A production application can offer a named function for every important type. A small generic macro is worthwhile when it makes those existing functions easier to call without hiding the required value conversions.

C.42 A complete three-file compilation exercise

The checksum module consists of:

- `examples/modules/checksum.h`, containing the public declaration;
- `examples/modules/checksum.c`, containing the definition;
- `examples/modules/main.c`, containing an executable test client.

The header must include `<stdbool.h>` for `bool` and `<stddef.h>` for `size_t`. It must not rely on the caller having included those declarations first.

A clean teaching build compiles both source files, then links their object files:

```
cc -std=c23 -Wall -Wextra -Wpedantic -c examples/modules/checksum.c
cc -std=c23 -Wall -Wextra -Wpedantic -c examples/modules/main.c
cc checksum.o main.o -o modules_demo
```

The commands are illustrative for a `cc` compiler and should be run in a clean scratch directory because their default object names may overlap other files.

Remove `checksum.o` from the final link. The caller still has a valid declaration, but the linker cannot find the definition. This demonstrates an unresolved external symbol, a different problem from an unknown function declaration.

Next deliberately change the function's output parameter type in the implementation without updating the header. Because the implementation includes the header, the compiler can diagnose the conflict.

Finally, modify the checksum algorithm to exclude the final byte. Both source files may compile and link successfully, but the executable test's expected checksum should now fail. That is an algorithmic regression detected at execution rather than a compile/link interface error.

C.43 Choosing evidence for the claim

A sorted-record example that passes deterministic tests provides evidence about its selected records and chosen ordering rules. It does not prove its comparator will never be given unterminated names by some future caller.

A generic macro that runs correctly for `int` and `long` does not establish safe behavior for every mixed-type call.

A successful two-file link does not establish that host and embedded targets use the same ABI or standard library.

When documenting any such example, separate the verified build and runtime observations from the assumptions under which the code remains correct. That distinction is particularly valuable when readers transfer a concise textbook example into a larger real-time or embedded application.

Appendix D. Glossary of Modern C

A concise language benefits from precise terminology. The words below are used consistently throughout the manuscript.

Abstract machine. The language-defined model against which program behavior is judged. A compiler may transform code while preserving required observable behavior.

Alignment. A requirement constraining where an object of a particular type may be stored. Misaligned accesses can have undefined behavior.

Array. A contiguous sequence of elements of one element type. In many expression contexts an array converts to a pointer to its first element; this does not make arrays and pointers identical types.

Automatic storage duration. Storage normally associated with a block invocation and released when its lifetime ends.

Binary interface (ABI). Platform-specific conventions governing calling conventions, object layout and linkage. A C source declaration does not fix one universal ABI.

Borrowed pointer. A pointer whose use depends on an object owned or managed elsewhere. The pointer does not itself extend the object's lifetime.

Buffer capacity. The number of elements or bytes that an allocated or provided buffer may hold. Distinguish it from the number currently containing meaningful data.

Callback. A function supplied to another component for invocation under a documented protocol. Callbacks may need a context pointer and lifetime rules.

Checked arithmetic. An operation that detects whether a mathematical result fits the destination type before relying on its value.

Compile-time assertion. A translation-time constraint written with `static_assert` or a compatible form.

Data race. Conflicting access to a shared memory location by concurrent threads without the synchronization required by C's memory model, where at least one access modifies it and the accesses are non-atomic. Such a race has undefined behavior.

Effective type. Rules determining which lvalue types may be used to access stored objects. Relevant to aliasing optimizations and representation reinterpretation.

Endianness. The byte ordering used for a multi-byte representation. File formats and wire protocols must define their own byte ordering rather than assume the host's.

Freestanding implementation. An implementation environment, often embedded, with different entry-point and library obligations from a hosted environment.

Half-open range. An interval that includes its start and excludes its end, conventionally `[begin, end)`. This representation makes length equal to end minus begin when both positions belong to a valid common domain.

Implementation-defined behavior. Behavior for which the implementation must choose and document a permitted alternative.

Invariant. A property preserved across the operations of an algorithm or data structure. For a vector, `length <= capacity` is fundamental.

Lifetime. The period during which an object exists. A pointer can retain an old bit pattern after its target object's lifetime has ended, but that does not make dereferencing it valid.

Lvalue. An expression designating an object or function under the rules of C. Not every lvalue is modifiable.

Object representation. The bytes of an object's stored representation. Copying them does not automatically define a portable external serialization format.

Opaque type. A type whose representation is hidden from the client, commonly through an incomplete structure declaration and operations accepting pointers.

Ownership. An application-level convention identifying which component is responsible for releasing a resource. C does not enforce a unique-owner discipline automatically.

Precondition. A condition the caller must satisfy before calling an operation, such as providing a valid pointer and sufficient length.

Postcondition. A property guaranteed when an operation finishes under its documented conditions, including what happens on failure.

Release/acquire synchronization. Atomic memory ordering used to publish data and establish visibility of earlier writes to a reader that observes the publication.

Restrict. A pointer qualification expressing an access promise enabling optimization when its conditions are respected; not a runtime alias check.

Sentinel. A distinguished value such as `EOF` indicating a condition outside the normal result domain.

Signed overflow. Evaluating a signed arithmetic result not representable in the result type. This remains undefined behavior in C23.

Static storage duration. Storage that persists for program execution, regardless of whether the identifier has file scope or block scope.

Stream. C's hosted abstraction for a sequence of input or output characters or bytes, often associated with a terminal or file.

Translation unit. The result of preprocessing one source file, including its headers, that is compiled as a unit.

Undefined behavior. A program action for which the standard imposes no requirements. An observed result in one build provides no portable contract.

Unspecified behavior. Behavior for which the standard allows more than one alternative without requiring the implementation to document its choice.

Volatile. A qualifier affecting accesses with observable side effects. It is not a general replacement for atomics or mutexes.

This glossary is deliberately selective. The ISO standard supplies precise normative definitions, sometimes with narrower conditions than a one-sentence teaching description can express.

Appendix E. Conformance and Migration Checklist

This appendix separates three questions commonly confused in modern C work: **What does C23 specify? What does the compiler implement? What does the target environment provide?**

E.1 Identify the build

Record the C compiler vendor, version, language-mode option, target triple, C library version, linker and startup/runtime configuration. A hosted GCC build on a development machine does

not establish the capabilities of an Arm cross-compiler paired with a reduced runtime library.

A language-version probe can inspect `__STDC_VERSION__`. Its C23 value is `202311L`, but this alone does not prove that every new library facility is available.

E.2 Core-language probes

Use tiny independent translation units to test the exact construct you plan to depend on. Features worth testing separately include:

- C23 boolean and null-pointer keywords: `bool`, `true`, `false`, `nullptr`.
- Empty initialization with `{}`.
- Binary integer literals with `0b` and quote digit separators.
- `auto` deduction, `typeof` and `typeof_unqual`.
- `constexpr` objects.
- `_BitInt(N)` within the required target width.
- Standard attribute syntax.
- The `#embed` directive for required binary resources.
- `#elifdef` and `#elifndef` conditions.

A successful syntax test is not an execution test. For numerical and representation-sensitive features, execute a small representative test on the target when possible.

E.3 Standard-library probes

C23's `<stdckdint.h>` and `<stdbit.h>` deserve independent include-and-link checks. A compiler may accept `-std=c23` while the selected C library does not provide them. The same caution applies to optional threading and atomic facilities in a freestanding environment.

For the checked arithmetic library, write a probe that calls `ckd_add` with representable and overflowing operands. For bit utilities, test a representative `stdc_count_ones` use. Avoid assuming availability because a header file with the same name happens to exist; compile and link the intended operation.

E.4 Representation assumptions

Record `CHAR_BIT`, integer widths, alignment requirements, and any assumptions about floating-point formats. Use static assertions only where the application requires the constraint. A text-only algorithm need not reject machines with unusual byte widths simply because a different example in the book requires eight-bit octets.

An external protocol should define field widths, byte order, and validation rules. Do not serialize native structs by writing `sizeof(struct)` bytes unless the external format explicitly specifies the exact local ABI as its contract.

E.5 A practical migration order

1. Build the existing code under its original supported standard and save test evidence.
2. Enable warnings and fix invalid declarations, mismatched prototypes, and ignored errors.
3. Audit signed arithmetic, narrowing conversions, pointer lifetime, and buffer-size calculations.
4. Remove reliance on old-style function declarations and other obsolescent assumptions.
5. Move to C23 mode without changing algorithms.
6. Introduce modern spelling and features selectively where the tested toolchains support them.
7. Run differential tests against the original behavior for valid inputs.
8. Validate on each real embedded target and document any implementation-defined dependencies.

A source-level conversion that compiles is not a complete migration. The program's runtime contract, memory usage and timing must still be verified.

E.6 Deployment acceptance checklist

Before accepting a release, confirm that the build uses the intended source commit; all shipped features passed their targeted probes; test inputs cover boundaries and failure cases; the compiler and linker warnings were reviewed; sanitizer results, where available, are recorded; and hardware-dependent code was exercised on real hardware.

The document build has its own acceptance criteria: all chapters and appendices appear, headings render in the correct hierarchy, the contents links work, code listings are legible, and the PDF's page count is measured directly. A PDF file existing on GitHub does not by itself prove that it corresponds to the latest manuscript commit.

E.7 Authoritative resources

- ISO/IEC 9899:2024, *Programming languages — C* (normative reference)
- https://en.cppreference.com/w/c/23 — overview and links to individual language/library topics
- https://gcc.gnu.org/projects/c-status.html — GNU C standard support
- https://clang.llvm.org/c_status.html — Clang language support

Library and compiler documentation change over time. Date your conformance matrix and retest after updating any part of a deployment toolchain.

Appendix F. Topic Finder

This topic finder is organized alphabetically for readers using the PDF or a printed draft. References name chapters rather than final page numbers, because pagination will change during the remaining editorial work. Subtopics are included when closely related concepts appear in different chapters.

For the precise normative meaning of any language rule, consult ISO/IEC 9899:2024. This index is a navigation aid, not an alternative language specification.

A

- **Abstract machine:** expression semantics, observable behavior, and optimization, Chapter 15.
- **Acquire ordering:** one-shot publication and SPSC queues, Chapter 16.
- **Alignment:** pointer access restrictions, Chapter 15; arena storage, Chapter 7; protocol decoding, Chapter 14.
- **Allocation:** `malloc`, `calloc`, `realloc`, and ownership, Chapter 7; resource failure tests, Chapter 13.
- **Arena allocator:** fixed-capacity bytes, reset semantics, and alignment limits, Chapter 7; worked solution, Appendix C.
- **Array decay:** array-to-pointer conversion and function parameters, Chapters 3–4.
- **Array indexing:** bounds, one-past pointers, counted slices, Chapter 4.
- **ASCII:** digit parsing and protocol tokens, Chapters 2, 3, and 18.
- **Atomics:** data races, ordering and lock-free limitations, Chapter 16.
- **Automatic storage duration:** object lifetimes, Chapters 3 and 15.

B

- **Binary literals:** C23 digit syntax, Chapter 9.

- **Binary protocols:** byte layouts and endianness, Chapters 10 and 14.
- **Bit-precise integers:** `_BitInt` and ranges, Chapter 9.
- **Bit rotation:** unsigned shifts, count bounds and tests, Chapter 14.
- **Bitwise operators:** masks, shifts, representation issues, Chapter 14.
- **Boolean values:** `bool` and C23 keyword spelling, Chapter 9.
- **Borrowed pointers:** views, lifetimes and ownership, Chapters 4 and 7.
- **Bounded input:** line reader, Chapter 6; TLV and frame decoders, Chapter 10; UTF-8 streaming, Chapter 18.
- **Buffer capacity:** slices, vectors, arenas, Chapters 4 and 7.

C

- **Callbacks:** function pointers and context, Chapter 17.
- **Character encodings:** UTF-8 and character count, Chapter 18.
- **Checked arithmetic:** addition, subtraction and multiplication, Chapters 2, 9 and 15; Appendix B.
- **CI:** compiler matrix, regression suites and reproducibility, Chapters 8 and 13.
- **Comma operator:** sequencing and expression rules, Chapter 15.
- **Comparator, three-way integer:** overflow-safe relational comparison and multi-key sorting, Chapter 17; Appendix C.
- **Comparators:** `qsort`, signed overflow and consistent ordering, Chapter 17.
- **Compound literals:** initialization and object lifetimes, Chapters 2 and 15.
- **`const` qualification:** pointer access and contracts, Chapter 4.
- **`constexpr`:** C23 constant objects and feature probes, Chapter 9.
- **Conversions:** promotions, narrowing, signed/unsigned arithmetic, Chapter 2.
- **Copying storage:** `memcpy` versus `memmove`, Chapter 4.
- **C23 migration:** language changes and toolchain probes, Chapter 9; Appendix E.
- **CRC and checksums:** format checks and limitations, Chapters 10 and 14.

D

- **Decoder, streaming frame:** four-state parser, bounded work and resynchronization, Chapter 10; worked solution, Appendix C.
- **Data races:** non-atomic shared objects and undefined behavior, Chapter 16.
- **Declarations:** pointers, functions, arrays and prototypes, Chapters 3–4; Appendix A.
- **Decoding:** counted input, TLV formats, UTF-8 state machines, Chapters 10 and 18.
- **Designated initialization:** structures, Chapters 5 and 9.
- **Device registers:** volatile access and hardware semantics, Chapter 14.
- **Diagnostics:** compiler warnings and sanitizers, Chapter 13.
- **Digit parsing:** unsigned and signed decimal boundaries, Chapters 2–3; Appendix C.
- **Dynamic memory:** ownership and allocation failures, Chapter 7.

E

- **Effective type:** aliasing and legal object access, Chapter 15.
- **Embedding resources:** `#embed` and toolchain support, Chapters 9 and 19.
- **Empty initialization:** C23 `{}`, Chapters 5 and 9.
- **Encoding:** UTF-8 and binary field formats, Chapters 14 and 18.
- **Endianness:** multi-byte values in external formats, Chapter 14.
- **EOF:** stream termination and incomplete records, Chapters 1 and 6.
- **Error statuses:** parser and I/O contracts, Chapters 3, 6 and 10.
- **Exercise solutions:** Appendix C.

F

- **Fixed slot pool:** generational handles, retirement and stale references, Chapters 5 and 7; worked solution, Appendix C.
- **Failure injection:** allocators and byte sinks, Chapters 7 and 13; Appendix C.
- **File I/O:** `FILE *`, buffering, short reads and writes, Chapter 6.
- **Flexible array members:** variable-length records, Chapter 5.
- **Floating point:** accuracy, range and NaNs, Chapter 2.
- **Frame decoder:** byte layouts, checksum and strong failure safety, Chapters 10 and 14.
- **Freestanding C:** library availability and MCU toolchains, Chapters 8, 9 and 16.
- **Function pointers:** callbacks and generic interfaces, Chapter 17.
- **Functions:** prototypes, out-parameters and linkage, Chapter 3.

G–H

- **`getchar`:** the `int` result and `EOF`, Chapter 1.
- **Generic macro dispatch:** typed maximum functions, side-effect evaluation and limitations, Chapter 19; Appendix C.
- **Generic selection:** `_Generic`, Chapter 19.
- **Headers:** declarations, include guards, standalone inclusion, Chapters 3 and 8.
- **Heap allocation:** `malloc`, `realloc`, capacity planning, Chapter 7.
- **Hosted implementations:** entry points, files and threads, Chapters 1, 6 and 16.

I–L

- **Implementation-defined behavior:** representation and portability, Chapters 2, 9, 15; Appendix A.
- **Invariants:** loop proofs, buffers, containers, Chapters 4, 7, 10 and 11.
- **Integer conversion:** representability before casts, Chapter 2.
- **Integer overflow:** signed undefined behavior, unsigned modular operations, Chapters 2 and 15.
- **Interrupts:** RTOS and atomic restrictions, Chapter 16.
- **I/O errors:** short transfers, buffering and cleanup, Chapter 6.
- **Lifetime:** automatic, static and allocated objects, Chapters 7 and 15.
- **Line input:** bounded line reader and resynchronization, Chapter 6.
- **Link-time errors:** missing definitions and multiple definitions, Chapter 8; Appendix C.
- **Linkage:** translation units, internal and external definitions, Chapters 3 and 8.
- **Lock-free algorithms:** progress properties, memory order and hardware constraints, Chapter 16.

M–N

- **Macros:** double evaluation, stringizing and token pasting, Chapter 19.
- **Memory layout:** arrays and pointers, Chapter 4; structures and padding, Chapter 5.
- **Memory ordering:** relaxed, release, acquire and sequential consistency, Chapter 16.
- **`memcpy`:** copying object representations, Chapters 4 and 15.
- **`memmove`:** overlapping buffers, Chapter 4.
- **Modules:** header plus implementation, Chapters 3 and 8.
- **Multi-threading:** C threads and resource cleanup, Chapter 16.
- **NaN:** floating equality and comparator behavior, Chapters 2, 5 and 17.
- **Narrowing conversions:** checked integer ranges, Chapter 2.
- **`nullptr`:** C23 null pointer constant, Chapter 9.
- **Numeric parsing:** grammar and overflow, Chapters 2, 3 and 18.

O–P

- **Object representation:** padding, serialization and aliasing, Chapters 5, 14 and 15.
- **One-past pointer:** valid construction versus invalid dereference, Chapter 4.
- **Opaque interfaces:** callback contexts and ownership, Chapters 3 and 17.
- **Ordering, evaluation:** function arguments and side effects, Chapters 3 and 15.
- **Obsolete handles:** logical lifetime versus physical storage, Chapters 5, 7 and 15.
- **Ownership:** borrowed and owning references, Chapters 4 and 7.
- **Padding:** structure size and equality, Chapter 5.
- **Partial writes:** callbacks and I/O, Chapters 6 and 17.
- **Pointer arithmetic:** valid arrays and bounds, Chapter 4.
- **Pointer provenance and lifetime:** object access restrictions, Chapter 15.
- **Portable serialization:** bytes versus native structures, Chapter 14.
- **Preconditions:** valid pointer ranges and object lifetime, Chapters 3 and 10.
- **Preprocessor:** inclusion, conditionals and macros, Chapter 19.

Q–R

- **`qsort`:** comparator contract and overflow-safe comparisons, Chapter 17.
- **`realloc`:** failure guarantees, invalidation and C23 zero-size behavior, Chapter 7.
- **Recursion:** base cases, limits and task stack use, Chapter 3.
- **Release ordering:** producer publication, Chapter 16.
- **Representation:** types, structure layout and wire formats, Chapters 5 and 14.
- **Resource cleanup:** partial acquisition and release, Chapters 3 and 7.
- **`restrict`:** aliasing promises and optimization, Appendix A.
- **Ring buffers:** fixed-capacity FIFO, Chapter 5; atomic SPSC, Chapter 16.
- **RTOS:** thread and interrupt portability, Chapters 8 and 16.

S

- **State-transition testing:** parser and pool validation, Chapter 13.
- **Sanitizers:** testing evidence and limitations, Chapter 13.
- **Sequencing:** unsequenced effects and evaluation ordering, Chapter 15.
- **Serialization:** field widths, endianness, padding, Chapters 5 and 14.
- **Signed integers:** C23 representation, conversion and overflow, Chapter 2.
- **Slices:** pointer/length views and range proofs, Chapter 4.
- **SPSC queue:** atomic producer-consumer exchange, Chapter 16.
- **`static`:** storage duration and internal linkage, Chapters 3 and 15.
- **`static_assert`:** translation-time assertions, Chapters 9 and 19.
- **Storage duration:** object lifetime, Chapter 15.
- **Streams:** text and binary modes, error checks, Chapter 6.
- **String literals:** static storage and mutability, Chapter 4.
- **`strtol`:** parser grammar, end pointers and range errors, Chapter 6.
- **Structures:** fields, padding and ownership, Chapter 5.

T–U

- **TLV format:** bounded type-length-value decoding, Chapter 10.
- **Thread lifecycle:** creation, partial failure and joining, Chapter 16.
- **Time bounds:** bounded memory versus bounded execution, Chapters 6, 10 and 16.
- **Toolchains:** host versus cross-compiler, Chapter 8; feature support, Chapter 9.
- **Translation units:** source/header compilation and linking, Chapter 8.
- **Type inference:** C23 `auto` and `typeof`, Chapter 9.
- **Undefined behavior:** overflow, invalid pointers, data races, Chapter 15.
- **Unicode:** scalar values, UTF-8 and normalization boundaries, Chapter 18.
- **Unspecified behavior:** evaluation order and portability, Chapter 15.

- **Unsigned arithmetic:** modular rules and checked sizes, Chapter 2.
- **UTF-8:** complete and incremental decoders, Chapter 18.

V–Z

- **`volatile`:** device access versus thread synchronization, Chapters 14 and 16.
- **Vector:** dynamic capacity, ownership, and failure guarantees, Chapter 7.
- **Warnings:** pedantic diagnostics and compiler differences, Chapters 9 and 13.
- **Wait-free algorithms:** progress guarantees, Chapter 16.
- **Word-index tokenizer:** ASCII assumptions, exact 31-byte limit and EOF behavior, Chapter 11.
- **Word index:** sorting, insertion invariants, errors and complexity, Chapter 11.
- **Zero initialization:** aggregate initialization versus raw bytes, Chapters 5, 7 and 9.
- **Zero-length requests:** allocation and slice policies, Chapters 4 and 7.

The final typeset edition should replace or supplement these chapter-level entries with verified page references. Automated PDF pagination is not stable enough during manuscript expansion to maintain a conventional fixed-page index by hand.